

POWER AMPLIFIER KALKULATOR SPREADSHEET USER GUIDE

FOR PAK-1v0-NM

(Non Macro version). User Guide compiled 1 November 2012

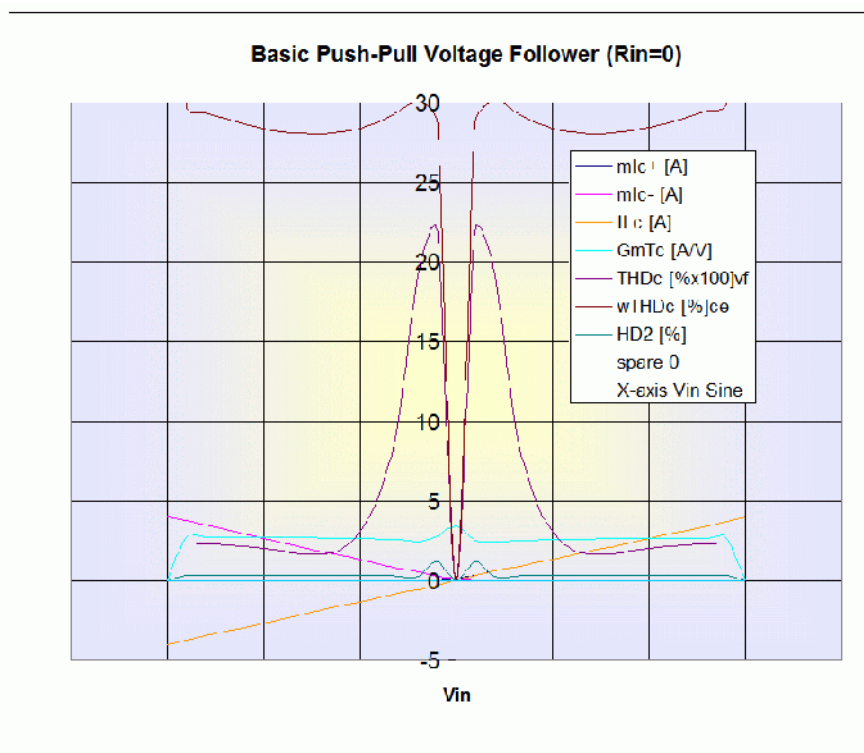
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An open source spreadsheet for evaluating transistor power amplifier circuits using large-signal explicit equations involving the Lambert W-function. Explicit solutions for currents, voltages and gains can be now written for many elementary circuits using the inverse of the Lambert W-function.

Example for a simple Push-Pull output stage using large-signal equations (Sheet 14)



Shows calculated currents, gains and distortions at optimum quiescent current of 90mA. Unlike SPICE Newton-Raphson is not used. PAK plots THD at each input voltage step (SPICE does not plot multiple THD's like this). Covered in [Section 15](#).

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You can download this User Guide free from [PakLaunchSite.jimdo.com/]. The editable .odt version is available on request from the author(s).

Disclaimer: While every effort has been made to validate the solutions in the PAK-1v0-NM spreadsheet and User Guide, neither the distributor nor the Author(s) and later contributors assume any liability for any injury and/or damage to persons or property resulting from any errors contained in this material or any of its derivatives.

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Feedback: Please direct any comments or suggestions about this document using the Contact form at PakLaunchSite.jimdo.com/ or [-add-another-].

Acknowledgements

I wish to thank the Editor of Linear Audio magazine for initially offering to post the PAK User Guide and spreadsheet. I am indebted to the pioneering work by Kevin Aylward [ref 8,9] and Thomas C Banwell [ref 1] for examples using the Lambert W-function to solve a few BJT circuits.

Modifications and updates

Version	Initials	Date	Brief description of change or link*
PAK-0v0-NM	IH	14 Nov 2011	Evaluation version
PAK-0v1-NM	IH	15 Aug 2012	Evaluation version 2
PAK-1v0-NM	IH	1 November 2012	First release at PakLaunchSite.jimdo.com/
PAK-1v_-NM			
PAK-1v_-NM			
PAK-_v_-NM			

*Details of changes should be placed either in Chapter 5 (new) or add a web link in 'Brief description' cell.

** [spare]

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Preface

The aim of this public project is to compile a library of royalty free transistor circuit solution including elementary current mirrors, current sources, differential amplifiers (LTP), push-pull output stages. Previously, the only way to find the current, gain and distortion in a transistor circuit was to solve the equations numerically using a simulator such as SPICE. But now using Thomas Banwell's examples, bipolar amplifiers can be solved analytically and large-signal currents, voltages and gains can be calculated directly rather than simulated. Distortion can also be estimated. Designers can see how a circuit works and behaves in symbolic form at the large signal level and can optimise circuits better for a lower distortion or a lower temperature drift.

It was originally hoped that the elementary circuits in the PAK spreadsheet can be used as building blocks for complete power amplifier circuits solutions. It was found that transistor circuits that use global negative feedback remains unsolvable in an exact form. Approximate solutions have been found in most cases that are sufficiently accurate for many situations. Recursion can then be invoked to reduce the residual error to any required level without fear of non-convergence. Finding good approximations is a time consuming task and this project should help to spread the workload.

This spreadsheet and User Guide could be developed into a web 'Wiki' if there is sufficient interest and support. If you can help see [Feedback](#).

This spreadsheet and User Guide can be integrated into teaching material without cost but copyright laws still apply under the terms of the Creative Commons share alike attribution license version 3.0 [<http://creativecommons.org/licenses/by/3.0/>] although a copyright waiver is granted to small portions ('small' being defined as no more than a total of one A4 page of normal print material). PAK equation solutions give valuable insights into SPICE models and their limitations for students and designers.

Why the '-NM' (Non Macro) version? It completely removes any danger of malicious macros being propagated through the spreadsheet. The '-NM' version requires more columns to calculate the equations than a macro version ('-ME' or Macro Enabled). Spreadsheet User Defined Function's (UDF's) allow in-line BASIC style of coding which makes a '-ME' spreadsheet compact. UDF's also allow recursion for more accurate result with approximated solutions. Recursion in a '-ME' spreadsheet is the next level of development for PAK. Anyone is invited to work on this aspect. It would be good if PAK-NM is developed as well as a virus-free environment for educational use.

A futuristic role for PAK is to eliminate the remaining non-convergence problems in SPICE related to exponential terms in diode and transistor models. The developers of SPICE did not have the Lambert W function solution available so their diode and transistor model equations involve exponential functions without the physical limitations on slope at high and low currents due to realistic series and shunt resistances. Now the diode and BJT models can be reformulated to eliminate most of the remaining non-convergence problems in SPICE.

To be clear, the PAK spreadsheet is just a framework to demonstrate the circuit equations by providing plots like a SPICE simulator. Worksheets can be distributed to others to 'Run' them similar to SPICE simulators. This User Guide contains Tutorial help to get started. Developers of new circuit equations can learning from the examples provided in the User Guide. Developers are encouraged to use the original PAK format to ensure their worksheets are compatible with other PAK users. See the User Guide for the format rules and branch versions.

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Chapter 1 Working with this spreadsheet

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1. QUICK HELP FOR PAK-1v0-NM SPREADSHEET

Table 1. A 'How To' summary for the PAK spreadsheet

View a circuit	Go to a circuit - Click bottom-left tab eg Sheet4
Run a circuit	Go to a circuit, Turn a circuit ON, type ON (u/c or l/c) in the cell, Enter.
View a plot	View the graph. Plots can be moved to anywhere convenient to see changes. Each plot should have a Title (eg in the X-axis caption area).
Move a plot	Click to select the plot. Drag with the mouse. When changing a parameter mode the plot to the same area to see the changes in the plot. You can also relocate the Plot Pointer array to under the plot (to turn ON/OFF any of the curves).
Move the PlotPointer	Select the Plot Pointer array and title. Cut (Ctrl+X) and paste in the first cell of the new position (usually under the plot).
Turn a curve OFF or turn another ON	The Plot Pointer array is the same order as the legend for each curve apart from the first cell which is the X-axis pointer. If you want to turn off the 3rd curve in the legend list then change to 4th pointer cell and enter the column with 'Spare'. (A pointer entry can also be deleted but Calc then displays 'Error 505').
Change curve to different series	Read the column letter for the new data series to display, then the first cell ref of that column, enter this cell reference in the Plot Pointer at the cell for the new data series (eg an empty pointer cell). More curves can be added -- see the User Guide.
Zoom and/or offset a curve	Right click on the X-axis or Y-axis main bar. Choose Format Axis, Scale, then enter Max and Min values to zoom and/or offset. Use Undo to return to the original view. Create a duplicate using Copy/Paste if you want to keep a custom zoomed view.
Copy Plot	Click to select plot. Copy & Paste to another sheet or document. If a document can't accept the format use 'PrtScr' key then Paste into a picture program eg MS Paint, select graph, then Paste (or Save As then import a Picture from File).
Modify a component value	To change all values in a column you only change the top number (under yellow row). A copy equation makes all values below the uppermost one the same value.
Modify a Step range	Stepping a component or voltage generates a plot. The yellow cell is set as the starting value, the next cell copy equation uses <Cell>+1/<N> to increment by a small amount. Copy down from the first cell to update each cell in the column. View the graph.
Step a different component or voltage	First turn off the presently stepped component. Remove 'Step' after the component name. Add an appropriate increment to the second cell then drag-copy down to update the column. Add "Step" after the component name for the plot Legend. Change the first X-axis pointer to the new stepped column. View plot and adjust the increment to suit the range.

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2. A tutorial for getting started

for Spreadsheet PAK-1v0-NM (Non Macro version)

1. Run the spreadsheet. A spreadsheet can be saved to open at the last worksheet area. This is useful if you want someone to look at a particular view such as a plot. To work through this tutorial Run the program and then choose Sheet3 (see **Figure 2.1**)

The Contents list appears the same on all sheets except the highlighting changes

To select a different sheet clicking on the Sheet tab at the bottom

	A	B	C	D	E	F	G	H
1	PAK-0v1-MN	PAK Power Amplifier Kalkulator Spreadsheet						
2	19-Jul-12							
3								
4	CONTENTS							
5	Sheet	Circuit Title	Variant(s)	Notes	Status			
6	1	Contents and Help	-	-	-			
7	2	Model library	-	-	-			
8	3	Diode & R circuits	Class-AB using D's					√
9	4	Single BJT circuits	CE+Rf	with Early effect & HD2,3				early
10	5	Complementary pair	CE+Rf					√
11	6	CE + CC pair (VAS+VF)	CE+Rf					UC
12	7	The Darlington pair						UC
13	8	The Differential pair		with Early effect & THD				early
14	9	Current Mirrors						UC
15	10	Current Sources						UC
16	11	Vbe multipliers						UC
17	12	Basic push-pull output						√
18	13	CFP push-pull output						√
19	14	Darlington PP output						early
20	15	Distortion estimates						early
21	16	HF frequency effects	-					-
22	17	[spare]	-					-

Sheet1 / Sheet2 / **Sheet3** / Sheet4 / Sheet5 / Sheet6 / Sheet7 / Sheet8 / Sheet9 / Sheet10

Figure 2.1. Contents is the same on all sheets except for the highlighting

2. CTRL+Home key returns you to the **Contents**. The Contents appears the same on all the sheets since it is copied using equations from Sheet1 Contents. The current worksheet is highlighted yellow.

The page to the Right of Contents is the circuit page, see **Figure 2.2** showing one of Sheet3 circuits. The on/off switch can be used to turn off large spreadsheets when not in use to prevent them slowing down other sheets. **Figure 2.3** Left screen shot shows AR and AS columns OFF and the columns show “n/a”. Right screen shot shows the cells ON. 2. The ON/OFF switch only disables the calculation cells used for the circuit, it does not blank out the data area. This can be useful if you want to see which is data separate from equations results. Notice also the calculation area header background is blue-grey and the data area header is grey.

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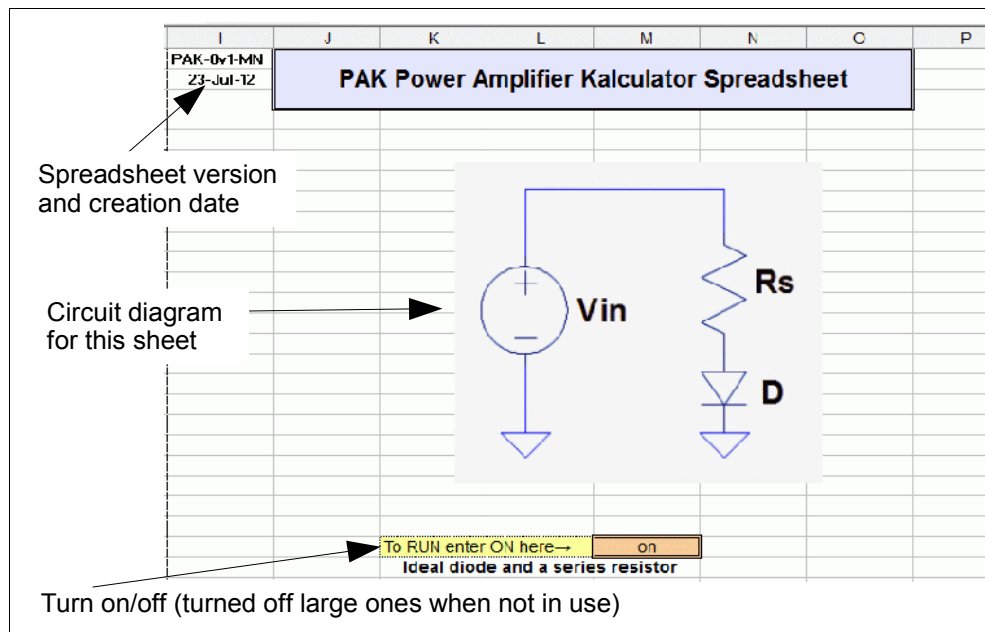


Figure 2.2. Circuit diagram and the equation On/Off switch

	AO	AP	AQ	AR	AS		AO	AP	AQ	AR	AS
29						29					
30						30					
31	Rjc	Rcs	Vto	ReTot+	ReTot-	31	Rjc	Rcs	Vto	ReTot+	ReTot-
32	[C/W]	[C/W]	at 27°C	[Ohm]	[Ohm]	32	[C/W]	[C/W]	at 27°C	[Ohm]	[Ohm]
33	0.1	0.5	0.02586	n/a	n/a	33	0.7	0.5	0.02586	0.38857143	0.35818182
34	0.7	0.5	0.02586	n/a	n/a	34	0.7	0.5	0.02586	0.38857143	0.35818182
35	0.7	0.5	0.02586	n/a	n/a	35	0.7	0.5	0.02586	0.38857143	0.35818182
36	0.7	0.5	0.02586	n/a	n/a	36	0.7	0.5	0.02500	0.30057143	0.35010102
37	0.7	0.5	0.02586	n/a	n/a	37	0.7	0.5	0.02586	0.38857143	0.35818182
38	0.7	0.5	0.02586	n/a	n/a	38	0.7	0.5	0.02586	0.38857143	0.35818182
39	0.7	0.5	0.02586	n/a	n/a	39	0.7	0.5	0.02586	0.38857143	0.35818182
40	0.7	0.5	0.02586	n/a	n/a	40	0.1	0.5	0.02586	0.38857143	0.35818182
41	0.7	0.5	0.02586	n/a	n/a	41	0.7	0.5	0.02586	0.38857143	0.35818182
42	0.7	0.5	0.02586	n/a	n/a	42	0.7	0.5	0.02586	0.38857143	0.35818182
43	0.7	0.5	0.02586	n/a	n/a	43	0.7	0.5	0.02586	0.38857143	0.35818182
44	0.7	0.5	0.02586	n/a	n/a	44	0.7	0.5	0.02500	0.30057143	0.35010102
45	0.7	0.5	0.02586	n/a	n/a	45	0.7	0.5	0.02586	0.38857143	0.35818182
46	0.1	0.5	0.02586	n/a	n/a	46	0.7	0.5	0.02586	0.38857143	0.35818182
47	0.7	0.5	0.02586	n/a	n/a	47	0.7	0.5	0.02586	0.38857143	0.35818182
48	0.7	0.5	0.02586	n/a	n/a	48	0.1	0.5	0.02586	0.38857143	0.35818182

Figure 2.2a. The ON/OFF switch disables the calculation cells but not the data area. Left shows AR and AS columns OFF showing “n/a”. Right shows the cells ON.

- Figure 2.3** shows various areas in the data entry section. Part of the plot area is obscured and to see the plot use the scroll bar or mouse-roller to move around. Change to Normal View for most things select View>Normal. (Before closing the spreadsheet it is helpful to set the view to Page Break View and place the cursor on the sheet number).

This circuit's input voltage V_{in} is swept from 0 to 1.2V.

Values for components are set or changed in the top one only – all the values in the column change once you press enter. This is done with an equation in the second row that copies the first row, then the second is copied down.

With mouse-over on a curve you can **read the value** at that point (x,y) and identify the parameter name easier than using the legend. The legend is useful to tell you what curves are plotted (it's not so easy to see which one is which).

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The TOP cell changes ALL values under it (enter data in the top row only)

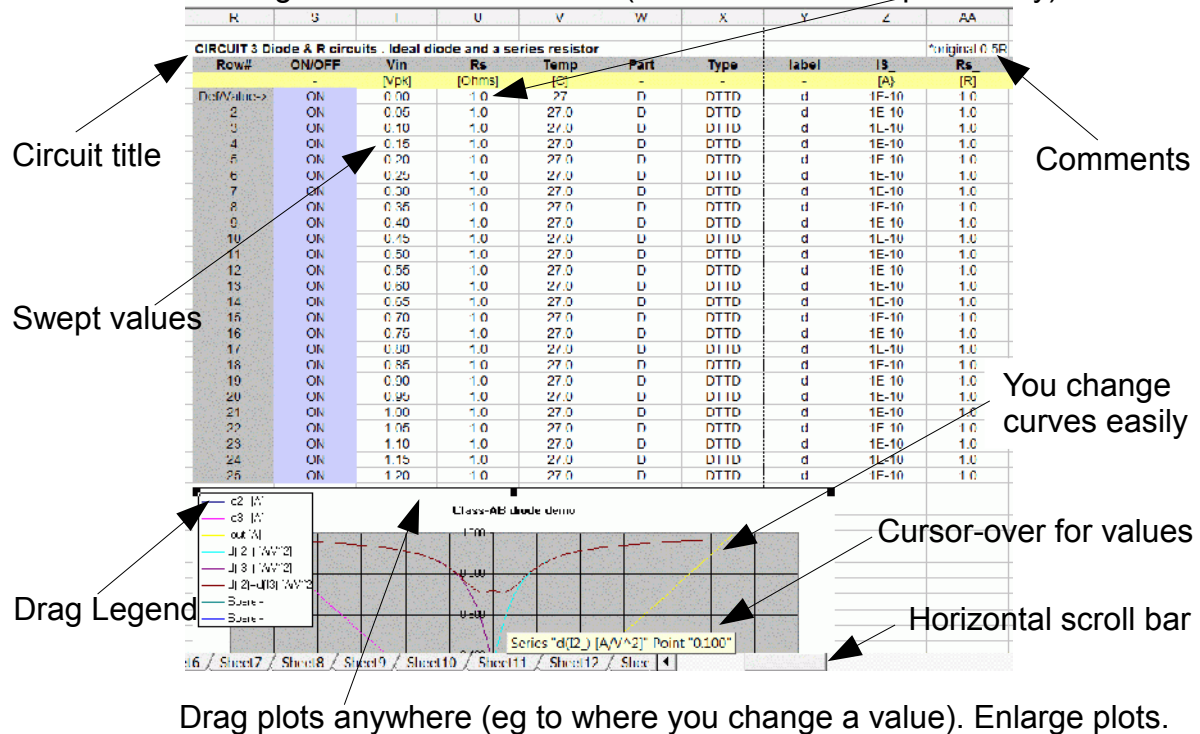


Figure 2.3. Various areas in the data entry area. Part of the plot is obscured. Use the scroll bar or mouse-roller to move around. Use View>Normal for most things and View> Page Layout when finished.

4. Use the spreadsheets **Undo** and **Redo** as a to repeat changes (mouse click on the icons) – look carefully at the plot area and you should see how curves change. Hint: drag the Plot to above where the area value is to be changed to see the curves move repeatedly using **Undo** and **Redo**.
5. Plots can be modified in many ways but it is not so easy to scale *just one curve* relative to the others. The easiest way is to copy a plot and paste it nearby, then remove the plots that are the largest so the smaller ones can be displayed (plots have Auto scaling on by default). Another option is to Insert a column next to the one to be scaled, then use an equation to scale the original one and change the plot pointer to the new scaled column. A temporary way to zoom is to **zooming-all** by right-click on the Y-axis and choose Format Y-axis, choose Scale and set the Y-Max and Y-Min to a suitable zoom range. When done use Undo to return to original scaling. It seems best if the scaling is left on Auto (ticked boxes) so changes in sweep ranges are automatically tracked.

Other things that can be changed in plots are the background shading can be turned off, the legend font type and size can be changed, the legend area can be moved or enlarged.

6. The plot-array sets which data appears in the plots (**Figure 2.6**). There are 8 user settable plots plus the x-axis. Normally the plot array is parked above the plotting columns at the right-hand end of the spreadsheet but it can be Cut/pasted to anywhere you like.

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AY	AZ	BA	BB	BC	BD	BE	BF	BG	BH	BI
PlotPointers>	T2	AJ0	AK0	AK0	AK0	AK0	AK0	AK0	AK0	AM0
X-axis	0	lin	dd(lin)	dd(lin)	dd(lin)	dd(lin)	dd(lin)	dd(lin)	dd(lin)	Spare
0	[Vpk]	[A]	[A/V]	[A/V^2]	[A/V^2]	[A/V^2]	[A/V^2]	[A/V^2]	[A/V^2]	-
-	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
-	0.050	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
-	0.100	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
-	0.150	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
-	0.200	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

X-axis pointer

Pointer array – change the letters to the column you want to plot

Headings are imported automatically from the column headings (once you enter the pointer cell)

Figure 2.6. The Plot Array (the top orange row). It sets which data gets into the plot area

The **Pointer Array** allows the plots to be changed easily without repeating the Chart Wizard. It allows one X-axis and up to eight Y-axis curves in X-Y 'scatter' format. Try changing the plot pointer H71 to GV71 and you will see the new heading appear below it and then in the plot itself. Note the Legend has been changed to the new heading.

7.  There is a **Quick Help guide** in the spreadsheet (Sheet1 below the Intro page). This area that can be printed from the spreadsheet (page 3 of sheet1 but always check Preview first). The same sheet is printable from this User Guide (see [Quick Help](#))
8. The PAK spreadsheet uses a subset of SPICE transistor and diode models (SPICE MOSFETs can be included too!). Figure 2.3 shows diode parameters Is and Rs. These parameters can be Copy/Pasted from Sheet2 Model Library. Notice in the Library there are columns of parameters. The first library 'block' is used by permission, courtesy of Bob Cordell from www.CordellAudio.com. You can also Copy/Paste any model variants that you want to keep from your worksheets back to the Model Library (Sheet2).

This introduction to PAK should allow you to find your way around and play with settings and plots. The next Section 3 provides a list of the circuits and their development status. [Section 4](#) of this chapter provides brief explanation of how PAK differs from a SPICE simulator. [Chapter 2](#) provides more help if you want to create your own Spreadsheet sheets and plot the results. [Chapter 3](#) provides help if you want to create your own equations and check them against LTspice plots. [Chapter 4](#) covers the derivation of the equations used in the LTspice circuits and in the PAK spreadsheet.

... continues to Section 3

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3. Table of Circuits in PAK-1v0-NM spreadsheet

Table 3. Circuit solutions in PAK-1v0-NM at 1 November 2012

Sheet	Topic	Variants			Future? (wish list)	
		See Footnote No's	In UG?	In SS?		
1	How PAK spreadsheet works	N/A	√	N/A	-	-
2	How to derive equations	N/A	√	N/A	-	-
3	Solving Diode & R circuits		√	√	2,8	
4	Solving single BJT circuits	1,3,4	√	√	2,5	
5	The Complementary pair	1,3,4	√	√	2,5	
6	The CE plus CC pair (VAS+VF)	1	√	√UC	2,3,4,5	
7	The Darlington pair		√	UC*	1,2,3,4,5	
8	The Differential pair		√	√*	1,2,3,4,5	
9	Current Mirrors		√	√	Wilson	
10	Current Sources		√	√	1,2,3,4,5	
11	Vbe multipliers		√	√	1,2,3,4,5	
12	Basic push-pull output stage	1,4	√	√	2,3,5,6	
13	CFP Push-Pull output stage	1	√	√	2,3,4,5,6	
14	Darlington PP output stage	1	√	√*	2,3,4,5,6	
15	Distortion estimates (CE)	1,7	√	√*	2,3,4,5,6	
16	HF frequency effects (CE)	-	-	-	SE & PP	
17	[Spare]					
18						
19						
20	SIT SE Common Source				pending v1.1	

Notes: SE Single Ended, CE Common Emitter, CC Common Collector (aka VF Voltage Follower), PP Push Pull, FB Feedback, NFB Negative Feedback. * earlier equations (usable – awaiting an update)

Variants: **1** Vce saturation, **2** includes Tj, **3** includes Early effect, **4** includes $R_{in} > 0$, **5** includes Beta-fall, **6** includes HF roll-off, **7** includes TLD (not self-heating), **8** includes self-heating, **9** [spare], **10** [spare].

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4. Overview of PAK-1v0-NM Spreadsheet

PAK-1v0-NM spreadsheet directly calculates currents in a circuit using the general solution equations for each type of circuit. The spreadsheet is a way to view the plots from these circuit equations. This User Guide [Chapter 4](#) provides details on the derivation of the equations used in the PAK spreadsheet in.

Each circuit type requires a custom equation set. The PAK spreadsheet allows X-Y oscilloscope like chart plotting that is user-selectable so you can change to another set of results without having to set up a new chart each time. (Spreadsheet programmers do not make it easy to assign an arbitrary range of cells to data labels in a chart, according to John Walkenbach's Excel 2002 Review at PC World www.pcworld.com/article/49607/first_look_excel_2002.html)).

General solutions for bipolar transistor circuits were not available until the year 2000 some 50 years after bipolar transistors were developed. Now, 10 plus years later, very few circuit solutions had been worked out. This spreadsheet covers a few of the basic building blocks for audio power amplifiers. This spreadsheet also provides a way to solve more of the elementary circuits in an open source framework.

The first version of PAK is limited to making solution details public through the User Guide but it could be done more quickly and efficiently using Wiki software [[Comparison of wiki software](#)]. Anyone interested in moderating and hosting a site should see [Feedback](#).

PAK-1v0-NM spreadsheet is arranged with each circuit type or topology on a separate worksheet. A worksheet from one persons spreadsheet can be imported to another using Copy/Paste to a blank sheet with the caveat that no cross-sheet linking is used (the one exception is the Contents list from Sheet1).

SPICE models are used in PAK. They are the DC parameters and a few AC parameters for later PAK versions that will include HF analysis. Temperature dependency parameters will also be included in later PAK versions therefore the PAK library of models should include temperature parameters from the beginning.

Within a worksheet several circuit variants can be run. Each variant 'block' has 10 or 20 rows with one variable stepped by incrementing values in that column. You can step any variable, but only one variable at a time in each block. Each block includes an ON/OFF switch to turn off blocks that are not needed and are slowing down calculations (with only a few sheets in a spreadsheet all circuits can be left ON). The [Tutorial](#) in the previous of this chapter demonstrated these features (Section 2).

PAK-1v0-NM spreadsheet is targeted for OpenOffice.org Calc. It can run under Excel without any changes. To convert the OpenOffice version use Save As Excel 97/2000/XP (.xls) file (being the latest option in OOo-Calc v3.3). [Chapter 2](#) lists known problems and things to avoid to ensure cross compatibility. The Excel version runs about 2 times faster than Calc and Excel has some useful editing features not available in Calc such as changing a chart legend order and global sheet changes. However, Open Office Calc editing of equations provides a readout of partial results for debugging functions within functions providing a tree listing.

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Chapter 2
Customising aPAK spreadsheet

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1. Guidelines for customising the PAK spreadsheet

Thank you (in anticipation) for contributing to PAK.

General guidelines for contributors:

- This is an Open Source project so your contribution must be given in accordance with the Open Source Licence. This Licence allows your additions to be distributed and/or modify under the terms of the Creative Commons Attribution License, version 3.0 [<http://creativecommons.org/licenses/by/3.0>] or later, and of course all trademarks mentioned belong to their legitimate owners.
- Anyone is welcome to add anything from PAK User Guide material to other Open Source projects such as Wikipedia with appropriate acknowledgement of the source and author(s) or by citing as a reference. I would be pleased to assist anyone wishing to include any of the PAK User Guide material in teaching aids such as textbooks or course material (depending on the availability of free time). Developers of simulators are also welcome to include any of the PAK equations or User Guide material in their software under the terms of the Creative Commons share alike attribution license, version 3.0 or later [<http://creativecommons.org/licenses/by/3.0>].
- By adding to this User Guide document or a PAK spreadsheet or providing links from this document to a website you are thereby agreeing to the Creative Commons Attribution License for all that material.
- If anyone is interested in starting and/or operating a User Group or a Blog page for PAK then please see [Feedback](#). A Google search may provide a quicker way to see new developments or new contact details.
- An editable OpenOffice .odt User Guide is available on request via [Feedback](#). Please include some information on any changes or additions you wish to make to the User Guide or to your version of the PAK spreadsheet. Please enter a contact address either in the added chapter or add a 'mailto' hyperlink in 'Brief description' cell of the 'Modifications and updates' section of page 2.
- Please use the standard formatting rules for PAK. These are outlined below (more may appear later). Format rules aim to preserve *forward* compatibility with similar spreadsheets. For example, *User A* with an old version of PAK can export to *User B* with a more recent PAK spreadsheet version.
- Format rules also aim to make PAK extensible (expandable) to allow a large (or small) number of circuits. One circuit topology is assigned to one Sheet. No inter-sheet cross referencing is to be used to other sheet to allow one sheets to be exported to other spreadsheets as a *stand alone* circuit solution. Sheets can be given meaningful names. Eg, "JDcct1" and as short as possible.
- Another aim of standard PAK formatting is make the spreadsheet "printer friendly" (if such a thing is even possible). When PAK is run under Excel 2002 (ver 10 for XP) or under OpenOffice Calc there are minor differences that can upset a page layout. Excel and Calc handle Header & Footer and margins slightly differently (see details later). The OpenOffice Calc default Row Height and fixed Column Width should be used for all cells in the spreadsheet and no column widths should be changed from the default (see below if you

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want wider columns using merging).

- Anyone can develop alternate (branch) spreadsheets that are not compatible with the original (or future) PAK versions. Incompatible versions with the original PAK-1v0-NM spreadsheet should not include “PAK” in their name to avoid confusion.

1.1. Excel and OpenOffice.org Page breaks for printing

Note: The printing margins in PAK are not the default settings for either Calc or Excel so they must be manually set if you create a spreadsheet from scratch. Assuming A4 paper use the following settings: Left & Right 1cm, Top & Bottom 1.5cm. Header and Footer in Excel are defined differently to Calc – in Excel Page Setup set Header and Footer to 1.5cm and Top & Bottom to 2cm. Calc uses the AutoFit default so Calc does not need non-standard settings for the Header and Footer section.

PAK spreadsheet uses the OOo-Calc default Row/Col spacings

PAK spreadsheet uses the OOo-Calc default Row/Col spacings to (try to) ensure the page view is consistent when converting to Excel. This is because Open Office Calc has the larger of the default column spacing.

Please do not change any Column spacings in the PAK spreadsheet

If you need a wider column to display text longer than 10 spaces then merge two adjacent column cells then to get a block of merged cells drag to copy down as many as are required. (The merge in Excel is the icon next to the Align Right icon). Similarly to get more height for larger fonts etc then merge two adjacent row cells. If you want to un-merge then select the area and click the merge icon again (it's the same procedure in OOo Calc).

Please use the standard formatting rules for PAK. These are outlined below (more may appear later). Format rules aim to preserve *forward* compatibility with similar spreadsheets. For example, *User A* with an old version of PAK can export to *User B* with a more recent PAK spreadsheet version.

Format rules also aim to make PAK extensible (expandable) to allow a large (or small) number of circuits. One circuit topology is assigned to one Sheet. No inter-sheet cross referencing is to be used linking to other sheets – this is for *stand alone* circuit solution to allow a circuit solution sheets to be exported to other spreadsheets “as is”. Please give shared Sheets a human readable (meaningful name), eg, “JDcct1” and try to make it as short as possible.

1.2. Formatting for Sheet1 – Contents & Help sheet

1. See **Figure 1.2** (or Sheet1 of the spreadsheet). Each Sheet in PAK starts with the same Contents information as Sheet1 Contents. The Contents page of Sheet1 contains the header “PAK Power Amplifier Kalkulator Spreadsheet” in Cell B1:G3. Cell B1:G3 is merged as one cell. Cell A1 is the current version “PAK-0v1-MN” and Cell A2 is the creation date.
2. The comment “Contents list is copied from Sheet1 – change Sheet1 ONLY” is a message intended to remind us that you must change the Contents page information by going to Sheet1. This is because the Contents cells in Sheet1 are copied into all other Sheets using equations (so only cells with information are copied and empty cells still show as empty).

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Columns J & K using the Center format for the entire Column L (see **Figure 1.3**)

I	J	K	L	M	N	O	P
PAK-0v1-MN	PAK Power Amplifier Kalkulator Spreadsheet						
29-Aug-12	Model Parameter Summary Table						
*Bob Cordell's model library http://www.cordellaudio.com							
* 2N3904C - created February 24, 2011 copyright Cordell Audio	IS	XTI					
.MODEL 2N3904C 2N3904C NI*NI	3.5E-15	3					
	*						
* 2N3906C - created February 24, 2011 copyright Cordell Audio	IS	XII					
.MODEL 2N3906C 2N3906C I*NI	1E-14	3					
	*						
* 2N3904C - created February 24, 2011 copyright Cordell Audio	IS	XII					
.MODEL 2N3904C 2N3904C NI*NI	3.5E-15	3					
	*						

Figure 1.3.

- If you have changed a Model's parameters and want to save your custom version then you can Copy/Paste them back to Sheet1 for archiving. If you want to create more space below the original model then select a block (eg J10:AO10) and Insert>Shift Down. Don't select the entire column to insert because this upsets the Contents page. The same applies to any sheet when Inserting a Row – only select a local block – never an entire Row.
- Parameter entries can be printed. If you only want to print the list of model names then either select the area and use Print>Selection.
- Future parameter lists may include non-SPICE parameter labels to suit Label recognition of equation variables by the spreadsheet. EG: Rb, Rc, Re may be changed to Rbp, Rcp, Rep in future to avoid confusion with external resistors labelled as Rb, Rc, Re (the suffix 'p' refers to 'parasitic' meaning internal resistances to the transistor). A spreadsheet may become available in future to import and rename such parameters for complete library files with minimal human input (Section 2.4 pending).

1.4. Formatting for Sheet3 and all others – Circuit Equations sheets

- As mentioned each Sheet in PAK starts with the same Contents information as Sheet1 Contents. See **Figure 1.4** for Sheet3 (or go to Sheet1 of the spreadsheet).

	A	B	C	D	E	F	G	H
1	PAK-0v1-MN	PAK Power Amplifier Kalkulator Spreadsheet						
2	29-Aug-12							
3								
4	CONTENTS	Contents list is copied from Sheet1 - change Sheet1 ONLY						
5	Sheet	Circuit Title	Variant(s)	Notes	Status			
6	1	Contents and Help	-	-	-			
7	2	Model library	-	-	-			
8	3	Diode & R circuits	Class-AB using D's		√			
9	4	Single BJT circuits	CE+Rf	with Early effect & HD2,3	early			
10	5	Complementary pair	CE+Rf		√			
11	6	CE + CC pair (VAS+VF)	CE+Rf		UC			
12	7	The Darlington pair			UC			
13	8	The Differential pair		with Early effect & THD	early			
14	9	Current Mirrors			√			
15	10	Current Sources			√			
16	11	Vbe multipliers			√			
17	12	Basic push-pull output			√			
18	13	CFP push-pull output			√			
19	14	Darlington PP output			early			
20	15	Distortion estimates			early			
21	16	HF frequency effects	-	-	-			
22	17	[spare]	-	-	-			
23	18	[template]	-	-	-			

Fig1.4

- The standard view is Page Break View. But to work on a sheet return to the Normal View (View>Normal, or speed keys Alt V N. To return to Page Break View Alt V P.
- Variants: A circuit sheet can contain any number of 'variants' of a topology. A 'variants' could be the same circuit but with a different parameter is stepped. These are standard for SPICE users. A 'variant' may include different model levels where some parameters are disabled to see

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their effect. Each variant is a separate block of equations, generally starting on the next page below to leave extra space if needed for a larger step range. Note in Fig 1.4 the Contents includes a column for the variants presently defined in each Sheet.

- Each circuit Sheet contains a circuit image on Page2 and further circuits are added below it (and then on the next page below Page2). There is no limit to the number of circuits that can be added to a Sheet – it is only constrained by system RAM and processing speed.
- Each circuit image can have any number of equation blocks (for parameter stepping). The first equation block starts at cell R3 being *one column* beyond the start of the 3rd sheet (see **Figure 1.4a**). Cell R2 is used for the Circuit Title which is generated using ="CIRCUIT "&A8&" "&B8&" - "&L26 where A8 and B8 are the Contents text and L26 is the text added just below the On/Off switch cell.

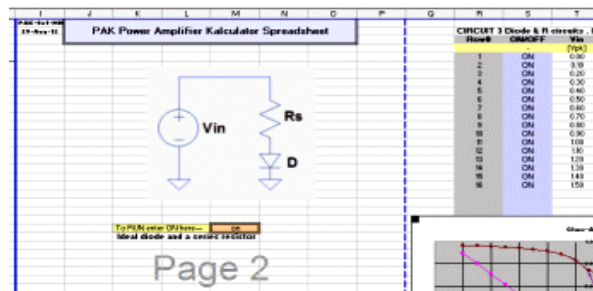


Figure 1.4a.

- A whole block can be copied to create another variant. You only need to copy the top 5 rows (because you can create all the others later by drag-copy down (see next)).
- In PAK you use drag-copy down to extending the parameter sweep range. You copy the 2nd row of equations because the first row is reserved for entering values. The second row copies the first row data with an equation, so the second row is the first row that is the same for all rows below it. (It all makes sense when you see how it's done and why).
- Figure 1.4b** shows the circuit equation area (for Sheet3). It appears after the components and models are defined. The Vt column is standard in circuit equations and is highlighted to show where the circuit equations start. The circuit equation section background is also changed (to Blue Grey).

AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN
Vt	Vk1	Lcf	X	We(X)	lin	d(lin)	dd(lin)	Spare	Spare	Don't ed
@Temp/	[V]			[A]	[A/V]	[A/V^2]				AutoTable
0.0259	0.607	0.999	-19.553115	3.2204E-09	0.000	0.000	0.000	eqn		
0.0259	0.607	0.999	-16.330635	8.0799E-08	0.000	0.000	0.000			

Figure 1.4b.

All equations in the circuit equation section include the common equation

"=IF(\$B33="OFF","n/a","eqn")" where \$B33 is the cell references ON/OFF column at the start of the block (which in turn references to the On/Off switch cell (eg Cell M25)). Note this equation is not used in any of the data entry cells. You enter your equation at the end in place of "eqn" (no equals is used since it is provided by the "=IF" part).

- Next is the plot area. The first part is greyed out because it should not be changed. It is a pointer table for the main plot area, generally used for generating the X-Y plots like an oscilloscope. See **Figure 1.4c**. A section is included on creating an modifying plots.

AN	AO	AP	AQ	AR	AS	AT	AU	AV	AW	AX	AY	AZ	BA	BB	BC	BD	BE	BF	BG	BH	BI
Don't edit	T2	AJ2	AJ3	AJ4	AJ5	AJ6	AJ7	AJ8	AJ9	AJ10	AJ11	AJ12	AJ13	AJ14	AJ15	AJ16	AJ17	AJ18	AJ19	AJ20	
AutoTable	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259	0.607	
	0.0259	0.607	0.999	0.0259	0.607	0.999	0.0259														

Figure 1.4c. Plot are used for User Selectable plots

10. The Number formats that were found useful were Vin & Vsp & Re's as 3 decimal places. For high value resistors in the k or Meg range the following Custom format is useful
 ##0.0E+0 eg 3k3 is 3.3e+3, or use ##0E+0 for no decimal places so 120k is 120e+3. Most of the general equation calculations can be done with general format (since very small numbers use exponents). For the final calculations such as IL or VL set the format to a more readable form. Note you can change your plots axis Number format to single digit increments once they are created.

2. Creating and editing X-Y Charts in PAK

Charts are quite complicated to edit the source data settings. Usually it is quicker and easier to create a new chart then delete the old one (if it looks as you want). But if you need to edit an existing chart then the following may be useful.

2.1. Have you added more rows and want to include them in an existing chart?

If you add more rows to an existing plot area then to include the new points in an existing chart you need to update the Source Data > Series and change the address range for all the X and Y series listed. If you have added more rows and want to increase the range for the plot data to include them then right click on the chart (double click first in Open Office Calc) then choose Source Data

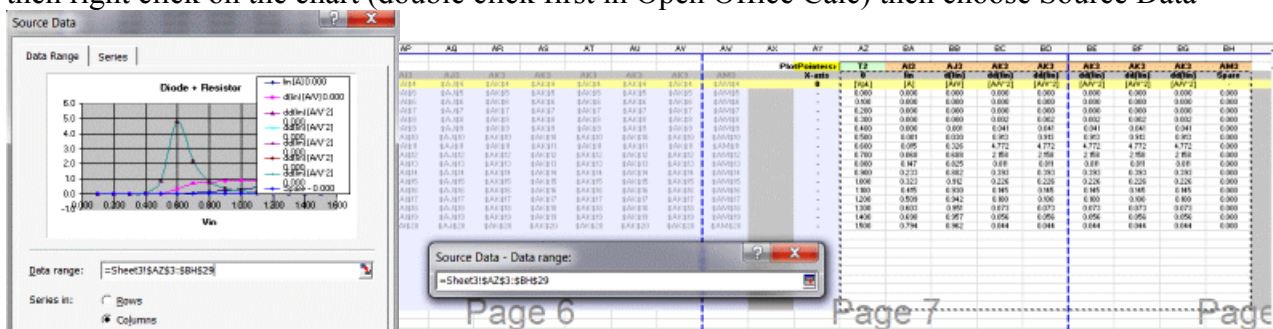


Figure 1.4d. Editing the Source Data range. You can see the present range selected.

Figure 1.4d shows the dialogue box. You click on the Data Range Icon (on the right) and you can view the present Data Range selected. You can drag the selection to where you want or you can type the new row number in the Data Range section. Click the X to close the smaller Data Range dialogue, and OK for the larger Source Data dialogue. View the plot. It should show all the new data.

2.2. Have you created a new block and need to add a Chart?

- Select the data area (AZ3:BH20 here). Notice this **does not** include the column with 'X-axis' (in this example exclude the AY column).
- Choose Insert Chart (or Chart icon) for the Wizard, choose 'XY (Scatter)' > sub-type 'Scatter with data points' > Next > Check Data Range in Columns > Go to Series and view the Legend, if OK click
- Enter a Chart Title (no need for X or Y titles), Grid-lines tick X-axis Major gridlines > Next check 'As object in Sheet#'. All done. The chart may need Y-axis scaling to give the best view (see the section **Sheet style format** on scaling Y-axis), eg, Format y-axis to Min 0, Max 20. If one plot is still too small then you need to insert a new column to create a scaled value to view it better in the plot with all the others (see the **Sheet style format** section on

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Scale or offsetting only one curve).

- If you want the X-axis to appear in the Legend then you need to add it as a Data Series. This means you can use the Plot Pointer to change which column of data is used as the X-axis (without having to edit the X-axis Title text). To start: Right Click on the chart area, select Source Data, click Add Series, click the icon in the name panel, select the cell with 'X-axis' (AY3 here) then select the cell below it (either hold the mouse and drag down or hold the Shift and second click on the cell below), click the X (close) and view the Legend (should show the X-axis as the last entry), click OK. Done. You can rearrange the Legend order in Excel (but not Calc).

2.3. Or do you want to create a new block

The easiest way to create a new block is to copy the first 5 rows of an existing block. You don't need to copy the entire block since all the equations can be generated below the 4th using drag-copy.

- Select a block of the first 5 rows all the way to the end of the plot area (R2:BI6 here). To do this click on Cell R2 then scroll to BI6 and hold down the Shift key, then click on BI6.
- Copy (Ctrl+C).
- Scroll to the cell where you want to start the new block, eg, click Q62 then Paste (Ctrl+V). You now have a new block ready to drag-copy.
- To drag-copy down, click on the first cell (Q66 here) then scroll to the last (BI66 here). Drag the corner tab down (at BI66 here) until you have enough rows. You may need to check the start to see whether the stepped parameter is large enough then return to the scroll tab and move up or down. You are now ready to add a chart – See the section on adding a chart.

2.4. Or do you need to create a chart for the first time?

The easiest way to add the equations that generate a chart area is to copy the first 5 rows of the plot area equations from another sheet (as seen in Figure 1.4d).

To create the chart area equations from scratch you only need to enter the equations for 3 key cells and all the others are created by drag-copy. In our example the first cell is in the LHS pointer table, say starting at CU31 which contains =DF30 (ie first cell in the Pointer array). CU31 can be drag-copied to DC31.

The second key cell is CU32 =ADDRESS(ROW(CU31)+1,COLUMN(INDIRECT(CU31))) . All other cells from CU2:DCxx can be drag-copied from cell HL72 (horizontally then vertically). This completes the pointer table. These are set to grey to remind users not to edit them (select the area, Format>Cells>Font>Colour eg Grey 40%).

The third key equation is in the plot table, say starting at DF31 enter=INDIRECT(DF30) . Cells from DF31:DN31 can be drag-copied , then select DF31:DN31 and drag-copy down.

Set up a pointer array DF30:DN30 above the plot area. Enter the cells for the data you want plotted. The first one DF30 is usually the X-axis which will be the swept parameter usually near the start of the block, eg you could use A31 with is the row number 1-n as the X-axis values, but C31 is usually the first possible column that can be stepped. For any of the plot columns you don't want to use enter a reference to a spare column, ie it has the text 'Spare' in entered – this gets displayed in the Legend and prevents an Error in the plot data (helpful for Calc users). If you don't want the '0' to appear in the legend (for a blank cell) then enter a ' (the single quote) in the cell (it will be invisible too, but visible in the edit bar if selected).

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The last thing is to add the text 'X-axis' in a spare column to the left of the pointer area (DE31 in this example). This appears on the Legend so the X-axis parameter can be selected using the plot array, in this case the first cell in the plot pointer. So the next cell below being DE32 has the equation =DF31 (first cell in the plot pointer). The remainder of this column can be left blank, or use '-' (single quote plus minus) to place a minus sign as text in each unused cell.

3. Known spreadsheet problems and solutions

3.1. Converting from Excel 'xls' versions to OpenOffice Calc 'ods' (vice versa)

If you are working with Excel and want to globally change all sheets at once



You can globally change all sheets at once but be careful and don't forget to turn off the global select after you have finished. To select All Sheets in a Excel, Right-click a sheet tab, and then click Select All Sheets on the shortcut menu.

Converting from Excel 'xls' versions to OpenOffice:



Excel allows empty cell ranges to be set using ={} but when converting to OpenOffice this converts to a null entry in the Data Range and when saved as an ods and reopened the plot crashes (it is converted to a copied plot table unrelated to the spreadsheet cells all but one plot is corrupted).

This was found to occur when the text "X-axis Vin" is imported into the Legend table (to indicate which variable is currently used for the X-axis) and this is a very useful feature for this pointer-reconfigurable plotting regime. Creating the spreadsheet in Excel then importing to OpenOffice has some benefits; for successful conversions back to Excel (which opens, saves and runs faster) and default plot settings are better in Excel reducing the need to change OpenOffice plots (default lines are too thick in OpenOffice) and you can change to Legend order too.

Work around: To prevent this plot crash enter the X-axis Range for the Legend for "X-axis xyz" (same as the other plot variable). Then for the Y-axis range set the range to the first cell below the "X-axis" "xyz" cells which is can be an empty cell of a "-" (dash using the minus key). If the Format Data Series>Options are set to Plot Missing Values is set to Gap then only one point appears on the plot while the Legend shows the line and marker. Any Spare plot lines can be reduced to a single point using this method. (Setting unwanted plot lines to be Invisible and No Symbol prevents the line from being selected to return it to visible! If this happens then Remove the Data Series (copy the Range for Name to clipboard first) and re-enter it.)

Converting from OpenOffice Calc to Excel xls:

An annoying thing when saving a Calc file in Xls (2000/XP) format is the Page Break Preview view loses the zoom setting. It is resets to 100% (but going the other way Excel to Calc retains the Excel zoom setting).

Work around: To restore Excel to the standard zoom for PAK, select all the sheets (right click mouse on a sheet tab) then View>Zoom>Custom 60%, then turn off the Grouping of all sheets by clicking on a different sheet tab. You can see if sheets are all selected in the Filename (title) bar where it adds "[Group]". If you need to reset the Normal view zoom, for PAK it is 85%.

When converting between these the scaling of the plots seems to reset to their original aspect ratio. Also the legend can change its position. Knowing this, it may be better to not change the aspect ratio of the plots.

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3.2. Excel & OpenOffice Calc spreadsheet rows & column default settings

	Col Width		Row width		Font size	# Char wide	Padding L=R
Excel	1.77 cm	8.11 units*	0.47 cm	13.2 U	10	6	1 char ea
Calc	2.27 cm	12.3 units*	0.43 cm	12.2 U	10	8	1mm ea

*Units are number of character spaces (including padding)

How Excel and Ooo-Calc calculate column spacings:

Excel Opt. Col Width 10 Font = $1.44 + (n-1) * 1.33$

Scale by Font No/10 gives padding of 1 character on each side

Default setting is cm 0.2cm padding (1mm each side)

OOo-Calc Opt. Col Width 10 Font = $0.2 + n * 0.1 * 2.54$

Default Excel col width gives 6 x 10 Font characters (Excel 8.11 units, OOo-Calc shows as 1.77cm)

Default OOo-Calc Col Width gives 8 x 10 Font Characters (OOo-Calc 2.27 cm, Excel shows as 12.3 units).

3.4. On Printouts

Excel and OpenOffice.org Page View and Page breaks (for printing)

PAK spreadsheet uses the OOo-Calc default Row/Col spacings

PAK spreadsheet uses the OOo-Calc default Row/Col spacings to (try to) ensure the page view is consistent when converting to Excel. This is because Open Office Calc has the larger of the default column spacing.

Please do not change any Column spacings in the PAK spreadsheet

If you need a wider column to display text longer than 10 spaces then merge two adjacent column cells then to get a block of merged cells drag to copy down as many as are required. (The merge in Excel is the icon next to the Align Right icon). Similarly to get more height for larger fonts etc then merge two adjacent row cells. If you want to un-merge then select the area and click the merge icon again (it's the same procedure in OOo Calc).

Open Office Calc has a slightly smaller default column spacing causing a difference of 1 row per page (60 rows/page for Excel and 59 for Ooo-Calc). Therefore a buffer of one row should be provided at the top of following sheets in case the page break settings return to their default in either Excel or Calc for some reason (they generally stay where they are manually set – seen as a solid page break blue line in page view).

Page settings for Excel and Open Office are shown in the table below. These set the default page break positions.

	L/R margin		Top/Bottom		Header/Footer		
Excel	1 cm	1 cm	2 cm	2 cm	1.5 cm	1.5cm	
Calc	1 cm	1 cm	1.5 cm	1.5cm	0.9 cm	0.9 cm	

Excel margin settings are viewed in File>Page Setup>Sheet.

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OpenOffice.org Calc in Format>Page>Page or Header or Footer.

Note: The printing margins in PAK are not the default settings for either Calc or Excel! They must be manually set if you create a new spreadsheet from scratch (to follow the original PAK page format setting). Assuming A4 paper use the following settings: Left & Right 1cm, Top & Bottom 1.5cm. Header and Footer in Excel are defined differently to Calc – in Excel Page Setup set Header and Footer to 1.5cm and Top & Bottom to 2cm. Calc uses the AutoFit default so Calc does not need non-standard settings for the Header and Footer section.

Printing on Letter (US standard size) has not been tested. Hopefully printer setup options can handle printing A4 on Letter size.

The print direction can be set as down then across or vice versa. PAK has all sheets set to the default of across then down. Use print preview before printing and be sure to specify a range to prevent large numbers of unwanted sheets being printed.

3.5. Excel equation label problems

3.5.1 The Excel Label interpreter can messes up equations sometimes

Using Search/Replace to convert R/C addresses to Labels sometimes creates bad label names. When the label interpreter looks at the mess it *suggests* a problem fix, often by adding single quotes to various labels. Accepting this suggestion can convert double quotes to single quotes which is unwanted. EG the equation

=IF(\$S5="OFF","n/a",Rc1_/Beta2+Re2_*(Beta2+1)/Beta2) gets converted to

=IF(\$S5='off',"n/a",'Rc1_/'Beta2'+Re2_*(Beta2+1)/Beta2)

where the “OFF” is converted to 'off'. This does not prevent the equation from generating the correct numbers but now it cannot be copied down by dragging the corner handle. Worse, in Win7 the program crashes when trying to edit this equation. Drag/Copy down gives (next cell down)

=IF(\$S6=#NAME?,"n/a",'Rc1_/'Beta2'+Re2_*(Beta2+1)/Beta2) and the error message

“#Name?” in each cell below the first. Excel crashed trying to edit this in the edit bar.

A tell-tale indicator of this is the top equation when selected and clicked in the edit bar to show which cells are referenced selects one lonely cell about 40 rows below it that is unrelated to the equation labels (a mystery).

To correct the equations fix the first equation with “OFF” then copy down. Check all other first equations for the same problem (either by copying down all in the first row equations, but only those with the IF(\$S\$5) test, or scan each cell in the edit bar using the right arrow for the next.

To prevent Search/Replace and the Label interpreter causing this problem in the first place add a single quote to the label in cell S3 to 'ON/OFF. This prevents ON/OFF from being interpreted as a Label name (without the quote the OFF is interpreted as one of two possible labels for that cell – may be useful sometime).

3.5.2. Label name issues

Labels are very useful in spreadsheets – it allows complicated equations to be human-readable. Also editing a label name in the header automatically appears in all the equations (sometimes you may need to wait till an equation is edited to see the changes). Labels make it easy to transfer an equation from LTspice ABM's directly into a spreadsheet equation if the label names are set up the

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same as the ABM equations (or very close apart from underscores).

But care is needed to ensure label names are not interpreted as specific cell references instead. To prevent say R2 from being interpreted as a cell use R2_ instead (the equations are still human-readable). Some are not so obvious. Re2 is outside the Excel column limit (256 cols, it's 1024 in OOo) but Re has a special significance for some reason. Adding an underscore Re2_ is not be interpreted as a cell but the Label interpreter does not recognise the label and needs 'Re2_' with single quotes to accept the equation. Also Rc (without numbers) needs to be quoted as 'Rc' but I don't know why (or as Rc_ without quotes). You can use the label editor to automatically insert the quotes for all equations at once if you change Rc_ to Rc in the label header (it knows if it needs quoting).

A similar complication can occur if a label name includes brackets like We(X1) and can be interpreted as a function and searches for a label 'We'. To prevent ambiguity add single quotes to these sorts of label names 'We(X1)'.

Label names that do not include a numeral are generally acceptable (apart from a few exceptions mentioned). Three letters (or more) followed by a number are no problem.

It is advised that two lettered labels with numbers that lie outside the present spreadsheets valid range of Excel which is 256 columns (last valid column is IV) should still include an underscore to make it compatible with future spreadsheets if/when the valid ranges are extended. Note OOo Calc with 1024 columns so to be compatible with Excel 2002 (for XP) you should not go beyond the 256 limit. OOo Calc has a valid range includes 3 letters plus a number (first is letter 'A') so beware that Excel Label interpreter may recognise AAA1 as a Label, but in OOo Calc it will think it is a cell reference so the equations will not work in OOo!

Notes: Neither Excel or Calc recognise upper/lower case distinctions in Label names.

Both Excel 2002 and Open Office 3.3 recognise Names as Addressing and this function is active by default. To turn this function off, choose Tools - Options - OpenOffice.org Calc - Calculate and clear the 'Automatically Find Column and Row Labels' check box.

See Help "Naming cells" excerpt:

"If you want a name to be automatically recognized by Calc, the name must start with a letter and be composed of alphanumeric characters. If you enter the name in the formula yourself, enclose the name in single quotation marks ('). If a single quotation mark appears in a name, you must enter a backslash in front of the quotation mark, for example, 'Harry\'s Bar'."

3.6. Excel equation Exp(x) range limit

The Exp(x) function gives #Num error if $x > 710$ or $x < -20$. The following can be used to hard limit the argument EXP(MIN(x,700)) to prevent an error. But this limits the range for the W(Exp(Y)) calculation and causes the calculated current to reach a limit below what it should actually be. The same limit also applies to OpenOffice Calc.

An unlimited result for W(Exp(Y)) is needed for calculating currents. LTspice had a similar Exp(x) range error (from $x > 70$) and the $W(Exp(Y)) = (Y-1) x > 70$ to give unlimited range (Section 12.2.2). The equations for calculating W(Exp(X1_)) for unlimited Y is:

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=IF(X1_<700, LN(1+EXP(X1_))* (1-LN(1+LN(1+EXP(X1_)))/(2+LN(1+EXP(X1_)))), X1_-1)

A similar error occurred as a divide by zero in long equations in Sheet 12 block 2 “Sw1_”.

=1/(1+1/LN(1+(1+0.5*EXP(MIN(x,500)))/(1+EXP(MIN(-MAX(x,-35),500))))))

where x more negative than -36 causes a #Div/0! error. This is prevented by the Min(x ,35) function above [implemented using double negatives in MIN(-MAX(x ,-35),500)].

=====<.>=====

4. A spreadsheet tool for importing SPICE models

[an 'importer' is under development] contact a developer, see [Feedback](#)

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Chapter 3 Working with circuit equations in LTspice

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1. Introduction – Running circuit files

LTspice circuit files are supplied with the PAK spreadsheet in the download ZIP. These circuit files were used to validate the equations for the PAK spreadsheet equations. The LTspice circuit files have several uses:

1. As a helpful Tutorial while working through the equations in Chapter 4. You can Run the circuits in LTspice and see the equations operating and generating the same plots as the user Guide. This interactive approach will help higher-level circuit analysis courses.
2. They demonstrate how you can develop your own circuit equations and then check them against LTspice.
3. They can be used to generate circuit drawings for importing into documents for course material or assignments and for sharing over the Internet.

If you have developed equations and want to add them to the PAK project then you can generate your own spreadsheet Sheet and if possible compile a new Section (or a new Chapter) for inclusion in a future version of the official the PAK User Guide. A developers section is planned at the PakLaunchSite.jimdo.com where work in progress can be viewed before an official public release (a login is required). For details visit the site.

The advantage of putting your equations in a spreadsheet is that they can be viewed by those who don't have LTspice (or don't know how to use LTspice). Spreadsheets are now a universal environment and can share files across nearly all computer platforms.

2. QUICK HELP – Getting started with LTspice ABM equations

The LTspice circuit files are supplied as standard with the PAK spreadsheet in the download ZIP file. You need to unzip these before running them. They do not need to be placed within the LTspice program directory tree as long as any model files are also present in your directory with the extracted files. Some of the circuits may require Cordell-Models.txt, and if this models file is not present then you can download it from www.cordellaudio.com.

To Run a file double click the ...asc file (not the ...plt file). This assumes you have already installed the LTspice program. LTspice can be obtained via BobCordell's site link – and while you are at Bob's site read about how to stop temporary files bloating your computer hard-disk, or get his Tutorial files if you want to learn how to use LTspice from scratch. Bob's tutorial circuits are best used with his book since several chapters are devoted to learning how to use LTspice from scratch (and some advanced things as well).

3. PAK Circuits Overview

Figure 3.1 shows the various areas of a typical circuit file for PAK.

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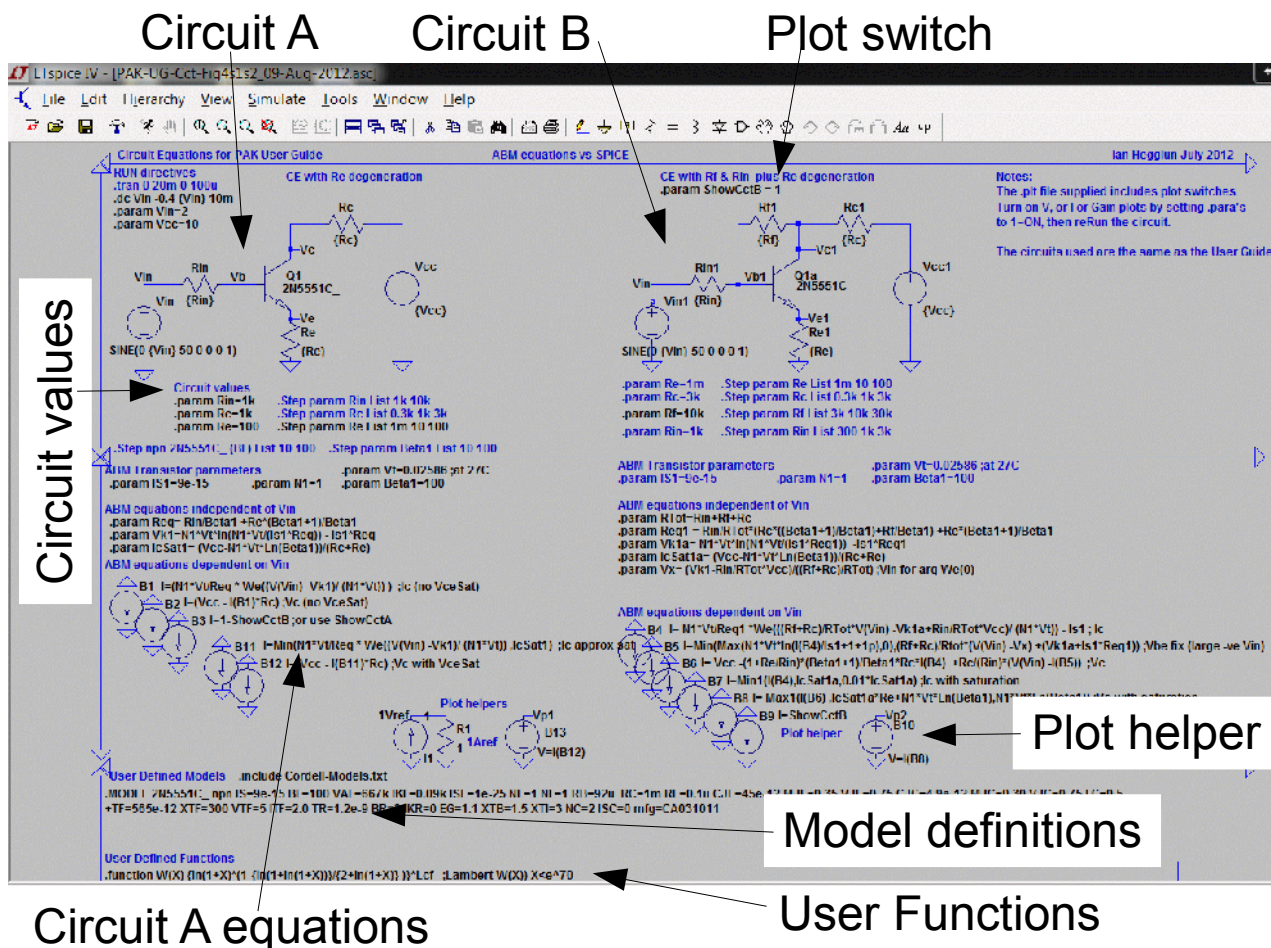


Figure 3.1 shows the various section of a typical circuit file for PAK

3.1 Why use circuit pairs?

Most of these circuits are drawn as pairs in the circuit files because they are usually similar where one circuit part is added to develop equations for a complete circuit. Often equations are built from the most basic form into a more complex version and this needs to be done step by step as more effects are included. Both circuits to be plotted together for comparisons and make it easier to develop equations for the next stage of equations.

Circuit pairs have plotting switches that allow various curves to be turned ON or OFF without needing to delete previous curves. Generally you turn one circuit ON and the other OFF. This also allows for first-time users to view the plots shown in the User Guide. This works if the '...plt' file included with the same plot variables as the User Guide plots used (and most are set the same).

3.2 Why include '.model ...' in circuits?

Circuits are placed at the top. Equations are in the middle area. Model definitions and User Functions are at the bottom area. When a lot of model definitions are needed then they can be placed on the RHS (for screens with a 'wide' aspect ratio). Alternatively they can be put into a special model file and called using the command line 'Include MyFile.txt'.

By having model definitions on screen means you can custom set the models quickly and easily to suit your equations. For example, normally Beta-fall and Early effect are disabled while developing

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lower level circuit equations.

3.3 Why use .Function?

User Functions enable complex formula that are used several times over to be defined only once. This reduces the number of Behavioural Current sources considerably. But there may be a limit to how many terms can be used in a Function because all the terms are 'flattened' into one equation before executing. You can see this if an error message is generated by bad syntax. One case did appear to overflow some limit, maybe it's a 256 character limit? If this happens then split the equation into parts with 2 ABM sources.

A useful function is `'.function Vk(Is,N,Req) {N*Vt*ln(N*Vt/(Is*Req)) -Is*Req}'`. V_k is calculated a lot and this is useful if Req becomes a function of V_{in} (as in the CFP with R_{be} added such as circuits "Fig5s4s5" and "Fig5s6"). But there's a catch – you can't define `'.param  $V_k=N*V_t*\ln(N*V_t/(I_s*Req)) -I_s*Req$ '` anywhere in the circuit when this function is defined because " V_k " is already allocated to the function (even if it is never invoked!). So you have to use another name like `' $V_{k1}=N*V_t*\ln(N*V_t/(I_s*Req)) -I_s*Req$ '` (or use `' $V_{kA}=...$ '` for circuit A).

3.4 Why use "Plot Helpers"?

Several plot helpers are included. A 1A current source (I_1) feeds a 1R resistor (label R_1) and these generate a 1A and 1V reference voltage for plotting. For example, you a Behavioural Current source equation generates a current and gets displayed on a plot with the units of Amps, but if the calculation is actually a transconductance gain then you want it displayed with units of Amps/Volt so it can appear with the same scaling as say a plot of $d(I_c(Q_1))$ when V_{in} is the X-axis.

Another example is if a Behavioural Current source equation generates a Beta value then you want it to appear with the same scaling as a plot of $I_c(Q_1)/I_b(Q_1)$. To do this divide the Behavioural Current source by $I(R_1)$ where R_1 carries 1A and you get a non-dimensional number for plotting.

Yet another is for the plot switch where a Behavioural Current source say $I(B_9)$ is set to either 1A or 0A, then a particular plot say $I_c(Q_1)$ is multiplied by $I(B_9)/I(R_1)$ giving the equation in the plot area of, eg, $I_c(Q_1)*I(B_9)/I(R_1)$. This will force the plot to 0 when $I(B_9)$ is switched to zero so you an uncluttered the plot area without deleting previously set up curves. This makes it easier for first time users to view the same plots as shown in the User Guide simply by running the circuit (assuming the circuit has the plot .plt file included as a set).

If Voltage, Current and gain plots are used then you can turn off the gain plots by multiplying the gain plots by 0 to turn them off when not wanted, eg, $I(B_9)/I(R_1)*0$ or $I(B_9)/I(R_1)*1$ to turn them on.

3.5 Exporting plots

When exporting plots (as .wmf files) for use in a User Guide etc you can avoid the clutter of the legend names being exported by deleting the curves that are not needed. But before you delete the curves save the standard plots first, that way you can get back to the full set again after exporting the plots and deleting the display window then re-running the circuit. Another tip: just before you export the plots check the X and Y scales are the right range.

3.6 Importing plots in Vector format

In OpenOffice you can import .wmf plot and circuit files into a document using

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Insert>Picture>From File then browse to get to the .wmf file and once converted it will appear in the document that fits the page width. Then right click and Anchor>As character to lock it to text. Then Ctrl+drag a corner to resize.

3.7 Importing circuits as Drawings in Vector format

If you import a circuit file and usually you want to be able to edit a circuit to change text or remove unwanted text so you need to import the file into a Drawing. In a drawing you can add arrows and more text or change line widths and Fonts etc. In a drawing all the imported components are retained in vector format which is good for publishing and retains sharpness of text and curves when zooming (great for PDF's).

Usually you can work from a document and Insert>Object>OLE Object>OpenOffice Drawing and this will open a Drawing window. You then use Insert>Picture>From File and get the file. Once inserted you usually right click >Break (to edit it). Once broken it remains selected so you can reposition and scale it (Ctrl+drag). A tip is to press F4 to get the Position and Size dialogue then enter 0 X offset and 0 Y offset and Enter and it will be located at the top LH corner, then Ctrl+drag to resize it to suit (it is better to resize it in the drawing than to resize after closing the drawing window).

A tip for editing text in the Drawing window – select a text box you want to edit then press F4 to get the Position and Size dialogue, in the bottom right turn on Adapt>Fit Width to Text and Fit Width to Text tick boxes then OK. Once this is set you can double click to edit the text and the box will expand to fit (otherwise more text gets squashed up or less text gets stretched!). When an item is selected you can use the keyboard arrow pad to move in major grid increments (left, right, up, down).

If you want the zoom ability for a drawing then you need to Select All (all your imported drawing contents), then open a Drawing using File>New>Drawing, then paste the contents. You can save it as a Drawing file or copy and paste it back to your drawing window. Alternatively, to get some zoom with smaller circuits in a Drawing and in a document you can enlarge the drawing window and select and enlarge the circuit (if you need more zoom then use the document View>Zoom or zoom slider at bottom RHS), when all changes are done then resize the circuit to the original and reset the window back to the original.

3.8 Importing LTspice equations into a spreadsheet

The equations used in LTspice can be Copy/Pasted into a spreadsheet equation with little modification if certain steps are taken to prevent format errors occurring. To make this work in the spreadsheet you use labels and the spreadsheet's auto label recognition is now turned on by default in OpenOffice Calc and MS Excel. So you want the equations in LTspice to be as close as possible to acceptable labels for the spreadsheet to recognise. But you also want to use equation terms that look good in an equation write-up in the User Guide. This means you need to add a few single quotes to the spreadsheet terms so the spreadsheet knows they are labels and not cell addresses. (A macro could be written to automatically reformat the equations if you need to do a lot of equations like this). More information on importing equations and label names is provided in [Known spreadsheet problems and solutions](#) (Ch 2 Section 3).

3.9 How to build a new circuit and equations

You can start with an existing multiple-circuit file, eg “Fig 3s1-4”, or a single-circuit file eg

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“Fig5s6”. The “Fig 3s1-4” file is a good one to start from because it includes several circuits that do not have the equations yet and it is set up ready to enter equations.

In a multiple circuit file you can use a general value for R_e then you can use $\{R_{s1}\}$ and $\{R_{s2}\}$ for the various series resistors in circuit 1, circuit 2 respectively. You assign `.param Rs1=Rs` and another `.param Rs2=Rs`. Now you can step `.Step param Rs List 10, 100` and both circuits will step together. If you want to offset one circuit then you can use `.param Rs2=Rs+2` then the circuit 1 will be 2 ohms more than circuit 2. This was used for offsetting V_{sp1} and V_{sp2} in the CPF push-pull circuit (eg Fig13s1).

Step1: Run the existing circuit. Then File>Save As> choose a name, copy the name to the clipboard, Save. Then click on the Plot and File>Save Plot As> and paste the name (remove .asc and put .plt).

Step2: You can now start building your equations. First add the curly brackets to the resistors and voltages. The components can have the same name as in the curly brackets but it may get confusing so name the parts with 1 and 2 suffixes. If you have two circuits each with 2 or more transistors then I call them Q1a, Q2a in Circuit A and Q1b, Q2b in Circuit B. Notice the Plot switches are also labelled as CircuitA and CircuitB. Having added the curly brackets enter the component values as `.para Re=10` as the default settings. Run the circuit to check the values are being recognised.

You can set your signal source to be programmable using curly brackets, eg, `.dc Vin {-VinPk} {VinPk} 1m` generates a DC sweep from $-V_{inPk}$ to $+V_{inPk}$ in 1mV increments, where `.param VinPk=1V` defines “ V_{inPk} ” as 1V. (You can now use this to step V_{inPk} , eg using `.Step param VinPk 1 2 4` or whatever).

Step 3: Next define the “Equations independent of V_{in} ”. These are equations that are NOT affected by V_{in} . Typical examples are R_{eq} equations (involving R_e , R_b , R_{in} and β) or V_k equations (involving I_S , N_F , R_{eq}). These equations are calculated only once over a swept input but are recalculated for Stepped parameters. Run to check for errors. Save. If you want to look at the value of a `.param` then enter the term into one of the Plot Helpers (for voltage, or a Current source for current but it's no big deal if the units are displayed wrong in the plot readout, just that the value looks about right). Run and click the probe on the plot helper output.

Step4: Define the “Equations dependent on V_{in} ”. These equations all involve V_{in} somewhere in the chain of ABM's. Add one equation at a time and Run to check it for errors. You can click on the source to plot the value. If you want to plot a voltage then enter the $I(B_n)$ n =number in the plot helper, Run and click the probe on the plot helper output.

You can build your equations until they are all defined. If you want to set the Voltage and Current plots to be controlled from the circuit settings then add the equations to the plots, as covered in Section 3.4 above (Why use “Plot Helpers”?).

Step5: Now with equations that Run and plot you can now step various parameters to see if the equation plots track the LTspice circuit plots. You can set up to 3 stepped parameters in LTspice. You can do a “rat's nest” stepping plot to show up any areas where the plots diverge, eg, see [Figure 5.3b](#).

3.9.1 The following now assumes you chose “Fig5s6”.

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Step1: First Run the existing circuit. Then File>Save As> choose a name, copy the name to the clipboard, Save. Then click on the Plot and File>Save Plot As> and paste the name (remove .asc and put .plt).

Now delete any of the old equations, models, etc. But leave the headings as a guide. Change title and name to suit. If you want to keep the line of ABM (current sources I(B1) to I(Bn)) for n elements, then edit their equations and set them to I=1 including the Plot Helpers to V=1.

Draw your circuit. You can start with fixed component values then later change them to use curly brackets such {Re} for an emitter resistor – that allows the parameter to be used in the equation, the value can then be stepped, and other components can be assigned the same term (or a scaled version to track other parts).

Now Run your circuit. You get error messages reporting where errors are. If they are in an ABM (current or voltage source) then you get an indication of where in the equation the problem is. But in .param equations there is no error reporting apart from the next equation that needs that value spits the dummy, so you need to look at what terms are used and then check each equation that generates those terms. If you get stuck then a tip is to copy the .param equation into an ABM (minus the .param bit) to get the interpreter to show where the error lies, then paste the fixed equation back into the .param line.

Once you get your circuit to run use the cursor to select a value to plot. You can delete the existing plot but most of the unknown terms will just disappear when you get it to run so just delete any remaining traces (F5 or delete key or right click and choose Delete this trace). Save the circuit and the plot.

Go to Step 2 to 5 above.

2. Overview of Circuit files with PAK-1v0-NM

Table 2 lists the circuits that are supplied with the PAK spreadsheet and User guide. They can be run in LTspice. Those still under construction are listed as “UC” where the circuits plot the LTspice values but the equations have yet to be defined. The footnotes show the known problems with operating equation circuits (most problems do not stop plots being obtained) and will be hopefully fixed for later versions (or fixes may be available later via [Feedback](#)). The Fig8-to-11-ABM file does not generate a plot but you can read voltages and currents using the probe after you close the '.op' results window.

Table 2. Circuits in PAK-1v0-NM

PAK-LTspice Circuit Files	Circuits	Status
PAK-UG-Cct-Fig3s1	Diode + Rs	√
PAK-UG-Cct-Fig3s1-PP	Diode Push-Pull demo	√
PAK-UG-Cct-Fig3s1PPs5pp	Diode Push-Pull (Rin no Rs)	UC

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PAK-LTspice Circuit Files	Circuits	Status
PAK-UG-Cct-Fig3s1-to-s4	Two diode combinations (4)	UC
PAK-UG-Cct-Fig3s5	1 Shunt 2 series diodes & Offset	UC
PAK-UG-Cct-Fig4s1s2	SE CE transistor & Rf	√
PAK-UG-Cct-Fig4s3s4-CC	SE CC/VF (2)	√
PAK-UG-Cct-Fig5s1s2s3	CE CFP (3)	√
PAK-UG-Cct-Fig5s4s5	CE CFP with Re & Rin (2)	√
PAK-UG-Cct-Fig5s6	CE with Re & Rf	√*
PAK-UG-Cct-Fig6s1s2	CE + CC (VAS+VF) OL*&CL	No CL
No Fig 7 circuits in V1.0	-	-
PAK-UG-Cct-Fig8s1-LTP	LTP	√
No Fig7 for Ver 1.0	SE Darlington	-
PAK-UG-Cct-Fig8-to-11-ABMs	LTP, Mirror, CCS (2)	√
PAK-UG-Cct-Fig8-to-11-LTP-mirror-CCS	Circuits only (no equations)	UC
PAK-UG-Cct-Fig12s1s2	Basic PP (2)	√‡
PAK-UG-Cct-Fig12s2	Basic PP (plots current drive)	√
PAK-UG-Cct-Fig12s3s4	Basic PP CC/VF	UC
PAK-UG-Cct-Fig13s1	CFP (V drive)	√†
PAK-UG-Cct-Fig14s1s2	Darlington pairs (2)	UC

* overflow problem needs something (plots look OK)

† see the W(exp(Y)) function, something needs a fixed (n earlier function is used for now).

‡ needs the Alternate SPICE solver, to change in LTspice: Simulate>Control

Panel>SPICE>Solver>Alternate.

4. [spare]

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Chapter 4 Inside PAK-1v0-NM spreadsheet

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1. How PAK-1v0-NM spreadsheet works

See Chapter 1 [Section 4](#) Overview of PAK-0v1-NM Spreadsheet for an earlier introduction. A list of the circuits in the spreadsheet and User guide can be viewed in the [Contents](#) page or [Table 3](#) in Chapter 1. Note there is a “[Contents](#)” link at the bottom of each page similar to [Home](#) links on websites.

This Chapter provides the derivation of the circuit equations used in the PAK spreadsheet. PAK a new approach. Numerical methods are not used like SPICE simulators. Equations are the only means for generating the plots.

For a brief introduction on how SPICE solves circuit equations see Paul D. Mitcheson's introductory lecture notes [Paul D. Mitcheson, Imperial College, London, Lecture 1 slides, SPICE algorithms and internals <http://www3.imperial.ac.uk/pls/portallive/docs/1/7292571.PDF>]. He said, “This single nodal KCL equation [for a diode and resistor] is a *transcendental* (meaning it is an equation involving terms other than simple polynomials which generally cannot be rearranged so that the variable becomes the subject of the equation) and **thus the equation can only be solved numerically**” [emphasis added].

Now we can solve many diode and transistor circuits in an exact form using the Lambert W-function [http://en.wikipedia.org/wiki/Diode_modelling#Explicit_solution]. The PAK spreadsheet demonstrates the circuits that have been solved to date and hopefully many more circuit solutions will be found.

Thomas Banwell published the first general solutions for a few elementary diode and transistor circuits in 2000 [ref 1 IEEE paper]. Kevin Aylward appears to have been the first to use the Lambert W-function “in its application to transistor circuits” [ref 9].

[Chapter 4](#) repeats some of those solutions and adds more elements into his circuits. Other elementary circuits are added, including the Basic Current Mirror, the Vbe multiplier, the Differential pair, the CPF and Darlington pairs, and various Push-Pull output stages.

Thomas Banwell's solutions did not include collector voltage saturation nor any second-order effects such as Beta-fall at high currents and the Early effect. He mentions that they can be added. A few circuits in PAK-0v1-NM spreadsheet demonstrate how these effects can be included, but not all circuits include them due to the need to release the first version. See [Table 3](#) in Chapter 1 for a summary of presently included effects.

One of the powerful features of circuit solutions using the Lambert W-function approach is the first derivative or 'gain' function can be directly expressed in an analytic form once the current equation is obtained. This is brilliant for circuit designers. Gain equations give a direct indication of nonlinearities which give rise to nonlinear distortion. So we can now analyse distortion from general equations and optimise circuit behaviour *in general* without the necessity of specifying all component values (which SPICE solutions demand).

With SPICE simulations distortion analysis in power amplifiers is tedious. Self-heating (aka electro-thermal) feedback effects such as Thermal Lag Distortion (TLD) in SPICE becomes very tedious due to the large number of cycles needed to span large thermal time constants. But with the Lambert W-function approach it is now possible to calculate initial and final limits (with the help of

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recursion if necessary) following conventional analysis with steady-state plus transient terms.

From the start, PAK equations are based on SPICE diode and transistor model parameters. Presently PAK only uses the DC parameters. AC analysis involving complex $R+jX$ and $R \angle \theta$ notation can be included later and the PAK model library already includes the main AC parameters for future AC calculations. The present DC solutions are applicable for gain nonlinearity and distortion analysis up to the middle audio frequency range before transistor capacitances become important.

This Chapter will be rather boring unless you have a keen interest in solving circuits that have never been solved before. This new approach offers treasures for those who dig. It will require study to climb the learning curve. I'd love to see what you come up with. All the best.

2. How to derive equations for elementary BJT circuits

The method used to rearrange the circuit equations in the following section is a bit like the derivation of the quadratic equation where you may recall completing the square [<http://www.purplemath.com/modules/sqrquad.htm> and http://en.wikipedia.org/wiki/Quadratic_equation]. In completing the square there is a balancing of each side to obtain a form that can then be resolved using the square-root function, and after that the equation is unpacked giving x equals ...

We will see this type of process below for solving exponential equations mixed with linear (non exponential) terms, by getting it into the form to allow the inverse of the Lambert W-function to provide the solution for I equals ...

Background on the Lambert W-function can be found at http://en.wikipedia.org/wiki/Lambert_W-function or <http://mathworld.wolfram.com/LambertW-Function.html>.

We will start with the simple diode plus resistor fed by a voltage source. But just before that, to refresh your memory on logs I'll mention " $\ln(a)$ " means the natural log of a with base "e". The often used exponential and log identities in this chapter are:

$$\exp(a) \cdot \exp(b) = \exp(a+b) \quad \& \quad \ln(a \cdot b) = \ln(a) + \ln(b) \quad \& \quad -\ln(a) = \frac{1}{\ln(a)}$$

but recall for $\ln(a+b)$ there's no simpler form.

If the above equations are gobbledegook then the following chapter is worse! You will need to do a refresher or crash course in algebra before you can make any sense of this chapter.



Before starting out you may like to look at the LTspice circuit files. They are supplied with the spreadsheet and User Guide download as a ZIP. As mentioned they use the same equations that are presented below. When you Run the circuits in LTspice you will see SPICE plots from the equations and SPICE plots. You will see that most plots are so close that one line obscures the other, so you can see that the equations solutions are as accurate as the SPICE solution. These equations have been transferred into the PAK spreadsheet from LTspice so the equations used in the spreadsheet can be used with confidence. Why not Run the circuits files now in LTspice.

3. Solving diode and resistor circuits

This section starts 1) with the most basic example which is needed for solving transistor circuits. Several other diode circuits are presented in this section: 2) for two non-identical diodes in series

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and with a series resistor, 3) two non-identical diodes in series and one diode has a parallel resistor, 3) two diodes in parallel where each diode has a series resistor. Solutions for 2), 3) 4) are also used later in transistor solutions. This is because the SPICE BJT for DC is accurately modelled by a number of diodes and resistors.

3.1. The solution for a diode and series resistor

Figure 3.1 shows the circuit of an ideal diode with a series resistor and fed by a voltage source.

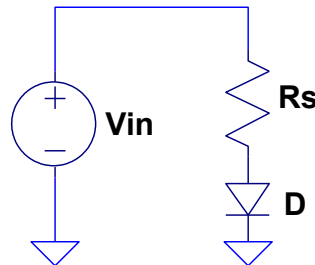


Figure 3.1. Ideal diode D fed by a voltage source Vin and a series resistor Rs

The equation for the current through an ideal diode is given by the Shockley diode equation

$$I_d = I_s \cdot \left(e^{\frac{V_{ak}}{N \cdot V_t}} - 1 \right) \quad (1)$$

where V_t is the 'thermal voltage' given by Boltzman's constant k_B , absolute temperature T , and q_e is the electron charge, and N is the diode non-ideality factor (1.0 for ideal). The V_t is

$$V_t = \frac{k_B T}{q_e} \quad \text{or } \sim 26\text{mV at } 27^\circ\text{C} \quad 2/(2)$$

Using the voltage loop method we write

$$V_{in} = I_d \cdot R + V_{ak} \quad (3)$$

We eliminate V_{ak} in Equation (1) using

$$V_{ak} = V_{in} - I_d \cdot R_s \quad 4/(4)$$

This gives

$$I_d - I_s = I_s \cdot \left(e^{\frac{V_{in} - I_d \cdot R_s}{N \cdot V_t}} \right) \quad 5/(5)$$

We want to solve Equation (5) for I_d in terms of the variable V_{in} .

Here's how the Lambert W-function provides a solution. The Lambert W-function is

$$w e^w = x \iff W(x) = w \quad 6/(6)$$

The inverse of the Lambert W-function allows us to find w in terms of x . $W(x)$ is the inverse function that satisfies $w e^w = x$. We can observe a few principle values of the inverse $W(x)$ from $0 e^0 = 1$ so we know $W(1)=0$. Also $1 e^1 = e$ so $W(e)=1$. Unfortunately, at this time most calculators and spreadsheets do not include the Lambert W-function as a standard function so we use an approximation involving multiple natural logs (more on this below).

To solve Equation (6) we work to rearrange it into the same form as Equation (6). In this case in the

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form

$$f_1(Id) e^{f_1(Id)} = f_2(Vin) \quad 7/(7)$$

where Id is only on the LHS involving $w e^w$ and Vin is only on the RHS. Comparing (6) and (7) we can identify that $w=f_1(Id)$ and $x=f_2(Vin)$. Note f_1 must be exactly the same function ahead of the “e” as to-the-power-of “e” to allow a solution for these types of circuits.

Now we have a roadmap for the solution lets start work on Equation (5) to get it into the form of Equation (7).

Step1. Start with Equation (5)

$$Id = Is \left(\exp \left(\frac{Vin - Id \cdot Rs}{N \cdot Vt} \right) - 1 \right) \quad (8)$$

Then move Is to LHS

$$(Id + Is) = Is \cdot e^{\left(\frac{Vin - Id \cdot Rs}{N \cdot Vt} \right)} \quad (9)$$

Use the identity $e^{(a+b)} = e^a \cdot e^b$

$$(Id + Is) = Is \cdot e^{\left(\frac{Vin}{N \cdot Vt} \right)} \cdot e^{\left(\frac{-Id \cdot Rs}{Vt} \right)} \quad (10)$$

Then move Id terms to LHS

$$(Id + Is) e^{\left(\frac{Id \cdot Rs}{N \cdot Vt} \right)} = Is \cdot e^{\left(\frac{Vin}{N \cdot Vt} \right)} \quad (11)$$

Balance $w e^w$ on LHS by multiplying both sides by Rs/Vt

$$(Id + Is) \frac{Rs}{Vt} e^{\left(\frac{Id \cdot Rs}{N \cdot Vt} \right)} = Is \frac{Rs}{Vt} e^{\left(\frac{Vin}{N \cdot Vt} \right)} \quad (12)$$

Introduce an IsR/Vt terms in the LHS exponent to completely balance $w e^w$ terms

$$(Id + Is) \frac{Rs}{N \cdot Vt} e^{\left(\frac{Id \cdot Rs}{N \cdot Vt} \right)} e^{\left(\frac{Is \cdot Rs}{N \cdot Vt} \right)} = Is \frac{Rs}{N \cdot Vt} e^{\left(\frac{Vin}{N \cdot Vt} \right)} e^{\left(\frac{Is \cdot Rs}{N \cdot Vt} \right)} \quad \text{Or} \quad 13/(13)$$

$$\frac{(Id + Is) \cdot Rs}{Vt} e^{\left(\frac{(Id + Is) \cdot Rs}{N \cdot Vt} \right)} = Is \frac{Rs}{Vt} e^{\left(\frac{Vin + Is \cdot Rs}{N \cdot Vt} \right)}$$

Step 2. We can now identify the two functions that are in Equation (7)

$$w = (Id + Is) \frac{Rs}{N \cdot Vt} \quad (14)$$

and can now solve for Id using the Lambert W inverse function $W(x)$ giving

$$Id = \frac{N \cdot Vt}{Rs} W(x) - Is \quad (15)$$

The notation w and $W(x)$ are synonymous. The latter form is used as a reminder that we have to put a number x into the Lambert W-function to obtain a number for Id .

Step 3. From the RHS of Equation (13) we can identify

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$$x = \frac{Is \cdot Rs}{N \cdot Vt} \exp\left(\frac{Vin + Is \cdot Rs}{Vt}\right) \quad (16)$$

The closed form equation for Id is finally

$$Id = \frac{N \cdot Vt}{Rs} W\left(\frac{Is \cdot Rs}{N \cdot Vt} \exp\left(\frac{Vin + Is \cdot Rs}{N \cdot Vt}\right)\right) - Is \quad 17/(17)$$

where the function W(x) can be calculated using the approximation

$$W(x) \simeq \ln(1+x) \left(1 - \frac{\ln(1+\ln(1+x))}{2+\ln(1+x)}\right) \quad 18/(18)$$

Notice on the LHS the “-Is” subtracted after all the other is calculated. It is therefore written well to the RHS to make it obvious that it is separated from all the rest.

Step 4. It is more convenient if we write Equation (17) in the form

$$Id(Vin) = \frac{N \cdot Vt}{Rs} W\left(\exp\left(\frac{Vin - Vk}{N \cdot Vt}\right)\right) - Is \quad \text{where} \quad 19/(19)$$

$$Vk = N \cdot Vt \cdot \ln\left(\frac{N \cdot Vt}{Is \cdot Rs}\right) - Is \cdot Rs \quad 20/(20)$$

The steps to obtain constant Vk, the 'knee voltage' (or why not the 'Omega voltage' V_Ω ?), are

$$\begin{aligned} x = e^{\ln\left(Is \frac{Rs}{N \cdot Vt}\right)} e^{\left(\frac{Vin + Is \cdot Rs}{N \cdot Vt}\right)} &\Rightarrow x = e^{\frac{(Vin + Is \cdot Rs)}{N \cdot Vt} + \ln\left(\frac{Is \cdot Rs}{N \cdot Vt}\right)} \Rightarrow x = e^{\frac{(Vin + Is \cdot Rs)}{N \cdot Vt} + \frac{N \cdot Vt \ln\left(\frac{Is \cdot Rs}{N \cdot Vt}\right)}{N \cdot Vt}} \\ &\Rightarrow x = e^{\frac{(Vin + Is \cdot Rs)}{N \cdot Vt} + \frac{N \cdot Vt \ln\left(\frac{Is \cdot Rs}{N \cdot Vt}\right)}{N \cdot Vt}} \Rightarrow x = e^{\frac{Vin + Is \cdot Rs + N \cdot Vt \ln\left(\frac{Is \cdot Rs}{N \cdot Vt}\right)}{N \cdot Vt}} \\ &\Rightarrow x = e^{\left(\frac{Vin + Is \cdot Rs - N \cdot Vt \cdot \ln\left(\frac{N \cdot Vt}{Is \cdot Rs}\right)}{N \cdot Vt}\right)} \end{aligned}$$

$$\text{and finally } x = \exp\left(\frac{Vin - Vk}{N \cdot Vt}\right) \quad \text{where } Vk = N \cdot Vt \cdot \ln\left(\frac{N \cdot Vt}{Is \cdot Rs}\right) - Is \cdot Rs$$

The knee voltage Vk is a very useful term for understanding circuit operation as well as simplifying the final equations. The knee voltage Vk is a similar term to the threshold voltage (V_{th}) of a FET. Vk is slightly more than the x-axis intercept on a linear-linear plot of Id versus Vin for the diode plus resistor circuit. The extra x-intercept voltage is about $2 \cdot N \cdot Vt$ volts or typically 60mV higher than Vk and the current is about 3 times higher. This is shown below with parameters for the ThermalTrak sensing diode.

Figure 3.1a shows the diode current plot for a diode with a 1 ohm series resistance, a ThermalTrak sensing diode is used ($Is=1e-10A$, $N=1.2$, $Rs=0.5$ ohms) and the series resistance is increased to 1 ohm for the sake of plotting the total conduction G, which ultimately is limited to 1A/V or 1/Rs. The calculated value of Vk from Equation (20) is 606.8mV. The x-axis intercept is about 680mV which is about 70mV higher than the calculated Vk value, or about $2.2 \cdot N \cdot Vt$. Note for this diode $N=1.2$ so the thermal voltage is 31mV at 27C in this case and not 26mV.

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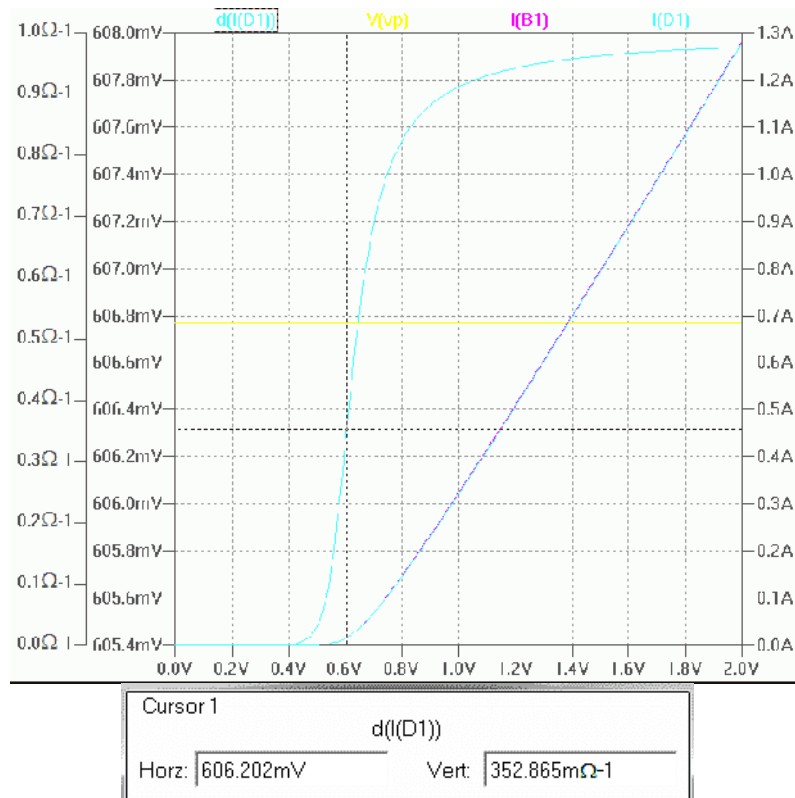


Figure 3.1a. Diode current and conductance plots for a diode with a 1 ohm series resistance. The cursor is at V_k of 606.2mV and conductance is measured as 353mS, or 35.3% of R_s .

Is there any meaningful relationship between V_k and the diode conductance or current levels? Placing a cursor at 606mV the total conductance is measured as 353mS, or 35.3% of G_s (where G_s is $1/R_s$).

Moving the cursor at the x -axis intercept voltage (680mV) and reading the conductance gives 626mS or 63% of G_s . The input voltage for 50% of G_s is 638mV, or 30mV *above* that V_k (about $1 \cdot NV_t$), or 40mV *below* the x -intercept (about $1.3V_t$). The currents at these 3 points are given in **Table 3.1a**. The currents increase by 1.7 times for each step in voltage or a factor of 3 times higher at V_k compared to the x -intercept voltage.

Table 3.1. Diode plus 1 ohms. Current and conductances at V_k

	V_k	Half G	X-int.
V_{in}	607mV	638mV	680mV
Current	18mA	31mA	55mA
Total conductance	0.353mS	502mS	641mS

The value of V_k is not a fixed internal voltage of a diode but V_k is determined by the diodes saturation current (I_s) and the external Thévenin resistance that the diode sees in a particular circuit (see http://en.wikipedia.org/wiki/Th%C3%A9venin%27s_theorem).

Looking further at Figure 3.1a for a significance of V_k voltage, the total conductance plot the V_k point appears to be the inflexion point, so a plot is generated of the second derivative to check this. Since LTspice can only generate the first derivative of a plotted value the second derivative was

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obtained by calculating the first derivative in the circuit then using LTspice to plot the derivative of the derivative. The first derivative is calculated using

$$\frac{dW(x)}{dx} = \frac{1}{x} \frac{W(x)}{(1+W(x))} \quad \text{so} \quad G = \frac{1}{R_s} \frac{W\{19\}}{(1+W\{19\})} \quad \text{where} \quad 21/(21)$$

$$W\{19\} = W\left(\exp\left(\frac{V_{in}-V_k}{V_t}\right)\right)$$

$W\{19\}$ means the Lambert Inverse part of Equation (19) (the equation that was used to calculate the current).

The value of G at V_k calculated using Equation (21) is $W(e^0)/(1+W(e^0))$ or $0.557/(1+0.557)$ or 358mS and the value of G measured using the cursor on LTspice $d(I_{d1})$ was 353mS so they do agree.

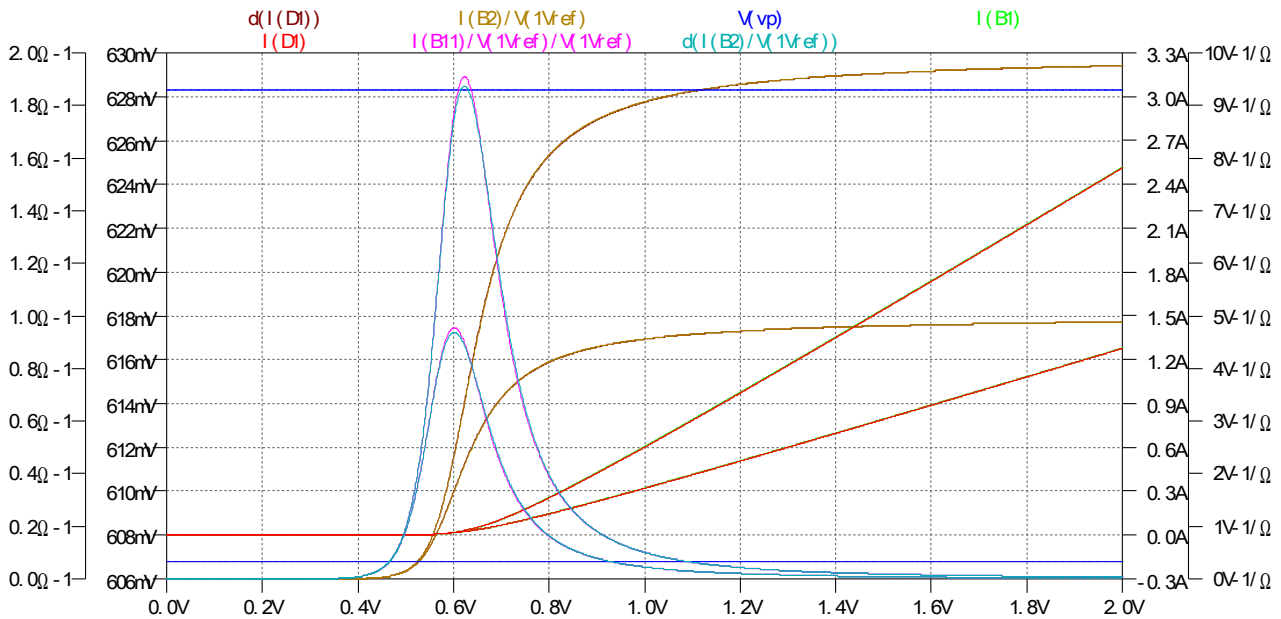


Figure 3.1c Plot of diode plus 0.5 ohm and then 1 ohm, showing plots of the first (brown) and second derivative (cyan). The peak demonstrates that inflexion *does* occurs at V_k (to within 5mV in this test). The peak is $4.66\text{A}/\text{V}^2$ at 602mV

A plot of both the first and second derivatives is shown in **Figure 3.1c**. The second derivative is calculated using

$$\frac{d^2W(x)}{dx^2} = \frac{1}{x^2} \frac{W(x)}{(1+W(x))^3} \quad \text{and} \quad \frac{dG}{dV_{in}} = \frac{1}{N \cdot V_t \cdot R_s} \frac{W\{19\}}{(1+W\{19\})^3} \quad 22/(22)$$

The peak value of the second derivative from Equation (22) is $1/0.031 \times 0.557/(1+0.557)^3$ or $4.77\text{A}/\text{V}^2$. Notice when R_s is changed the first and second derivatives agree. The second derivative is calculated by the LTspice plot function and this agrees with the second derivative from Equation (22) when R_s is varied. Small differences in I_d (and therefore derivatives) are assumed to be due to the Lambert W -inverse being approximate to around 1% on average. Higher derivatives can also be calculated directly and the general formula for the n th derivative is given by Corless Eq 3.5 & 3.6 [ref 6 Corless] (but I haven't figured out how to convert it for a diode circuit).

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What have we achieved by solving the current in this simple circuit?

Previously, placing a diode in a linear circuit with Resistors and Voltages (and/or Currents) meant we had to use a simulator to solve it numerically because there *was* no known analytical solution. Now we have an equation to solve the current through the diode given R_{Th} and with *no analytical approximations*. The result from this calculation will be same as that using SPICE with the same model and parameters. **Figure 3.1e** shows the SPICE solution and the equation solutions together as $I(D1)$ and $I(B1)$. The plots are on top of each other and any differences are not visible. With an inbuilt Lambert W-function inverse the SPICE result will be less accurate because SPICE only iterates to a moderate accuracy; the default RelTol in LTspice is 0.1% (or AbsTol is 1e-12).

As alluded to above, R_s in Figure 3.1 could represent a Thévenin (or Norton) *equivalent resistance* for a more complex circuit around the diode. It means we can now solve currents in more complex circuits incorporating *one* diode so far).

It also opens up a new world to solving circuit equations for transistor circuits in the large signal domain when previously only the small signal domain could be analysed in detail. Bipolar transistors are modelled on the diode, so now transistor models can include the effects of series resistance in the actual model equations. Previously this has not been possible and the effect of series resistances were always solved outside the model by the numerical SPICE engine.

Since we are considering the practical relevance of V_k and the use of the diode plus resistor to model a transistor a two diode circuit is now used to represent a two transistor Class-AB output stage. **Figure 3.1d** show the basic circuit. Each diodes has a bias voltage set to V_k (606.8mV) and the diode currents are differenced for plotting a representation of the collector currents if two transistors were actually used. Here they are represented by a Beta of one. If you want a more realistic plot them multiply the curve by say 100.

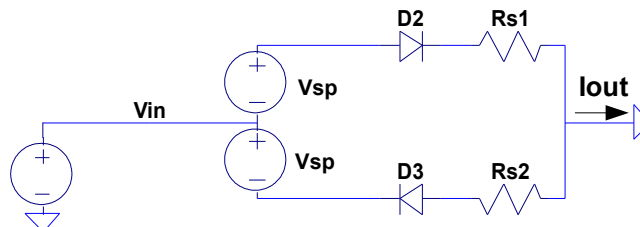


Figure 3.1d. Simple demo of Class-AB using the diode plus resistor equations

Figure 3.1e shows the plots using LTspice to solve the circuit and the plots from the calculated currents and gains using the above diode equations. The calculated currents and gains are very close to the SPICE results.

We are interested here in the significance of biasing at V_k . The gain plot or “wingspread” plot when V_k is used as the biasing voltage looks slightly under-biased for minimum crossover distortion. The dip in the gain at $V_{in}=0$ is measured on the LTspice as 722.6mA/V or 72% of the ultimate value at high output current. This 72% value is twice the 36% seen above for one diode at $V_{in}=V_k$ since there is a bias voltage added now to force V_k across each diode when $V_{in}=0$.

When the bias voltage is increased by one NV_t the wingspread curve looks more like optimum bias for minimum crossover distortion since any higher bias gives a larger overshoot in gain and distortion increases with one more NV_t bias voltage than this. Recall from Table 3.1 one NV_t above

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V_k we get half the maximum conductance, so with two diodes we get 100% total conductance when $V_{in}=0$ and we see this in Figure 3.1e.

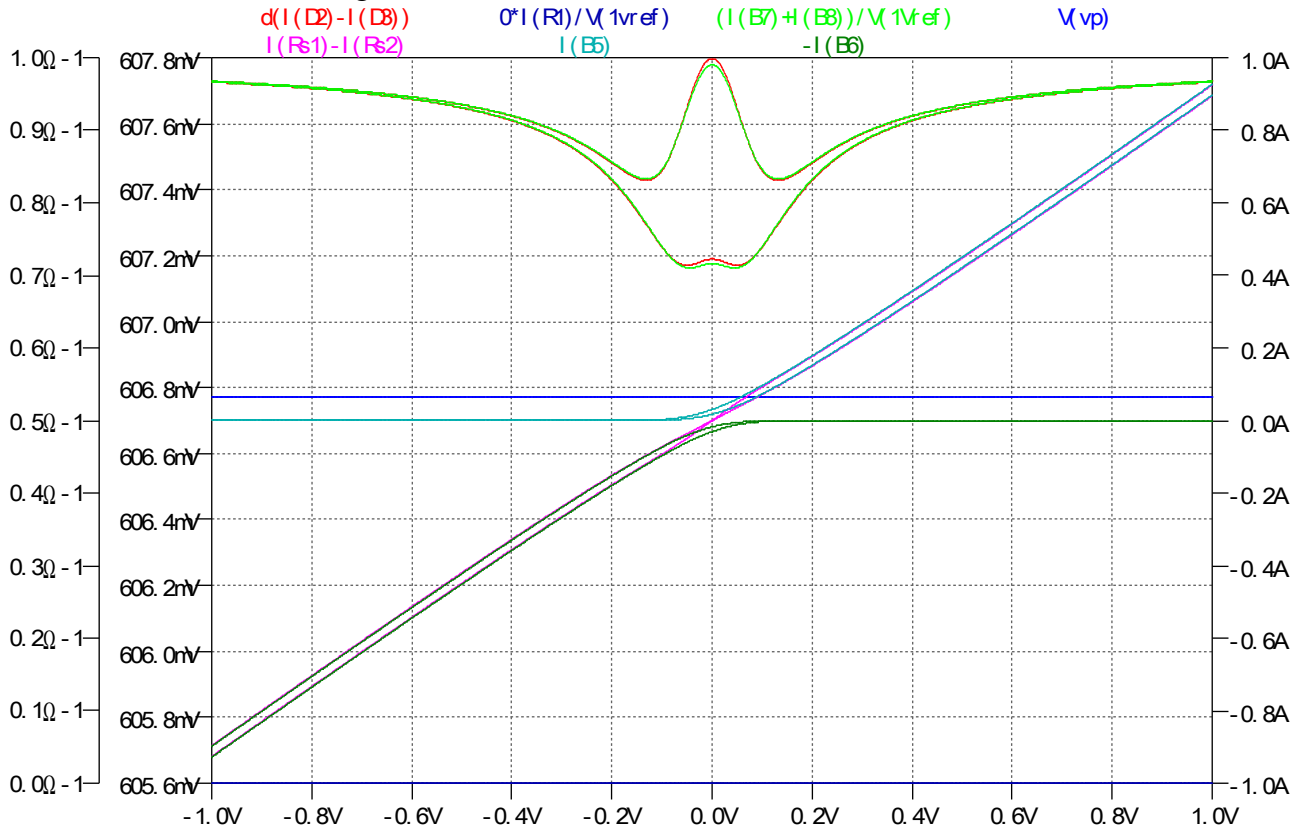


Figure 3.1e. Class-AB using diode plus resistor equations. LTspice circuit (red/pink) and equations (light and dark green). The bias spread voltage is stepped as V_k (607mV lower) and $V_k+1 \cdot NV_t$ (640mV upper)

Another observation for Class-AB is the conductance curve in Figure 3.1d passes through a region where the curve *appears* to be straight. The second derivative shows the slope is never constant. However a small range of input voltages can be identified where the slope is within say 10% of its maximum. The voltages where the slope is more than 90% of the peak are 574mV and 634mV (centred about 604mV or V_k).

What is interesting is a constant slope or gain ramp represents is a Square-Law relationship between V_{in} and I_d . We can see an almost Square-Law exists over this 60mV region. This can be called the “Square-Law Class-AB” region. **Figure 3.1f** zooms in on this range and it can be seen that the gain variation is very small in this region. If the two diode slopes were straight in this region then there would be a constant gain and the distortion would be zero. But there is a small variation in gain because the diode slopes are varying from 90% of the peak to the peak then back again to 90% of the peak. With larger input swings the gain eventually begins to rise monotonically toward the limit determined by R_s . So this is another insight into the practical relevance of V_k in circuit design.

Designing a power amplifier to exploit this low distortion Square-Law Class-AB region has lead to a number of designs [ref Olsson Dec 1994, Hegglun EW 199 & L|A Vol.4]. The above diode circuit (which is indicative of bipolar transistor) can be exploited by reducing the values of the emitter resistors in series with the power transistors and making them as small as possible while still being able to stabilise their quiescent current as their junction temperatures change, then to obtain even

lower series resistance several output pairs can be paralleled (each with their own safe-value emitter resistors).

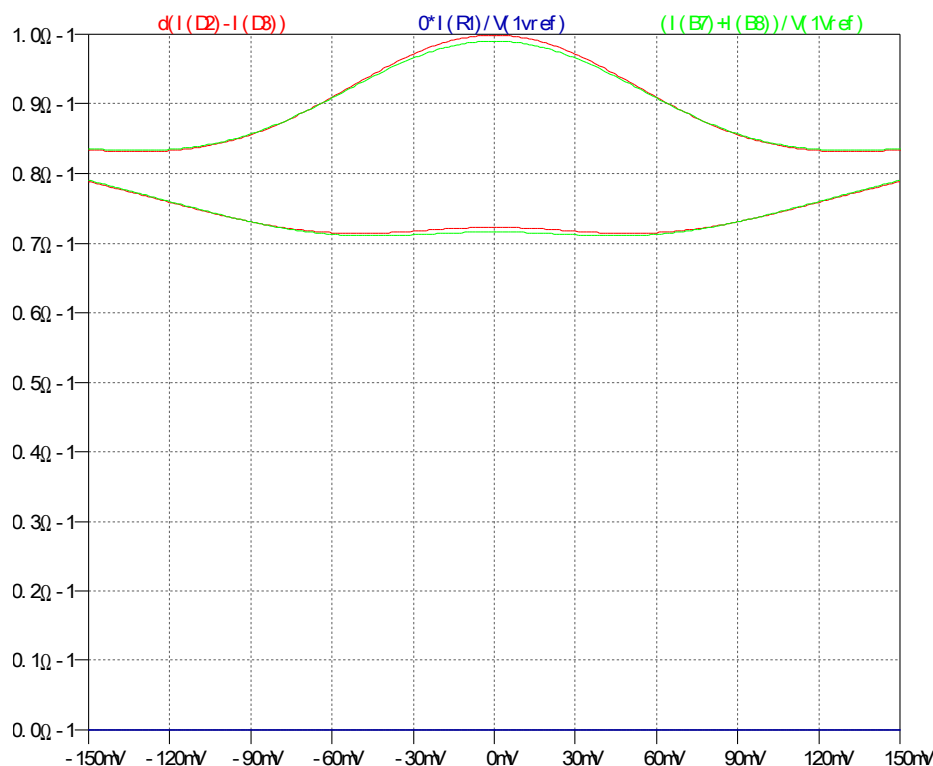


Figure 3.1f. A good approximation to a Square-Law Class-AB region can be obtained when the bias voltage is V_k (607mV lower curves) and the input voltage is plus or minus one NV_t (30mV). Red is LTspice, $R_e=1R$, Upper curve $V_{sp}=V_k+NV_t$ (640mV).

The diode equation solution can give valuable insights for circuit designers of transistor power amplifiers. The 'knee voltage' V_k is a key term in Class-AB power amplifiers and optimum biased Class-AB is normally biased at around one NV_t more than V_k for minimum distortion, while Square-Law Class-AB aims for bias at precisely V_k voltage for the best possible crossover function from each transistor. This gives an approximate Square-Law for plus or minus one NV_t of input signal variation.

Incidentally, at $V_{in}=V_k$ the inverse Lambert value is $W(1) = \Omega$ or the **Omega constant** with a value of approximately 0.5671432904097838729999686622, because an alternate name for the Lambert W-function is the *Omega function*. [http://en.wikipedia.org/wiki/Omega_constant] Omega is the intersection of $y=e^x$ and $y=x$ (or $y=\ln(x)$ and $y=-x$). The approximation used above returns $W(1)=0.557$ but the Omega constant is better stated as 0.567 to 3 significant figures (and 567 easier to remember).

So, maybe the knee voltage “ V_k ” should be called the 'Omega voltage' “ V_Ω ” instead? And, maybe, the BJT form of approximated Square-Law Class-AB which aims for bias at precisely the knee voltage V_k or Omega voltage “ V_Ω ” should be called Class- Ω (or Class-Omega)?

3.1.1 Alternative diode plus resistor solution

Back now to the task of how to derive circuit equations. Next we need to take a look at a second way to arrive at the same solution. This time starting with the diode voltage equation instead of the

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current equation. It is helpful to be able to use either approach.

Recall in the solution above we used the form $w e^w = x$ and $W(x)$ is the inverse function – given x we can find w . An alternative form of the Lambert W-function is

$$w = \ln\left(\frac{x}{w}\right) \quad 23/(23)$$

obtained from two simple steps, by dividing both sides by w then taking the $\ln$ of both sides.

This form may be easier for some if diode voltages involving logs are used rather than diode currents involving exponentials because more complex circuits usually use meshing KVL voltage loop equations and in this form diode and transistor junction voltages pop up naturally in these equations. This will be seen later.

So lets repeat the diode and resistor solution using the diode voltage log form rather the exponential current form. In this case we want to get an equation of the form of $w = \ln(x/w)$ where V_{in} only appears on the RHS, **and Id terms on the LHS must match the Id terms in the denominator of the log**, and the log has no terms in front of it.

Rewriting the Shockley diode equation in the log form we start with

$$V_{ak} = V_t \ln\left(\frac{I_d}{I_s} + 1\right) \quad (24)$$

Using Equation (4) $V_{ak} = V_{in} - I_d \cdot R_s$ we can eliminate V_{ak} , giving

$$V_{in} - I_d \cdot R_s = V_t \ln\left(\frac{I_d}{I_s} + 1\right) \quad (25)$$

First, remove the V_t in from of the log. Then move V_{in} to the RHS, giving

$$s \frac{-I_d \cdot R}{V_t} = \ln\left(\frac{I_d + I_s}{I_s}\right) - \frac{V_{in}}{V_t} \quad (26)$$

We look at the w term inside the log and need to invert it to get I_d in the denominator (recall $-\ln(x/y) = \ln(x/y)^{-1} = \ln(x/y)$) so we multiplying both sides by minus 1, giving

$$\frac{I_d \cdot R_s}{V_t} = \ln\left(\frac{I_s}{I_d + I_s}\right) + \frac{V_{in}}{V_t} \quad (27)$$

We note I_d in the denominator inside the log has a plus I_s term so we change the LHS to have the same form. So we add $I_s R_s / V_t$ to both sides, giving

$$\frac{(I_d + I_s) \cdot R_s}{V_t} = \ln\left(\frac{I_s}{I_d + I_s}\right) + \left(V_{in} - \frac{I_s \cdot R}{V_t}\right) \quad (28)$$

We also need R_s / V_t in the denominator inside the log so we multiply the numerator and denominator by R_s / V_t inside the log, giving

$$\frac{(I_d + I_s) \cdot R_s}{V_t} = \ln\left(\frac{I_s \cdot R_s / V_t}{(I_d + I_s) \cdot R_s / V_t}\right) + \left(V_{in} - \frac{I_s \cdot R_s}{V_t}\right) \quad (29)$$

Now move the V_{in} parts inside the log, noting $\ln(\exp(x)) = x$ and $\ln(a) + \ln(b) = \ln(ab)$, giving

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$$(Id + Is) \frac{Rs}{Vt} = \ln \left(\frac{\frac{Is \cdot Rs}{Vt} \exp \left(\frac{Vin - Is \cdot Rs}{Vt} \right)}{(Id + Is) \frac{Rs}{Vt}} \right) \quad (30)$$

We now have the form required for $w = \ln(x/w)$. We identify $w = (Id + Is) \frac{Rs}{Vt}$ which is the same as

we obtained above (Eq.14) and $x = \frac{Is \cdot Rs}{Vt} \exp \left(\frac{Vin + Is \cdot Rs}{Vt} \right)$ so we can write

$$x = \exp \left(\frac{Vin - V_k}{Vt} \right) \text{ where } V_k = Vt \cdot \ln \left(\frac{Vt}{Is \cdot Rs} \right) - Is \cdot Rs$$

Both paths lead to the same solution. Sometimes it is easier to use the voltage equation for a diode (as we will see later). When solving bipolar transistor circuits there are a number of two diode circuits that are helpful to solve first.

3.2 A Solution for two diodes in series with different IS's and equal NF's

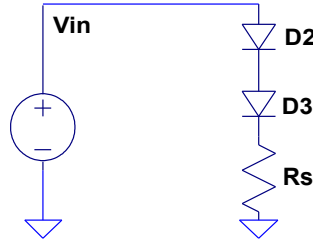


Figure 3.2. Two series diodes and Rs

The derivation for the current through two diodes in series with an equivalent resistance is similar to [Section 1.1] but we need to back up one step first and use the diode voltage equation

$V_{D1} = Vt \cdot \ln \left(\frac{Id}{Is1} + 1 \right)$ for D1 and $V_{D2} = Vt \cdot \ln \left(\frac{Id}{Is2} + 1 \right)$ and where Id flows through both diodes

being in series. The voltage loop equation for two diodes in series plus a resistor is

$Vin = V_{D1} + V_{D2} + Id \cdot Rs$ and substituting gives $Vin = Vt \cdot \ln \left(\frac{Id}{Is1} + 1 \right) + Vt \cdot \ln \left(\frac{Id}{Is2} + 1 \right) + Id \cdot Rs$ so

$Vin = Vt \cdot \ln \left(\frac{Id^2}{Is1 \cdot Is2} \right) + Id \cdot Rs$ or $Vin = Vt \cdot \ln \left(\left(\frac{Id}{\sqrt{Is1 \cdot Is2}} \right)^2 \right) + Id \cdot Rs$ or

$Vin = 2 \cdot Vt \cdot \ln \left(\frac{Id}{\sqrt{Is1 \cdot Is2}} \right) + Id \cdot Rs$ and $Vin = Vt_{12eq} \cdot \ln \left(\frac{Id}{Is_{12eq}} \right) + Id \cdot Rs$. where $Vt_{12eq} = 2 \cdot Vt$ and

$Is_{12eq} = \sqrt{Is1 \cdot Is2}$. This has reduced the two diodes to one equivalent diode with an equivalent saturation current of $\sqrt{Is1 \cdot Is2}$ and an equivalent thermal voltage of $2Vt$. The solution for different NF's is not provided here.

A LTspice plot is given later in **Section 3.4**. The above equations have not been tested in LTspice and no equations for this circuit appear in the PAK Version 1.0 spreadsheet.

3.3 Solution for two non-identical diodes in series and one diode has a parallel resistor [under construction]

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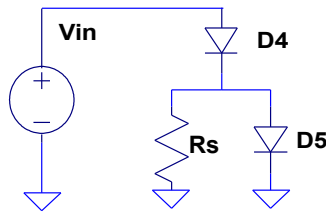


Figure 3.3. Two series diodes with one shunt R_s

No equation solutions are given in the LTspice circuit files or the PAK Version 1.0 spreadsheet for this circuit.

3.4 Solution for two diodes in parallel where each diode has a series resistor.

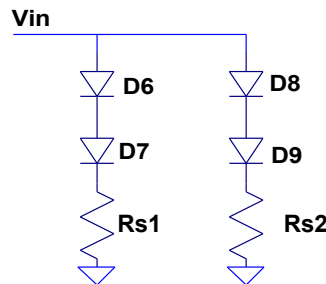


Figure 3.4. Two 2 series parallel diodes and different R_s 's.

Figure 3.1-4 shows a combined LTspice plot for Sections 3.1-4. No equation solutions are given in the LTspice circuit files or the PAK Version 1.0 spreadsheet for this circuit.

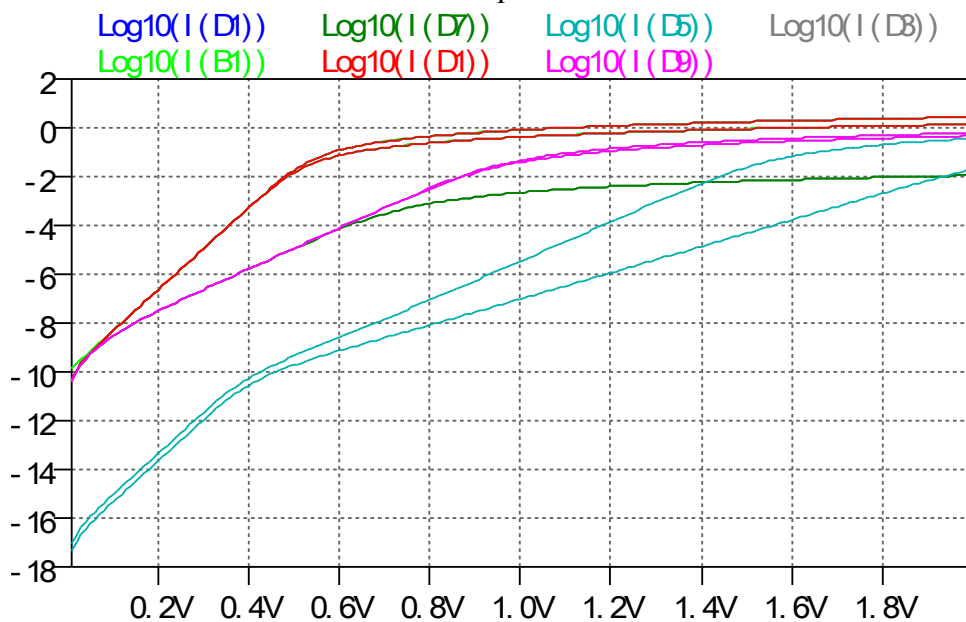


Figure 3.1-4. $\ln(I_d)$ plots for 4 circuits with R_s stepped $0.5R$ and $1R$: Sections 3.1 Single diode+ R as I_{d1} ; Sections 3.2 as I_{d3} ; Sections 3.3 as I_{d5} ; Sections 3.4 as I_d & I_{d9} .

Notice in **Figure 3.1-4** that I_{d5} (Figure 3.3. Two series diodes with one shunt R_s) gives a knee like high current Beta-fall. This is discussed further in Section 4.1.3.

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3.5 Solution for two diodes in anti-parallel with spread voltages and Rin (no Rs)

The solution of **Figure 3.5** is relevant to solving the Basic Push-Pull output stage Rin is large. In this circuit the diode series resistances (Rs) are removed so the effect of Rin can be solved without the complication of Rs.

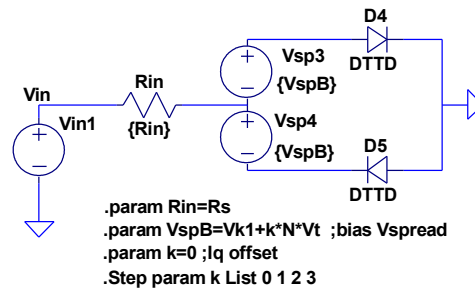


Figure 3.5. Circuit of two anti-parallel diodes with Vspread, no Rs for each diode and Rin provides the series resistance.

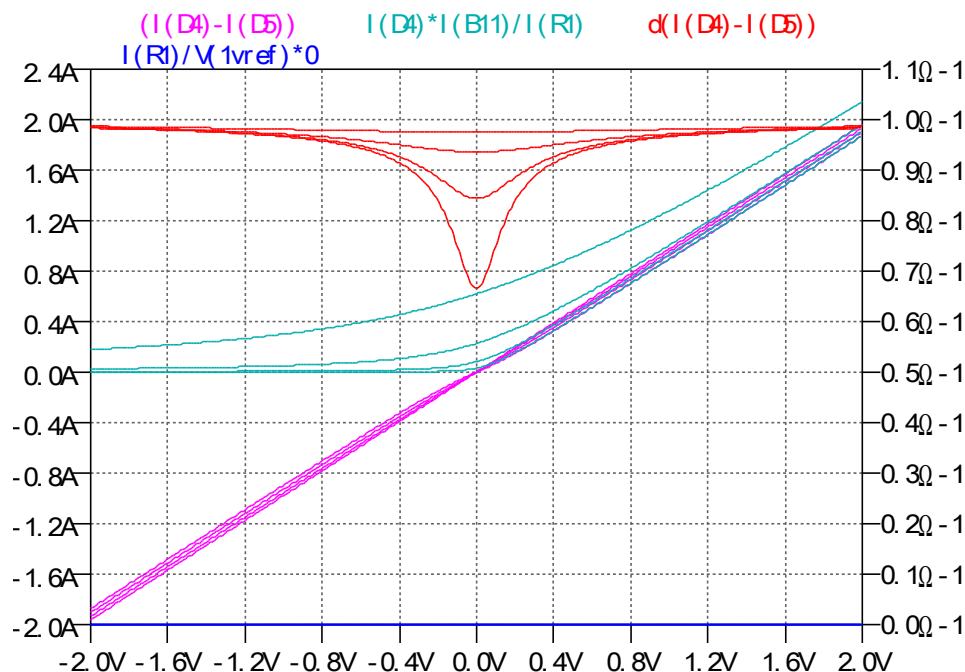


Figure 3.5a. 'Wingspread' Plot for two anti-parallel diodes with Vspread, no Rs for each diode. Rin=1R, IS=1e-10, N=1.2.

4. Solving single BJT circuits

The bipolar junction transistor (BJT) is usually modelled using the Ebers & Moll (EM) model from 1954 and enhanced to the Gummel & Poon (GP) model in 1970 – both model the collector current in the normal forward mode on the intrinsic (internal) base to emitter voltage (Vbe) – intrinsic means after the parasitic resistance voltage drops due to connections have been stripped away.

In the EM and GP models the equation for the primary collector current is

$$I_c = I_s \cdot \exp\left(\frac{V_{be}}{N \cdot V_t}\right) \quad (31)$$

where I_s is the collector-side saturation current and NV_t is thermal voltage Equation (2) $V_t = KT/q$ multiplied by a diode ideality slope factor N .

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The collector side saturation current appears to have been chosen because it makes it easier to formulate the collector current equation free from the second order effects of base leakage current which are significant at low input voltages. But the use of collector side saturation current is counter-intuitive; people normally assume that a base-emitter voltage develops a base current and the base current then causes the collector current. But in the EM & GP models it is the base-emitter voltage that directly generates the collector current and the base current is a smaller fraction of the collector current. The generation of collector current directly as a result of the base-emitter voltage in the EM and GP models is primarily a field-effect device, a transconductance meaning a voltage to current conversion.

When using the collector side approach the base current is I_c/Beta and the base current is calculated using

$$I_b = \frac{I_s}{\text{Beta}} \cdot \exp\left(\frac{V_{be}}{N V_t}\right) \quad (32)$$

The models used in the PAK spreadsheet usually use the collector side I_s following these standard SPICE models. This means resistances on the base side of a transistor circuit are referred back to the collector side as in the following example to allow the Lambert W-function to solve the collector current. Once the collector current is found then the base current and other circuit voltages can be calculated as well.

4.1 Common Emitter BJT with emitter degeneration and R_{in} & R_c

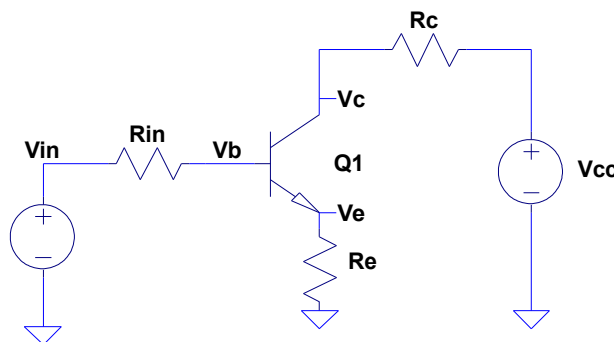


Figure 4.1. Single Ended Common Emitter amplifier

Figure 4.1 shows a Single Ended BJT Common Emitter amplifier with an emitter degeneration resistor R_e and a base resistor R_{in} and a collector load resistor R_c . R_{in} could be a base stopper resistor or it can be thought of as an equivalent output resistance for a signal generator or a preceding stage. If the preceding stage is actually a current source then R_{in} can be used as the Thévenin resistance that the current source develops an equivalent input voltage across (See section 4.4).

The next **Figure 4.1a** shows a BJT modelled by a base diode with a series base resistance R_{in} and the emitter resistance R_e . The R_e can be a parasitic (internal terminal) resistance or may be an external resistor such as an emitter degeneration resistor. The collector resistor resistance R_c is left out for now.

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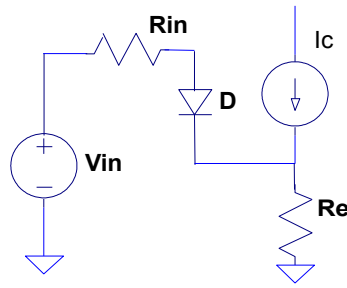


Figure 4.1a. Common Emitter BJT with R_{in} emitter degeneration by R_e . The basic BJT is modelled by a diode and controlled current source.

Incidentally, in circuit simulators the parasitic resistances R_{eP} , R_{bP} and R_{cP} are all 'external' to the EM & GP BJT model equations and these models rely on the 'SPICE engine' to solve their effects in the circuit. Here we will merge these parasitic resistances (and the other circuit resistances) into an explicit equation solution for the circuit. When this is achieved we don't need a 'SPICE engine' to solve our circuit – and if we do need to generate sweep plots then we can use a simple spreadsheet and our equation solutions.

To solve the circuit in Figure 4.1a we first combine R_e and R_{in} to simplify the circuit to one diode and one equivalent resistor (R_{eq}) (see **Figure 4.1b**) using

$$R_{eq} = \frac{R_{in}}{\beta} + R_e \left(\frac{1 + \beta}{\beta} \right) \quad (33)$$

This is called *referring* the base resistance to the collector side. R_{in} is divided by β to bring it to the collector side since $I_c = \beta \cdot I_b$. But why use the $(\beta + 1)/\beta$ for R_e ? Because resistor R_e carries emitter current (not collector current) so

$$I_e = I_c + I_b \quad \text{so} \quad I_e = I_c + \frac{I_c}{\beta} = I_c \left(1 + \frac{1}{\beta} \right) \quad \text{so} \quad I_e = I_c \left(\frac{\beta + 1}{\beta} \right) \quad (34)$$

Why the collector side and not the emitter side? The Lambert W inverse solution is for I_c . If we need I_e instead of I_c then use the equation above to convert I_c to I_e .

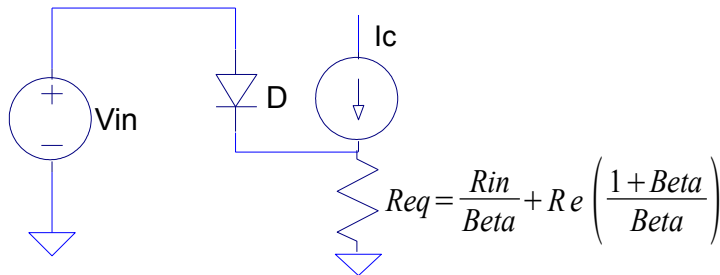


Figure 4.1b. Simplified BJT model with R_{in} brought into R_{eq} on the collector side.

We can now use our previous diode plus resistor solution to define the current as

$$I_c = \frac{N \cdot V_t}{R_{eq}} W \left(\exp \left(\frac{V_{in} - V_k}{N \cdot V_t} \right) \right) - I_s \quad \text{where} \quad V_k = N \cdot V_t \cdot \ln \left(\frac{N \cdot V_t}{I_s \cdot R_{eq}} \right) - I_s \cdot R_{eq} \quad \text{and} \quad (35)$$

$$R_{eq} = \frac{R_{in}}{\beta} + R_e \cdot \left(\frac{1 + \beta}{\beta} \right)$$

where I_s is the collector side saturation current as used in SPICE models. The “ $W(\exp((V_{in} -$

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$V_k)/(NV_t))$ ” part is the Lambert W-inverse function – it can be calculated using the approximation given in Equation (18). Incidentally, the base diode D in Figure 4.1b is not the actual diode we associate with R_{eq} . We need to imagine that R_{eq} is in series with a virtual diode “Deq” on the collector side and having a saturation current of I_s (which is actually Beta times the saturation current of the base diode D). This virtual diode arises from the collector-side I_c equation adopted in the EM and GP models. If however, the emitter resistance had been referred to the base side (giving R_{eq}') then it makes more sense (to most people) because R_{eq}' is then series with the base diode D and the equations above would give a base current which then *produces* the collector current when multiplied by the Beta. Alternate views are discussed further in Section 4.1.3 (High-current Beta-fall).

Anyhow, we *have* succeeded in solving a BJT circuit that includes emitter degeneration and a base resistance. We can now write equations for a Common Emitter amplifier with a collector resistor. The Common Collector circuit solution is essentially the same except the collector resistor is not used and the output is taken at the emitter instead of the collector.

The CE solution the output voltage V_c is calculated using I_c from above with

$$V_{out} = V_c = I_c \cdot R_c \quad (36)$$

Figure 4.1c shows I_c and output voltages V_c for the CE case. No V_{ce} saturation is included so I_c and V_{ce} can exceed their physical limits. LTspice (red/pink) and ABM equations (cyan/gold). $V_{cc}=10V$, $R_{in}=1k$, $R_c=1k$, $R_e=1mR$, $10R$ and $100R$.

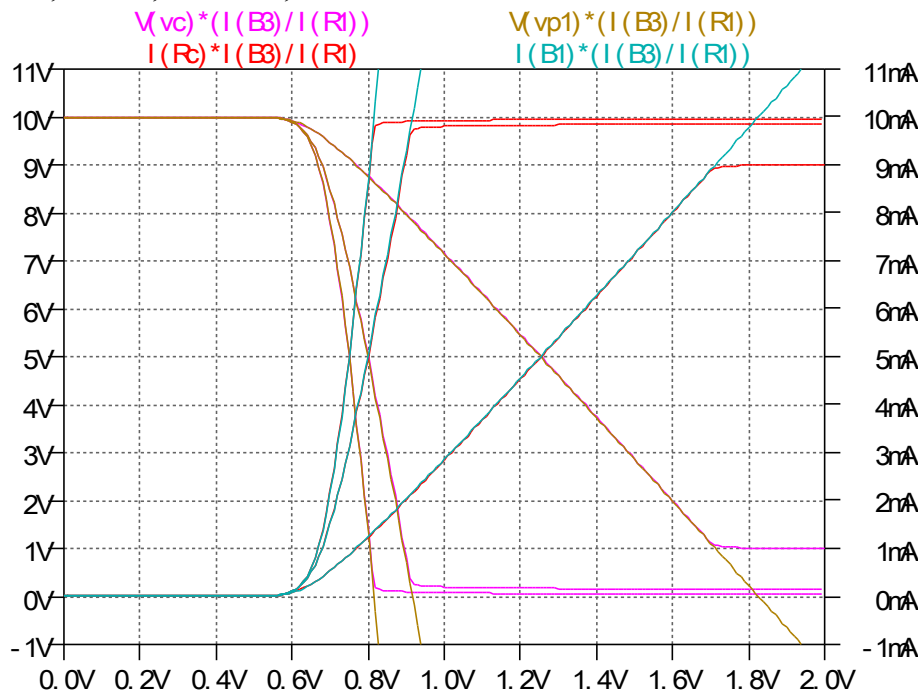


Figure 4.1c. Single Ended Common Emitter amplifier. No V_{ce} saturation effect is included so I_c and V_{ce} can exceed their physical limits. LTspice (red/pink) and ABM equations (cyan/gold). $V_{cc}=10V$, $R_{in}=1k$, $R_c=1k$, $R_e=1mR$, $10R$ and $100R$. Beta=100.

4.1.1 SE CE amplifier with V_{ce} saturation

Saturation can be approximated by restricting the maximum collector current using

$$I_c = \text{Max}(I_{c_{Gen}}, I_{c_{Sat}}) \quad \text{where} \quad I_{c_{Sat}} = \frac{V_{cc} - V_t \cdot \ln(\text{Beta})}{R_c + R_e} \quad (37)$$

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Where $I_{c_{Gen}}$ is an ideal current source that has no restrictions on the voltage across it. The Max function provides the hard limit at I_{cSat} .

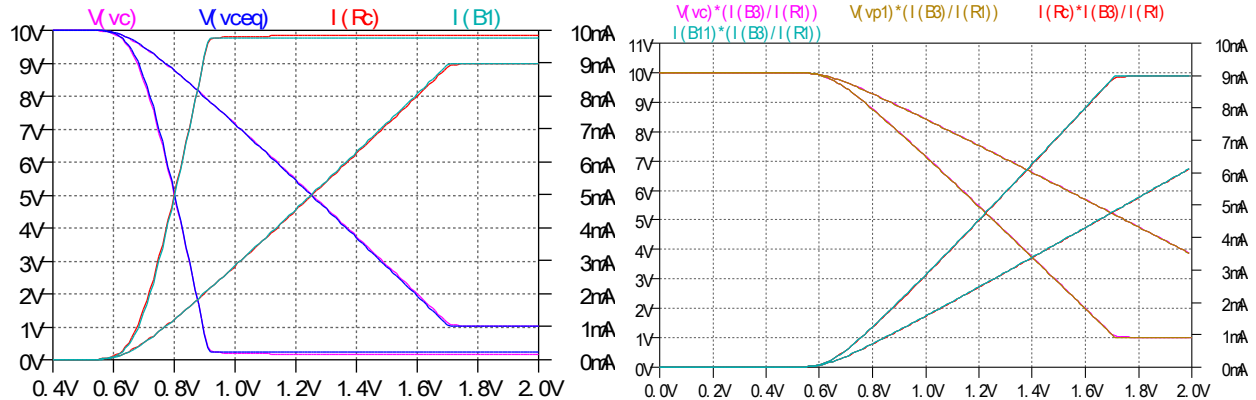


Figure 4.1d. Single Ended Common Emitter amplifier. I_c saturation using $\text{Max}(I_c, I_{cSat})$ Left: $V_{cc}=10V$, $R_{in}=1k$, $R_c=1k$, $R_e=10R$ and $100R$, $\beta=100$. Right: β stepped 10 and 100, $R_{in}=R_c=1k$, $R_e=100R$.

The current limit used here is based on a V_{ce} saturation voltage of $V_t \cdot \ln(\beta)$ since it is supposed that the base-collector diode carries β times the excess base current while the base-emitter junction carries I_c/β as usual. This means in moderate saturation

$$\frac{I_{bc}}{I_{be}} = \beta \quad \text{and} \quad \frac{NF}{NR} \cdot V_t \cdot \ln(\beta) \quad (38)$$

where NF and NR are the ideality factors which are assume to be the same for convenience here. Some rounding can be included using an alternative user defined Max function with 0.1mA of rounding for a maximum current of 10mA in this example.

A more accurate saturation solution is on the wish-list for a later PAK version. A better approach may be to collapse β_{eff} from β to zero in the limit as V_{in} becomes large causing I_c to reach a saturation limit and then I_c is used in $V_{cc} - I_c \cdot R_{Tot}$ to give the output voltage. The $\tanh(x)$ function can be used as a switch since it involves the Exponential function and gives a similar slope to actual transistors going into saturation.

4.1.2 SE CE amplifier with the Early effect

Adding second order effects such as the Early effect to a Lambert W solution adds unwanted complications to otherwise relatively simple solutions. But these effects are desirable to include because they *can* contribute a significant amount of additional nonlinearity and therefore distortion. Also it is helpful to include them, to show that their effect is insignificant in certain cases and can be neglected, to allow simpler and easier equations solutions.

The Early effect is where collector to emitter voltage adds a nonlinear current component collector current source. Since the Early effect current varies with applied collector-emitter voltage it is modelled as a shunt resistance across the current source that passes the collector current which is free of the Early effect. This shunt resistance is really a variable conductance represented by the slope dI_{ce}/dV_{ce} . The average slope of a dI_{ce}/dV_{ce} curve is a straight line passing through a common point on the $-V_{ce}$ x-axis called the Early voltage $-V_A$. Since dI_{ce}/dV_{ce} is approximated to a straight line the Early voltage $-V_A$ is independent of I_c and the slope is directly proportional to the collector current. This means the shunt conductance for the Early effect is directly proportional to the collector current, or the shunt conductance is given by $g_o = I_{co}/V_A$.

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This equation allows the Early effect to be added to the transistor equation model. If there was no emitter resistance then I_{co} is independent of the current through the shunt conductance g_o . This allows the shunt conductance g_o to be included in the next stage as a source resistance to allow the solution for output current of the second stage drive by the first stage.

Emitter resistance provides some local feedback which reduces the effective shunt conductance $g_o = I_{co}/V_A$ (it increases the shunt resistance). If we assume that I_{co} is not affected by the current through the shunt conductance g_o then local feedback from R_e raises the effective Early voltage and a general solution is possible using the general feedback equation with loop gain $g_m \cdot R_e$ where g_{m0} is calculated along with the Lambert W solution for I_{co} (with the Early effect added to g_{m0}). This gives a higher effective Early voltage

$$V_{a_{eff}}(R_e) = V_A \left(1 + R_e \cdot g_{m0} \left(1 + \frac{V_{ce}}{V_A} \right) \right) \quad (39)$$

The collector current with the Early effect and R_e local feedback is

$$I_c(R_e, V_A) = I_{co} \left(1 + \frac{V_{ce}}{V_{a_{eff}}} \right) \quad 40/(40)$$

for a constant V_{ce} (no resistors in the collector and the subtractive effect of V_{Re} on V_{ce} can be neglected).

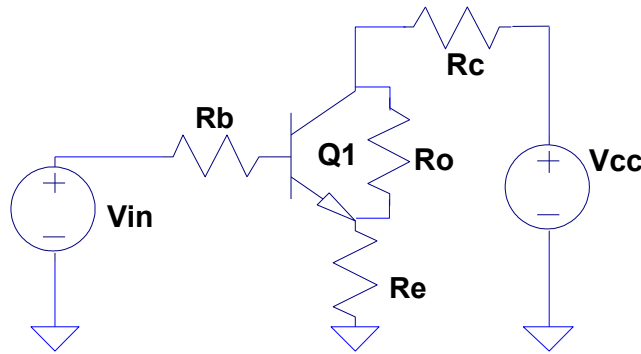


Figure 4.1e. Common Emitter amplifier. The equivalent shunt resistance across collector-emitter is due to the Early effect.

Next, V_{ce} will vary over a wide range with I_c . Since $I_c(V_A)$ depends on V_{ce} and V_{ce} depends on $I_c(V_A)$ another equation must be solved involving the effect of the load resistance R_c and R_e on V_{ce} . With **Figure 4.1e** we use Equation (40) above and write

$$V_{ce} = V_{cc} - I_c(V_A, R_e) \cdot R_c - I_c(V_A, R_e) \left(\frac{\beta + 1}{\beta} \right) \cdot R_e \quad 41/(41)$$

where the current through R_o is included in $I_c(V_A)$. Substituting V_{ce} gives

$$I_c(R_e, V_A) = I_{co} \left(1 + \frac{V_{cc}}{V_{a_{eff}}} \right) - I_c(V_A, R_e) \cdot I_{co} \left(\frac{R_c}{V_{a_{eff}}} + \left(\frac{\beta + 1}{\beta} \right) \cdot \frac{R_e}{V_{a_{eff}}} \right) \quad \text{and collecting terms}$$

gives

$$I_c(\text{Re}, V_a) = \frac{I_{co} \left(1 + V_{cc} / (V_{a_{\text{eff}}}) \right)}{1 + \frac{I_{co}}{V_{a_{\text{eff}}}} \left(R_c + R_e \left(\frac{\text{Beta} + 1}{\text{Beta}} \right) \right)} \quad \text{Where} \quad (42)$$

$$V_{a_{\text{eff}}}(\text{Re}) = V_a \left(1 + g_{m_0} R_e \left(1 + \frac{V_{ce_{\text{est}}}}{V_a} \right) \right) \quad \& \quad V_{ce_{\text{est}}} \approx V_{cc} - I_{co}(R_c + R_e)$$

where g_{m_0} is calculated along with the Lambert W solution for I_{co} . Note the R_e in the $I_c(\text{Re}, V_a)$ equation does not include the base resistance referred to the emitter (since the equation is derived from the voltage drop across R_e due to I_e).

We can check this, first setting $R_c=0$ and $R_e=0$ giving the expected result

$$I_c(V_a) = I_{co} \left(1 + V_{cc} / (V_{a_{\text{eff}}}) \right) \quad . \text{ If we set } R_e=0 \text{ and } R_c=R_o (=I_{co}/V_{a_{\text{eff}}}) \text{ then}$$

$I_c(V_a) = \frac{1}{2} I_{co} \left(1 + V_{cc} / (V_{a_{\text{eff}}}) \right)$ which is expected since R_o drains half the current away from R_c (effectively forming a current divider).

The above solution can be used for the Voltage Follower by setting $R_c=0$ and R_e is now R_L (alternatively, R_e may appear in parallel with R_e).

The above can also be used for the Common Emitter stage (with emitter degeneration and/or base resistance). The VAS stage in a power amplifier is an example. **Figure 4.1f** shows an additional loading stage with resistance R_L and terminated to an arbitrary voltage V_o . This circuit can be solved using superposition where

$$V_{cc_{\text{eff}}} = V_{cc} \left(\frac{R_L}{R_L + R_c} \right) + V_o \left(\frac{R_c}{R_L + R_c} \right) \quad \& \quad R_{c_{\text{eff}}} = R_c \parallel R_L \quad (43)$$

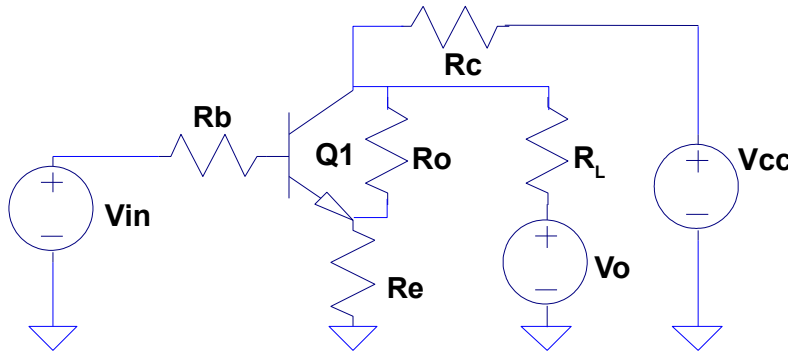


Figure 4.1f. Common Emitter amplifier with the Early effect with a generalised load R_L

4.1.3 High current Beta-fall [under construction]

Power amplifiers commonly use a CC Voltage Follower output stage and therefore Beta variations in the output transistors can affect the loading on the VAS stage. This means equations for output stages should include the high current Beta-fall effect, if only to see if Beta-fall is significant or not.

Figure 4.1g shows an Gummel plot for a power transistor with $I_{kf}=10A$ and $B_f=100$. The usual effects of R_b and R_e at high current have been made very small so we can see the intersection of the base current line and the collector current line at high current at “A”. The ratio of collector current

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to base current (effective Beta) is shown as a third line (blue). The effective Beta peaks at around 85% of the value set in the model (here it is 100).

The bell shape curve for the effective Beta is created by base leakage current at the low current LHS. “B” marks the intersection where the leakage current equals the collector current. The leakage current looks like a second diode in parallel with the main base diode (the one that gives a $B_f=100$) and this is how the EM and GP models show it in an equivalent circuit to represent the model equations.

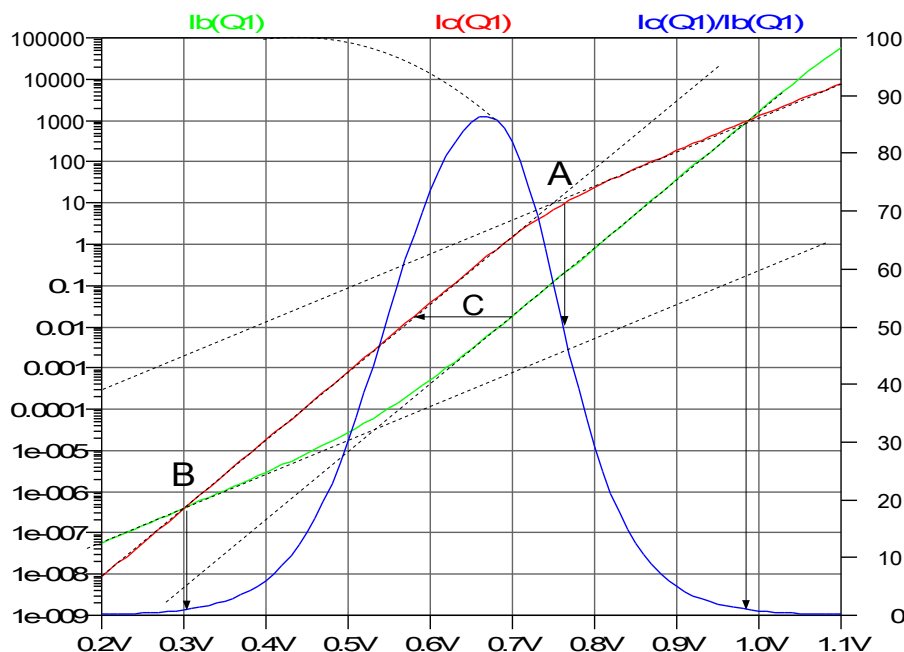


Figure 4.1g. Beta derived from Gummel plot with $I_{kf}=10A$ and R_b and R_e effects have been made close to zero effect. Red is collector current and green is the base current. Blue is the calculated current gain (I_c/I_b).

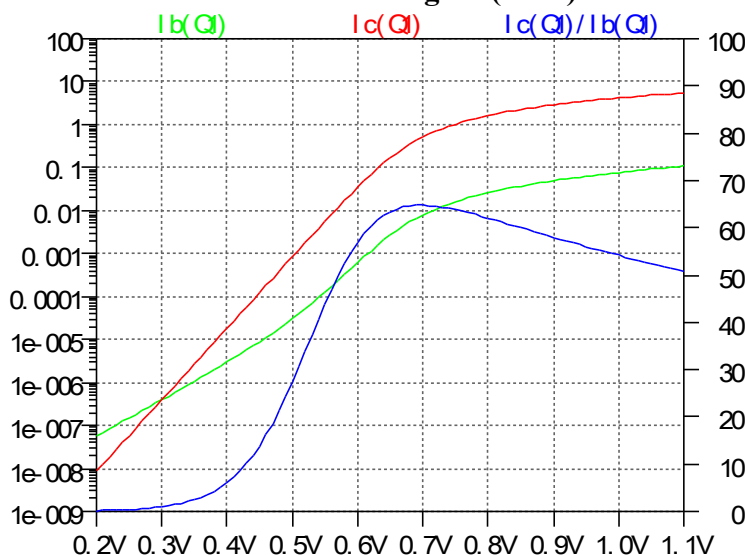


Figure 4.1h. Beta derived from Gummel plot with typical MJL21194C R_b and R_e values. Note how the base voltage falls over for V_{in} above 0.7V due to R_b and R_e degeneration effects.

Interestingly, at high currents a Gummel plot with effects of R_b and R_e stripped away shows the

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base current versus base-emitter voltage continuing rise after we see the collector current change to another slope above the I_{kf} knee. It shows that the base current remains “ideal” and is not affected by what is happening in the collector part. This can be used as evidence that the collector current is not produced by the base current *intrinsically* but instead the collector current arises from the base-emitter voltage as is assumed by EM and GP model equations.

As mentioned above, this leads to the concept of a *virtual diode* at the collector side with a larger saturation current than the base diode (B_f times) and it is this *virtual diode* that changes its ideality factor N_f (or slope) from its normal value at low currents to a higher value at high currents (lower slope). It means the junction transistor effectively operates as a field controlled device. One possible explanation is an *offset voltage* to V_{be} that moves the virtual diode plot to the left by $N_f \cdot V_t \cdot \ln(B_f)$ volts as shown by “C” (120mV for $B_f=100$). This offset voltage is related to the base width; the thicker the base the less offset voltage is 'seen' by the collector.

EM and GP noticed that generally the slope at high current halves so N_f eventually becomes $2 \cdot N_f$. This is generally obscured on a typical Gummel plot and requires special effort to see this effect. Figure 4.1h when compared to Figure 4.1g shows how the base voltage falls over for V_{in} above 0.7V due to R_b and R_e degeneration effects and obscured the internal base-emitter voltage relationship from being noticed. It may be thought that R_b and R_e swamp the effect of Beta-fall and we can ignore it but R_b and R_e degeneration do not affect the base current relationship with the collector current and this is seen in the beta curve still showing fall in current gain even at moderate currents of 5A here where I_{kf} is 14A so we are still well below the knee current.

The Ebers-Moll BJT model used a parameter θ to model the knee point for the transition and Gummel-Poon BJT used a similar term “ I_{kf} ”. The change of ideality slope of N to $2N$ represents a change from the normal diode current relationship with V_{be} to a square-root of the normal current. The transition from one to the other turns out to be a solution of a quadratic equation which can be written using the quadratic formula. Recall, $N=1$ means the slope of I_c vs V_{be} gives a change of 1 decade of current for 60mV, so a half slope with $N=2$ requires 120mV per decade of current.

Gummel-Poon BJT model high current Beta-fall is modelled using

$$I_b = \frac{I_c}{\frac{B_f}{(1 + I_c / I_{kf})}} \quad (44)$$

where I_{kf} is the collector current where Beta-falls to 50% of Beta with no Beta-fall. Since I_c appears in this equation twice the collector current must be found using the quadratic equation

$$I_c = \frac{\sqrt{1 + 4 I_{co} / I_{kf}} - 1}{2 / I_{kf}} \quad \text{where} \quad I_{co} = B_f \cdot I_b \quad (45)$$

I_{co} is the collector current calculated from the ideal diode equation with no Beta-fall. This shows that I_{co} follows the square-root of I_{co} at high currents ($I_{co} \gg I_{kf}$) as shown on the Gummel plot. But when I_{co} is $\ll I_{kf}$ the binomial expansion gives $I_c \sim I_{co}$ which is a straight line on the Gummel plot.

Including I_{kf} in the Lambert W solution for the CE stage with R_{in} and R_e

Introducing a series resistance the solution for high current Beta-fall involves a quadratic solution of

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$$I_c(I_{in})_{Q10} = \frac{NVt}{\frac{R_{in}}{Beta} + R e \cdot \left(\frac{Beta + 1}{Beta} \right)} W(x) \quad \text{where } Beta = \frac{BF}{1 + I_c / IKF} \quad (46)$$

giving

$$I_c^2 \left(\frac{R_{in}}{BF \cdot IKF} \right) + I_c \left(R e' + \frac{R_{in}}{BF} \right) - Vt \cdot W(x) = 0 \quad (47)$$

The solution is $I_c = \frac{\sqrt{(b^2 - 4ac)} - b}{2a}$ where a,b,c can be identified in the previous quadratic equation.

The Gm is obtained by differentiating the quadratic equation with respect to 'c' using the chain rule

$$\text{giving } Gm = \frac{\left(1 + \frac{Rp}{((Bf + 1) \cdot R e')} \right) \frac{W(x)}{1 + W(x)}}{\sqrt{\left(R e' + \frac{R_{in}}{(BF + 1)} \right)^2 + 4 \left(\frac{Rp \cdot Vt \cdot W(x)}{(BF + 1) \cdot IKF} \right)}}$$

Since the square-root expression is the discriminant $\sqrt{(b^2 - 4ac)}$ an alternative form can be calculated from the quadratic equation $\sqrt{(b^2 - 4ac)} = b + 2a I_c$ giving

$$Gm = \frac{\left(1 + \frac{Rp}{((BF + 1) R e')} \right) \frac{W(x)}{1 + W(x)}}{\left(R e' + \frac{R_{in}}{(Bf + 1)} \right) \left(1 + \frac{2 I_c}{IKF} \right)} \quad \text{where } I_c \text{ has usually already been calculated.}$$

The above equations were tested in earlier LTspice ABM's but they are not included in the LTspice circuit files or the PAK Version 1.0 spreadsheet.

An alternative approach to model high Current Beta-fall

In **Figure 3.1-4** it was noted that the Id5 plot for two series diodes with a shunt Rs across one of the diodes gives a high current Beta-fall knee. It may provide an equation solution and may enable to an alternate SPICE model for the BJT which includes parasitic resistances in the equations to reduce or eliminate annoying non-convergence problems of the present SPICE BJT models. It also means a SPICE subcircuit can be formed to add various non-idealities of a BJT such as high current (and low current Beta-fall in next section).

4.1.4 Low current Beta-fall

In the Gummel-Poon model the base current is non-ideal and is modelled effectively by two diodes in parallel – one of these diodes carrying current in relation to I_c and the other diode a leakage current unrelated to I_c . The 'leakage diode' is modelled as having a high series resistance so its current reaches a limit of a percent or less of the first diode when the base-emitter voltage is above the knee voltage of approximately 0.6V. This combination generates dual-slopes on the base current curve in the Gummel plot. The 'leakage diode' has a higher slope modelled by NE (where $NE > NF$ and typically $NE \sim 2$).

This effect is demonstrated in circuit **Figure 4.1i** (L) using two diodes in series D3 and D4, with

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D2 representing the collector current part. **Figure 4.1j (R)** shows the individual currents in each branch and the total input current. With these settings the low current Beta knee occurs at around 0.25V.

No equation solutions are given in the LTspice circuit files or the PAK Version 1.0 spreadsheet for low current Beta-fall.

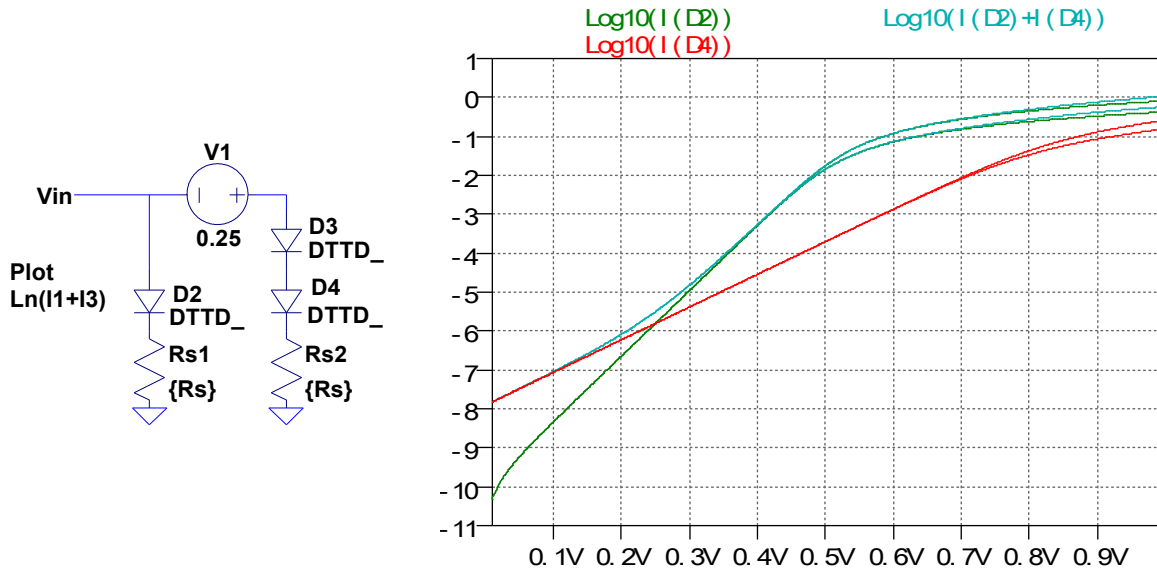


Figure 4.1i (L). Circuit for generating low current Beta-fall for the base-emitter diode leakage using D3 and D4 with D2 representing the collector current part. Figure 4.1j(R) the currents. Rs is stepped 0.5R and 1R

This circuit may provide a way to model low current Beta-fall in future equation solution and may enable to an alternate SPICE model for the BJT.

4.2 SE CE with Shunt Voltage Feedback

In this circuit a feedback resistor is added from collector to base for a Common Emitter amplifier as shown in **Figure 4.2**. The derivation of the solution published by Thomas Banwell [ref 1 Banwell Fig.12 Eq.21] but without emitter degeneration (R_e). The solution below now includes R_e .

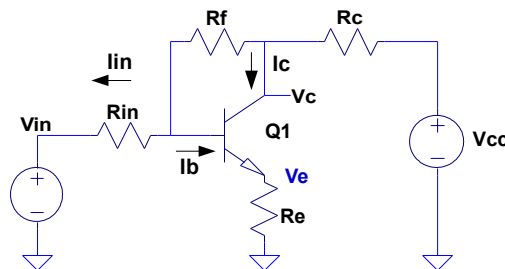


Figure 4.2. Common Emitter amplifier with base to collector and emitter resistors

Three equations are written. First the input voltage to the base node:

$$\text{For } R_{in} \text{ write } (V_{be} + V_e) - V_{in} = I_{in} \cdot R_{in} \text{ where } V_e = I_e \cdot R_e \quad (48)$$

Note I_{in} is defined as flowing from the power source through R_c and R_f . Although voltage loops can be defined arbitrarily, this an intuitive approach (based on the power source). Repeating this for the voltage across R_f due to current I_{in} plus I_b :

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$$\text{for } R_f \dots V_c - (V_{be} + V_e) + I_{in} \cdot R_f + \frac{I_c}{\beta} \cdot R_f \quad (49)$$

$$\text{for } R_c \dots V_{cc} - V_c = I_{in} \cdot R_c + I_c \cdot R_c + \frac{I_c}{\beta} \cdot R_c \quad (50)$$

We first eliminate V_c using (38) and (39) then eliminate I_{in} using (37). Rearrange (38) and (39)

$$V_c = (V_{be} + V_e) - I_{in} \cdot R_f - \frac{I_c}{\beta} \cdot R_f \text{ and } V_c = V_{cc} - I_{in} \cdot R_c - I_c \cdot R_c - \frac{I_c}{\beta} \cdot R_c$$

$$\text{equating } (V_{be} + V_e) - I_{in} \cdot R_f - \frac{I_c}{\beta} \cdot R_f = V_{cc} - I_{in} \cdot R_c - I_c \cdot R_c - \frac{I_c}{\beta} \cdot R_c$$

$$\text{collecting terms } \dots I_c \left(R_c + \frac{R_f + R_c}{\beta} \right) = V_{cc} - (V_{be} + V_e) - I_{in} (R_f + R_c)$$

Substitute $R_c' = R_c + (R_f + R_c) / \beta$. Now eliminate I_{in}

$$I_c R_c' = V_{cc} - (V_{be} + V_e) - \left(\frac{R_f + R_c}{R_{in}} \right) ((V_{be} + V_e) - V_{in})$$

Since V_e is $I_e R_e = I_c R_e (\beta + 1) / \beta$ we can now collect V_e into I_c . First get V_e to LHS

$$I_c R_c' + V_e \left(1 + \frac{R_f + R_c}{R_{in}} \right) = V_{cc} - \left(1 + \frac{R_f + R_c}{R_{in}} \right) V_{be} - \left(\frac{R_f + R_c}{R_{in}} \right) V_{in}$$

then substitute V_e and substitute $R_{Tot} = R_{in} + R_f + R_c$

$$I_c R_c' + I_c R_e \left(\frac{\beta + 1}{\beta} \right) \left(\frac{R_{Tot}}{R_{in}} \right) = V_{cc} - \left(\frac{R_{Tot}}{R_{in}} \right) V_{be} - \left(\frac{R_f + R_c}{R_{in}} \right) V_{in}$$

$$I_c \left(R_c' + R_e \frac{R_{Tot}}{R_{in}} \left(\frac{\beta + 1}{\beta} \right) \right) = V_{cc} - \left(\frac{R_{Tot}}{R_{in}} \right) V_{be} - \left(\frac{R_f + R_c}{R_{in}} \right) V_{in} \text{ so simplify to}$$

$$I_c R_c'' = V_{cc} - \left(\frac{R_{Tot}}{R_{in}} \right) V_{be} - \left(\frac{R_{Tot}}{R_{in}} \right) V_{in} \text{ where } R_c'' = R_c' + R_e \frac{R_{Tot}}{R_{in}} \left(\frac{\beta + 1}{\beta} \right) \quad (51)$$

Next, V_{be} can be eliminated using the log form of the diode Equation (23), giving

$$I_c R_c'' = V_{cc} - \left(\frac{R_{Tot}}{R_{in}} \right) N V_t \cdot \ln \left(\frac{I_c}{I_s} + 1 \right) - \left(\frac{R_f + R_c}{R_{in}} \right) V_{in} \quad (52)$$

We now proceed to get Equation (41) into the Lambert W-function form of $w = \ln(x/w)$ with I_c or w terms on the LHS and $\ln(x/w)$ terms on the RHS where the log denominator w must match the LHS w precisely. Moving terms in front of the log gives

$$I_c R_c'' \left(\frac{R_{in}}{R_{Tot} N V_t} \right) = - \ln \left(\frac{I_c}{I_s} + 1 \right) - \left(\frac{R_{in}}{R_{Tot} N V_t} \right) \left(\left(\frac{R_f + R_c}{R_{in}} \right) V_{in} + V_{cc} \right)$$

$$I_c \frac{R_c'' R_{in}}{R_{Tot} N V_t} = \ln \left(\frac{I_s}{I_c + I_s} \right) - \left(\left(\frac{R_f + R_c}{R_{Tot} N V_t} \right) V_{in} + \left(\frac{R_{in}}{R_{Tot} N V_t} \right) V_{cc} \right)$$

Add I_s to both sides

$$(I_c + I_s) \frac{R_c'' R_{in}}{R_{Tot} N V_t} = \ln \left(\frac{I_s}{I_c + I_s} \right) - \left(\left(\frac{R_f + R_c}{R_{Tot} N V_t} \right) V_{in} + \left(\frac{R_{in}}{R_{Tot} N V_t} \right) V_{cc} \right) + I_s \frac{R_c'' R_{in}}{R_{Tot} N V_t}$$

So we can write this as

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$$(I_c + I_s) \frac{Req}{NV_t} = \ln \left(\frac{\left(I_s \frac{Req}{V_t} \right)}{\left(I_c + I_s \right) \left(\frac{Req}{V_t} \right)} \right) - \left(\left(\frac{R_f + R_c}{R_{Tot} NV_t} \right) V_{in} + \left(\frac{R_{in}}{R_{Tot} NV_t} \right) V_{cc} \right) + I_s \frac{Req}{NV_t} \text{ where} \quad (53)$$

$$Req = \frac{R_{in} R_{c''}}{R_{Tot}} \quad \text{or} \quad Req = \frac{R_{in}}{R_{Tot}} \left(R_c \left(\frac{Beta + 1}{Beta} \right) + \frac{R_f}{Beta} \right) + R_e \left(\frac{Beta + 1}{Beta} \right)$$

Since we have matched LHS & RHS w terms we can now bring the remaining RHS terms inside the log

$$(I_c + I_s) \frac{Req}{NV_t} = \ln \left(\frac{\left(I_s \frac{Req}{NV_t} \right) \exp \left(\left(\left(\frac{R_f + R_c}{R_{Tot} NV_t} \right) V_{in} + \left(\frac{R_{in}}{R_{Tot} NV_t} \right) V_{cc} \right) + I_s \frac{Req}{NV_t} \right)}{\left(I_c + I_s \right) \left(\frac{Req}{NV_t} \right)} \right) \quad (54)$$

and write the inverse as

$$(I_c + I_s) \frac{Req}{NV_t} = W \left(\left(I_s \frac{Req}{NV_t} \right) \exp \left(\left(\left(\frac{R_f + R_c}{R_{Tot} NV_t} \right) V_{in} + \left(\frac{R_{in}}{R_{Tot} NV_t} \right) V_{cc} \right) + I_s \frac{Req}{NV_t} \right) \right)$$

then solving for Ic gives

$$I_c = \frac{NV_t}{Req} W \left(\left(I_s \frac{Req}{NV_t} \right) \exp \left(\left(\left(\frac{R_f + R_c}{R_{Tot} NV_t} \right) V_{in} + \left(\frac{R_{in}}{R_{Tot} NV_t} \right) V_{cc} \right) + I_s \frac{Req}{NV_t} \right) \right) - I_s \quad (55)$$

Expressing in terms of V_k and a common denominator gives

$$I_c = \frac{NV_t}{Req} W \left(\exp \left(\frac{\left(\frac{R_f + R_c}{R_{Tot}} \right) V_{in} - V_k + \frac{R_{in}}{R_{Tot}} V_{cc}}{NV_t} \right) \right) - I_s$$

56/(56)

where $V_k = NV_t \cdot \ln \left(\frac{NV_t}{I_s \cdot Req} \right) - I_s Req$ and

$$Req = \frac{R_{in}}{R_{Tot}} \left(R_c \left(\frac{Beta + 1}{Beta} \right) + \frac{R_f}{Beta} \right) + R_e \left(\frac{Beta + 1}{Beta} \right)$$

LTspice plot **Figure 4.2a** show this solution is amazingly accurate. Although the derivation is tedious we can now use it as a template to write a solution to similar circuits (such as the CFP later).

This solution is similar to Thomas Banwell's but this version now includes local degeneration from R_e . As noted by Thomas Banwell the seemingly complicated interactions in this circuit actually reduce to a few key resistor combinations: Req , $(R_f + R_c)/R_{Tot}$, R_{in}/R_{Tot} and R_f/R_{in} . (The ratio R_f/R_{in} is mentioned in *Section 4.2b* below). From this exact solution we can now understand how this circuit works in *large signal* terms – this has been impracticable with simulators or small signal analysis.

The PAK spreadsheet project is for sharing tedious circuit solutions like this and hopefully avoiding

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the repeating the same solutions unnecessarily.

4.2a Calculating the output voltage

Having solved one current I_c we can now find any other current, voltage, gain, power or whatever. V_{out} is now found by back substitution into the original equations. For the output voltage we can use

$$V_{out} = V_c = V_{cc} - I_{in} \cdot R_c - I_c \cdot R_c - \frac{I_c}{\beta} \cdot R_c \quad \text{where} \quad I_{in} = \frac{V_{be} + V_e - V_{in}}{R_{in}} \quad \text{and}$$

$$V_{be} = NVt \ln \left(\frac{I_c}{I_s} + 1 \right) \quad \text{and} \quad V_e = I_e \cdot R_e = I_c R_e \left(\frac{\beta + 1}{\beta} \right)$$

which gives

$$V_{out} = V_{cc} - \frac{R_c}{R_{in}} NVt \ln \left(\frac{I_c}{I_s} + 1 \right) - \frac{R_c R_e}{R_{in}} \left(\frac{\beta + 1}{\beta} \right) I_c - \frac{R_c}{R_{in}} V_{in} - R_c \left(\frac{\beta + 1}{\beta} \right) I_c$$

$$V_{out} = V_{cc} - \left(1 + \frac{R_e}{R_{in}} \right) \left(\frac{\beta + 1}{\beta} \right) R_c \cdot I_c - \frac{R_c}{R_{in}} \left(V_{in} + NVt \ln \left(\frac{I_c}{I_s} + 1 \right) \right) \quad 57/(57)$$

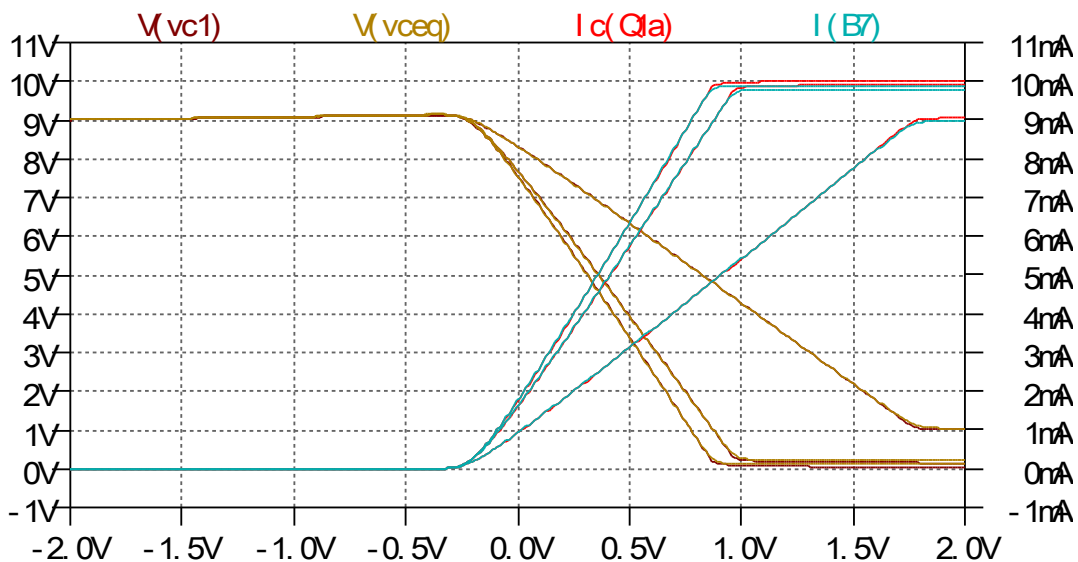


Figure 4.2a Common Emitter amplifier with R_f & R_{in} with R_e stepped 1mR, 10R and 100R. LTspice is red & brown, while ABM equations are gold & cyan. $V_{cc}=10V$, $R_c=R_{in}=1k$, $R_f=10k$. $\beta=100$. Some small differences are visible at saturation.

Figure 4.2a shows a plot for I_c and V_{out} with R_e stepped that includes I_c and V_{ce} saturation. Equation (57) was modified to allow a large reverse input voltage. The log term in Equation (57) runs out of numeric range when V_{in} is more negative than -1.2V. An equation that fixes this is

$$V_{be} \approx \text{Min} \left(\text{Max} \left(NVt \cdot \ln \left(\frac{I_c}{I_s} + 1 \right), 0 \right), \frac{R_f + R_c}{R_{Tot}} (V_{in} - V_x) + (V_k + I_s R_{eq}) \right) \quad 58/(58)$$

$$\text{where} \quad V_x = \frac{(V_k I - (R_{in} / R_{Tot}) V_{cc})}{(R_f + R_c) / R_{Tot}}$$

where V_x is the input voltage where the $W(x)$ argument of Equation (58) is $W(e^0)$ or $W(1)$. The inner $\text{Max}(x, y)$ function stops using the $\ln(f(I_c))$ when V_{be} passes through zero and the $\text{Max}(x, 0)$

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function keeps it at zero rather than switching to an unwanted large -ve limit when overflow occurs. The other function (a straight line) then takes control using the outer *Min* function. The plot in Figure 4.2a is extended to -2V to show that V_c is now correctly calculated beyond -1.2V. Saturation for I_c and V_c was obtained using

$$I_c = \text{Min1}(I_c, I_{cSat}, 0.01 \cdot I_{cSat}) \quad \& \quad I_{cSat} = \frac{(V_{cc} - V_t \cdot \ln(\text{Beta}))}{R_e + R_c} \quad (59)$$

$$V_c = \text{Max1}(V_c, I_{cSat} \cdot R_e + V_t \cdot \ln(\text{Beta}), V_t \cdot \ln(\text{Beta})) \quad (60)$$

where the above $\text{Min1}(x,y,e)$ and $\text{Max1}(x,y,e)$ are User Defined functions in LTspice using

$$. \text{function } \text{Max1}(X, Y, e) \{ 0.5 \cdot (X + Y + \sqrt{((X - Y) \cdot (X - Y) + 4 \cdot e \cdot e)}) \} \quad (61)$$

$$. \text{function } \text{Min1}(X, Y, e) \{ 0.5 \cdot (X + Y - \sqrt{((X - Y) \cdot (X - Y) + 4 \cdot e \cdot e)}) \} \quad (62)$$

Note two curly brackets are required around the function, also put the dot (period) in front of 'Function' to make it a SPICE command.

4.2b Calculating the differential voltage gain

The differential voltage gain $\frac{dV_c}{dV_{in}}$ can be calculated for any V_{in} by differentiation of Equation (57) with respect to V_{in} , giving

$$A_{v_{CL}} = \frac{dV_c}{dV_{in}} = - \left(\frac{R_f}{R_{in}} - \frac{1}{1 + W(57)} \frac{R_{in} + R_f}{R_{in}} \frac{R_f + R_c}{R_{Tot}} \right) \quad 63/(63)$$

where $W(57)$ is the Lambert argument used in Equation (57).

Interestingly, the voltage gain is not defined explicitly by R_e as we might expect since it alters the gm of the transistor and therefore the internal voltage gain, but it is affected by $W(57)$. R_f/R_{in} is the standard result for the gain limit of a transresistance stage and an inverting opamp.

4.2c Estimates of Distortion HD2 and HD3

The section above shows that we can now directly calculate the differential current gain and voltage gain of a complete BJT circuit. In the above CE example the accuracy is limited only by the models. Given the differential gain equation we can now make estimates of second harmonic distortion and third harmonic distortion (HD2 & HD3) levels.

SPICE calculates nonlinear harmonic distortion levels using FFT processing from a number of sinewave cycles in the Transient simulation mode. This involves a huge computational effort. Often we are after rough estimates of second harmonic distortion and third harmonic distortion from trigonometric relationships. The following uses Ed Cherry's equations [ref Cherry JAES 2000, EW Jan 1995]. It yields analytic approximations (rather than specific operating point distortions of FFT using SPICE) which are more helpful to designers showing how distortions arise and how they are related to circuit choices and temperature.

The 2nd and 3rd harmonic distortion figures to be estimated for any AC level and bias voltage using

$$HD2 = \frac{\delta G1 - \delta G2}{8} \cdot 100\% \quad HD3 = \frac{\delta G1 + \delta G2}{24} \cdot 100\% \quad \text{where} \quad (64)$$

$$\delta G1 = \frac{G1 - G0}{G0} \quad \delta G2 = \frac{G2 - G0}{G0} \quad (65)$$

where $G1$ and $G2$ are the gains at the dip and peak of the AC cycle and $G0$ is the gain at the

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zero crossing. These are calculated using Equation (63) where G1 is obtained with $V_{in}=V_{dc}+V_{pk}$ and G2 is obtained with $V_{in}=V_{dc}-V_{pk}$ for an AC sinewave superimposed on a DC bias point. Then a plot can display HD2 and HD3 as V_{dc} is varied over a range of bias settings.

The model used in these equations ignores saturation of the transistor. Section 4.1.1 showed how saturation can be included to limit the output voltage swing to the available supply voltage and this includes limiting the collector current to reflect the V_{ce} minimum voltage limit of a BJT. For THD estimates a test for saturation is necessary over the full range of input signal; if there is no saturation at any point in the cycle then the above equations estimates are valid, but the slightest saturation will invalidate the distortion estimates.

See [Section 15](#) for estimation of THD for Class-AB crossover distortion.

4.3 SE CC stage (Common Collector) or Voltage Follower

This section covers the Voltage Follower (VF) as a special case of the CE stage with no resistor in the collector and the output voltage is across the emitter resistor.

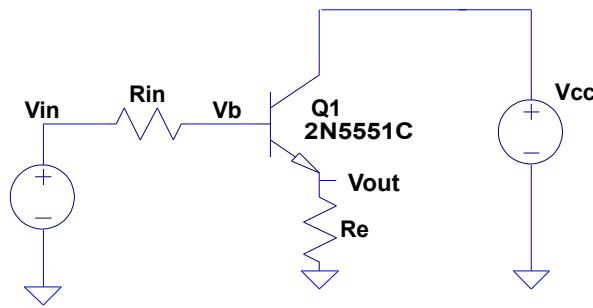


Figure 4.3. SE Common Collector stage (CC) or Voltage Follower (VF)

The CC solution for the output voltage across R_e ($V_{out}=V_e$) is easily calculated using I_c from the above CE solution with emitter degeneration and $R_c=0$ (external R_c removed but may still include the R_{cP} parasitic term). We get I_c from

$$I_c = \frac{N \cdot V_t}{R_{eq}} W \left(\exp \left(\frac{V_{in} - V_k}{N \cdot V_t} \right) \right) - I_s \quad \text{where} \quad V_k = N \cdot V_t \cdot \ln \left(\frac{N \cdot V_t}{I_s \cdot R_{eq}} \right) - I_s \cdot R_{eq} \quad \text{and} \quad (66)$$

$$R_{eq} = \frac{R_{in}}{\beta} + R_e \cdot \left(\frac{1 + \beta}{\beta} \right)$$

then

$$V_{out} = I_c \cdot \left(\frac{\beta + 1}{\beta} \right) \cdot R_e \quad (67)$$

If an additional load resistor is placed across R_e then we calculate the new R_{eq} with R_L in parallel with the previous R_{eq} giving

$$R_{eq} = \frac{R_{in}}{\beta} + R_e \cdot \left(\frac{1 + \beta}{\beta} \right) \parallel R_L \quad \text{and then use} \quad I_L = \frac{V_{out}}{R_L} \quad (68)$$

The CC solution is different to the CE stage in that the CE solution is not affected by R_c and external loading at the collector. But in the CC the external load affects R_{eq} and therefore is part of the Lambert W inverse solution. This is an important practical issue when designing power

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Plots are shown in **Figure 4.3a** with Vce saturation and no added RL. The RHS plots are Vout and d(Vout)

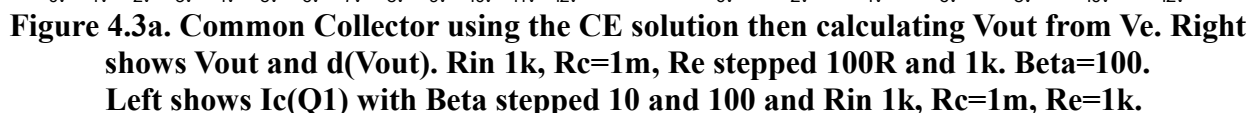


Figure 4.4 shows a general circuit when current drives a CE stage with I_{in} rather than a voltage source V_{in} . This is quite common in amplifiers when a CE stage drives another stage and the first stage provides a current source along with a shunt resistance. Here the shunt resistance is represented by R_{in} and connected to common. Later in the CE+CC we have R_{in} connected to V_{cc} , and later in the CFP R_{in} is the base-emitter resistor R_{be} for the second transistor.

$$I_{c1} = \frac{N1 \cdot V_t}{Req} W \left(\exp \left(\frac{I_{in} \cdot R_{in} - V_k}{N1 \cdot V_t} \right) \right) - I_{s1} \text{ where } V_{k1} = N1 \cdot V_t \cdot \ln \left(\frac{N1 \cdot V_t}{I_{s1} \cdot Req} \right) - I_{s1} \cdot Req \quad (69)$$

$$\text{and } Req = \frac{R_{in}}{\beta_{a1}} + R_e \cdot \left(\frac{1 + \beta_{a1}}{\beta_{a1}} \right)$$

Figure 4.4b,c shows the Common Emitter current (Ic) and voltage (Vc) plots with current drive using a current source of 1mA/V.

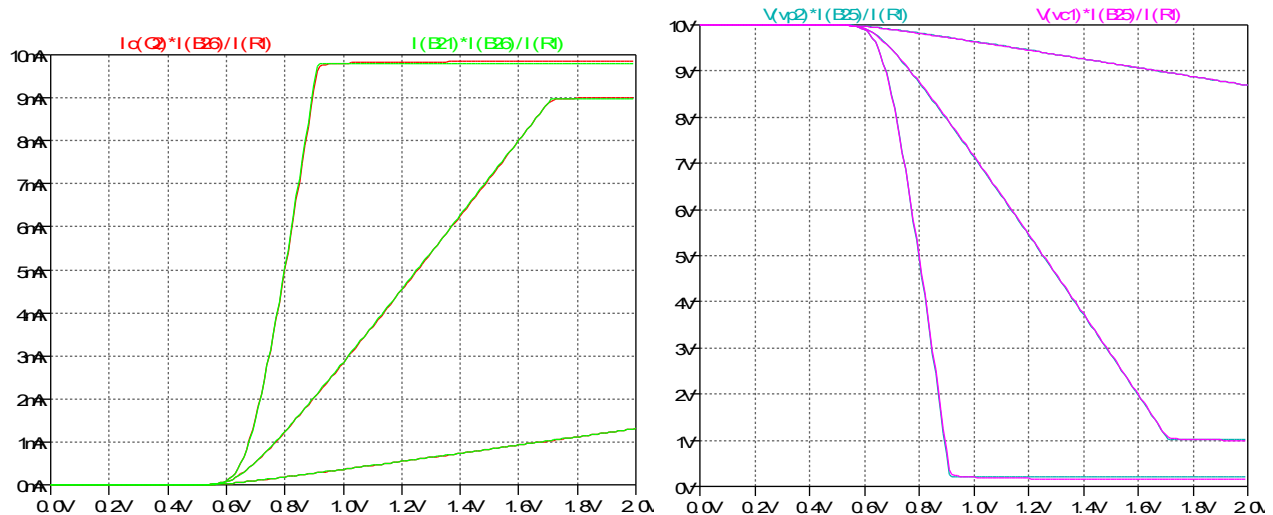


Figure 4.4b,c shows the Common Emitter current (Ic) and voltage (Vc) plots with current drive using a current source of 1mA/V. Rin=1k, Rc=1k and Re stepped 10R, 100R, 1k, Beta=100.

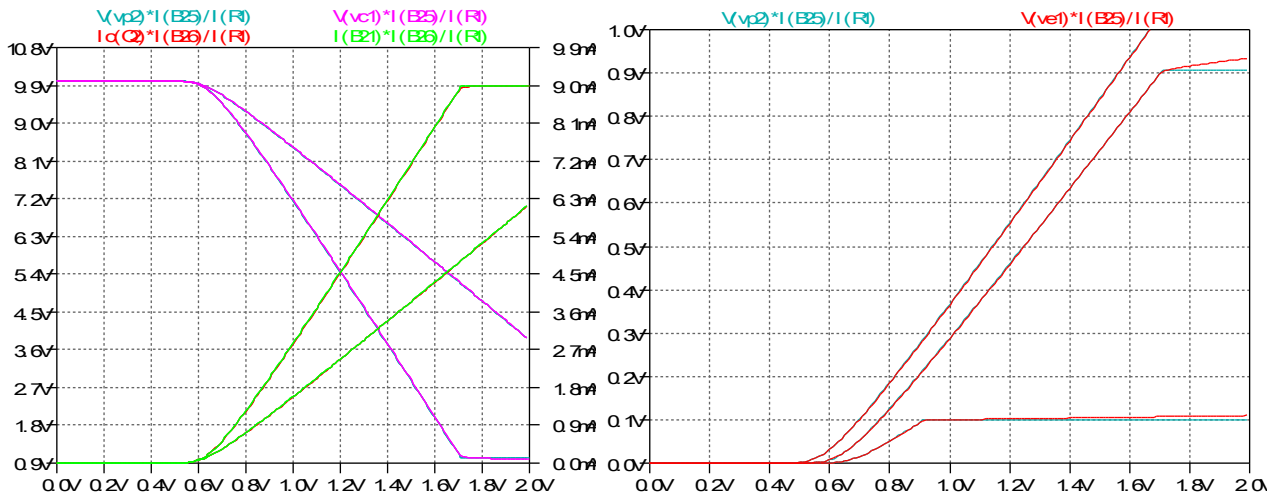


Figure 4.4d Left shows the Common Emitter Ic and Vc when Beta is stepped 10 & 100 with Rin=1k, Rc=1k and Re=100, Figure 4.4e Right shows Vc with Re stepped 10R, 100R, 1k and Rin=1k, Rc=1k.

It may seem strange that only $I_{in} \cdot R_{in}$ appears the Ic equation since there is nonlinear loading of the transistors base. But this shows the utility of the Lambert W solution where you only need to know two terms: first the equivalent circuit resistance Req (which is effectively where the base-emitter diode has zero resistance which is approached at very high current), and second the input voltage which is simply the driving current source multiplied by the input resistance (and ignoring the nonlinear part due to the transistor base-emitter circuit). The Lambert W inverse function takes care

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of the nonlinear loading of the transistor junction for you.

5. SE CE Complementary Feedback Pair

This section develops a solution for the two transistor Complementary Feedback Pair (CFP) also known as the Szakli pair. This is slightly easier to solve than the Darlington pair so the CFP is covered first.

5.1 Open Loop SE CE Complementary Pair (no Rbe)

The complementary pair can be configured in a number of ways. The easiest configuration to solve first is when Q2 does not provide current feedback to Q1 via an emitter resistor for Q1 with link X1 closed and X2 open as shown in **Figure 5.1**. Also the complementary pair is easier to solve when the base-emitter resistor (Rbe) for Q2 is absent (normally present as a speed up resistor).

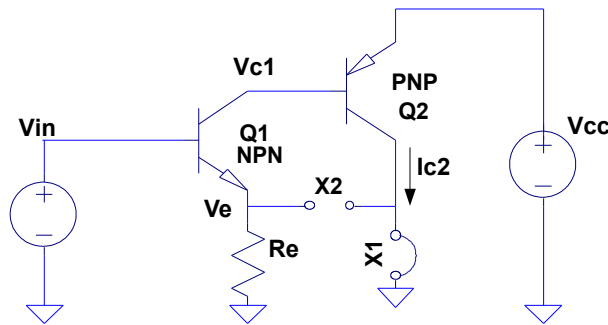


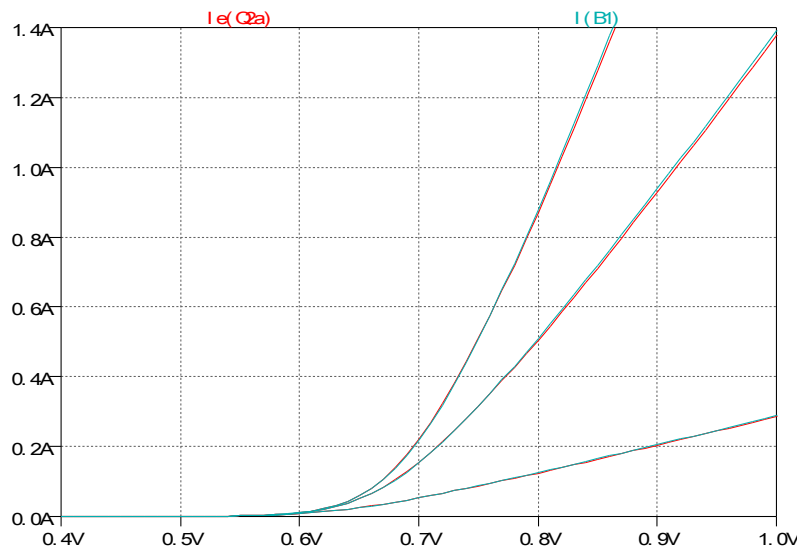
Figure 5.1. Simple Complementary Feedback Pair without Rbe for Q2. Link X1 is closed and X2 is open for an open loop solution.

In open loop we can write the equation for I_{c1} independent of I_{c2} and I_{c2} is just β_2 times I_{c1} . We can write a solution based on Equation (56) as

$$I_{c2} = \beta_2 \cdot I_{c1} \text{ where } I_{c1} = \frac{N_1 \cdot V_t}{R_{eq}} W_1 \left(\exp \left(\frac{V_{in} - V_{k1}}{N_1 \cdot V_t} \right) \right) - I_{s1} \text{ where} \quad (70)$$

$$V_{k1} = N_1 \cdot V_t \cdot \ln \left(\frac{N_1 \cdot V_t}{I_{s1} \cdot R_{eq}} \right) - I_{s1} \cdot R_{eq} \text{ and}$$

$$R_{eq} = R_e + \frac{R_{in}}{\beta_2 + 1} \quad (71)$$



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Figure 5.1a. Open loop SE CE Complementary Feedback Pair without Rbe. LTspice Ic2 (red) and ABM Ic2 (cyan). Re stepped 10R and 100R. Beta1=Beta2=100, Rin=0, RL=0.

5.2 Closed loop Complementary Feedback Pair (no Rbe)

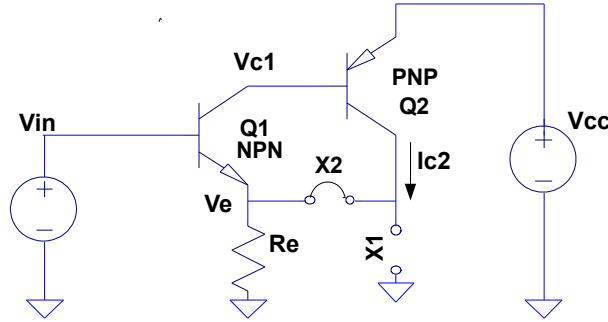


Figure 5.2. Complementary Feedback Pair, Link X2 is closed and X1 is open for a closed loop.

To simplify the analysis R_{in} is removed. We have $I_{c2} = \beta_2 \cdot I_{c1}$ where I_{c2} feeds back to Q1 adding an emitter voltage component of $I_{c2} \cdot R_e$ which subtracts from V_{in} . First the solution for Q1 is

$$I_{c1} = \frac{N I \cdot V_t}{R_e} W \left(\exp \left(\frac{V_{in} - I_{c2} R_e - V_{k1}}{N I \cdot V_t} \right) \right) - I_{s1} \quad (72)$$

We note I_{c2} is a pure current source so other terms are the same as open loop which includes V_{k1} . We need to remove $I_{c2} \cdot R_e$ from the RHS. But how. If we go back one step to the log form $w = \ln(x/w)$ form, then we can move the $I_{c2} \cdot R_e$ to the LHS, we find,

$$(I_{c1} + I_{s1}) \frac{R_e}{N I \cdot V_t} = \ln \left(\frac{\exp \left(\frac{V_{in} - I_{c1} \cdot \beta_2 \cdot R_e - V_{k1}}{N I \cdot V_t} \right)}{(I_{c1} + I_{s1}) \frac{R_e}{N I \cdot V_t}} \right) \quad \text{where} \quad (73)$$

$$V_{k1} = N I \cdot V_t \cdot \ln \left(\frac{N I \cdot V_t}{I_{s1} \cdot R_e} \right) - I_{s1} \cdot R_e$$

and then moving the RHS " $I_{c1} \cdot \beta_2 \cdot R_e$ " to the LHS giving,

$$(I_{c1} + I_{s1}) \frac{R_e \cdot (\beta_2 + 1)}{N I \cdot V_t} = \ln \left(\frac{\exp \left(\frac{V_{in} - V_{k1}}{N I \cdot V_t} \right)}{(I_{c1} + I_{s1}) \frac{R_e}{N I \cdot V_t}} \right) \quad (74)$$

then balancing the equations by multiplying the logs' denominator and numerator by $(\beta_2 + 1)$ we arrive at

$$(I_{c1} + I_{s1}) \frac{R e \cdot (Beta2 + 1)}{N I \cdot V_t} = \ln \left(\frac{(Beta2 + 1) \left(\exp \left(\frac{V_{in} - V_{k1}}{N I \cdot V_t} \right) \right)}{(I_{c1} + I_{s1}) \frac{R e \cdot (Beta2 + 1)}{N I \cdot V_t}} \right)$$

$$(I_{c1} + I_{s1}) \frac{R e \cdot (Beta2 + 1)}{N I \cdot V_t} = \ln \left(\frac{\left(\exp \left(\frac{V_{in} + N I V_t \ln(Beta2 + 1) - V_{k1}}{N I \cdot V_t} \right) \right)}{(I_{c1} + I_{s1}) \frac{R e \cdot (Beta2 + 1)}{N I \cdot V_t}} \right) \quad (75)$$

$$\text{So } I_{c1_{CL}} = \frac{N I \cdot V_t}{R_{eq}} W \left(\exp \left(\frac{V_{in} - V_{k1_{CL}}}{N I \cdot V_t} \right) - I_{s1} \right) \text{ where} \quad (76)$$

$$V_{k1_{CL}} = N I \cdot V_t \cdot \ln \left(\frac{N I \cdot V_t}{I_{s1} \cdot R_{eq}} \right) - I_{s1} \cdot R_{eq} \text{ and } R_{eq} = R e \cdot (Beta2 + 1)$$

To get Ic2 (or Ie2) use

$$I_{c2} = Beta2 \cdot I_{c1_{CL}} \text{ or } I_{e2} = (Beta2 + 1) \cdot I_{c1_{CL}} \quad (77)$$

Next, Rin is introduced by guessing that it adds a component of resistance to Req of Rin/(Beta1+1). This was confirmed by simulation and is shown by **Figure 5.2a**. The derivation with Rin included in the above equations have not been done. With Rin present use the updated form for Req as

$$R_{eq} = R e \cdot (Beta2 + 1) + \frac{R_{in}}{Beta1 + 1} \quad (78)$$

Figure 5.2b shows the plots for Re is stepped in closed loop and a resistor RL to Vcc. Results agree well with LTspice while other parameters were stepped as well.

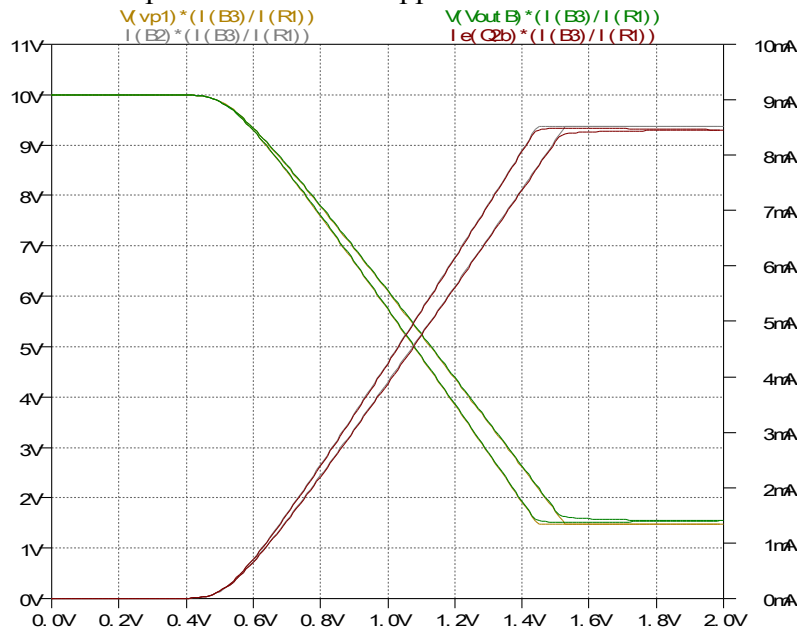


Figure 5.2a. Closed loop SE CE Complementary Feedback Pair without Rbe. Vce (Left) and Ie2 (Right). Beta1=Beta2=100, Re=100, RL=1k. Rin stepped 1k and 100k.

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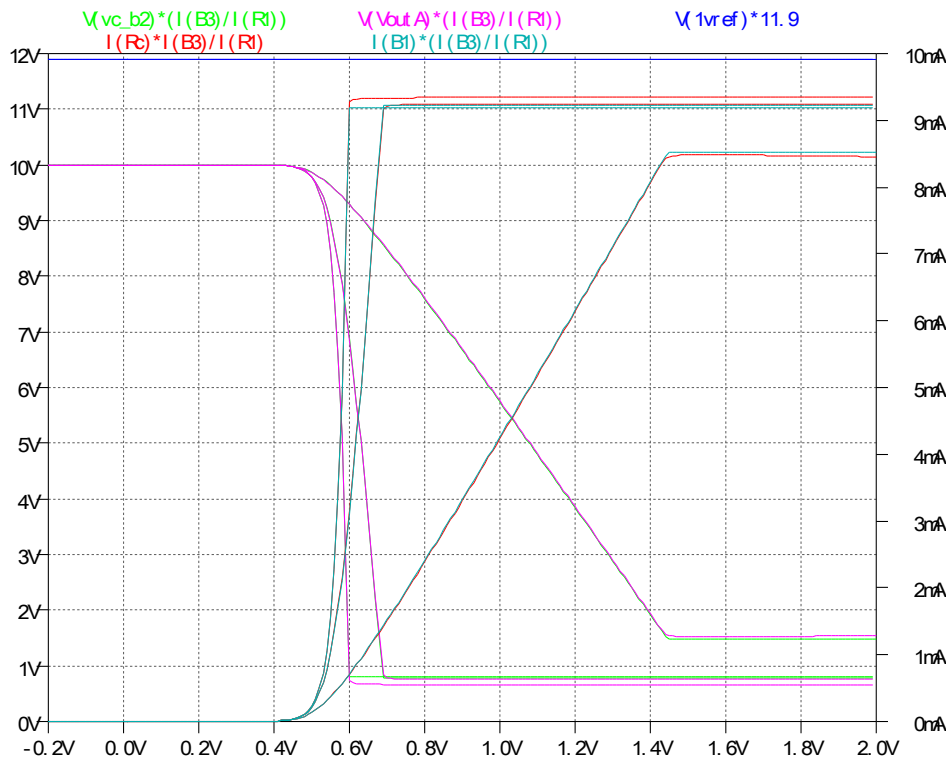


Figure 5.2b. Closed loop SE CE Complementary Feedback Pair without Rbe. Vce (Left) and Ie2 (Right). Beta1=Beta2=100, Rin=RL=1k. Re stepped 1mR, 10R, 100R.

Tracing the steps the feedback voltage subtracting from Vin due to Ic2 (or Beta times Ic1) is transferred to the LHS where it adds to Ic1, then to balance the change due to feedback we need to scale the log denominator by the same factor as used for Ic1 on the LHS and this requires the numerator to be scaled by the same factor. Then the numerator factor can be merged into an offset voltage, which appears a new reduced knee voltage in closed loop called $V_{k_{CL}}$.

Now we can see analytically the effect of NFB on the large signal response. Firstly, when we compare the open loop case to the close loop case we see the feedback current increases the equivalent series resistance R_e by Beta2 times, so including the original R_e gives a total loop gain of $(\text{Beta2}+1) \cdot R_e$. This higher resistance means the closed loop gain I_{c2}/V_{in} is reduced by the factor $\text{Beta2}/(1+\text{Beta2})$ – the expected effect for series voltage negative feedback.

Secondly, there's the expected reduction in the open loop knee voltage. This is reduced by $N1 \cdot V_t \cdot \ln(10)$ volts; it's the standard 60mV per decade figure. With a typical Beta of 200 this involves a reduction of about 140mV due to NFB. Notice that in the (theoretical) limit the effective $V_{k_{CL}}$ does go to zero as expected; in other words we can say an infinite loop gain reduces the input offset voltage V_k to zero.

5.3 SE CE Complementary Feedback Pair with Rin & Rf (no Rbe).

Having solved the closed loop complementary pair with Rbe absent we can try solving **Figure 5.3** with Rin and Rf by substitution of Beta's in the above final equations.

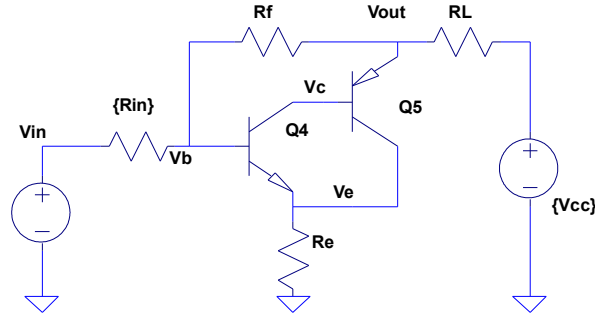


Figure 5.3. SE CE Complementary Feedback Pair with Rf, Rin (no Rbe)

In this circuit RL has been renamed R_L because it now appears at the emitter of Q2 and replaces R_C in the previous equations with R_L . Similarly Vout replaces V_C . In shorthand form we can write the changes as $R_C \rightarrow R_L$ and $V_C \rightarrow V_{out}$ and $\beta \rightarrow \beta_1 \cdot (\beta_2 + 1)$. Notice the emitter current of Q2 is $(\beta_2 + 1) \cdot I_{C1}$. (I_{E2} is also $I_{C1} + I_{C2}$, useful for later).

Beginning with I_{C1}

$$I_{C1} = \frac{NV_t}{R_{eq}} W \left(\exp \left(\frac{\left(\frac{R_f + R_C}{R_{Tot}} \right) V_{in} - V_k + \frac{R_{in}}{R_{Tot}} V_{CC}}{NV_t} \right) \right) - I_s \quad (79)$$

$$\text{Where } V_k = NV_t \cdot \ln \left(\frac{NV_t}{I_s \cdot R_{eq}} \right) - I_s R_{eq} \quad \& \quad R_{Tot} = R_{in} + R_f + R_L \quad \&$$

$$R_{eq} = \left(\frac{R_{in}}{\beta_1} \cdot \left(\frac{R_C \cdot (1 + \beta_1 \cdot (\beta_2 + 1))}{R_{tot}} + \frac{R_f}{R_{tot}} \right) + R_e \frac{(\beta_1 \cdot (\beta_2 + 1) + 1)}{\beta_1} \right) \quad (80)$$

Having obtained I_{C1} we can calculate the load current using I_{E2} where

$$I_{E2} = I_{C2} \frac{\beta_2 + 1}{\beta_2} \quad \text{which is} \quad I_{E2} = I_{C1} \cdot (\beta_2 + 1) \quad (81)$$

The output voltage can now be calculated by substitutions $V_C \rightarrow V_{out}$ and $\beta \rightarrow \beta_1 \cdot (\beta_2 + 1)$ and $I_C \rightarrow I_{E2}/(\beta_2 + 1)$ for V_{be1} , giving

$$V_{out} = V_{CC} - \left(1 + \frac{R_e}{R_{in}} \right) \left(\frac{\beta_1 (\beta_2 + 1) + 1}{\beta_1 (\beta_2 + 1)} \right) R_L \cdot I_{E2} - \frac{R_C}{R_{in}} \left(V_{in} + NV_t \ln \left(\frac{I_{C1}}{I_s} + 1 \right) \right) \quad (82)$$

Figure 5.3a shows the plots when R_e is stepped and results agree well with LTspice. β_2 was set to 1 and β_1 to 10, R_{in} and R_L were stepped and plot matched as well in all cases. **Figure 5.3b** shows the plots when R_f , R_{in} and R_e are all stepped so any regions of error can be spotted.

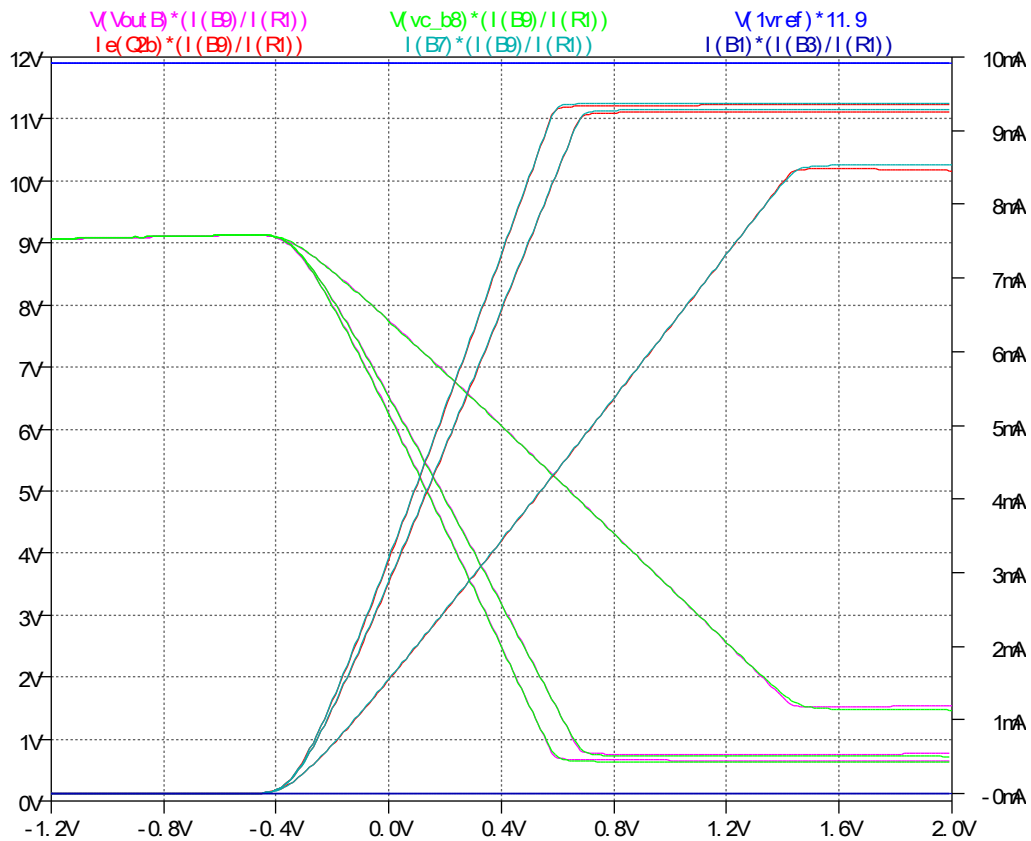


Figure 5.3a. SE CE Complementary Feedback Pair with R_{in} & R_f (no R_{be}). V_{ce} Left and I_{e2} Right. $\beta_{a1}=\beta_{a2}=100$, $R_f=10k$, $R_{in}=R_L=1k$, R_e stepped 1mR, 10R, 100R.

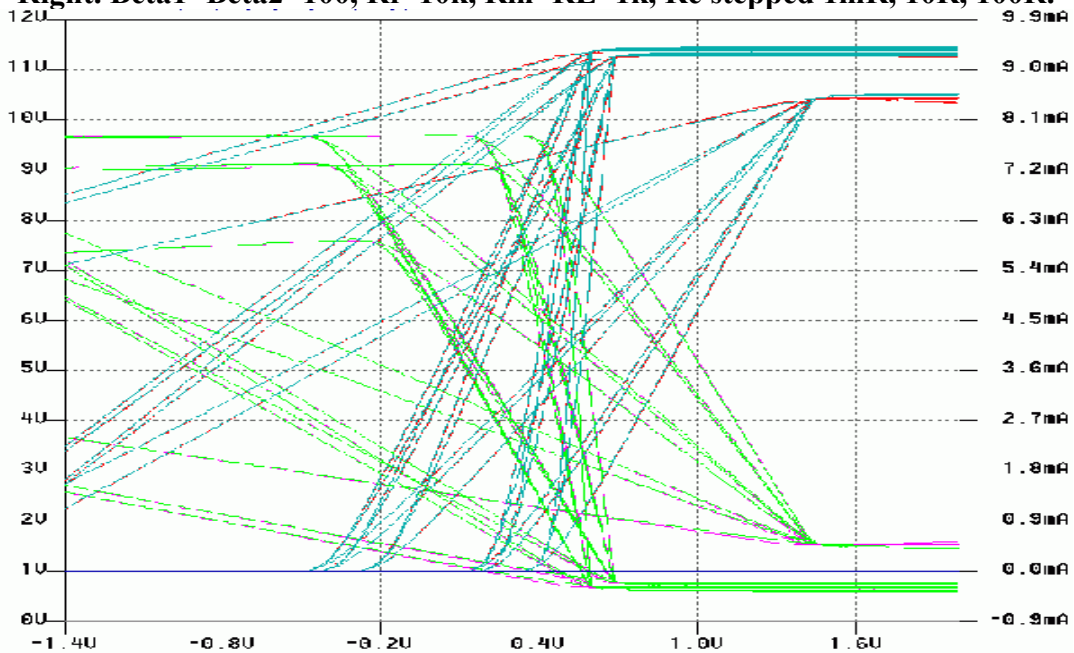


Figure 5.3b. As above but stepped R_{in} with stepped R_f and stepped R_e (27 cases) – a “rat's nest”. All plots cover the red LTspice plots – all solutions are shown to be accurate.

5.4 SE CE Complementary Feedback Pair with R_{be} (no R_{in} or R_f)

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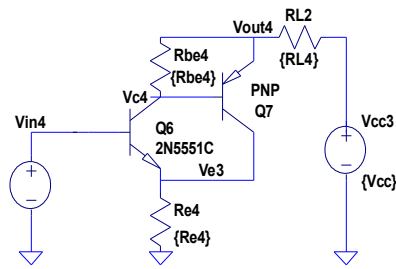


Figure 5.4. Common Emitter Complementary Feedback Pair with Rbe (no Rin or Rf).

Adding a base-emitter resistor (R_{be}) bypasses some of the base current from Q2 and this has the effect of progressively reducing the *net* Beta of Q2 as Q1 current falls. R_{be} provides a speed up path to drain the stored charge in the base of Q2. Without this resistor the base junction charge is used by base current to support the collector current. If R_{be} is too small the *net* Beta for Q2 is too small and the driver transistor must dissipate much of the power that Q2 otherwise dissipates. Notice also in Equation (70) when R_{be} is very small the circuit Q2 provides no current gain and the circuit reverts to the single transistor solution. But if R_{be} is too large the turn off time Q2 is not reduced by much and the usefulness of R_{be} is lost. There is a similar effect with R_{be} extending the breakdown voltage $V_{ce_{max}}$, where too large R_{be} does not improve $V_{ce_{max}}$ by much and the usefulness of R_{be} is lost.

To solve **Figure 5.4** with R_{be} added we are actually trying to solve a global feedback loop around *two* nonlinear BJT-diode equations. A solution for Figure 5.3 was done by omitting R_{be} which meant Q2 was a *linear* current amplifier and there was therefore only *one* diode equation to solve and that we can solve with global feedback present. We can also write the solution with two cascaded transistors *without* overall feedback as a product of two independent diode solutions assuming Q1's Early effect is neglected. But now we want a solution for *two* diode/transistors with global feedback. Is this possible?

We can infer from Equation (76) that an exact explicit solution is *not* possible in terms of V_{in} because the *effective* Beta2 due to R_{be} appears in the w term for I_{c1} . We can write $w = R_{eq}/V_t$ where R_{eq} is dependent on Beta2 which varies with I_{c1} and hence V_{in} . It was shown earlier that balancing the Lambert W-function must remove any dependency on V_{in} for the w term. Since V_{in} necessarily appears in the w term in Equation (76) we can infer that the normal Lambert W-function cannot provide an *exact* solution for a CFP circuit with R_{be} . Appendix [A3](#) provides a proof of this by attempting to balance the Lambert W equations from first principles with R_{be} present and it cannot be balanced with R_{be} present.

It is surmised that most circuits that use two or more transistors with global feedback are not solvable in an exact form using the Lambert W-function approach above. Maybe there is another function that can provide a solution? In the mean time it appears that the only circuits that can be solved *with global feedback* in exact form are elementary single transistor circuits or single diode circuits or complementary pairs with no R_{be} resistors. Open loop circuits devoid of feedback around *more than one stage* can still be expressed in an exact form; the complication is solving the global feedback around *more than one diode/transistor stage*. Of course there are still approximate solutions that give very good results. The following example demonstrates one of the better approximations for the effect of R_{be} in the CE Complementary Feedback Pair.

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An approximated solution for Figure 5.4 circuit can be obtained using Equation (76) by formulating an approximation for the effective Beta of Q2 as a function of Vin. Recursion can also be used to improve the accuracy of a single approximated equation if more accuracy is needed. It is a time consuming process to find a good approximation that do not require recursion for most cases. The following approximation presented for Figure 5.4 illustrates these goals for formulating approximations. We use simple approximations based on the underlying effects, usually using the first derivative to formulate a Newton Raphson improved solution, and this usually avoids empirical fitting factors. Sometimes empirical fitting factors are useful to tweak the improved solution.

The approximation chosen for solving Figure 5.4 uses an approximation for effective Beta2 (Beta2_{EFF}) that is later used in the Lambert W-function solution by replacing Beta2 with Beta2_{EFF} which varies with Vin. A switch function is used to turn on Beta2_{EFF} when the voltage across Rbe reaches approximately Vk2. The switching point can be calculated reasonably accurately because we know Beta2 is effectively zero before Q2 starts to conduct and the input voltage required to obtain Vk2 across Rbe can be calculated using

$$V_{sw} = V_t \frac{V_{k2}}{V_t} Re4 \left(\frac{1}{Rbe4} + \frac{1}{Beta1 \cdot Beta2 \cdot Re} \right) + V_t \cdot \ln \left(\frac{V_{k2}}{V_t} \cdot Re4 \left(\frac{1}{Rbe4} + \frac{1}{Beta1 \cdot Beta2 \cdot Re} \right) \right) + V_{k1} \quad 83/(83)$$

The term Beta1·Beta2·Re caters for very high Rbe values where the Q2 base current determines the input voltage needed for Vbe2 to reach Vk2.

Current Ic1 can be calculated at Vin=Vsw using

$$I_{sw} = \frac{NIV_t}{Re4} W \left(\exp \left(\frac{V_{in} - V_{k1}}{NIV_t} \right) \right) \quad (84)$$

From these two values we build an approximation for Beta2_{EFF} based on straight line approximations for Ic1 and Ic2 after Vsw (once Q2 turns on). Ic1 starts from I1sw at Vsw and is approximated by a line with a constant slope of 1/(Beta2Re) due to Ic2 feeding current into Re. Ic2 starts from zero at Vsw using the Max function and has a slope of 1/Re. The ratio of these two currents Ic2/Ic1 gives the correct limits for Beta2_{EFF} with large input voltages, at Vsw, and the correct general shape between these limits.

$$Beta2_{Eff} \simeq \frac{Beta2}{\left(1 + \frac{I_{sw} \cdot Beta2 \cdot Re + V_o}{Max(V_{in} - V_{sw}, 0)} \right)} \quad 85/(85)$$

where Vo is a constant fitting term that improves the approximation for Beta2_{EFF}.

But Beta2_{EFF} rises quicker toward 100% of Beta2. Including the offset 'tweak' or fitting term Vo gives a good fit to the required LTspice Beta2_{EFF} curve (IcQ2/IcQ1) as shown in **Figure 5.4a**. This tweak removes the need for recursion in most cases.

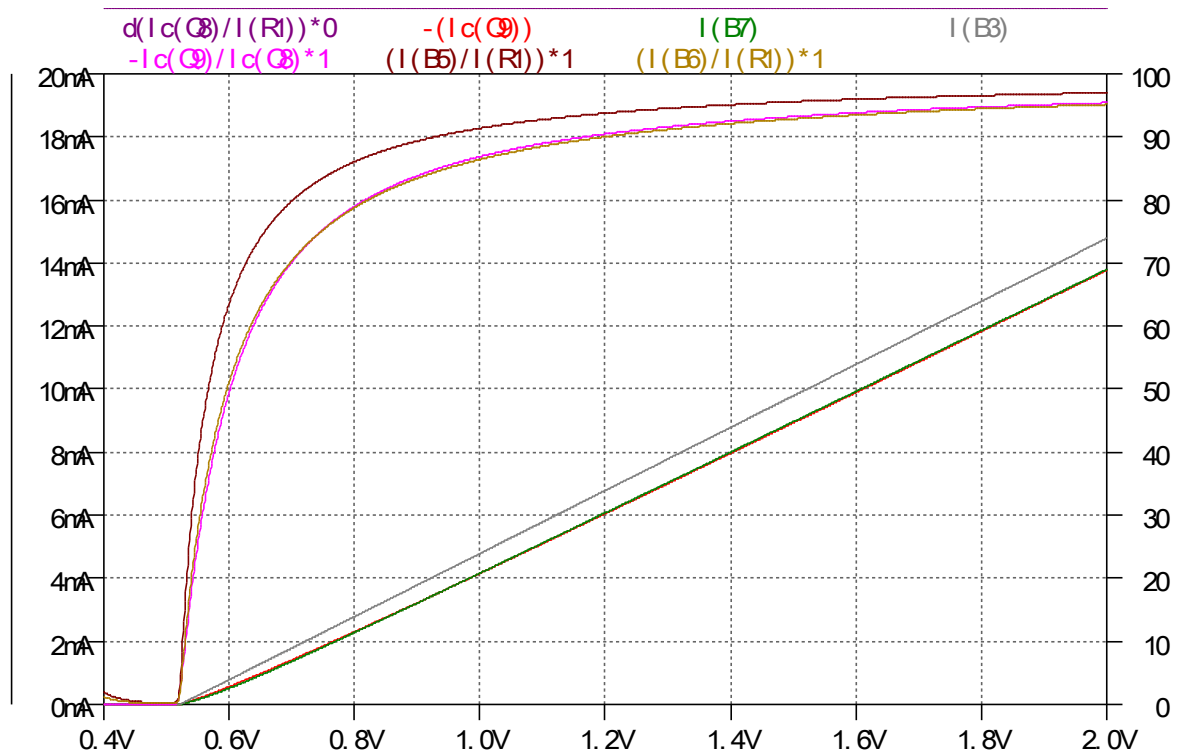


Figure 5.4a. Shows $\text{Beta2}_{\text{Eff}}$ from LTspice (pink) and $\text{Beta2}_{\text{Eff}}$ (brown) from the linear approximation for I_{c1} and I_{c2} shown as $I(B3)$. With fitting term V_o (30mV) the $\text{Beta2}_{\text{Eff}}$ and $I_{c2_{\text{App}}}$ are now very close to LTspice circuit curves.

$R_{be}=100k$, $R_e=100R$, $\text{Beta1}=\text{Beta2}=100$, no Beta-fall effects or Early effect.

Varying the fitting term V_o for the best fit to the LTspice curve requires 30mV when R_{be} is 100k and R_e is 100R. $\text{Beta2}_{\text{Eff}}$ and $I_{c2_{\text{App}}}$ very closely follow the LTspice curves using this very simple approximation. A function was formulated for V_o is to cover a wide range of values for R_{be} . It is also possible to calculate V_o for any R_{be} once the solution for I_{c1} has been validated.

An empirical approach can be used to find a function for V_o by plotting the optimum choice of V_o against $(R_{be}/\text{Beta2})/R_e$, where V_o is found using the circuit in with ABM Equations (65)-(67) above. The following is provided as an example. **Table 5.4** shows the data and resulting values using two fitting curves.

Table 5.4. CFP using $\text{Beta1}=\text{Beta2}=100$ and $R_e=100R$						
R_{be}	100	1k	10k	100k	1Meg	10Meg
$(R_{be}/\text{Beta2})/R_e$	0.01	0.1	1.00	10	100	1000
V_o (target)	10	1.0	0.20	30m	5m	1m
V_{o1} (fn 1)	20	2.0	0.20	20.7m	2.7m	0.8m
V_{o2} (fn 2)	8.0	1.3	0.20	32m	5.0m	0.8m

$$V_{o1} = 0.2V \cdot \left(\frac{1}{R_{be}(\text{Beta2} \cdot R_e)} + \frac{1}{300} \right) \quad (86)$$

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$$V_{o2} = 0.2V \cdot \left(\frac{R_{be}}{\beta_{2} \cdot R_e} \right)^{-0.8} \quad (87)$$

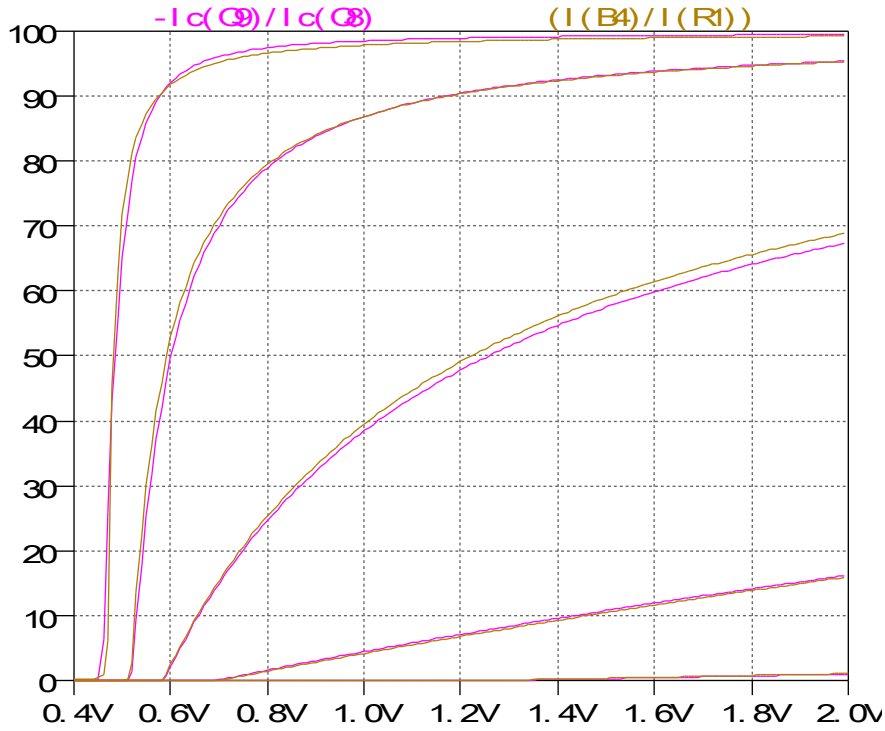


Figure 5.4b Shows $\beta_{2_{eff}}$ approximate function with V_{o2} (cyan) and LTspice effective β from I_{c2}/I_{c1} (red). R_{be} stepped 100R to 1Meg, $R_e=100R$, $\beta_1=\beta_2=100$, no β -fall effects or Early effect.

Writing the solution for I_{c1} and I_{c2}

1. Now we can solve I_{c1} for Figure 5.4 by employing $\beta_{2_{eff}}$ with Equation (76) and V_{o2} equation giving

$$I_{c1} = \frac{N I \cdot V_t}{R_e \cdot (\beta_{2_{eff}} + 1)} W \left(\exp \left(\frac{V_{in} - V_{k1}}{N I \cdot V_t} \right) - I_{s1} \right) \quad \text{and} \quad (88)$$

$$V_{k1} = N I V_t \cdot \ln \left(\frac{N I \cdot V_t}{I_{s1} \cdot R_e (\beta_{2_{eff}} + 1)} \right) - I_{s1} R_e$$

Note I_L is $I_{c1} + I_{c2}$ since I_{c1} splits through R_{be} and recombines with I_{c2} at the load, giving

$$I_{c2} = I_{c1} \cdot \beta_{2_{eff}} \quad \text{and} \quad I_L = I_{c1} \cdot (\beta_{2_{eff}} + 1) \quad (89)$$

Figure 5.4c shows the load current for LTspice for a range of R_{be} values from 100R to 1Meg. The error is only a few percent over 4 decades of R_{be} using a single approximate solution and no recursion.

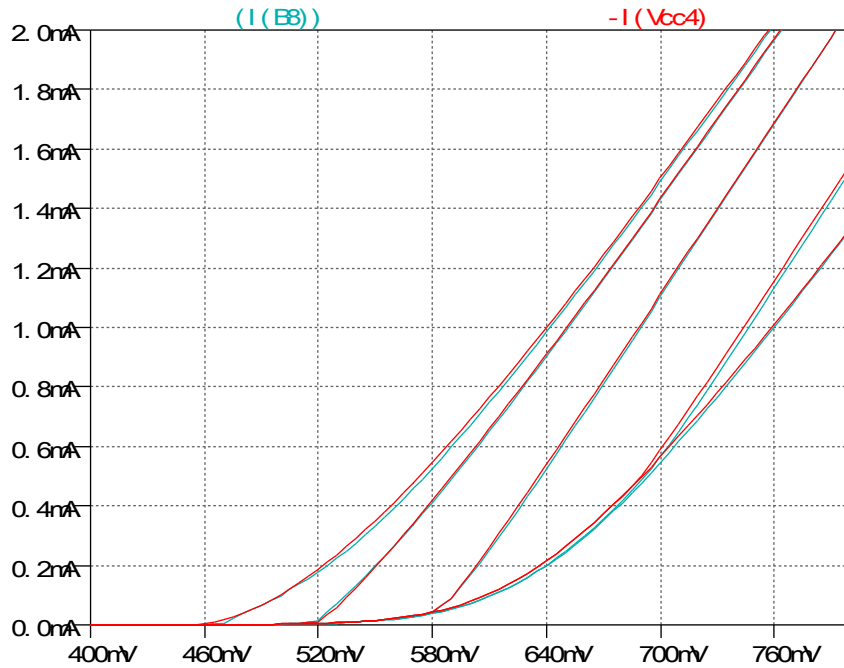


Figure 5.4c. Shows supply current. LTspice (red) and $I_{c1}+I_{c2}$ using $\text{Beta2}_{\text{Eff}}$. R_{be} stepped 100R to 1Meg, $R_e=100R$, $\text{Beta1}=\text{Beta2}=100$, no Beta-fall effects or Early effect.

Calculating V_o

As mentioned it is possible to calculate V_o for any R_{be} after a solution for I_{c1} is known for certain. The empirical approach does not guarantee that a variable has been omitted in the formulation (which is only a guess). Calculating V_o involves all the relevant terms that influence V_o . But note calculating V_o does not remove all the error in the approximation for $\text{Beta2}_{\text{Eff}}$. It pins the curve to one more point, previously it was pinned at start and end values.

The Lambert- W-function can be used to first calculate I_{c1} when $\text{Beta2}_{\text{Eff}}$ is half the maximum value. This 50% of Beta2 point appears to be the only point that permits a balancing of the equations. Solving Q2 with I_{c1} splitting half through R_{be} and half to Q2 base gives

$$\frac{I_{c1h}}{2} R_{be} = N_2 \cdot V_t \ln \left(\frac{\text{Beta2} \cdot I_{c1h}/2}{I_{s2}} + 1 \right) \quad (90)$$

Then arranging in the $w=\ln(x/w)$ form

$$\begin{aligned} \frac{\left(\frac{\text{Beta2} \cdot I_{c1h}}{2} \right) \frac{R_{be}}{\text{Beta2}}}{N_2 \cdot V_t} &= -\ln \left(\frac{I_{s2}}{\frac{\text{Beta2} \cdot I_{c1h}}{2} + I_{s2}} \right) \\ \left(\frac{\text{Beta2} \cdot I_{c1h}}{2} + I_{s2} \right) \frac{R_{be}/\text{Beta2}}{N_2 \cdot V_t} &= -\ln \left(\frac{I_{s2}}{\frac{\text{Beta2} \cdot I_{c1h}}{2} + I_{s2}} \right) + \frac{I_{s2} \cdot R_{be}/\text{Beta2}}{N_2 \cdot V_t} \\ \left(\frac{\text{Beta2} \cdot I_{c1h}}{2} + I_{s2} \right) \frac{R_{be}/\text{Beta2}}{N_2 \cdot V_t} &= -\ln \left(\frac{\frac{I_{s2} R_{be}/\text{Beta2}}{N_2 V_t} \exp \left(\frac{-I_{s2} \cdot R_{be}/\text{Beta2}}{N_2 \cdot V_t} \right)}{\left(\frac{\text{Beta2} \cdot I_{c1h}}{2} + I_{s2} \right) \frac{R_{be}/\text{Beta2}}{N_2 V_t}} \right) \end{aligned} \quad (91)$$

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The minus sign can be moved to the LHS and balanced in the log giving

$$-\left(\frac{\text{Beta}2 \cdot \text{Ic}1h}{2} + \text{Is}2\right) \frac{\text{Rbe} / \text{Beta}2}{N2 \cdot Vt} = \ln \left(\frac{-\frac{\text{Is}2 \text{ Rbe} / \text{Beta}2}{N2 Vt} \exp\left(\frac{-\text{Is}2 \cdot \text{Rbe} / \text{Beta}2}{N2 \cdot Vt}\right)}{\left(\frac{\text{Beta}2 \cdot \text{Ic}1h}{2} + \text{Is}2\right) \frac{\text{Rbe} / \text{Beta}2}{N2 Vt}} \right) \quad (92)$$

Now the solution can be written for the Lambert W-function as

$$\begin{aligned} \text{Ic}1h &= \frac{2 \cdot N2 \cdot Vt}{\text{Rbe}} W \left(\exp \left(\frac{V_{k2}}{N2 \cdot Vt} \right) \right) - \text{Is}2 \quad \& \\ V_{k2} &= N2 \cdot Vt \cdot \ln \left(\frac{N2 \cdot Vt}{\text{Is}2 \cdot \text{Rbe} / (\text{Beta}2 + 1)} \right) - \text{Is}2 \cdot \text{Rbe} / (\text{Beta}2 + 1) \end{aligned} \quad (93)$$

Vk2 was used with Rbe/(Beta2+1) rather than Rbe/Beta2 as above but this does not significantly affect the fit when Beta2 is relatively large (say >10). The +Vk appears since there is a double negative. The factor of 2.7 was found to work slightly better than “2” [why?].

Having found Ic1_{50%} at half the full Beta2 value we can work backwards from the Ic1 solution [Equation 76] to find Vin_{50%}. Solving for Vin_{50%} gives

$$V_{in_H} = V_{k1} + N1 Vt \left(\frac{(\text{Ic}1h - \text{Is}1) N1 \cdot Vt}{\text{Re} \cdot (\text{Beta}2 / 2 + 1)} + \ln \left(\frac{(\text{Ic}1h - \text{Is}1) N1 \cdot Vt}{\text{Re} \cdot (\text{Beta}2 / 2 + 1)} \right) \right) \quad (94)$$

lastly calculate Vo from the Beta2_{Eff} Equation (85) giving

$$V_o = V_{in_H} - V_{sw} - \text{Is}w \cdot \text{Re} \quad 95/(95)$$

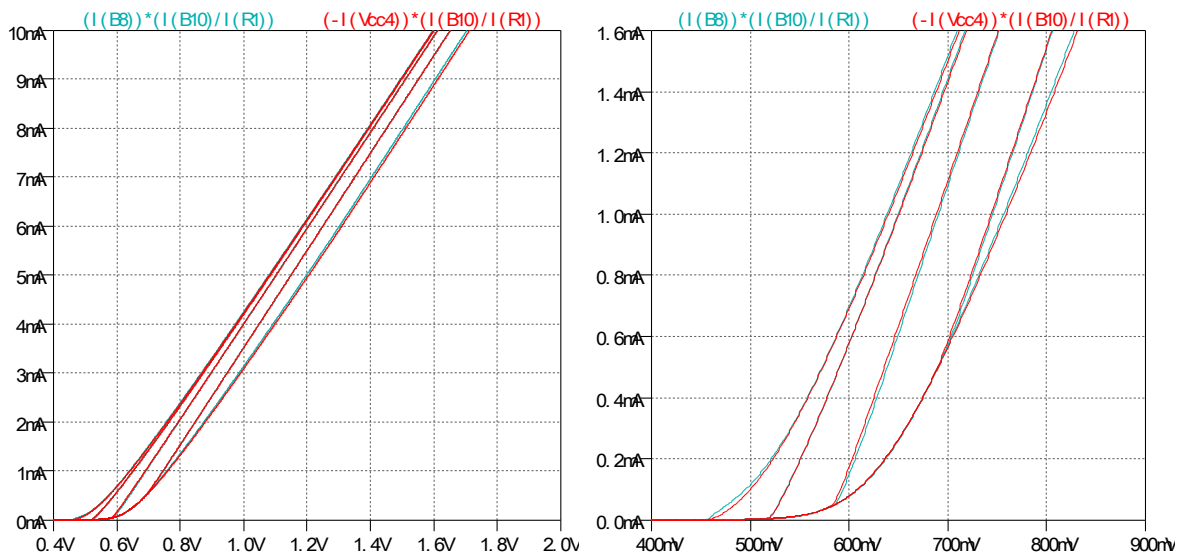


Figure 5.4d shows the plots using calculated Vo at half of Beta2 with Rbe stepped from 1Meg to 1k and Re=100R. Q1 and Q2 Beta=100.

Figure 5.4d shows the plots using calculated Vo from for an effective Beta2 of 50%. The plots are very similar to Figure 5c which used an empirical function. The use of Equation (95) to calculate the offset voltage Vo for the approximate Beta2_{Eff} function has the advantage of no longer needing manual fitting which is a big advantage.

It was also found that the 'Beta1 · Beta2 · Re' term (Equation (83) which caters for very high Rbe

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values) could be set to a very high value of times (100Meg) when Vo was calculated. This allows a simpler form for Vsw $V_{sw} = V_t \frac{V_{k2}}{V_t} \frac{Re4}{Rbe4} + V_t \cdot \ln \left(\frac{V_{k2}}{V_t} \cdot \frac{Re4}{Rbe4} \right) + V_{k1}$ when Vo is calculated as above.

To sum up, it is very pleasing to see that the approximate solution used here does provide a very close fit to the SPICE circuit behaviour using the same model parameters. The effective Beta for Q2 with a base emitter resistor can now be represented with good accuracy and without manual fitting factors (when the offset constant Vo is calculated). This method can be applied to other transistor pairs involving a resistor (Rbe) across a second transistor in a compound pair.

The next section completes the CFP by adding shunt feedback. But an intermediate step is needed where Rin is added to Q1.

5.5 SE CE Complementary Feedback Pair with Rin and Rbe (no Rf)

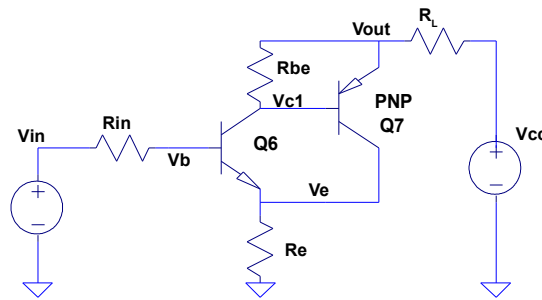


Figure 5.5. Common Emitter Complementary Feedback Pair with Rin and Rbe (no Rf)

It was originally hoped that Rf and Rin could be introduced simply by substituting Beta2_{Eff} for Beta2 in the previous section. This worked well in the previous section allowing an approximated solution when Rbe was added. But when Rin is added it was found that Vsw needed an additional offset term (Vk2_{Rin}) to track the effect of Rin on the knee voltage for the Beta2_{Eff} function. When this was added the Beta2_{Eff} function gave a reasonably good approximation with high Rin values (around Beta1 · Beta2 · Re) and the current was accurate also.

The equations used to allow for Rin are:

$$I_{e2}' = (Beta2_{Eff} + 1) \cdot \frac{NIV_t}{Req(fVin)} \cdot W \left(\exp \left(\frac{Vin - V_{k1}(fVin)}{NIV_t} \right) \right) - Is1 \quad (96)$$

$$\text{where } V_{k1}(fVin) = NIV_t \cdot \ln \left(\frac{NIV_t}{Is1 \cdot Req(fVin)} \right) - Is1 \cdot Req(fVin) \quad \text{where}$$

$$Req(fVin) = \frac{Rin}{Beta1} + (Beta2_{Eff} + 1) \left(\frac{Beta1 + 1}{Beta1} \right) \quad (97)$$

and the Beta2_{Eff} function is calculated using an offset voltage in Vk2_{Rin} in Vsw and Isw

$$Beta2_{Eff} \simeq \frac{Beta2}{\left(1 + \frac{I_{sw} \cdot Beta2 \cdot Re0 + Vo}{Max(Vin - V_{sw}, 0)} \right)} \quad (98)$$

Where Re0 is the equivalent emitter resistance before Q2 starts conducting (no feedback)

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$$R_{e0} = \frac{R_{in}}{\beta_{e1} \cdot (\beta_{e2} + 1)} + \frac{R_{e0} \cdot (\beta_{e1} + 1)}{\beta_{e1}} \quad (99)$$

and

$$V_{sw} = V_{k2} \cdot R_{e0} \left(\frac{1}{R_{be}} + \frac{1}{\beta_{e1} \cdot \beta_{e2} \cdot R_{e0}} \right) + V_t \cdot \ln \left(\frac{V_{k2} \cdot R_{e0} \left(\frac{1}{R_{be}} + \frac{1}{\beta_{e1} \cdot \beta_{e2} \cdot R_{e0}} \right)}{V_t} \right) + V_{k10} + V_{k2_{Rin}} \quad (100)$$

and

$$I_{sw} = \frac{N I V_t}{R_{e0}} \cdot W \left(\exp \left(\frac{V_{in} - V_{k10} - V_{k2_{Rin}}}{N I V_t} \right) \right) \quad (101)$$

$$V_{k10} = N I V_t \cdot \ln \left(\frac{N I V_t}{I_{s1} \cdot R_{e0}} \right) - I_{s1} \cdot R_{e0}$$

and

$$V_o = 0.2V \cdot \left(\frac{R_{be}}{\beta_{e2} \cdot R_{e0}} \right)^{-0.8} \quad \text{and} \quad V_{k2_{Rin}} = V_{k2} \frac{R_{in}}{R_{e0} \cdot (\beta_{e1} + 1) \cdot \beta_{e2}} \quad (102)$$

Where the 0.2V is a 200mV constant found by empirical fitting as above. Alternatively, the value for V_o can be calculated using the 50% beta method above.

Figure 5.5a shows R_{in} is stepped from 1k, 100k, 1Meg, 2Meg and the knee voltage shift to the right can be seen with the higher input resistance. There is also a reduction in slope. Interestingly, with R_f absent the equations do not require a scaling of V_{in} (see below); it might be expected that V_b is the voltage to use rather than V_{in} , but the Lambert W-function solution does not need this in the simple case with R_{in} as seen in the simple CE solution (Figure 4.1a) where R_{in} is lumped into R_{eq} for a solution. However, notice in the single transistor circuit (Figure 4.2) with R_f present there is a voltage divider effect on V_{in} in the equation for I_c Equation (56).

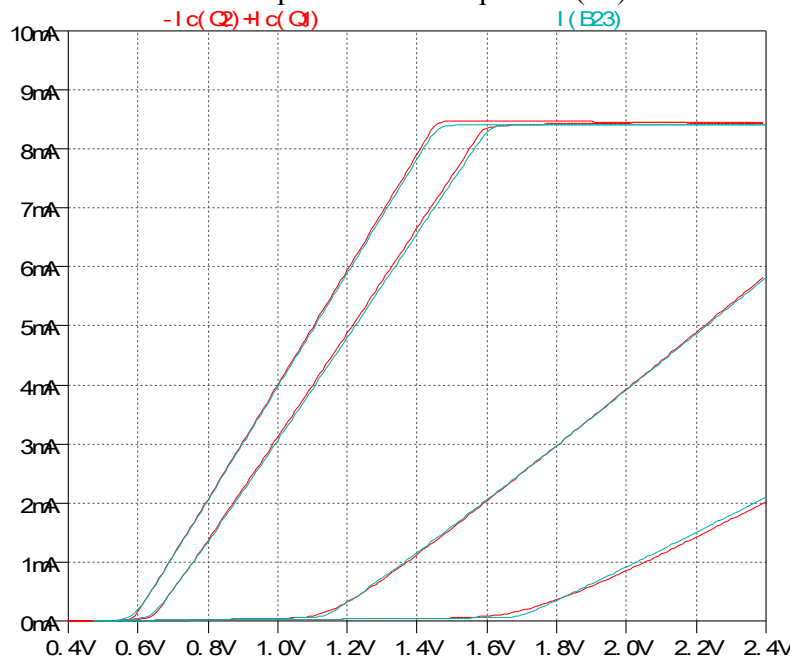


Figure 5.5a. Common Emitter Complementary Feedback Pair with R_{be} and R_{in} (no R_f). R_{in} is stepped 1k, 100k, 1Meg, 2Meg showing the offset in the knee voltage and a reduction in

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slope. Red is LTspice. Q1 and Q2 Beta=100.

5.6 SE CE Complementary Feedback Pair with Rbe, Rf & Rin

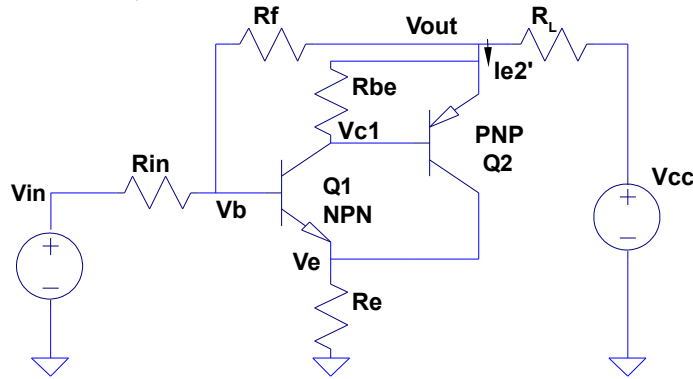


Figure 5.6 Common Emitter Complementary Feedback Pair with Rbe, Rf & Rin

We can now solve currents I_{c1} and I_{c2} in the circuit of **Figure 5.6** with R_f and R_{be} by employing $\text{Beta}2_{\text{Eff}}$ with Equation (85). The combined currents through R_{be} and Q_2 's emitter $I_{e2'}$ as noted above is $(I_{c1} + I_{c2})$, which equals $I_{c1} \cdot (\text{Beta}2_{\text{Eff}} + 1)$, which conveniently allows us to substitute $\text{Beta} \rightarrow \text{Beta}1(\text{Beta}2_{\text{Eff}} + 1)$. The updated Equations become

$$I_{e2'} = (\text{Beta}2_{\text{Eff}} + 1) \cdot \frac{NIVt}{\text{Req}(fV_{in})} \cdot W \left(\exp \left(\frac{\left(\frac{R_f + R_L}{R_{\text{Tot}}} \right) V_{in} - V_{k1}(fV_{in}) + \frac{R_{in}}{R_{\text{Tot}}} V_{cc}}{NIVt} \right) \right) - I_{s1} \quad (103)$$

$R_{\text{Tot}} = R_{in} + R_f + R_L$

$$\text{where } V_{k1}(fV_{in}) = NIVt \cdot \ln \left(\frac{NIVt}{I_{s1} \cdot \text{Req}(fV_{in})} \right) - I_{s1} \cdot \text{Req}(fV_{in}) \quad \text{and}$$

$$\begin{aligned} \text{Req}(fV_{in}) = & \frac{R_{in}}{\text{Beta}1} \left(\frac{R_f}{R_{\text{Tot}}} + \frac{R_L (\text{Beta}1 (\text{Beta}2_{\text{Eff}} + 1) + 1)}{R_{\text{Tot}}} \right) \\ & + \frac{R_e}{\text{Beta}1} \cdot (\text{Beta}1 (\text{Beta}2_{\text{Eff}} + 1) + 1) \end{aligned}$$

and the output voltage can now be calculated from

$$\begin{aligned} V_{out} = & V_{cc} - \left(1 + \frac{R_e}{R_{in}} \right) \left(\frac{\text{Beta}1 (\text{Beta}2_{\text{Eff}} + 1) + 1}{\text{Beta}1 (\text{Beta}2_{\text{Eff}} + 1)} \right) R_L \cdot I_{e2'} - \\ & \frac{R_c}{R_{in}} \left(V_{in} + NIVt \ln \left(\frac{I_{e2'} (\text{Beta}2_{\text{Eff}} + 1)}{I_{s1}} \left(\frac{\text{Beta}2_{\text{Eff}} + 1}{\text{Beta}2_{\text{Eff}}} \right) + 1 \right) \right) \end{aligned} \quad (104)$$

The $\text{Beta}2_{\text{Eff}}$ term is calculated from modified equations below substituting $V_{in} \rightarrow V_{in}' = V_{in} \cdot (R_f + R_L) / R_{\text{Tot}}$. Note V_b due to V_{in} is reduced by the voltage divider action of $R_f + R_L$ with R_{Tot} as suggested in Equation (56). When R_f is added it was found that V_{sw} needs a further offset term to track the effect of R_f on the knee voltage for the $\text{Beta}2_{\text{Eff}}$ function so this offset voltage was subtracted from $V_{k2_{\text{Rin}}}$, giving

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$$Beta2_{Eff} \simeq \frac{Beta2}{\left(1 + \frac{I_{sw}' \cdot Beta2 \cdot Re0 + Vo'}{Max(Vin' - V_{sw}', 0)}\right)} \quad \text{where} \quad Vin' = Vin \cdot \frac{Rf + RL}{RTot} \quad (105)$$

and

$$Re0 = \frac{Rin}{Beta1 \cdot (Beta2 + 1)} + \frac{Re \cdot (Beta1 + 1)}{Beta1} \quad (106)$$

and

$$V_{sw}' = V_{k2} \cdot Re0 \left(\frac{1}{R_{be}} + \frac{1}{Beta1 \cdot Beta2 \cdot Re} \right) + V_t \cdot \ln \left(\frac{V_{k2} \cdot Re0 \left(\frac{1}{R_{be}} + \frac{1}{Beta1 \cdot Beta2 \cdot Re} \right)}{V_t} \right) + V_{k10} + V_{k2_{Rin}} \quad (107)$$

and

$$I_{sw}' = \frac{NIV_t}{Re0} \cdot W \left(\exp \left(\frac{Vin' - V_{k10} - V_{k2_{Rin}}}{NIV_t} \right) \right) \quad \text{and} \quad (108)$$

$$V_{k10} = NIV_t \cdot \ln \left(\frac{NIV_t}{Is1 \cdot Re0} \right) - Is1 \cdot Re0 \quad \text{and}$$

$$Vo2' = 0.2V \cdot \left(\frac{R_{be}}{Beta2 \cdot Re0} \right)^{-0.8} \quad \& \quad (109)$$

$$V_{k2_{Rin}} = V_{k2} \frac{Rin}{Re0 \cdot (Beta1 + 1) \cdot Beta2} - \frac{Rin}{R_{Tot}} \cdot V_{cc}$$

In summary the calculations proceed as follows: Calculate constants Vo2' & Vsw' and Isw'. Then calculate Beta2Eff at each Vin and then Req(fVin). Then calculate Ie2'. Calculate Vout from Ie2' first without saturation. Add Vce saturation to Ie2'. Calculate Vout.

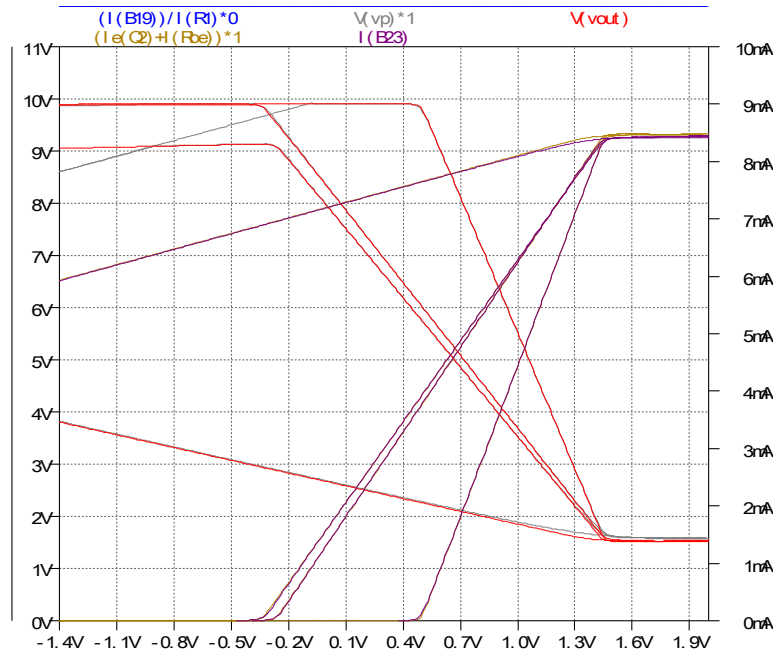


Figure 5.6a Common Emitter Complementary Feedback Pair with Rbe, Rf & Rin. Currents

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rise from left to right and voltages fall. Red is LTspice. R_{in} is stepped 1k, 10k while R_f is stepped 10k, 100k. $R_e=100$ and $R_L=1k$. Q_1 and Q_2 Beta=100. The divergence for $R_{in}=1k$ and $R_f=10k$ is caused by an underflow error of $\ln(x)$.

Figure 5.6a shows the CE CFP with all the components R_{be} , R_f & R_{in} now added. The currents rise from left to right and voltages fall. R_{in} is stepped 1k, 10k while R_f is stepped 10k, 100k, $R_e=100r$ and $R_L=1k$. The divergence (for $R_{in}=1k$ and $R_f=10k$) is caused by an underflow log error previously mentioned (this is left for later to debug). Overall currents are quite accurate, the errors are due mainly to V_{ce} saturation which uses a very simple approximation.

All the calculated currents appear to be quite accurate with the only visible differences at V_{ce} saturation because of the very simple V_{ce} saturation model. This is a very pleasing outcome for an approximated solution. The V_o parameter used the empirical formula but V_o could be calculated and remove the need for the manual fitting terms (left for later).

6. Common Emitter + Common Collector pair (VAS+VF)

This section covers a Common Emitter stage with a Voltage Follower **Figure 6.2** with shunt voltage feedback. This circuit is a simpler version of the Lin Class-AB amplifier **Figure 6.0** with a “VAS” (Voltage Amplification Stage) stage Q_1 . In **Figure 6.2** Q_2 is a Single Ended near-unity gain Voltage Follower stage that operates in Class-A stage. In **Figure 6.0** the output stage in the Lin amplifier is a push-pull Class-AB stage that acts as a Voltage Follower.

Shunt feedback is used in the Lin amplifier. If we can solve **Figure 6.2** then we can use the same approach to solve the Lin amplifier with a push-pull stage. The more common 3 stage power amplifier topology adds a differential input stage to the original Lin topology.

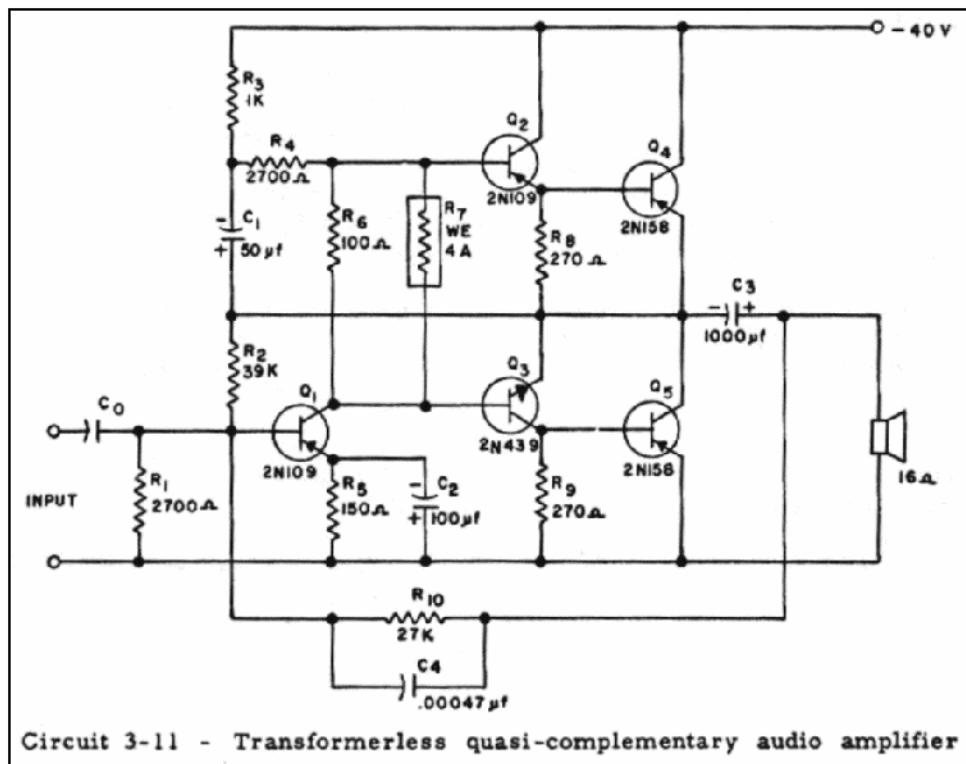


Figure 6.0. Dr. Hung Chang Lin “Jimmy Lin” amplifier

[http://www.semiconductormuseum.com/Transistors/RCA/OralHistories/Lin/Lin_Page7.htm]

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First the open loop solution for the VAS+VF in Figure 6.1.

6.1 CE + CC pair no Rf (VAS+VF in open loop)

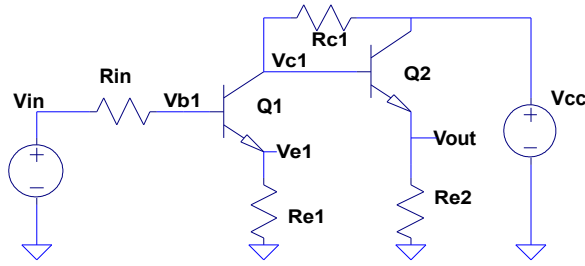


Figure 6.1. Common Emitter plus a Common Collector pair (VAS+VF in open loop)

In Open loop the solution for I_{c2} can be found from I_{c1} and I_{c1} does not depend on I_{c2} since there is no overall feedback in this version. I_{c1} is the solution of the Common Emitter circuit Figure 4.1. Q2 is a Voltage Follower and collector current of Q2 (I_{c2}) can be calculated using another CE solution. This was covered in Section 4.4.

Figure 6.1a shows Q2 as being current driven by Q1 (I_{c1}) with R_{c1} as “ R_{in} ” for a CE stage. R_{c1} is connected to V_{cc} so we can use $V_{cc} - I_{c1} \cdot R_{c1}$ as the input voltage “ V_{in} ” with a CE stage Lambert W solution.

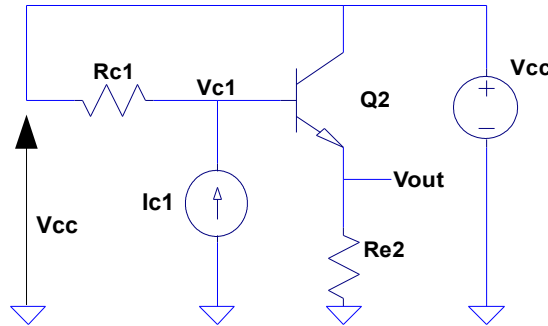


Figure 6.1a. Redrawn Q2 CC as a CE. The second stage is current driven by Q1 (I_{c1}) and V_{cc} supplies current through R_{c1} to the CE stage

For the first stage we write

$$I_{c1} = \frac{N1 \cdot V_t}{Req1} W \left(\exp \left(\frac{V_{in} - V_{k1}}{N1 \cdot V_t} \right) \right) - I_{s1} \quad \text{where} \quad 107(110)$$

$$V_{k1} = N1 \cdot V_t \cdot \ln \left(\frac{N1 \cdot V_t}{I_{s1} \cdot Req1} \right) - I_{s1} \cdot Req1 \quad Req1 = \frac{R_{in}}{\beta_1} + R_{e1} \cdot \left(\frac{1 + \beta_1}{\beta_1} \right) \quad \text{and}$$

and

$$I_{c2} = \frac{N2 \cdot V_t}{Req2} W \left(\exp \left(\frac{V_{cc} - I_{c1} \cdot R_{c1} - V_{k2}}{N2 \cdot V_t} \right) \right) - I_{s2} \quad \text{where} \quad (111)$$

$$V_{k2} = N2 \cdot V_t \cdot \ln \left(\frac{N2 \cdot V_t}{I_{s2} \cdot Req2} \right) - I_{s2} \cdot Req2 \quad \text{and} \quad Req2 = \frac{R_{c1}}{\beta_2} + R_{e2} \cdot \left(\frac{1 + \beta_2}{\beta_2} \right)$$

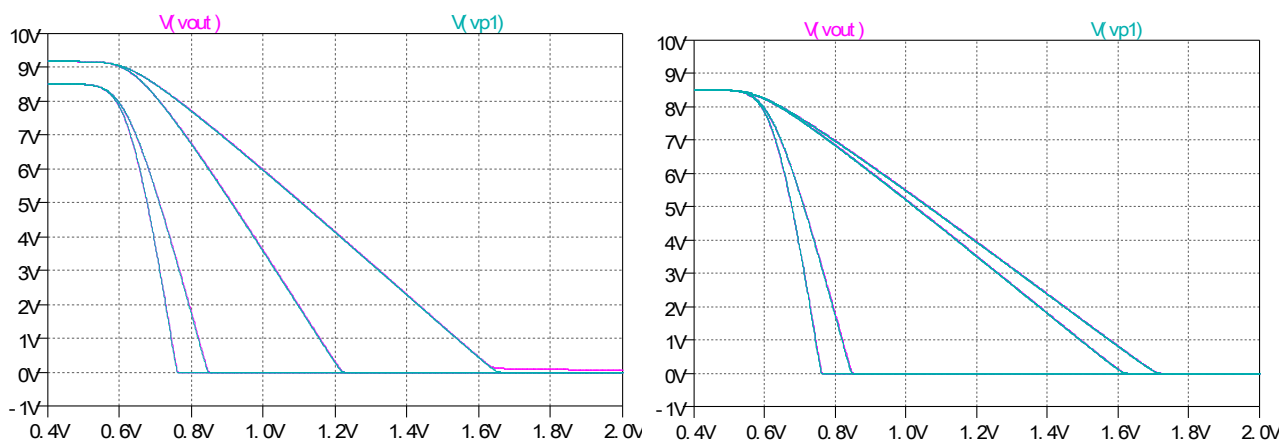


Figure 6.1b. Left is open loop VAS+VF Vout plots with Rc1 stepped 2k & 10k then Re1 stepped 10R & 100R. (Re2=1k, Rin=10k, Beta1=Beta2=100).

Figure 6.1c. Right is Vout in open loop stepped Beta1=10 and 100 then Re1 stepped 10R & 100R (Re2=1k, Rin=10k, Rc1=10k)

Figure 6.1b shows Vout in open loop. Figure 6.1c shows Vout with Beta1 stepped 10 and 100.

6.2 CE + CC pair with Rf (VAS+VF closed loop) [under construction]

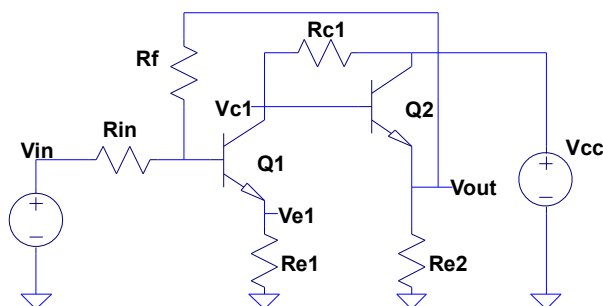


Figure 6.2. Common Emitter plus CC pair (VAS+VF) with global shunt feedback

A closed loop solution with Rf in **Figure 6.2** is not possible as an exact solution. If we can represent Q2 as an effective Beta ($\text{Beta}_{2\text{Eff}}$) as we did in the CFP then we can obtain an approximate solution.

Using Figure 6.1a we can treat Q2 as a Common Emitter stage with Rc1 like “Rin” and connected to an input voltage of Vcc as shown in **Figure 6.2a**.

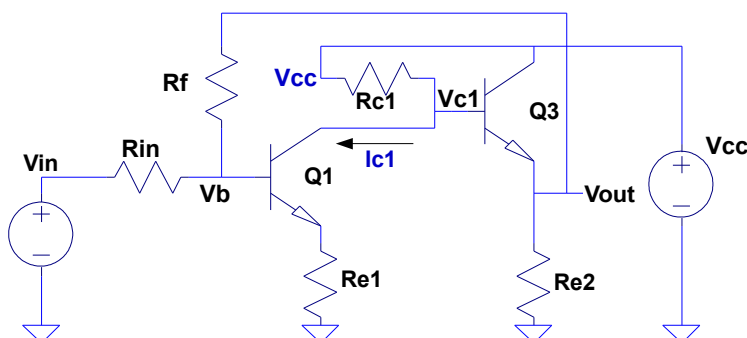


Figure 6.2a. CE+CC redrawn as a CE+CE -- second stage is current driven by Q1

We now run and plot the ratio of I_{c2}/I_{c1} and try to formulate an equation for $\text{Beta}_{2\text{Eff}}$ to allow a

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closed loop solution. **Figure 6.1b** shows the exploratory plot to view the form of a $Beta2_{Eff}$ function by plotting $Ic2/Ic1$ and the current Isc through $Rc1$ due to Vcc . The equation used is

$$Beta2_{Eff} = \frac{Ic2}{Isc - Ic1} \quad \text{where} \quad Isc = \frac{Vcc}{Rc1} \quad (112)$$

Figure 6.1b shows the same characteristic curve as the CFP when Rbe was added but it is now curving back to the left. There are the abrupt knee voltages (Vsw) seen at 835mV and 890mV. There is the plateau to a maximum effective Beta. The maximum Beta occurs when Q1 is completely off so the above equation limits to $Beta2_{Eff} = Ic2_{Max}/Isc$. Notice $Ic2_{Max}$ can be calculated using the Lambert solution

$$Ic2_{Max} = \frac{N \cdot Vt}{Req2} W \left(e^{\left(\frac{Vcc - Vk2}{N \cdot Vt} \right)} \right) \quad \text{where} \quad Vk2 = N2 \cdot Vt \cdot \ln \left(\frac{N2 \cdot Vt}{Is2 \cdot Req2} \right) - Is2 \cdot Req2 \quad \text{and} \quad (113)$$

$$Req2 = \frac{Rc1}{Beta2} + \left(\frac{1 + Beta2}{Beta2} \right) Re2 || (Rf + Rin)$$

and so we can calculate the plateau $Beta2_{EffMax}$ using

$$Beta2_{EffMax} = \frac{\frac{N \cdot Vt}{Req2} \cdot W \left(\exp \left(\frac{Vcc - Vk2}{N \cdot Vt} \right) \right)}{(Vcc / Rc1)} \quad (114)$$

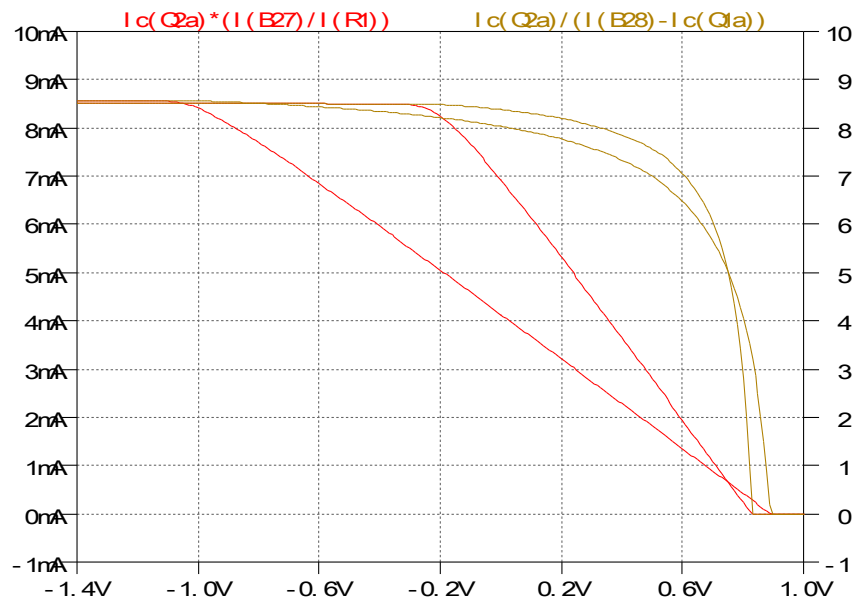


Figure 6.1b. Plot of $Ic2$ (red) and $Beta2_{Eff}$ (gold). Rf is stepped 50k and 100k, $Rin=Re=1k$, $Rc1=10k$. $I(B27)$ calculates offset current $Isc=Vcc/Rc1$.

Next, the Vsw and Isw values are calculated for the knee.

$$e = 1 \quad (115)$$

7. Single Ended Darlington stage

7.1 Single Ended CE Darlington (no Rbe)

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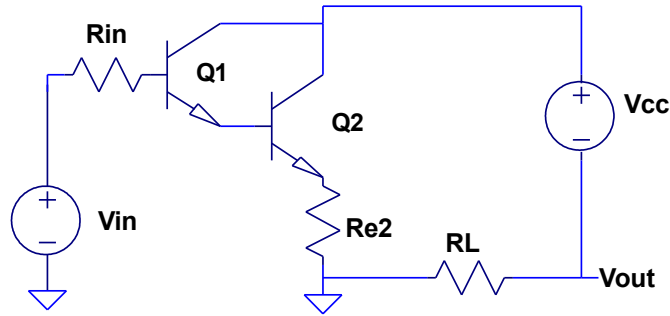


Fig 7.1. Single Ended Common Emitter Darlington stage (no Rbe)

To solve the Darlington Q1 with Q2 for the simplest case where Re1 is omitted, it is first necessary to find the solution or a resistor in series with two base-emitter junctions, effectively two dissimilar diodes in series with an equivalent resistance.

A solution is given in Section 3.3. The solution reduces two diodes to one equivalent diode with an equivalent saturation current of $\sqrt{(Is1 \cdot Is2)}$ and an equivalent thermal voltage of $2 \cdot Vt$.

Plots with a very large Re1 (or Rbe2) are provided in Section 7.3.

7.2 Single Ended Emitter Follower Darlington (no Rbe) [under construction]

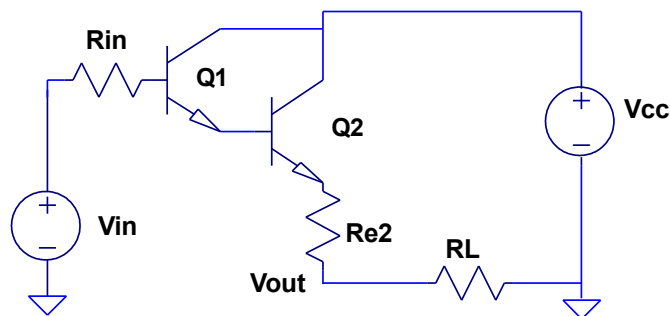


Fig 7.2. Single Ended Emitter Follower Darlington stage(no Rbe)

7.3 Single Ended CE Darlington with Rbe

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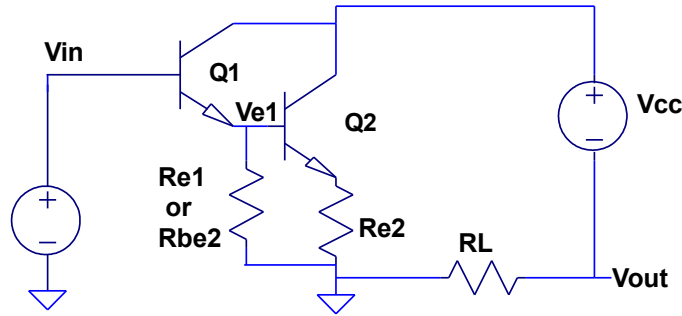


Fig 7.3. Single Ended Emitter Follower Darlington stage with Rbe (no Rin)

Figure 7.3 shows the Common Emitter Darlington stage. Equations are solved assuming constant Beta's and no Early effect. Re1 is treated as a base-emitter resistor for Q2 (or Rbe2). With Re1 present we can assume Q1 gain rises to near unit before Q2 begins conduction therefore we can calculate Ic2 then Ic1, and finally IL as Ic1+Ic2.

Ic2 for Q2 was calculated using the diode plus resistor solution on the emitter side

$$I_{c2} = \frac{N \cdot V_t}{Req} W \left(\exp \left(\frac{0.98 (V_{sp} + V_{in} - V_{k2})}{N V_t} \right) \right) \text{ where } V_{k2} = V_t \ln \left(\frac{V_t}{I_{s12} \cdot Req} \right) \text{ and}$$

$$Req = \frac{R_{in}}{Beta1 \cdot Beta2} + Re2 \left(\frac{Beta2 + 1}{Beta2} \right) \text{ and } N=2 \text{ for two diodes. The scale factor } 0.98 V_{in} \text{ accounts}$$

for the slightly less than unity gain of Q1 once Q2 starts conduction. This gain can be estimated from Q1's intrinsic emitter resistance (V_t/I_e) at the knee voltage (typically 0.65V) where

$$I_e \sim 0.65V/Re1 \sim V_{k2}/Re1. \text{ This gives the gain } AI \approx \frac{Re1}{Re1 + V_t \frac{Re1}{2 V_{k1}}} \text{ so}$$

$$AI_{Rin=0} \approx \frac{1}{\left(1 + \frac{V_t}{2 \cdot V_{k1}} \right)} \text{ where the factor of 2 allows for twice the current assuming Q2 has just}$$

turned on. The factor of 2 can be altered so the final gm of Q2 reaches the same gm as the simulated circuit and this makes the two I_L plots identical. The use of this fixed gain factor approximation simplifies the equations but is not a good general solution approach when Re1 is very large.

The required transition from 2V to 1Vt as Re1 as well as the transition of Is12 changing from $\sqrt{(Is1 \cdot Is2)}$ to Is2 when Re1 is very small (eg $Re1 \sim Re2$) is handled by a simple empirically relation

$$N = \frac{2-x}{1-x} \text{ which changes N from 2 to 1 as x changes from 0 to infinity. The simplest form is}$$

$$x = k \cdot \frac{Beta2 \cdot Re2}{Re1} \text{ where k is a fitting constant that was found when } Re1 = Beta2 \cdot Re2$$

$$\text{requiring } k \sim 2. \text{ This gives } N = \frac{(2 + 2 \cdot Beta2 \cdot Re2 / Re1)}{(1 + 2 \cdot Beta2 \cdot Re2 / Re1)} \text{ and so } V_{k12} = 2 \cdot V_t \cdot \ln \left(\frac{V_t}{I_{s12N} \cdot Re2} \right) + V_o$$

$$\text{where } I_{s12N} = \sqrt{(Is2(Is2(2-N) + Beta2 \cdot Is1(N-1)))} \text{ and } V_o = \frac{4.4 \cdot V_t}{1 + 2 \cdot Re1 / (Beta2 \cdot Re2)}. \text{ The}$$

term Vo is an additional fitting factor that gives the best fit as Re1 becomes very small which approaches 4.4Vt when $Re1 \sim Re2$.

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This approach worked reasonably well where V_{k12} used the fixed scaling of $2V_t$ rather than NV_t . But it involves several fitting terms (later versions may be able to eliminate these like was done for the CFP). An alternative approach that gave accurate values for V_{k12} at $N=1$ and 2 was

$$V_{k12} = 2 \cdot V_t \cdot \ln \left(\frac{V_t}{I_{s12N} R_{e2}} \right) + (2 - N) V_{k1} \quad \text{which involves solving a quadratic equation for } V_o.$$

7.4 Single Ended Common Emitter Darlington with R_{be}

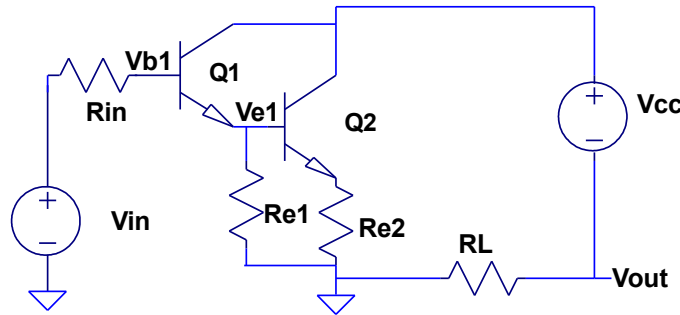


Figure 7.4 shows the Common Emitter Darlington stage with R_{be1} and R_{in} .

One way to approximate the effect of including R_{in} is to use

$$R_T = R_{e2} + \frac{1}{\left(\frac{1}{R_{in}/(\beta_{e1} \cdot \beta_{e2})} + \frac{1}{(R_{e1}/\beta_{e2})} \right)} \quad \text{which is recognisable as } R_{e2} \text{ plus the parallel}$$

resistances R_{in} and R_{e1} (both are referred back to the emitter side of Q_2). R_T replaces R_{e2} in the earlier empirical V_{k12} equation. The equation for N deliberately does not use R_T (despite R_{e2} appearing in the equation for N).

An additional voltage offset voltage V_{o2} is needed when R_{in} becomes large as in

$$I_{c2} = \frac{N \cdot V_t}{R_T} W \left(\exp \left(\frac{A1(V_{sp} + V_{in} - V_{k2}) - V_{o2}}{NV_t} \right) \right) \quad \text{where } A1 \text{ is the voltage divider ratio formed by}$$

R_{in}/β_{e1} and R_{e1} or $A1 = \frac{R_{e1}}{R_{e1} + R_{in}/\beta_{e1}}$ and $V_{o2} = 700m \ln \left(\frac{1/A1}{1 + \ln(1/A1)} \right)$. Notice that $A1$ does not involve R_{e2} – this is because Q_1 will always start to turning on before Q_2 (since R_{e1} is present).

To complete the Darlington the contribution from the driver transistor is important to the supply current and the loading at the input due to Q_1 base current.

The collector current for Q_1 with R_{in} is

$$I_{c1} = \frac{V_t}{R_{e1} + R_{in}/\beta_{e1}} W \left(\exp \left(\frac{2 \cdot V_{sp} + V_{in} - V_{k1}}{V_t} \right) \right) + \frac{I_{c2}}{\beta_{e2}} \quad \text{where}$$

$$V_{k1} = V_t \cdot \ln \left(\frac{V_t}{I_{s1}(R_{e1} + R_{in}/\beta_{e1})} \right) .$$

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Notice I_{c1} is made up of two parts: firstly the current due to R_{e1} using V_{k1} , and second the current that Q1 must supply to drive Q2 which is I_{c2}/β_{a2} . This approach works well for typical values of R_{e1} and R_{e2} . It assumes that Q1 has switched its gain to nearly 100% its limiting value by the time Q2 starts conducting and does not need to appear as a variable in the calculation of I_{c2} , only as a fixed gain of 0.99 (now 1.0 for best fit when R_{in} is large, ie $R_{in} \gg \beta_{a1} \cdot \beta_{a2} \cdot R_{e2}$).

The power rail current is I_{c1} plus I_{c2} and is the same as the load current. The total emitter current ($I_{R_{e1}} + I_{e2}$) is slightly more than the power rail current and load current by the base current of Q1. Parasitic emitter and base resistances can be included where $\beta_{a1} \cdot R_{e1_p} + R_{b1_p}$ can be added to R_{in} while and $R_{e2_p} + R_{b2_p}/\beta_{a2}$ added to R_{e2} .

Gm's for Q1 and Q2 are calculated separately using $W(x)$ with R_{e1} and $W(y)$ with R_T . A component of Q2's gm being g_{m2}/β_{a2} is also added to Q1 gain to account for some I_{c1} driving Q2. Gm for the supply rail current and hence the load is the sum of Q1 and Q2 gains. The total gm seen at the load can be used to estimate THD.

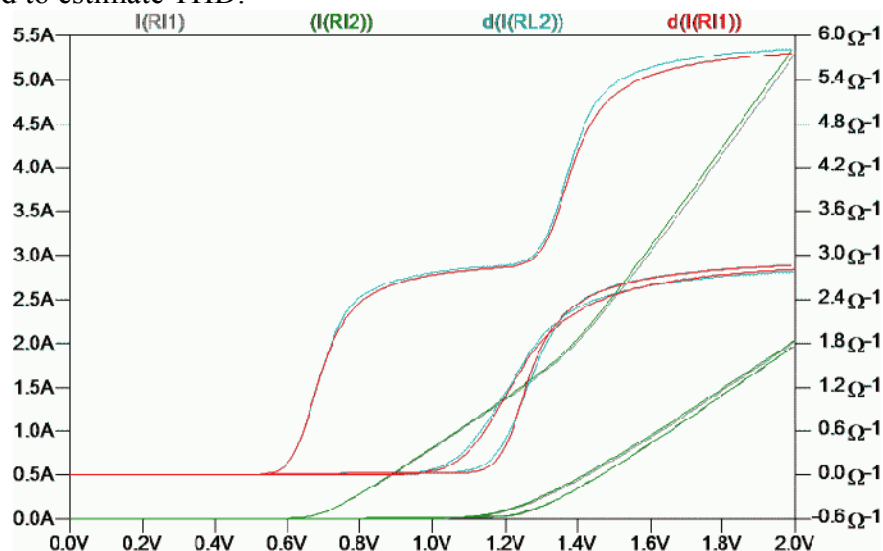


Figure 7.4a. SE CE Darlington with R_{e1} stepped 0.33R (Highest gain and current), 33R and 33k (lower slope) ($R_{e2}=0.33R$, Red is SPICE)

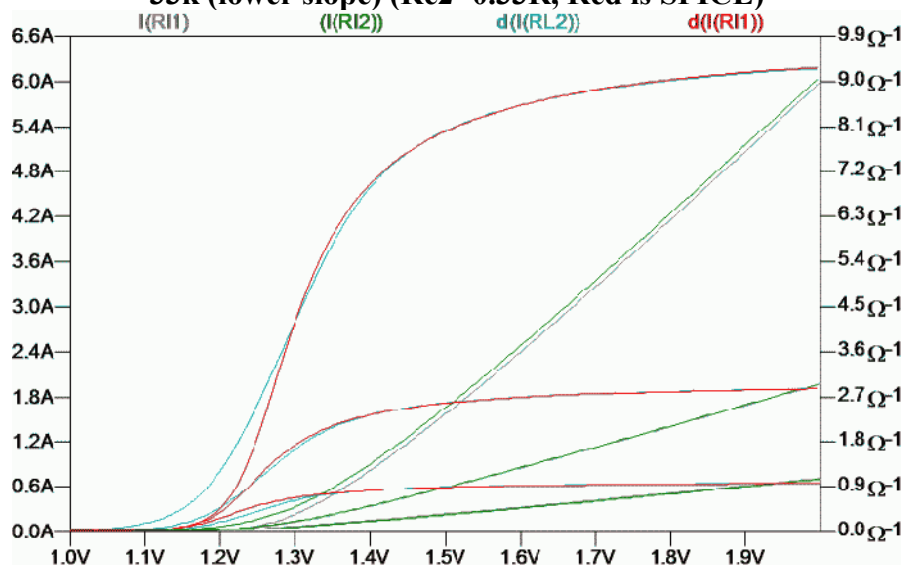


Figure 7.4b. SE CE Darlington with R_{e2} stepped 0.1R (highest Gm & Ic), 0.33R and 1.0R ($R_{e1}=33R$, Red is SPICE)

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Accuracy in **Figure 7.4a** where Re1 is stepped from a very low value to a very high value (0.33R to 33k) and with Re2 fixed at 330mR is very good. Note the slope change for 33k (the lower slope curve). In **Figure 7.4b** where Re2 is stepped while Re1 is fixed at 33R (a typical value since Beta2*Re2=33) the accuracy is not maintained around the turn on region for a low Re2 of 100mR. This seems to be due to the empirical approximation which does not account for Q2's loading on Q1 when Q2 starts to turn on when Re2 is a low value, if so then making the fixed constant 0.98, a variable involving Q2's during turn on may improve the accuracy when Re2 is small. But with the present solution Ic2 a small offset voltage can be applied to shift Ic2 to the right to improve accuracy in Ic2 is Re2 is constant at 100mR.

Figure 7.4c shows the SE CE Darlington with Rin stepped from 33mR to 330k while Re1 is fixed at a typical value of 33R and Re2 is 330mR. Increasing Rin causes the knee voltage to increase slightly. (Note the Gm reduces as Rin is increased).

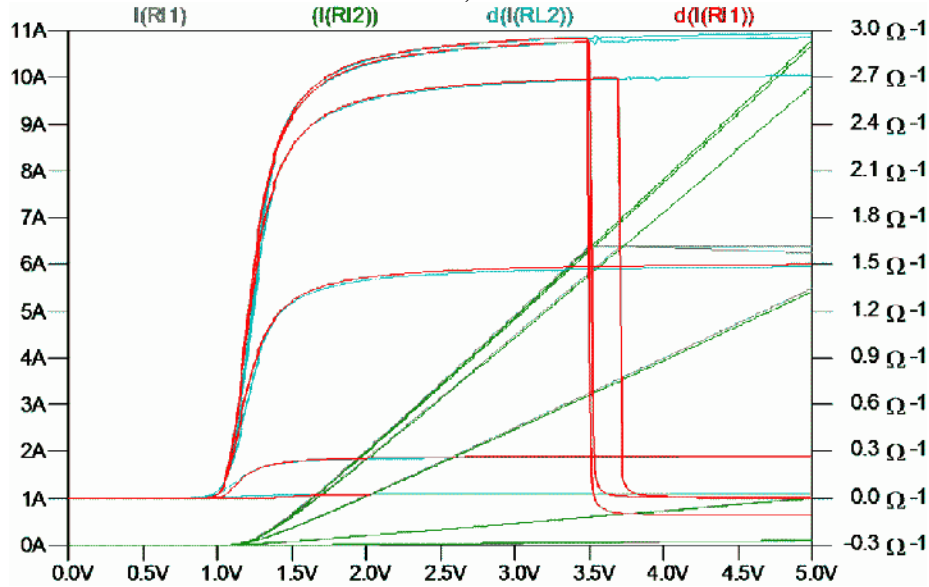


Figure 7.4c. SE CE Darlington with Rin stepped from 33mR to 330k (Re1=33R, Re2 330mR, Red is SPICE)

Solution with a high Rin and current drive

In a typical amplifier if the preceding stage is common emitter then it will have a relatively high output resistance and this can be represent by Rin shunted to common and a current source driving to the base of Q1. Current source drive can be provided using the Norton equivalent $I_{in} = V_{in}/R_{in}$

and with Rin shunted to common. The level of peak current needed is $I_{in_{pk}} = \frac{I_{c2_{pk}}}{\beta_{a1} \cdot \beta_{a2}}$.

The above equations for Ic1 and IC2 with current drive become

$$I_{c2} = \frac{N \cdot V_t}{R_T} W \left(\exp \left(\frac{A1(V_{sp} + I_{in} \cdot R_{in} - V_{k2}) - V_{o2}}{N V_t} \right) \right) \text{ and}$$

$$I_{c1} = \frac{V_t}{R_{e1} + R_{in} / \beta_{a1}} W \left(\exp \left(\frac{2 \cdot V_{sp} + I_{in} \cdot R_{in} - V_{k1}}{V_t} \right) \right) + \frac{I_{c2}}{\beta_{a2}}$$

where terms for Ic1 and Ic2 have been defined above for voltage drive.

Figure 7.4a shows the SE CE Darlington with current drive and Rin stepped 3k3, 33k and 330k

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($R_{e1}=33k$, $R_{e2} 330mR$, Red is SPICE). Here R_{e1} is deliberately very high for a negligible effect. The linearising effect can be clearly seen when current drive is used. Notice that when the shunt resistance is increased under current drive the nonlinearity is decreased and the knee voltage is reduced. The change in gain from 1A to 5A with $R_{in} = 3k3$ is 6%, with $R_{in}=33k$ it is 1% and 0.08% with 330k. When R_{e1} is the usual 33R there is no useful improvement in linearity with current drive since R_{e1} is effectively in parallel with R_{in}/β_{a1} so R_{e1} does not allow the high input resistance of R_{in} to be seen by Q2.

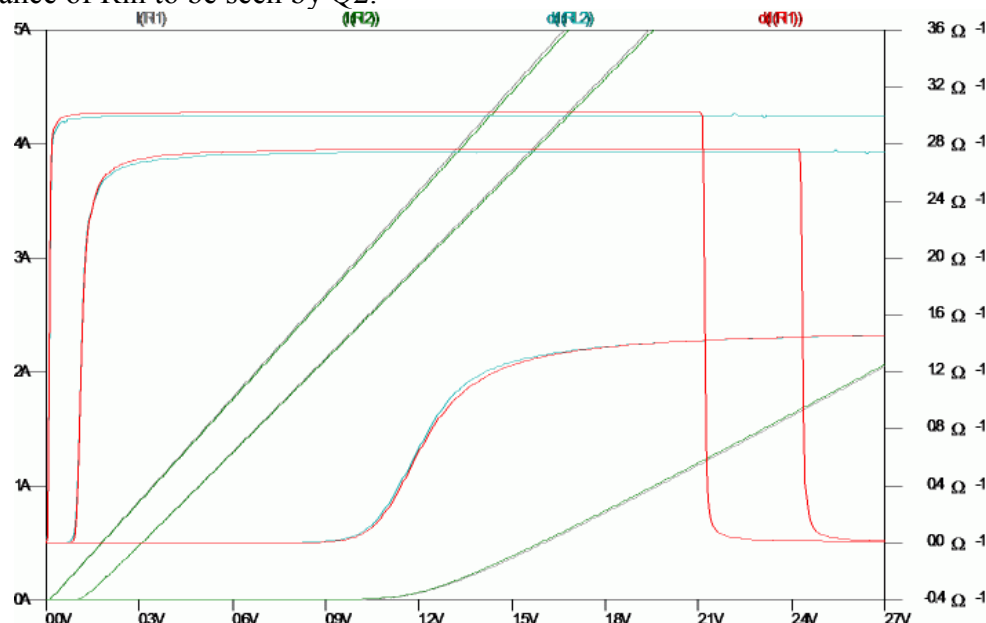


Figure 7.4a. Darlington pair current and gain with a current source. R_{in} stepped 3k3, 33k and 330k. $R_{e1}=33k$ (for negligible effect) and $R_{e2} 330mR$. Red is LTspice.

7.5 Single Ended Emitter Follower Darlington with R_{be} and R_{in} [under construction]

Figure 7.5 shows an Emitter Follower (EF) Darlington stage. Emitter resistors are R_{e1} for driver Q1 and R_{e2} for Q2. Equations are solved assuming constant Beta's and no Early effect.

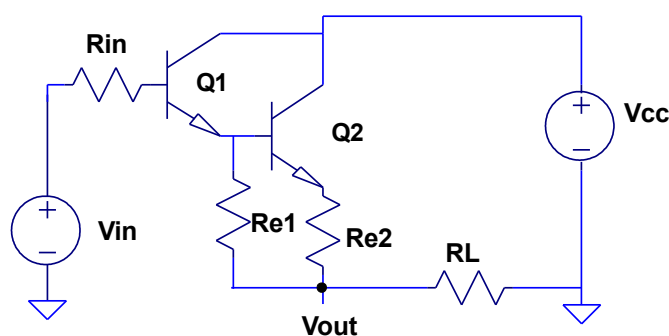


Fig 7.5. Single Ended Emitter Follower Darlington stage with R_{be}

8. The Differential pair

8.1 The differential input stage

Figure 8.1 shows a Differential input stage or the Long Tail Pair (LTP) shown with a 10mV input

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voltage and the output current is summed using an equation in ABM B1.

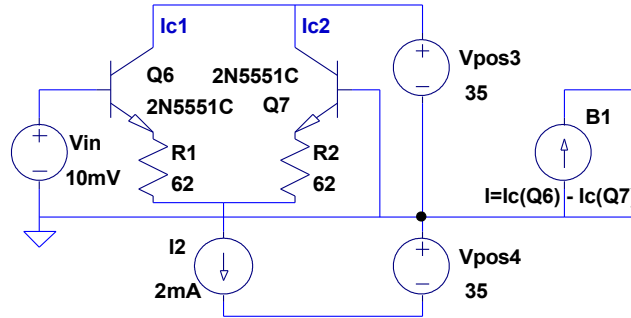


Figure 8.1. Differential input stage (aka Long Tail Pair or LTP)

When no emitter resistors are present and the transistors parasitic base and emitter resistors are negligible the current in each transistor is known to follow the Tanh(x) function [ref 9 Alyward] of

$$I_{c1} = \frac{I_o}{2} \left(\tanh\left(\frac{V_{in}}{2V_t}\right) + 1 \right) \left(1 + \frac{1}{\text{Beta}} \right) \quad \text{where } I_o \text{ is the tail current and Beta is the average value}$$

for Q1 and Q2. The factor $1+1/\text{Beta}$ arises from the tail currents adding to I_o and the collector currents are reduced by the base current related to $1/\text{Beta}$.

It was assumed that a solution with emitter resistors must involve the Lambert W-function of a diode plus resistor and the solution had to give the Tanh(x) function in the limit for R_e approaching zero. Since the Lambert diode plus resistor solution gives improving exponential behaviour below the knee voltage it is reasonable to guess that using $W[\exp((V_{in}+V_b)/V_t)]$ in place of the $\exp(V_{in}/(2V_t))$ function used in the $\tanh(V_{in}/(2V_t))$ since $\tanh(x) = \frac{\exp(x) - \exp(-x)}{\exp(x) + \exp(-x)}$.

The above equation for I_{c1} without R_e is $I_{c1} = \frac{I_o}{2} \left(\tanh\left(\frac{V_{in}}{2V_t}\right) + 1 \right) \left(1 + \frac{1}{\text{Beta}} \right)$ which can be

written as $I_{c1} = \frac{I_o}{2} \left(\frac{\exp(x) - \exp(-x)}{\exp(x) + \exp(-x)} + 1 \right) \left(1 + \frac{1}{\text{Beta}} \right)$ or

$$I_{c1} = \frac{I_o}{2} \left(\frac{2 \cdot \exp(x)}{\exp(x) + \exp(-x)} \right) \left(1 + \frac{1}{\text{Beta}} \right) \quad \text{so} \quad I_{c1} = I_o \left(\frac{\exp(x)}{\exp(x) + \exp(-x)} \right) \left(1 + \frac{1}{\text{Beta}} \right).$$

Substituting the equations (yet to be stated) for $W1$ and $W2$ for the Lambert solution for Q1 with $R1$ and Q2 with $R2$ we can guess that $I_{c1} = I_o \left(\frac{W1(x)}{W1(x) + W2(-x)} \right) \left(1 + \frac{1}{\text{Beta}} \right)$. The formulation for $W1$ and $W2$ that gave good accuracy for the collector over a wide range of V_{in} and R_e and were found to be

$$W1(x) = W \left(\exp \left(\frac{V_{in} + V_b}{V_t} \right) \right) \quad \text{where} \quad V_b = I_o \frac{(R1 + R2)}{2} + V_t + 0.8 \ln \left(\frac{((R1 + R2) + V_t/I_o)}{(V_t/I_o)} \right).$$

formulation for I_{c2} is $I_{c2} = I_o \left(\frac{W2(x)}{W1(x) + W2(-x)} \right) \left(1 + \frac{1}{\text{Beta}} \right)$. The gm for I_{c1} can be calculated

using $G_{m1} = \frac{1}{R1 + R2 + V_t/I_o} \frac{W1}{(1 + W1)} \frac{W2}{(1 + W2)}$. Note G_{m2} will be the same as G_{m1} which is expected since the tail source has negligible conductance or leakage of the emitter currents. It also follows that G_{m1} gain fall for $+V_{in}$ must be the same as $-V_{in}$ with a pure tail current source.

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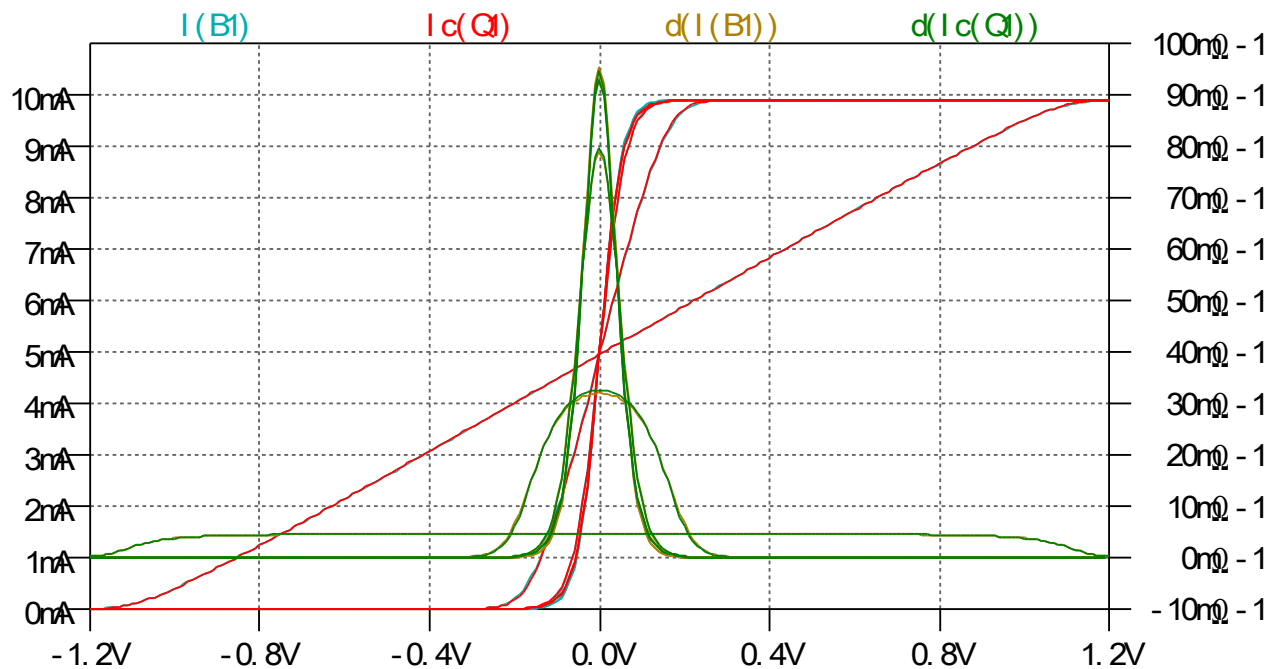


Figure 8.1a. LTP approximate solution compared to SPICE with R_e stepped 1m, 100m, 1R, 10R, 100R ohms, $\beta=100$, $I_s=9e-15A$, $I_o=10mA$.

Fig 8.1a LTP approximate solution compared to SPICE with R_e stepped 1m, 100m, 1R, 10R, 100R ohms, $\beta=100$, $I_s=9e-15A$, $N_f=1$, $I_o=10mA$. Beta-fall and Early effect parameters were disabled for this plot. Accuracy is maintained over a wide range of emitter resistance demonstrating the validity of the approximation.

9. Current Mirrors

9.1 Basic current mirror

Figure 9.1 shows a Basic BJT current mirror. The Lambert W solution is similar to the Widlar current mirror published by Banwell [ref 1] and Atloward [ref 10] but this basic current mirror has resistor R_1 added to emitter of the input transistor Q_1 which effectively operates as a diode .

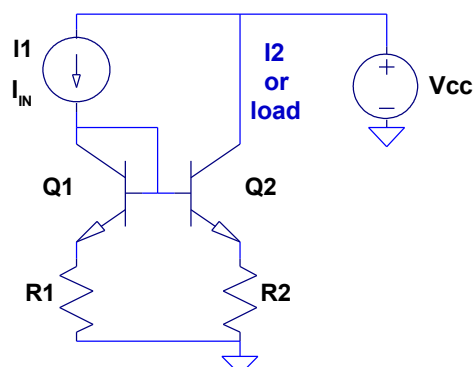


Figure 9.1. Basic bipolar current mirror

The ratio of the areas of Q_1 and Q_2 denoted by $k=A_2/A_1$ determines the mirror current gain I_2/I_1 provided resistors R_1 and R_2 are also in the same ratio. The finite beta's of the transistors drains

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some current from the input giving a mirror current gain that is slightly less than the ratio of the transistor areas. The area ratio k for discrete devices is assumed to be the ratio of saturation currents where $k=I_{s2}/I_{s1}$ for Q2 and Q1 respectively (but differences at low currents can arise from transistor base leakage effects (modelled by I_{se} and N_e)).

Resistor R2 provides emitter degeneration for Q2 which is needs to be a minimum value to prevent thermal runaway; the minimum value depends on V_{ce} of Q2 and collector current I_{c2} . R1 provides linearisation of the mirror current ratio at high currents and is shown below that it requires $R1 \approx k \cdot R2$.

To start the analysis assume the beta's are very high and the base currents can be neglected. The voltage loop equation around the two emitters and R1 and R2 gives

$$I_2 \cdot R_2 - I_1 \cdot R_1 = V_t \ln \left(\frac{I_1}{I_{s1}} + 1 \right) - V_t \ln \left(\frac{I_2}{I_{s2}} + 1 \right) \quad \text{and usually } I_2 \gg I_{s2} \text{ giving}$$

$$I_2 \cdot R_2 - I_1 \cdot R_1 = V_t \ln \left(\frac{I_1}{I_2} \cdot \frac{I_{s2}}{I_{s1}} \right) = V_t \ln \left(\frac{k \cdot I_1}{I_2} \right) .$$

Since I_1 is the input variable we need the form $w e^w = x$ to find $I_2(I_1)$ where x is the input variable and w is fixed by circuit parameters only. Taking exponential gives

$$\exp \left(\frac{I_2 R_2}{V_t} \right) \cdot \exp \left(\frac{-I_1 R_1}{V_t} \right) = \frac{k \cdot I_1}{I_2} \Rightarrow \exp \left(\frac{I_2 R_2}{V_t} \right) = \frac{k \cdot I_1}{I_2} \exp \left(\frac{I_1 R_1}{V_t} \right) \Rightarrow$$

$$I_2 \cdot \exp \left(\frac{I_2 R_2}{V_t} \right) = k \cdot I_1 \exp \left(\frac{I_1 R_1}{V_t} \right) . \text{ Now multiply both sides by } R_2/V_t \text{ to make the LHS of the}$$

$$\text{form } w e^w = x \text{ giving } \frac{I_2 R_2}{V_t} \exp \left(\frac{I_2 R_2}{V_t} \right) = \frac{k \cdot I_1 R_2}{V_t} \exp \left(\frac{I_1 R_1}{V_t} \right) \Rightarrow$$

$$\frac{I_2 R_2}{V_t} \exp \left(\frac{I_2 R_2}{V_t} \right) = \exp \left(\frac{I_1 R_1 - V_t \ln(V_t / (k I_1 R_2))}{V_t} \right) \text{ so } W = \frac{I_2 R_2}{V_t} \text{ and}$$

$$x = \exp \left(\frac{I_1 R_1 - V_t \ln(V_t / (k I_1 R_2))}{V_t} \right) . \text{ We can now find } I_2 \text{ using}$$

$$\frac{I_2 R_2}{V_t} = W \left(\exp \left(\frac{I_1 R_1 - V_t \ln(V_t / (k I_1 R_2))}{V_t} \right) \right) \text{ or finally}$$

$$I_2 = \frac{V_t}{R_2} \cdot W \left(\exp \left(\frac{I_1 R_1 - V_t \ln(V_t / (k \cdot I_1 \cdot R_2))}{V_t} \right) \right) \quad (116)$$

$$\text{where } W(x) \simeq \ln(1+x) \left(1 - \frac{\ln(1+\ln(1+x))}{2+\ln(1+x)} \right) .$$

To make the mirror linear for all input currents (Betas neglected) we can use the above expression

$$I_2 \cdot R_2 - I_1 \cdot R_1 = V_t \ln \left(\frac{I_1}{I_2} \cdot \frac{I_{s2}}{I_{s1}} \right) = V_t \ln \left(\frac{k \cdot I_1}{I_2} \right) \text{ and we want to force the nonlinear RHS to be}$$

ideally zero, so $V_t \ln \left(\frac{k \cdot I_1}{I_2} \right) = 0$ which means $\left(\frac{k \cdot I_1}{I_2} \right) = 1$ or $\frac{I_2}{I_1} = k$ and the LHS is

$$I_2 \cdot R_2 - I_1 \cdot R_1 = 0 \text{ and solving for } R_1 \text{ gives } R_1 = \frac{I_2}{I_1} \cdot R_2 \text{ which means choosing } R_1 = k \cdot R_2$$

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to be linear at high current as well as low currents.

When Betas are included there is base current to Q2 of I_{c2}/β_{a2} and base current to Q1 of I_{c1}/β_{a1} . Since I_1 is dependent on I_2 and we need to know I_1 before we can find I_2 in our solution above so an approximation is used assuming R_1 is $k \cdot R_2$ and $I_{b_{1+2}} \approx \frac{I_{c2}}{\beta_{a2}} + \frac{(I_{c2}/k)}{\beta_{a1}}$ then

calculate I_2 using $I_1' \approx I_1 \left(1 - \left(\frac{k}{\beta_{a2}} + \frac{1}{\beta_{a1}} \right) \right)$ giving

$$I_2 \approx \frac{V_t}{R_2} \cdot W \left(\exp \left(\frac{I_1 R_1 \left(1 - \left(\frac{k}{\beta_{a2}} + \frac{1}{\beta_{a1}} \right) \right) - V_t \ln \left(\frac{V_t}{I_1 R_2 \left(1 - \left(\frac{k}{\beta_{a2}} + \frac{1}{\beta_{a1}} \right) \right)} \right)}{V_t} \right) \right) \quad (117)$$

A simulation with $R_1 = 6.2R$ and $R_2 = 6.2R$ and $I_1 = 10\text{mA}$ gave $I_{c2} = 9.84266\text{ mA}$. The transistor used $\beta_a = 125$, $I_s = 9e-15$, $N_f = 1$ and parasitic R_e and R_b were set to very small values and V_{AF} and I_{KF} were very large. The ABM equations are

B5: $I = (I_1) \cdot R_5 \cdot (1 - (k/\beta_{a2} + 1/\beta_{a1})) - V_t \cdot \ln(V_t / (I_1) \cdot R_6 \cdot (1 - (k/\beta_{a2} + 1/\beta_{a1}))) / V_t$ then

B6: $I = V_t / R_6 \cdot \ln(1 + \exp(I(B5))) \cdot (1 - \ln(1 + \ln(1 + \exp(I(B5)))) / (2 + \ln(1 + \exp(I(B5))))$ where

.param $k = I_{s2}/I_{s1}$. This gives $I_{c2} = 9.83425\text{ mA}$ and the error is 0.1%. Another simulation with the same parameters as above and same ABM equations but $R_1 = R_2 = 62R$ gave $I_{c2} = 9.84253\text{ mA}$.

The ABM result for $I_{c2} = 9.86215\text{ mA}$ and the error is 0.2%.

9.2 The Wilson current mirror [under construction]

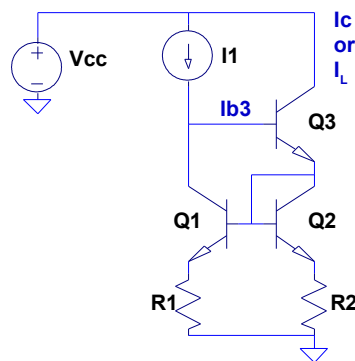


Figure 9.2. The Wilson Current Mirror

$$e = 1 \quad (118)$$

10. Current sources

10.1 Basic constant current source

Figure 10.1 shows a Basic constant current source where an LED diode D1 is used as a voltage

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reference for Q1. The diode knee voltage is around 1.7V for standard LED's and is reported to be more temperature stable than 3 silicon diodes in series to give the same reference voltage.

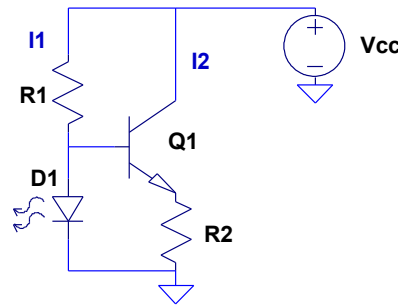


Figure 10.1. Basic constant current source using an LED voltage reference

This can be solved like the basic current mirror except the diode transistor here has a beta of zero (or the collector is open circuit). The LED has a finite equivalent series resistance and this becomes the same as R1 in the basic current mirror.

Following the same path as for the current mirror we neglect the beta of Q1 is at first. The voltage loop equation around the LED and Q1 emitter via LED Rs1 and Q1 R2 gives

$$I2 \cdot R2 - I1 \cdot Rs1 = N1 \cdot Vt \ln \left(\frac{I1}{Is1} + 1 \right) - N2 \cdot Vt \ln \left(\frac{I2}{Is2} + 1 \right) \quad \text{where } Is1 \text{ and } N1 \text{ are for the LED and}$$

N2 and Is2 are for Q1. Combining logs and with $I1 \gg Is1$ and $I2 \gg Is2$ and assuming $N2 \sim 1.0$ gives

$$I2 \cdot R2 - I1 \cdot Rs1 = Vt \ln \left(\left(\frac{I1}{Is1} \right)^{N1} \right) - N2 \cdot Vt \ln \left(\frac{I2}{Is2} \right) \Rightarrow \exp \left(\frac{I2 \cdot R2 - I1 \cdot Re1}{Vt} \right) = \frac{I1^{N1}}{I2} \cdot \frac{Is2}{Is1^{N1}} \quad \text{and}$$

separating I1 and I2 leads to

$$I2 = \frac{Vt}{R2} \cdot W \left(\exp \left(\frac{I1 \cdot Rs1}{Vt} - \ln \left(\frac{Vt}{(I1^{N1} \cdot R2 \cdot Is2 / Is1^{N1})} \right) \right) \right) \quad \& \quad I1 = \frac{(Vcc - V_{LED})}{R1} \quad (119)$$

where V_{LED} can be assumed to be an approximate value of 1.7V or for more accuracy calculated using a recursive approach such as $V_{LED} = N1 \cdot Vt \ln \left(\frac{I1_{app}}{Is1} \right) + I1_{app} \cdot Rs1$ where

$$I1 = \frac{(Vcc - V_{LED})}{R1} \quad . \quad \text{The reduction of } I1 \text{ to the LED for Q1's base current with a finite beta can be}$$

accounted for by $I1' \approx I1 \left(1 - \frac{Is2 / Is1^{N1}}{\beta_{21}} \right)$ as explained in the current mirror above.

A simulation with $R1 = 33k$ and $R2 = 620$ and $Vcc = 35V$ gives $I2 = 1.9701 \text{ mA}$ and $I1 = 1.00283 \text{ mA}$ and $V_{LED} = 1.90662 \text{ V}$. The transistor used $\beta = 125$, $Is = 9e-15$, $Nf = 1$ and parasitic Re and Rb were set to very small values and VAF and IKF were very large. The Green LED is modelled Using Bob Cordell's Green LED [www.cordellaudio.com] where $Is = 1e-19 \text{ A}$, $N1 = 2.0$, $Rs1 = 1.5 \text{ R}$ and $Eg = 2.23 \text{ V}$. The ABM equations are B7: $I = (((V(Vcc) - Vled) / R1) \cdot Rs1 \cdot (1 - 1/\beta_{21}) - Vt \cdot \ln(Vt / (((V(Vcc) - Vled) \cdot (1 - 1/\beta_{21}) / R1) \cdot Is2 / (Is1^{N1})))) / Vt$ then B8: $I = Vt / R2 \cdot \ln(1 + \exp(I(B7))) \cdot (1 - \ln(1 + \ln(1 + \exp(I(B7)))) / (2 + \ln(1 + \exp(I(B7)))))$ where $.param \text{ Vled} = N1 \cdot Vt \cdot \ln((35 - 1.7) / (Is1 \cdot R1))$. This gives $I2 = 1.98662 \text{ mA}$ with error of 0.8%, and $I1 = 0.986959 \text{ mA}$ with error of 1.6%, and $V_{LED} = 1.9059 \text{ V}$ with error of 0.03% using 1.7V as the initial estimate. (The parameter

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Eg=2.23 V was not used in these equation).

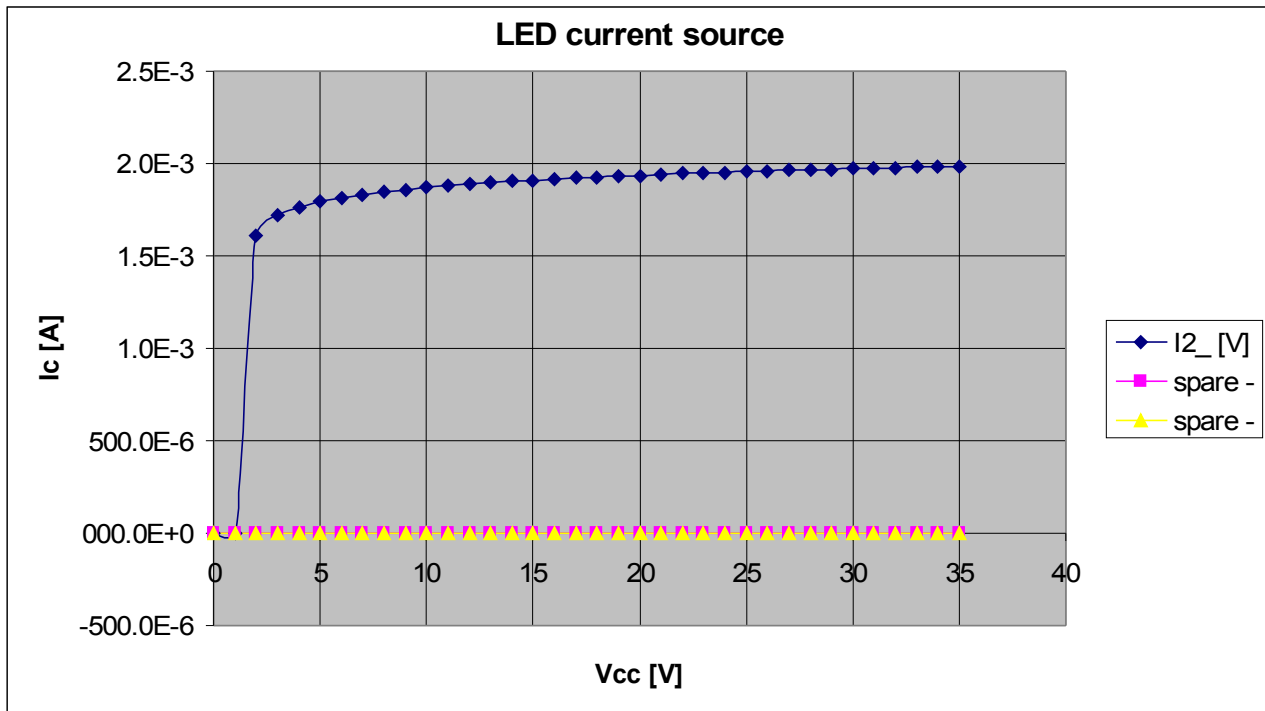


Figure 10.1a. Plot from PAK with Vcc stepped. R1 = 33k and R2=620. At Vcc=35V I2 = 1.987mA. LTSpice I2=1.9701mA (equation error -0.9%).

In **Figure 10.1a** the Early effect has been disabled and the slope is due to the changing current through R1. This slope does not indicate the output resistance of the current source but rather the PSRR of the overall circuit.

10.2 Constant current source with feedback

Figure 10.2 shows a Constant current source with feedback where the current through R1 is multiplied by the Beta of Q2 but when the voltage across R2 is sufficient to start Q1 conducting then base current is robbed from Q2 until an equilibrium is obtained where both are satisfied.

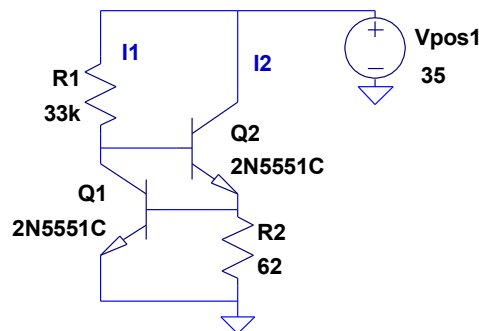


Figure 10.2. Constant current source with feedback

This equation for Ic Q2 in terms of input current I1 via R3 was solved by first assuming the Beta of Q2 is very large so the primary relationship between I2 and I1 can be found. The loading effect of Q2 beta can be added later.

Assuming the Beta of Q2 is very large then we know $V_{be1} = V_t \ln \left(\frac{I_{c1}}{I_{s1}} + 1 \right)$ and since

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$$V_{be1} = \left(I_2 - \frac{I_{c1}}{\beta_{a1}} \right) R_2 \quad \text{and combining these we get the first solution}$$

$$I_2 = \frac{I_{c1}}{\beta_{a1}} + \frac{V_t}{R_2} \ln \left(\frac{I_{c1}}{I_{s1}} + 1 \right) \quad \text{provided } \beta_{a2} \text{ is neglected.}$$

Q1 base current removes some current from I1 via R1 so we need to increase I1 by a sufficient amount. Since a designer does not usually know β_{a2} value a worst case and some more is assumed. Since there is usually not a problem increasing I1 we can choose $I_1/I_2=0.1$ which is roughly the inverse of the square-root of the typical β_{a2} value.

If we define the working factor $m = I_2/I_1$ then we can find the fraction of current that Q1 collector must sink relative to incoming I1 so Q1 has sufficient base current to generate I2. Let $k=I_{c1}/I_1$ then

$$\frac{I_2}{\beta_{a2}} = (1-k) \cdot I_1 \quad \text{so} \quad \frac{I_2}{I_1} = (1-k) \cdot \beta_{a2} = m \quad \text{giving} \quad k = \left(1 - \frac{m}{\beta_{a2}} \right) \quad . \quad \text{Since } I_{c1} = k \cdot I_1 \text{ and}$$

the current through R2 is now $I_e = I_2 \left(\frac{\beta_{a2}}{1 + \beta_{a2}} \right)$ and neglecting the +1 term in the log gives

$$I_2 = \left(\frac{\beta_{a2}}{1 + \beta_{a2}} \right) \left(\frac{k \cdot I_1}{\beta_{a1}} + \frac{V_t}{R_2} \ln \left(\frac{k \cdot I_1}{I_{s1}} \right) \right) \quad \text{and}$$

$$I_2 = \left(\frac{\beta_{a2}}{1 + \beta_{a2}} \right) \left(\frac{I_1}{\beta_{a1}} \left(1 - \frac{m}{\beta_{a2}} \right) + \frac{V_t}{R_2} \ln \left(\frac{I_1}{I_{s1}} \left(1 - \frac{m}{\beta_{a2}} \right) \right) \right) \quad .$$

The first term $I_a = \left(\frac{\beta_{a2}}{1 + \beta_{a2}} \right) \left(\frac{I_1}{\beta_{a1}} \left(1 - \frac{m}{\beta_{a2}} \right) \right) \approx \frac{I_1}{\beta_{a1}} = \frac{I_2}{m \cdot \beta_{a1}}$ will generally be very small or around 0.1% of I2. Therefore finally we can generally use

$$I_2 = \left(\frac{\beta_{a2}}{1 + \beta_{a2}} \right) \frac{V_t}{R_2} \ln \left(\frac{I_1}{I_{s1}} \left(1 - \frac{m}{\beta_{a2}} \right) \right) \quad \text{where} \quad m = \frac{I_2}{I_1} \quad 120/(120)$$

Note $\left(1 - \frac{m}{\beta_{a2}} \right) > 0$ requires $m < \beta_{a2}$ (as expected).

In a calculation we first estimate V_{be1} from V_{k1} using

$$V_{be1_{Est}} \approx V_{k1} = N_1 \cdot V_t \cdot \ln \left(\frac{N_1 \cdot V_t}{I_{s1} \cdot R_2} \right)$$

This then allows an estimate for $I_{c2_{Est}} = V_{be1_{Est}}/R_2$ and then an estimate for $m \sim I_{c2_{Est}}/I_{1_{Est}}$ where $I_{1_{Est}} \sim (V_{cc} - 2V_{be1_{Est}})/R_1$. Now we can use Equation (120) to calculate I2.

Using LTspice ABM with $I = (\beta_{a2}/(1 + \beta_{a2})) \cdot V_t/R_4 \cdot \ln((V(V_{cc}) - 2 \cdot V_{be})/R_3)/I_{s1} \cdot (1 - m/\beta_{a2})$ where $m=10$, $\beta_{a1}=\beta_{a2}=125$, $I_s=9e-15$, $N_f=1$, $R_3=33k$, $R_4=62$, and $V_t=25.86$. This gives $I_2=10.4967mA$ while the simulation (with negligible R_e , R_b and V_A , IKF disabled) gave $10.5057mA$ which is to within 0.1%. The value used for V_{be} was $0.717V$ from $V_t \cdot \ln(10m/I_s)$ and the simulated value was $0.723V$. This difference hardly affect the accuracy of I2 since V_{b1} essentially determines I2 and V_{be2} only affects the drop across R_1 and hence the accuracy of I1.

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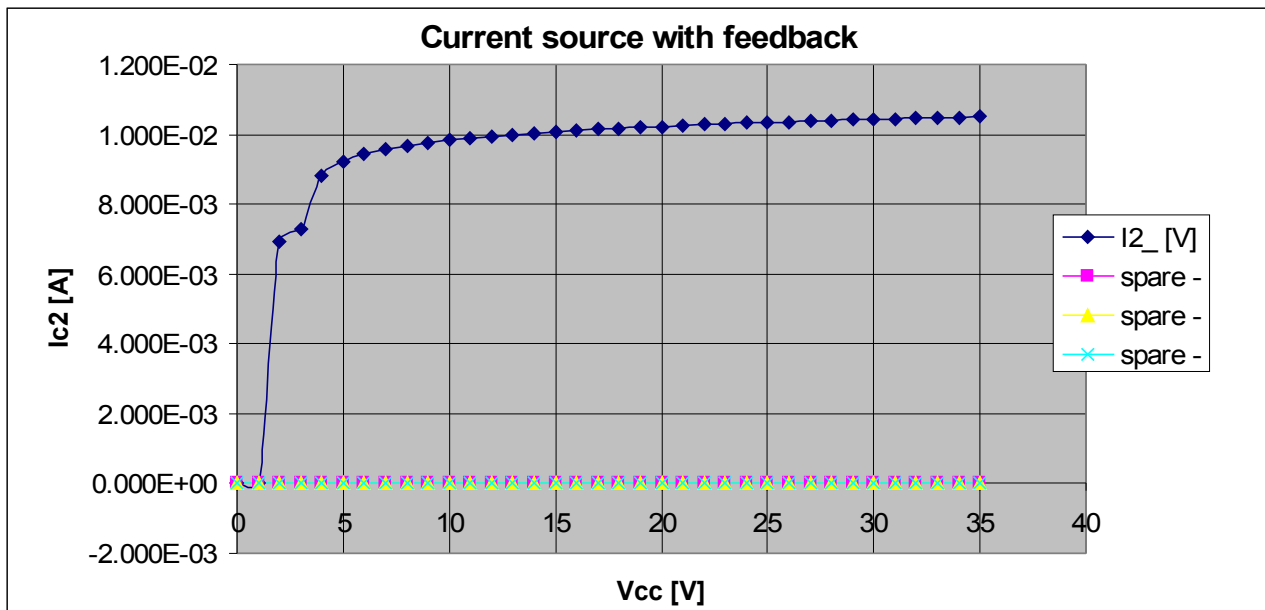


Figure 10.2a. Plot from PAK with Vcc stepped. R1 = 33k and R2=62. At Vcc=35V I2 = 10.50mA. LTSpice I2=10.513mA (equation error -1.2%).

In **Figure 10.2a** the Early effect has been disabled and the slope is due to the changing current through R1. This slope does not indicate the output resistance of the current source but rather the PSRR of the overall circuit.

To design a practical feedback current source you first decide what source current I2 you want, then choose a workable value for $m \sqrt{(\text{Beta}2_{typ})}$ then calculate $R1 = \frac{(V_{cc} - 2 V_{be_{approx.}})}{I2/m}$. You then

calculate R2 using $R2 = \left(\frac{\text{Beta}2}{1 + \text{Beta}2} \right) \frac{V_t}{I2} \ln \left(\frac{I2/m}{I_s} \left(1 - \frac{m}{\text{Beta}2} \right) \right)$. If you don't need such high

accuracy for R2 then simply use $R2 = I2/V_{be}$ where V_{be} is around 640mV to 660mV at room temperature for 1mA with a 100mA transistor and changes by 60mV for every decade of current change.



Tip: A multimeter with a diode tester can provide V_{be} (usually at 1.5mA) if you connect the base and collector together and read the voltage. If you want V_{be} at a very high current then measure just the base emitter voltage at 1.5mA (giving its V_{be} at $I_c \sim \text{Beta}/3 \cdot 1.5\text{mA}$ where $\text{Beta}/3$ roughly allows for some Beta-fall at high current).

11. Vbe multipliers

11.1 The Vbe multiplier

The V_{be} multiplier also called the multiplied diode is used in Class-B power amplifiers for temperature information feedback, usually the temperature of the heatsink, to maintain roughly the same quiescent current with ambient temperature variations, and also to stabilise the quiescent current after turn otherwise the quiescent would slowly increase over a long time and eventually destroy the output transistors.

The V_{be} multiplier is explained compactly in Bob Cordell's book [ref 11 p40]. The V_{be} multiplier

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is effectively a two terminal device that behaves like a diode with a variable knee voltage determined by the divider ratio of Rf and Rbe which is conveniently a trimpot. It is known that it generates a variable 'knee voltage' of slightly over (1+Rbe/Rf)Vbe with series resistance that is then shunted by Rf plus Rbe.

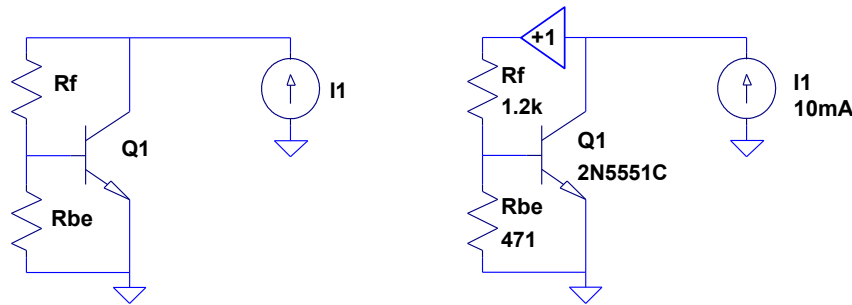


Figure 11.1. The Vbe multiplier. Figure 11.1a. Buffered version

To derive an equation for Vce in terms of an input current I1 in **Figure 11.1** (left) it is made easier if the set-up current drawn by Rf and Rbe can be neglected, made possible by buffering as in **Figure 11.1a** (right). The collector side Collector current can be expressed in terms of an assumed Vce where

$$I_c = \frac{NV_t}{R_{eq}} W \left(\exp \left(\frac{V_{ce} \left(\frac{R_{be}}{R_{be} + R_f} \right) - V_k}{NV_t} \right) \right) - I_s \quad \text{where } R_{eq} = \frac{(R_{be} || R_f)}{\beta} + \frac{R_{b_p}}{\beta} + R_{e_p} \left(\frac{1 + \beta}{\beta} \right)$$

and $V_k = NV_t \cdot \ln \left(\frac{NV_t}{I_s \cdot R_{eq}} \right) - I_s \cdot R_{eq}$. R_{e_p} and R_{b_p} are the emitter and base parasitic resistances.

$W(x)$ can be calculated using an approximation such as $W(x) \approx \ln(1+x) \left(1 - \frac{\ln(1+\ln(1+x))}{2+\ln(1+x)} \right)$

We want Vce as a function of Iin, so we need to invert the equation above by returning it to the form

$w e^w = x$ to give $\frac{I_c R_{eq}}{NV_t} = W \left(\exp \left(\frac{V_{ce} \left(\frac{R_{be}}{R_{be} + R_f} \right) - V_k}{NV_t} \right) \right) - I_s$ and we identify $w = \frac{I_c R_{eq}}{NV_t}$

and $x = \exp \left(\exp \left(\frac{V_{ce} \left(\frac{R_{be}}{R_{be} + R_f} \right) - V_k}{NV_t} \right) \right)$. Taking logs of $w e^w = x$ gives $\ln(w) + w = \ln(x)$ and

substituting w and x gives $\ln \left(\frac{I_c R_{eq}}{NV_t} \right) + \frac{I_c R_{eq}}{NV_t} = \frac{V_{ce} \left(\frac{R_{be}}{R_{be} + R_f} \right) - V_k}{NV_t}$. From this

$$V_{ce} = \left(1 + \frac{R_f}{R_{be}} \right) \left(V_k + NV_t \ln \left(\frac{I_c R_{eq}}{NV_t} \right) + I_c R_{eq} \right) \quad \text{where } R_{eq} \text{ and } V_k \text{ are given above.}$$

We can now see that the voltage divider requires that Vce is increased by the inverse factor $(1+Rf/Rbe)$ to sink all the input current I1. This solution looks a bit like an ideal diode of $V_D = V_t \cdot \ln(I_c/I_s + 1)$ except the resistors that set the base voltage enter into the equation as an equivalent emitter resistance, and this resistance provides emitter degeneration local type negative feedback. The Vbe diode volt drop is scaled up by the inverse of the voltage divider ratio. This is the familiar situation to the gain of a non-inverting opamp where $A_v = 1 + R_f/R_{in}$ where the feedback around the opamp inverts the voltage divider ratio of the feed back path.

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What about the equivalent series resistance or dynamic resistance, is it affected by the divider ratio?

The derivative dI_c/dV_{ce} can be found using our first equation where $\frac{dI_c}{dV_{ce}} = \frac{1}{R_{eq}} \frac{W}{1+W}$

allowing the equivalent resistance collector to emitter to be found using

$$R_{ce} = \frac{dV_{ce}}{dI_c} = R_{eq} \left(\frac{1+W}{W} \right) = R_{eq} \left(1 + \frac{1}{W} \right) . \text{ Since } I_c = I_1 \text{ with the buffer and } I_1 \text{ is the input}$$

variable we found above that $w = \frac{I_c R_{eq}}{V_t}$ giving $R_{ce} = R_{eq} \left(1 + \frac{NV_t}{I_c \cdot R_{eq}} \right)$ and thus

$$R_{ce} = R_{eq} + \frac{NV_t}{I_c} . \text{ The term } V_t/I_c \text{ is the familiar intrinsic emitter resistance (26 ohms for 1mA or}$$

2.6 ohms at 10mA). This shows that the equivalent series resistance is not affected by the divider ratio. The resistors only involve the net parallel resistance of the divider (not the divider fraction).

Next, additional loading of the divider can be included as in Figure 11.1 (left). The divider string

consumes some of the input current where $I_c = I_1 - \frac{V_{ce}}{R_f + R_{be}}$. Noting that V_{ce} can be

approximated to a linear form with I_c such as $V_{ce} = \left(1 + \frac{R_f}{R_{be}} \right) (V_k + I_c R_{eq})$ where $I_c \cdot R_{eq}$ is for

the values in Figure 11.1 and at 10mA it gives a voltage of 34mV which is slightly more than $1 \cdot V_t$ making the $\ln(1)$ which gives almost zero additional voltage to the V_{ce} equation. Substituting V_{ce}

into the I_c equation gives $I_c \approx I_1 - \frac{(V_k + I_c R_{eq})}{R_{be}} = I_1 - \frac{V_k}{R_{be}} - \frac{I_c R_{eq}}{R_{be}}$ and

$$I_c \approx \frac{I_1}{1 + \frac{R_{eq}}{R_{be}}} - \frac{V_k / R_{be}}{1 + \frac{R_{eq}}{R_{be}}} \text{ or } I_c \approx \frac{I_1 / R_{be}}{R_b + R_{eq}} - \frac{V_k}{R_{be} + R_{eq}} . \text{ This current is now used in the}$$

original V_{ce} equation involving I_1 with the buffer, and now without the buffer

$$V_{ce} = \left(1 + \frac{R_f}{R_b} \right) \left(V_k + NV_t \ln \left(\frac{I_c R_{eq}}{NV_t} \right) + I_c R_{eq} \right) \text{ where } I_c \approx \frac{I_1 / R_b}{R_{be} + R_{eq}} - \frac{V_k}{R_{be} + R_{eq}} \text{ where}$$

$$R_{eq} = \frac{(R_{be} || R_f)}{\beta} + \frac{R_{b_p}}{\beta} + R_{e_p} \left(\frac{1 + \beta}{\beta} \right) \text{ and } V_k = NV_t \cdot \ln \left(\frac{NV_t}{I_s \cdot R_{eq}} \right) .$$

A simulation with $R_f = 1k2$ (R8 in LTspice) and $R_{be} = 471$ (R9 in LTspice) and $I_1 = 10mA$ gave $V_{ce} = 2.61022$ V and $I_c = 8.41894$ mA. The transistor used $\beta = 125$, $I_s = 9e-15$, $NF = 1$ and parasitic R_e and R_{be} were set to very small values and VAF and IKF were very large. The ABM equations are: $I_c = I(I_3) \cdot R_{10} / (R_{10} + R_{e8}) - V_k / (R_{10} + R_{e8})$ and V_{ce} :

$I = (1 + R_9 / R_{10}) \cdot (NF \cdot V_k + V_t \cdot \ln(I(B_3) \cdot R_{e8} / (NF \cdot V_t)) + I(B_3) \cdot R_{e8})$ where R_{eq} (R_{e8} in LTspice) is $\text{.param } R_{e8} = (R_9 \cdot R_{10} / (R_9 + R_{10})) / \beta_{a1}$ and $\text{.param } V_k = NF \cdot V_t \cdot \ln(NF \cdot V_t / (I_s \cdot R_{e8}))$. This gives $V_{ce} = 2.60997$ mA and $I_c = 8.4312$ mA. The error in V_{ce} is -1% and I_c is +0.5%.

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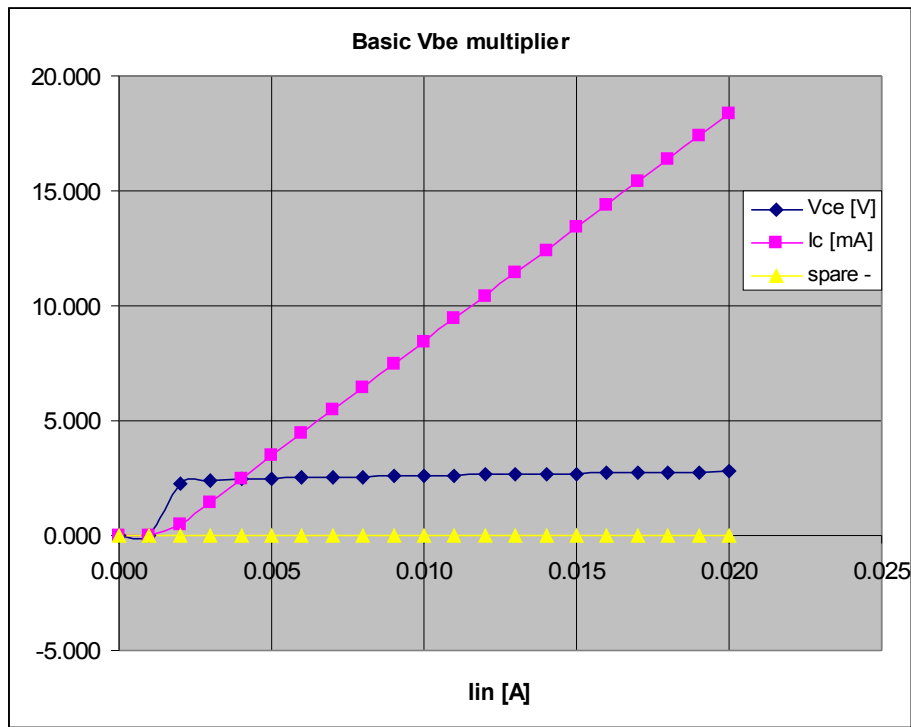


Figure 10.3a. Plot from PAK with I_{in} stepped. I_c is also shown. $R_f = 1k\Omega$ and $R_{be} = 471\Omega$. At $I_{in} = 10mA$ $V_{ce}=2.610V$. LTspice $V_{ce}=2.654V$ (equation error -1.0%).

The error at 10mA is acceptable. However, the dynamic resistance is calculated from in **Figure 10.3a** results at $I_{in}=10mA$ and then at 20mA $R_{ce}=(2.777-2.610)/(0.02-0.01)=94\Omega$. LTspice gives $(2.895-2.654)/10mA=24\Omega$. This large difference in R_{ce} is mainly due to the -1% errors in V_{ce} .

12. The Basic Push-Pull output stage

12.1 Basic CE push-pull output stage

Figure 12.1 shows a basic common emitter (CE) amplifier with bias voltage generators (V_{sp1} & V_{sp2}) and emitter resistors (R_1 and R_2). Floating power supplies allow the output current to be summed and fed into a ground referenced load resistor. This gives the simplest possible equation solution for a push-pull output stage. The load current is $I_L=I_{c1}-I_{c2}$ where I_{c1} flows into the collector of Q_1 while I_{c2} flows out of the collector of Q_2 . Initially, the equations are solved for $R_{in}=0$ and the current gains (Beta) of Q_1 and Q_2 are constant, their gains are assumed to be the same and the Early effect is not included.

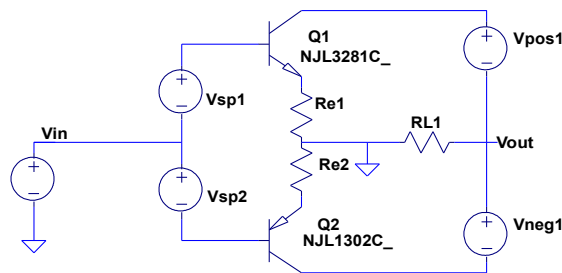


Fig 12.1. The Basic Push-Pull Common Emitter bipolar output stage

Figure 12.1a shows a family of 'wingspread' g_m and I_L curves generated by LTspice when the bias sources (V_{sp1} & V_{sp2}) are stepped from 480mV to 880mV in 100mV steps. The quiescent current

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varies from 0.46mA to 638mA. The 3rd curve is an intermediate value of 630mV for a quiescent current of 65mA close to the 'optimum bias' (more on THD later). The above equations are implemented in LTspice using ABM current sources and the same model parameters are used in the equations where Beta's=100, Is=5x10⁻¹²A and Re = 330mR. Other secondary SPICE parameters such as IKF, ISE, VAF and VAR were disabled by setting them 1000 times beyond to their normal values, ie, IKF=50k, ISE=20x10⁻²², VAF=1000k, VAR=4.3k. Notice VAR does have a noticeable effect on the forward current and it needs to be disabled as well as VAF to see the primary currents and gains in this most basic form.

These wingspread curves are therefore created by the diode plus series resistance nonlinearity with a pure voltage source drive and before many of the other secondary effects modify the plots.

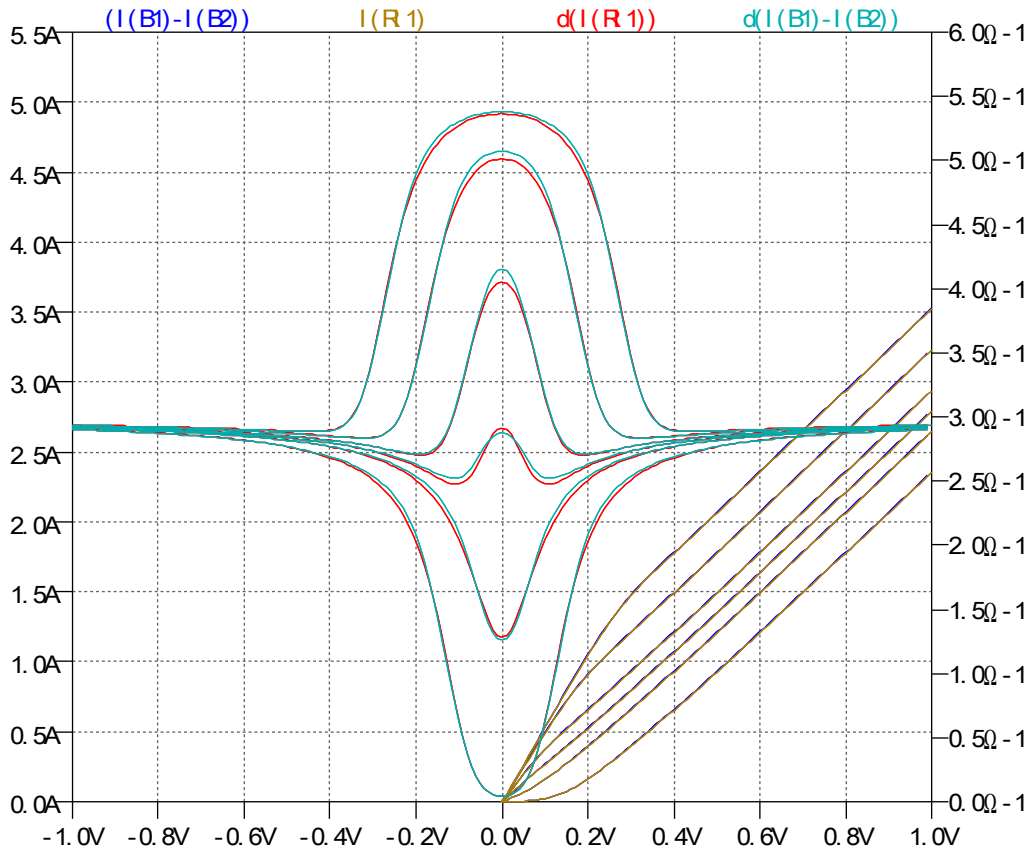


Figure 12.1a. Wingspread plots using ABM equations concurrent with the LTspice circuit. Load current (bottom right) and gm with each Vsp stepped (480mV, 580mV, 630mV, 680mV, 780mV, 880mV) (Red = SPICE, Rin=1mR, Re=330mR, no IKF & Ise & V_A).

The accuracy using the 3 term Lambert W function approximation gives around +/-1% maximum error for Ic and gm over a wide range of current and bias settings. No additional curve fitting scaling factors were used. The gm equation solution is calculated directly by an ABM.

Using the diode plus resistor solution on the collector side for Q1 when Rin =0 is

$$I_{c1} = \frac{NI \cdot V_t}{Re q1} W \left(\exp \left(\frac{V_{in} + V_{sp1} - V_{k1}}{NI \cdot V_t} \right) \right) \text{ where } V_{k1} = NI \cdot V_t \ln \left(\frac{NI \cdot V_t}{Is1 \cdot Re q1} \right) \text{ and} \quad 127(121)$$

$$Re q1 = \frac{Rb1_p}{Beta} + (Re + Re1_p) \cdot \left(\frac{Beta + 1}{Beta} \right) \text{ and } W(x) \approx \ln(1+x) \left(1 - \frac{\ln(1+\ln(1+x))}{2+\ln(1+x)} \right)$$

where Re_p and Rb_p are Q1's parasitic emitter and base resistances. Q2 has a similar set of equations

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except the sign for V_{in} is reversed. As already mentioned, I_{c2} is positive for a PNP transistor; I_{c2} is the magnitude of collector current. The load current is therefore calculated from $I_L = I_{c1} - I_{c2}$. Saturation can be added to I_{c1} using a simple approximation like

$$I_{c1_Sat} = \text{Min} \left(I_{c1}, \frac{(V_{pos} - N1 \cdot V_t \cdot \ln(Beta1))}{R_L + R_{e1} + R_{e1_p}} \right) \quad (122)$$

also added to I_{c2} gives approximate saturation to the load voltage. The individual gms for I_{c1} is also readily calculated using $G_{m1} = 1/R_{eq1} \{W127\}/(1 + \{W127\})$ and a similar expression for I_{c2} .

Figure 12.1b shows the G_m total plot for $G_{m1} + G_{m2}$ from I_{c1} and I_{c2} equations. The equation gain plots (green) lie on top of the LTspice gain plots (red obscured) and are more accurate than the gains derived by LTspice from I_{c1} and I_{c2} equations (cyan). This is because the errors in G_{m1} and G_{m2} cancel when they are added.

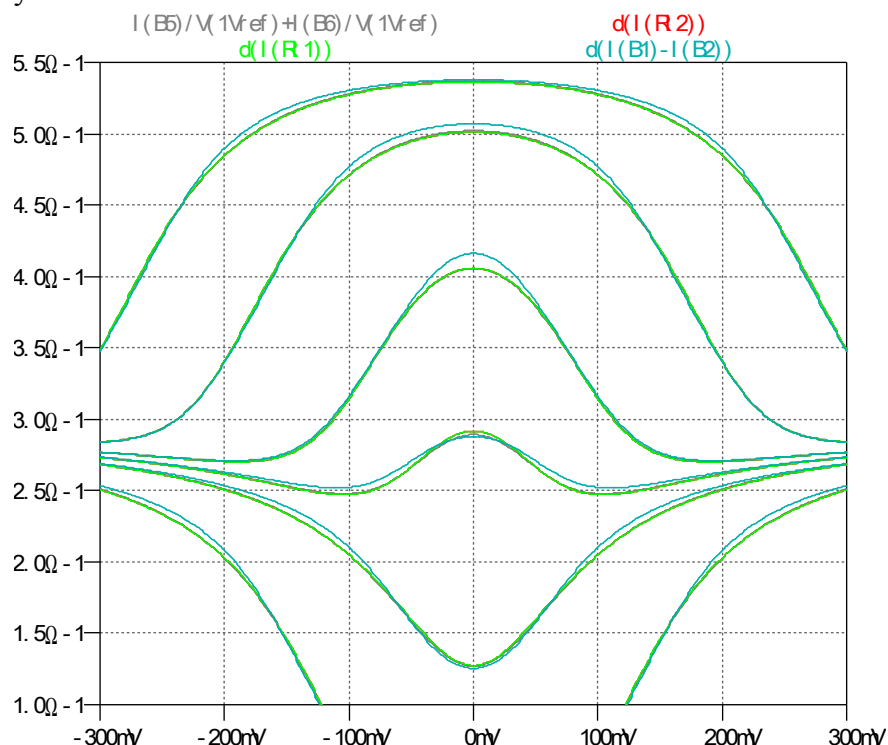


Figure 12.1b. Shows the equations for G_{m_Tot} (green) obscures the LTspice G_m (red is obscured). G_{m_Tot} is more accurate than I_L using these equations.

12.1.1. On optimum biasing for Class-B (aka Class-AB)

Section 3.1 (Fig's 3.1d-f) introduced the use of the Diode plus Resistor Lambert W solution for representing push-pull. It was stated that optimum bias in a BJT Class-B push-pull circuit occurs at approximately one NV_t higher than the knee voltage V_k . The knee voltage is determined by the transistor area (through the parameter ' I_s ') and resistances R_e and the source resistance R_{in} making up the equivalent total R_{eq} . The section shows how to derive the 'one NV_t ' voltage.

Interestingly, when pure voltage drive is used as in Figure 12.1 the Beta's for Q1 and Q2 do not have much effect on the shape of the curves. The only effect is the $R_{bp}/Beta$ part of R_{eq} . In terms of symmetry of the +ve and -ve sides the matching of Beta's with voltage drive has little effect. It is possible to get symmetry with different Beta's by adding some base resistance to the higher Beta transistor. Alternatively, the emitter resistance for the higher Beta transistor could be reduced. If the internal emitter resistances R_{ep} are not matched then the external R_e 's can be trimmed for

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symmetry. Even high current Beta-fall can be trimmed for symmetry with voltage drive.

Similarly, with voltage drive 'IS' mismatch doesn't affect symmetry – IS's affects V_{k1} and V_{k2} for V_{sp1} and V_{sp2} respectively so differences in IS's can be trimmed out by an offset voltage in the bias circuit. This gives a big advantage for voltage drive of a BJT output stage over non-voltage drive. But how low does R_{in} need to be to constitute voltage drive? One way is to increase R_{in} until a second order effect gives say 10% asymmetry that was otherwise suppressed by pure voltage drive. This is checked in the following section where R_{in}>0.

Calculating the bias point for 'optimum bias' is roughly where the total gm is half the large signal gm, that is $gm(0) = gm_{Lim}$. Since both devices are conducting when $V_{in}=0$ and assuming they are matched their gains double giving $gm_{Tot} = gm_{Lim}$. Since $gm(0) = \frac{1}{Re} \frac{W(x)}{1+W(x)} = \frac{1}{2}$ we can solve for x to find the bias voltage V_{sp} from Re and Is. Solving for W(x) gives W(x)=1. To find x we need to invert this relation, the inverse of an inverse, so we simply use the original definition for W(x). In this case $1e^x = x$ giving $x=e$ or ~2.71. Since $x = \exp\left(\frac{V_{sp}-V_k+0}{NV_t}\right) = e^1$ we get $\frac{(V_{sp}-V_k)}{NV_t} = 1$ so (V_{sp}-V_k)=NV_t. Therefore the bias spread voltage V_{sp} needs to be V_{sp}=V_k plus one NV_t. This gives half the large signal gain at $V_{in}=0$ which near optimum bias for minimum distortion.

Interestingly, this bias voltage is transferred to the emitter resistor with a gm of half the full gain at high currents. From the Gm equation we know W(x)=1, so $I_{c1}(V_{in}=0) = (V_t/Req) \cdot W(x) = V_t/Req$ which means one NV_t is across Req. The actual voltage measured across the external Re is a voltage divider involving Re and Req giving $V_{Re} = NV_t \cdot (Re/Req)$.

Optimum bias requires approximately one NV_t across each emitter resistor and is referred to as “Oliver's condition” [Ref 11 Cordell cites 'BM Oliver HP Jn. Feb 1971'].

Optimum bias for minimum THD is usually more complicated due to the base resistance varying with base current and high current Beta-fall. The best bias setting also depends whether unweighted or weighted THD is used [ref 19 L|A Vol 4], and whether slight over bias is exploited to pre-empt thermal lag distortion (TLD).

Very low external emitter resistances can be used to raise the optimum bias current and this reduces the proportion of high order harmonics which our hearing is very sensitive to. In this situation it is better to aim for bias at V_k to exploit the inflection point mentioned in relation to Figure 3.1f – this further reduces high order distortion compared to V_k plus one NV_t biasing and the quiescent heat dissipation is nearly halved.

12.2 Basic Common Emitter Push-Pull output stage with large R_{in}

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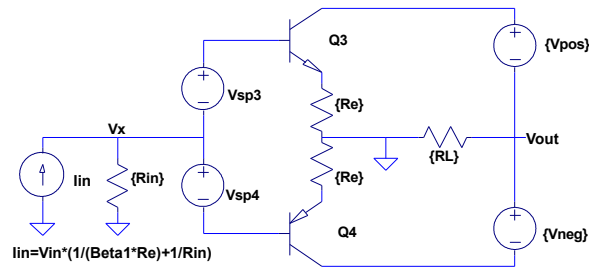


Figure 12.2. Circuit for current drive with a high Rin

Increasing the drive source resistance R_{in} in circuit **Figure 12.2** has the effect of reducing non-linearity in the crossover region provided Beta-fall is negligible. When Beta-fall is disabled, as in the next two plots, we can observe the effect of current drive, rather than voltage drive in the previous section, on the other nonlinearities generated by the basic bipolar transistor push pull output stage.

The equation for the driving current source allow R_{in} to be varied from a small (but non zero value) to very large values relative to a nominal input resistance of the transistors is

$$I = V_{in} \cdot \left(\frac{1}{\text{Beta}1 \cdot \text{Re}1} + \frac{1}{R_{in}} \right) \quad (123)$$

For current drive to be noticeable for reducing crossover distortion the driver source resistance needs to be at least 10 times more than the nominal input resistance – namely 330R to 3k3 for the Basic CE configuration (8k to 80k for a Basic Voltage Follower). Current drive using this equation gives normalised plots with the same limiting gains independent of stepping R_{in} over many decades. In a practical amplifier circuit the driver stages ' R_{in} ' would be the Norton equivalent with R_{in} to ground and current source being the VAS stage.

Figure 12.2c and **Figure 12.2d** shows increasing R_{in} linearises the crossover region. This occurs with both over-biasing (Figure 12.2c) and under-biasing (Figure 12.2d). But this is only possible if IKF for high current Beta-fall and V_A for the Early effect are disabled.

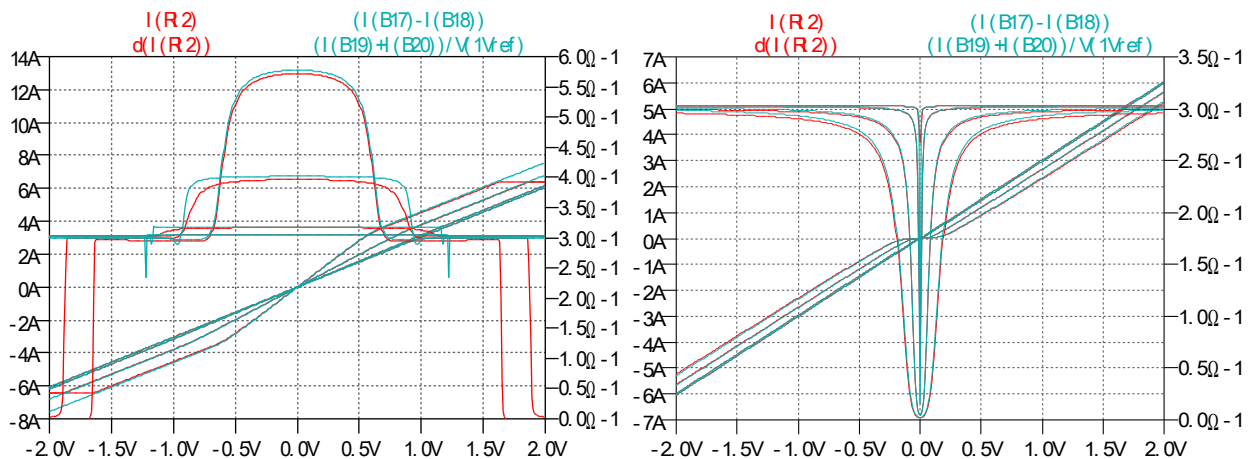


Figure 12.2c (L). Wingspread plots with over-bias $V_{sp}=1220\text{m}$ (giving $I_q=1.7\text{A}$) and Figure 12.2d (R) $V_{sp}=460\text{m}$ ($I_q=0.25\text{mA}$). R_{in} stepped 33mR, 33R, 3k3. (Red = LTspice). Note the linearisation effect with a high R_{in} (only possible if IKF and V_A are disabled).

If weighted distortion is used to assess the relative merits of the voltage drive and current drive

approaches to minimising the audibility of crossover distortion then it would seem from Figure 12.2d (under-biased) that current drive generates more high order distortion than voltage drive given the same number of stages to accomplish current drive or voltage drive. Additional simulations or bench tests are needed with weighted distortion to confirm this [ref 19 L|A Vol 4].

Interestingly, Figure 12.2c shows slight over-biasing with current drive does not produce very sharp gain changes as for under biasing with current drive. So current drive with slight over-biasing may not be that much different to voltage drive.

12.2.1 Equation solutions for the Basic PP output stage when current driven with a high Rin

The equations for $R_{in} > 0$ requires the effective input voltage at the base node (V_x) needs to be known since the Lambert solution involves an input voltage to the base-emitter diode plus an equivalent resistor in series with this diode.

Since the voltages to the bases are not proportional to V_{in} any more the crossover region exhibits different equivalent resistances depending on whether one or two transistors are conduction. This suggests that a switch is needed to model the equivalent series resistance to find the current in one transistor.

In circuit Fig 12.1 with high R_{in} the net impedance of node V_x to common is a nonlinear function of V_{in} due to the two current dependent conductances of the base-emitter paths to common. The overall conductance at node V_x is the sum of the two base-emitter conductances which must then be paralleled with R_{in} . That is $G_x = G_1 + G_2 + 1/R_{in}$, where G_1 is conductance of Q1 base-emitter plus R_{e1} (and G_2 is likewise for Q2 plus its R_{e2}). The input resistance at this node is $1/G_x$. (Capital G is used for the large signal function namely $G_x(V_{in})$).

The node resistance $1/G_x$ defines the voltage gain V_x/V_{in} for either the voltage driven case or the current driven one. It was found that the Lambert solution incorporating a switch functions for the equivalent resistances seen by each base-emitter diode gave close enough approximations over a wide range of bias points and input resistances. The resulting switch solution shown in Figure 12.2c,d give a typical accuracy of around $\pm 5\%$ for currents and gains. This is considered adequate for comparing relative distortion figures and relative sensitivities to temperature.

12.2.2 First approach

At high V_{in} and with only one transistor conducting, the solution for the collector current following Ebers-Moll and Gummel-Poon approach so R_{in} is referred to the collector side. The equations

become $I_c(I_{in})_{Q10} = \frac{V_t}{R_{e_{Tot}}} W(x)$ where $x = \exp\left(\frac{(I_{in} \cdot R_{in}' - V_k)}{V_t}\right)$ and $V_k = V_t \ln\left(\frac{V_t}{I_s \cdot R_{e_{Tot}}}\right)$ &

$$W(x) \approx \ln(1+x) \left(1 - \frac{\ln(1+\ln(1+x))}{2+\ln(1+x)}\right)$$

R_{in}' is the net parallel input node resistance which is determined by the bias point (R_p in the above equations) and $Sw11$ (IB11 above) which is ON ($Sw=1$) for the crossover region and OFF ($Sw=0$) when Q10 turns off. (The $I_s \cdot R_{e_{Tot}}$ term has been dropped from I_c since I_s is typically 4-6 orders of magnitude smaller than I_b).

The formulation chosen for $R_{e_{Tot}}$ is $R_{e_{Tot}} = R_e + \frac{R_{in}'(1+Sw11)}{\beta}$. Interestingly, the term $(1+Sw)$

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effectively doubles $R_{in'}$ within the crossover region which occurs during over biasing with both transistors conducting. This may be thought of as the two transistors sharing R_{in} , so each transistor sees $(2 \cdot R_{in} / \beta + R_e)$ for the net diode series resistance within the crossover region and both transistors conducting. This observation may be helpful for solving other circuits.

The switch functions B1 and B11 use a Lambert solution with both transistors ON. This uses the two term approximation which is sufficiently accurate. The input parameter I_{in} is $I(B21)$. The switch function is simply the Gm Lambert value for I_c without the scaling with $1/R$ and this gives the normalised and non-dimensional Gm 'switch' function.

The term $V_k = V_t \ln \left(\frac{V_t}{I_s \cdot R_{e_{Tot}}} \right)$ is split into $V_{k_{Tot}} = V_t \ln \left(\frac{V_t}{I_s \cdot R_e} \right) + V_o + I_o \cdot R_{in'}$ to separate out the terms independent of R_{in} and those affected by R_{in} . The I_o term is the offset that involves the Switch plus an offset needed to ensure the current just before the switch turns OFF will match the current when the switch has turned OFF. This is found using the inverse Lambert function (similar to the CFP solution in Section 13). I_o also includes a 'small' fixed current that is added as a 'fix' to ensure that the quiescent current remains reasonably constant for large values of R_{in} compared to $R_{in}=0$.

The joining offset is calculated when I_{in} (or V_{in}) satisfies the condition

$I_c(I_{in})_{Q10} = \frac{V_t}{R_{e_{Tot}}} W(\exp(0))$ which is a convenient reference point since $I_c(I_{in})_{Q10} = \frac{V_t}{R_{e_{Tot}}} \cdot 1$ and this current must be the same for the switch ON and OFF at the particular I_{in} , which is or $V_{in} = V_k$ ($I_{in} \cdot R_{in'} - V_k = 0$ the knee point).

Solving for I_o offset using the relation $w \cdot e^w = x$ to give the same x values before and after switching.

After the switch turns OFF we have W_0 in the relation $W_0 \exp(W_0) = x = \exp \left(\frac{I_o \cdot R_{in'} - V_k}{V_t} \right)$

giving $V_o = V_k + V_t \frac{(W_1 + \ln(W_1))}{(W_0 + \ln(W_0))}$ where W_1 is at the same x value as W_0 but with the switch turned ON.

Plots were provided above for current drive in Figure 12.2c for over-biased and Figure 12.2d for under-biased cases.

In LTspice the numerical overflow limit for the $\exp(x)$ function can easily be exceeded before the full output current was reached. This occurs in the basic output stage when R_{in} is large (and it also occurs with the CFP with low R_{in}). When the limit is exceeded in LTspice a zero result is displayed while the simulation of other circuit currents continues.

To obtain larger range in the ABM equations the Lambert approximate solution is switched to the approximation $W(x) \sim (1 - 1/\ln(x)) \cdot \ln(x)$ for $x > \exp(70)$. The upper limit value is first computed as $((1 - 1/(70 \cdot 70)) / W(\exp(70)))$ as 'Lcf' (Lambert correction factor) which is also used to improve the 3 term approximation for large x accuracy by about 1%. The limit applies to $x \gg 10^{20}$ (and is chosen for <1% error). This uses the limit $W(x) \rightarrow \ln(x)$ for large x . Our diode plus resistor (and transistor) circuit equations use $W(\exp(Y))$ so we get $W(\exp(Y)) \rightarrow Y - 1$ when Y is large and $Y > 100$ the error for $W(\exp(Y)) = Y - 1$ falls below 1%.

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The ABM equation for the 3 term Lambert approximation is usually of the form: `.function We(Y) {(ln(1+Exp(Y))*(1-(ln(1+ln(1+Exp(Y))))/(2+ln(1+Exp(Y)))))*Lcf}` for $Y < 70$ in LTspice, and Lcf can be generated as a constant using `.param Lcf=(70-(1-1/70)*ln(70))/(ln(1+exp(70))*(1-ln(1+ln(1+exp(70)))/(2+ln(1+exp(70)))))`.

Since unlimited range is needed the extra over-range function is added (shown in bold)
`.function We(Y) {(ln(1+Exp(Y))*(1-(ln(1+ln(1+Exp(Y))))/(2+ln(1+Exp(Y)))))*Lcf*(1+Sgn(70-Y)/2+(Y-(1-1/Y)*Ln(Y))*(1+Sgn(Y-70))/2)}`

This appears to give a glitch free transition. Equations in Excel and Calc spreadsheets do not appear to need the approximation to handle typical large values although the Lcf factor is usually included to improve the 3 term Lambert approximation.

In the ABM circuit equations a calibration factor for R_p is used. This is determined empirically by stepping R_{in} with R_{in} very large until the crossover wobble due to equation switching is minimised. This is an interesting effect can be observed where an upper limit to the amount of improvement that current drive can give. This upper limit changes mainly with R_{in} but also depends the bias setting. For example, the improvements were limited to 3% gain variation in Figure 12.2d (under-biased) but only 0.3% for Figure 12.2c (over-biased). This shows that over-biased (gm-doubling) and with current drive where R_{in} is very high is linearised ten times better than underbiased case. For over-bias the limit occurred when R_{in} was around 3k3 or 100 times the nominal input resistance of $\beta \cdot R_e$ or 33 ohms. The underbiased case required ten times higher source resistance of 33k before a negligible improvement was seen.

This suggests current drive reduces crossover distortion far better when over-biased than under-biased; in this example by two orders of magnitude or 100 fold. This will only apply to a basic output stage, when driver transistors are added there are other limitations that reduce the upper limit on the amount of improvement possible using current drive. However, this analysis (with no β -fall and no Early effect) is useful as it gives an insight into the theoretical upper limit to what current drive can achieve in Class-B power amplifier circuits.

The following sections cover a solution for a bipolar Common Emitter Push-Pull output stage. It appears an exact solution is not possible since it involves two diode/transistor resistor solution and the standard Lambert W-inverse method does not allow an exact solution. An approximate solution is covered later. It was found that this solution is not suitable for modelling low current β -fall and a better set of equations that allow the addition of low current β -fall is under-way.

12.2.3. Second approach for the basic Push-Pull output stage with large R_{in}

An alternative approach for developing equations for the basic Push-Pull output stage with large R_{in} is presented here. The First approach gives good plots for the total Gm and the total output current over a wide range of R_{in} values but individual gains and individual currents are not very accurate with very large R_{in} and it also prevents low current β -fall from being represented due to such large errors in the individual currents.

This second approach starts with the simplest possible circuit with R_{in} and that is when the R_e 's are removed giving an equivalent circuit of Figure 3.5. **Figure 3.5a** shows the Gm when R_e 's are removed. Notice how the plots are different to normal 'wingspread' plots (such as Figure 3.1e with voltage drive – no R_{in}). It is found that increasing V_{spread} with $R_e=0$ does not lead to a peaking of the gain at $V_{in}=0$ and the gain variations are much wider ($10V_t$ or $20V_t$) and therefore has mostly

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low order harmonic distortion. This is a better form of Class-AB than Figure 3.1f (bias at $V_{sp}=V_k$) since there is less gain variation when transitioning to the high currents. Notice the gain variation becomes very small when V_k+3V_t is used and the quiescent current is about 600mA with $R_{in}=1R$ which gives a non-switching situation where the current is still 40mA with 10V input and the other diode is almost 10A.

Interestingly, higher bias current does not give higher distortion so there is no optimum bias for minimum distortion just a trade-off of quiescent dissipation and linearity. This is an ideal situation and not achievable in practice but it does give an idea of what the form of an equations might represent this behaviour. The $d(I_{d4})$ plot looks like the $\tanh(x)$ function but not perfectly symmetric. The current is therefore close to a $\text{Log}(\cosh(x))+C$ function (which is the integral of $\tanh(x)$).

The next stage is to find a way to morph the $\text{Log}(\cosh(x))+C$ function for I_{d4} to the Lambert W-solution for I_{d4} as V_{spread} is reduced from say V_k+3V_t to zero, similar to Section 8.1 for the differential pair. Then R_e can be reintroduced and hopefully the solution will give better individual I_c solutions for including the low current Beta-fall effect. If you develop this approach then please send a solution (see [Feedback](#)).

12.2.4. On the practical use of current drive for linearising power output stages

This method of linearisation has been previously used first by J Faran [ref IRE Jul 1962 Fig.7] then Peter Blomley [ref WW Feb 1971] and then Paul Mills with Malcolm Hawksford [JAES Oct 1989]. **Figure 12.2e** shows the circuit used by Faran to reduce crossover distortion. Faran did not use any quiescent bias voltage to the output power transistors and still the output waveforms appeared to be surprisingly linear (Faran's Fig 3 vs Fig 4). Thermal-lag distortion (TLD) cannot happen with this approach.

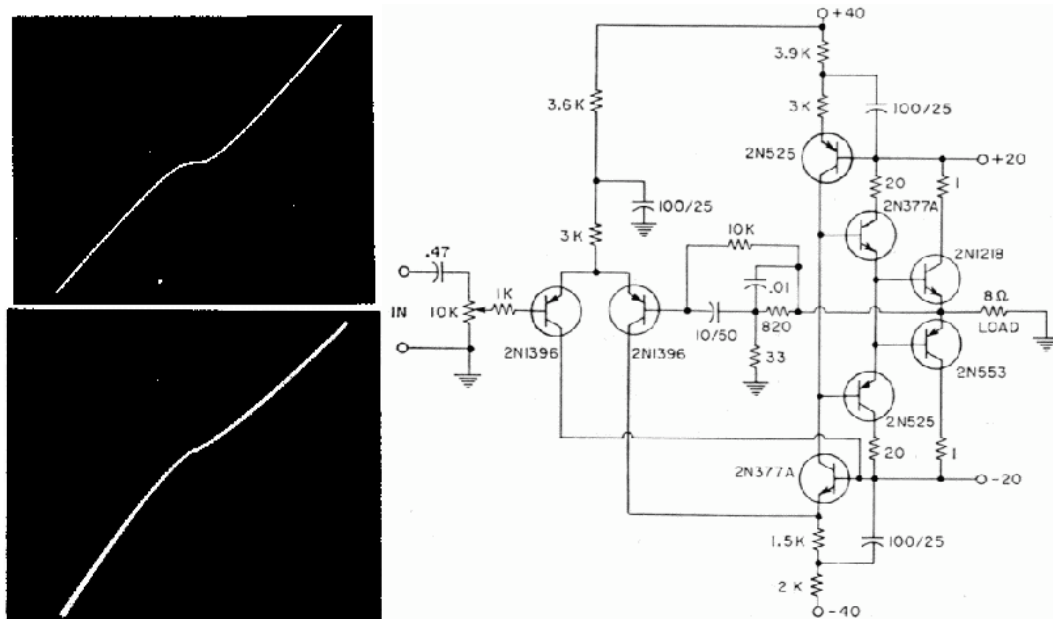


Figure 12.2e. Top Left is voltage drive (no bias), Bottom Left current drive (Beta asymmetry and low current Beta-fall are visible) . Right High impedance drive of an early bipolar output stage by J Faran [ref IRE Jul 1962 Fig7].

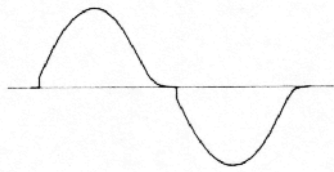


Fig. 4—Crossover distortion waveform with current drive.

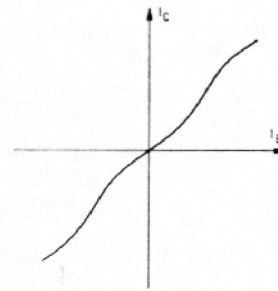


Fig. 5— I_C/I_B characteristics showing the nonlinearity due to h_{FE} variations with I_B or I_C .

Figure 12.2f. Waveforms from Blaser 1963. Fig 4 shows slow cut-off and turn on due to Cbe. Fig 5 shows beta variations with I_C (the steeper part is due to beta peaking).

Peter Blomley found that it was helpful to add some bias voltage to the crossover circuit, in his case thermal lag distortion was not a problem despite the addition of a quiescent current because the crossover took place in a low power stage where junction temperature changes were insignificant and the power output halves used CFP arrangement provides good thermal stability.

Malcolm Hawksford's common base current driven output stage **Figure 12.2g**. This is similar to Faran common base using a driver transformer except Hawksford uses a high current opamp as current source. Note the driving opamp must supply the full output current at a low voltage of under 1V peak to peak. Hawksford's output stage is also equivalent to Blomley's except that Blomley uses a low current common base splitter and CFP power halves. Hawksford's practical CB output stage includes some bias voltage and emitter resistors as does Blomley's. The traces in Figure 12.2e show the distorted input voltage to the common base output stage due to the current drive and the resulting linear output.

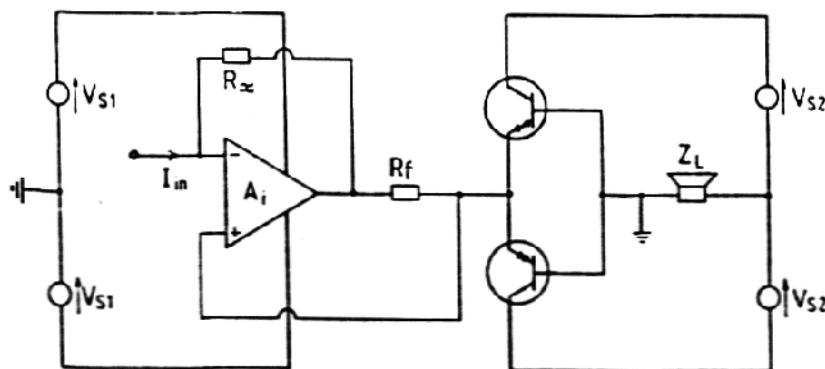


Fig. 4. Alternative configuration for current gain.

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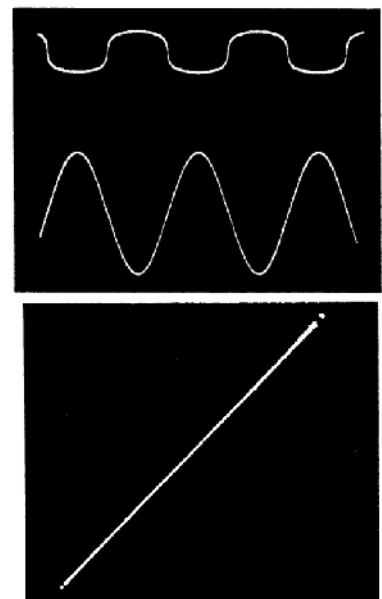


Figure 12.2g. Left Malcolm Hawksford's common base current driven output stage. Right Faran's common base waveforms (be upper, VL mid, X-Y lower)

Both Bob Cordell and Douglas Self aim to drive the output stage with as low source resistance as possible by using a triplet Darlington so the VAS output impedance is small relative to the Darlington triplet input impedance [ref 11 Cordell book p64-67]. My simulations of Bob Cordell's

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triplet Darlington circuits suggest there is a useful linearity gain by adding the extra driver transistors of a Triple stage to provide better voltage drive of the output stage. This is probably due to the 'shielding' from the effects of Beta-fall and beta asymmetry. The additional phase contribution of the extra driver stage does not seem to introduce a significant stability problem (at least in simulations) probably because the driver transistor FT's are higher than the driver and power transistors.

John Curl who designs all the ParaSound power amplifiers uses a complementary MOSFET driver stage for bipolar power transistors to prevent the variable speaker loading affecting the loading of the VAS stage [ref 12]. This can achieve the same isolation effect as the Triple Darlington but with one less stage. Incidentally, John Curl found MOSFET output transistors were less rugged than bipolar transistors. This may be due to long run inductance effects which can be overcome by decoupling physically close pairs of output MOSFET's then paralleling those pairs. Anthony Holton [www.ausiaudio.com] solved this problem by mounting the upper-MOSFET's on one side of the heatsink and the lower-MOSFET's on one side (BTW Anthony uses Bengt Olsson's totem-pole output stage from Electronics World Dec 1994 and see Letters Oct 1995 p886-7).



Figure 12.2h. Totem-pole transistors on opposite sides and decoupled locally solved the problem of earlier versions.

12.3 The Voltage Follower Basic Push-Pull output stage [under construction].

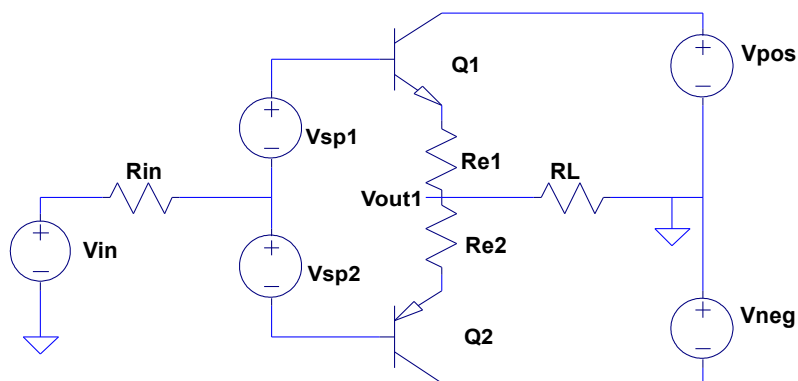


Figure 12.3. Basic Voltage Follower PP output stage with voltage drive and R_{in}

Figure 12.3 shows the basic push-pull Voltage Follower bipolar output stage. The only difference between the CE circuit above is the load common has been moved from the emitter resistors node to the power supply centre rail and the Vout take-off point is swapped.

A solution for this circuit is not included in the circuits file but an approximation is used in the spreadsheet based on the CE solution where the output current and distortion is scaled by $1/(1+G_m \cdot R_L)$. A more appropriate solution would be to incorporate R_L into R_{in} , such as

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$R_{in_{eff}} \approx R_{in} + \beta \cdot R_L$, and using the pending solution mentioned in Section 12.1.2. This works for matched Beta's and no Beta-fall. It should be possible to modify the solution for unmatched Beta's and with Beta-fall.

12.4 Basic Voltage Follower PP output stage with current drive

[under construction]

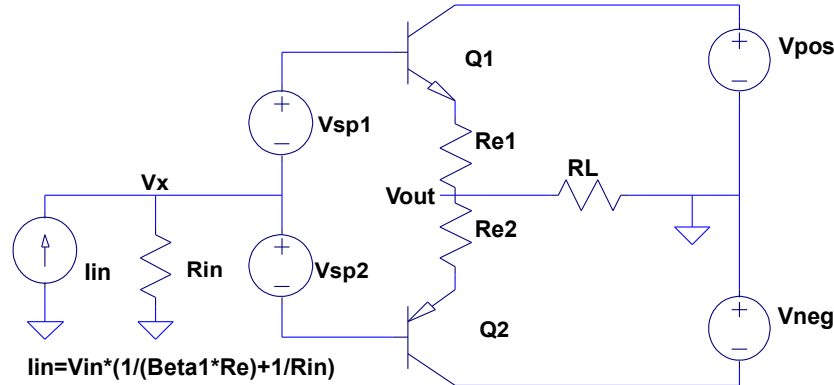


Fig 12.4. Basic Voltage Follower PP output stage with current drive

Figure 12.4 shows the basic Voltage Follower Push-Pull bipolar output stage with current drive into a termination resistor R_{in} . The equation for I_{in} normalises the upper limit G_m to $1/R_e$ independent of R_{in} allowing more relevant plot comparisons when R_{in} is varied over many decades (as used in Figure 12.2c,d).

$$I_{in} = V_{in} \cdot \left(\frac{1}{\beta \cdot R_e} + \frac{1}{R_{in}} \right) \quad (124)$$

12.5 Basic Push-Pull amplifier with the Early effect

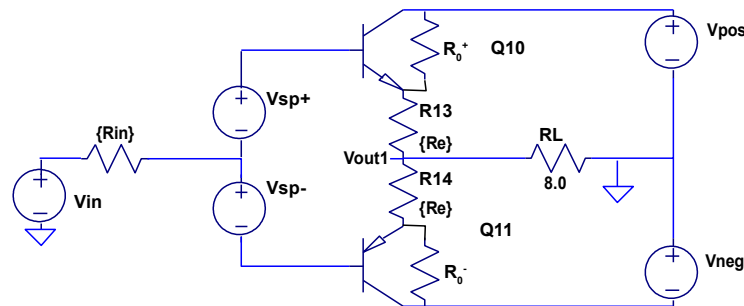


Figure 14.3. The basic push-pull VF bipolar output stage with the Early effect

The Early effect was added previously to the Single Ended CE stage in Section 4.1.2. The Early effect is added here to a Push-pull class-B VF output stage shown in **Figure 14.3**. This is similar to the single ended common emitter stage except that I_L is the sum of the two transistors collector currents which are modulated by their collector to emitter voltages.

The Early voltage is again modified by the local feedback from the emitter resistors, for Q10 we use the “+” superscript $I_c^+(V_{a_{eff}}^+) = I_{co}^+ \left(1 + \frac{V_{ce}^+}{V_{a_{eff}}^+} \right)$ where

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$$Va_{eff}^+(Re) = Va^+ \left(1 + Re^+ \cdot gm_0^+ \left(1 + \frac{Vce_{est}^+}{Va^+} \right) \right) \quad \text{where } gm_0^+ \text{ is calculated along with the Lambert W}$$

solution for Ico^+ . The collector to emitter voltage for Q10 is

$$Vce^+ = Vpos - IL(RL + Re^+ + Rep^+ + Rcp^+) \quad . \text{ Similar equations can be written for Q11 with “-”}$$

$$\text{superscripts} \quad Ico^-(Va_{eff}^-) = Ico^- \left(1 + \frac{Vce^-}{Va_{eff}^-} \right) \quad \text{and} \quad Vce^- = Vpos + IL(RL + Re^- + Rep^- + Rcp^-) \quad . \text{ The}$$

load current is the difference of the two collector currents $IL = Ico^+(Va_{eff}^+) - Ico^-(Va_{eff}^-)$ so

$$IL = Ico^+ \left(1 + \frac{Vce^+}{Va_{eff}^+} \right) - Ico^- \left(1 + \frac{Vce^-}{Va_{eff}^-} \right) \Rightarrow$$

$$IL = Ico^+ \left(1 + \frac{Vpos - IL(RL + Re^+ + Rep^+ + Rcp^+)}{Va_{eff}^+} \right) - Ico^- \left(1 + \frac{Vpos + IL(RL + Re^- + Rep^- + Rcp^-)}{Va_{eff}^-} \right)$$

. Collecting IL terms on the LHS gives

$$IL \left(1 + \frac{Ico^+(RL + Re^+ + Rep^+ + Rcp^+)}{Va_{eff}^+} + \frac{Ico^-(RL + Re^- + Rep^- + Rcp^-)}{Va_{eff}^-} \right) = Ico^+ \left(1 + \frac{Vpos}{Va_{eff}^+} \right) - Ico^- \left(1 + \frac{Vneg}{Va_{eff}^-} \right)$$

$$\text{finally} \quad IL = \frac{Ico^+ \left(1 + \frac{Vpos}{Va_{eff}^+} \right) - Ico^- \left(1 + \frac{Vneg}{Va_{eff}^-} \right)}{\left(1 + \frac{Ico^+(RL + Re^+ + Rep^+ + Rcp^+)}{Va_{eff}^+} + \frac{Ico^-(RL + Re^- + Rep^- + Rcp^-)}{Va_{eff}^-} \right)} \quad \text{where}$$

$$Va_{eff}^+(Re) = Va^+ \left(1 + Re^+ \cdot gm_0^+ \left(1 + \frac{Vce_{est}^+}{Va^+} \right) \right) \quad \text{and} \quad Va_{eff}^-(Re) = Va^- \left(1 + Re^- \cdot gm_0^- \left(1 + \frac{Vce_{est}^-}{Va^-} \right) \right)$$

which requires an estimates for Vce^+ and Vce^- with Ico for Q10 and Q11 using

$$Vce_{est}^+ = Vpos - (Ico^+ - Ico^-)(RL + Re^+ + Rep^+ + Rcp^+) \quad \text{and}$$

$$Vce_{est}^- = Vpos + (Ico^+ - Ico^-)(RL + Re^- + Rep^- + Rcp^-) \quad .$$

The Lambert W collector current solutions for Q10 and Q11 (with no Early effect) can be post-processed, and the solution is exact for $Re=0$. With typical emitter resistance values there is only a small error due to an approximation for Vce (to calculate the effective Early voltage modified by the local NFB due to Ro via Re).

The denominator involves the sum of the collector currents (magnitudes). Interestingly, in Class-A (also Class-AB within the gm -doubling region) the sum of the collector currents can be fairly constant, where one transistors current ramps up while the other ramps down. In this situation the Early effect does not introduce any significant additional distortion.

Also note that under all biasing situations the numerator does not introduce any nonlinearity from the Early effect when Vce varies since the numerator has fixed voltages of $Vpos$ and $Vneg$. The only distortion effect the numerator can add is gain asymmetry where the collector currents can be scaled by different amounts, and this occurs if the Early parameters are asymmetric (not matched). If Q10 and Q11 have significantly different Early parameters (and yet all other complementary parameters are symmetric) then any voltage variations in supply voltages ($Vpos$ and $Vneg$) will intermodulate into the output signal. This can occur when one rail sags due to the collector current, and since a capacitor is across the supply rails then this intermodulation distortion will be seen to

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increase at lower frequencies due to Early parameter asymmetry.

13. Push-pull Complementary Feedback pair (CFP)

The solution for the Single Ended CFP or Sziklai pair was given in **Section 5**. The first section covers the Push-Pull case with voltage drive ($R_{in}=0$) and then R_{in} is added to cover current drive. Equations are first solved assuming all Beta's are constant, also the Early effect is not included at this stage.

Figure 13.1 shows a CFP (Sziklai pair) output stage with bias voltage generators (V_{sp}) and emitter resistors R_1 and R_2 . Floating power supplies allow the output current to be summed and fed into a ground referenced load resistor.

The load current is $I_L = I_{e3} + I_{c1} - (I_{c4} + I_{c2})$ following the same approach as Section 5.5 where it was noted that the sum of I_{c1} plus I_{c3} combine and flow through V_{pos} so the current through R_{be1} does not need to be mentioned. Emitter resistors for the output transistors $Q_{3,4}$ are not shown but they can be easily included if needed when paralleling output transistors (along with R_{ep} etc).

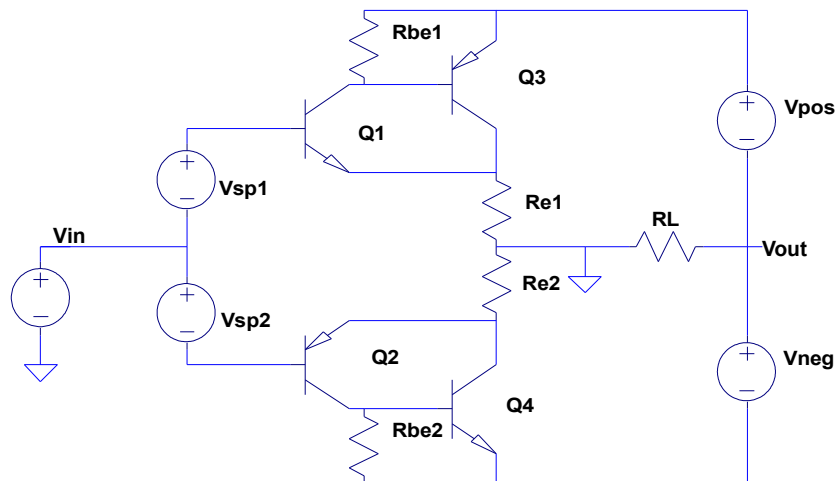


Figure 13.1 Common Emitter Push-Pull CFP output stage (no R_{in})

The equations for Figure 13.1 are two duplicate sets of the Single Ended equation in Section 5.5 where R_{be} is included. The spreadsheet uses the same empirical fitting constant V_o for $\beta_{3_{eff}}$ and $\beta_{4_{eff}}$. The top half equation labels are appended with 'P' (for Positive) and the lower equations with 'N' for Negative). The equation sets are not provided here but can be viewed in Sheet13 where labels are used which means the equations are human readable.

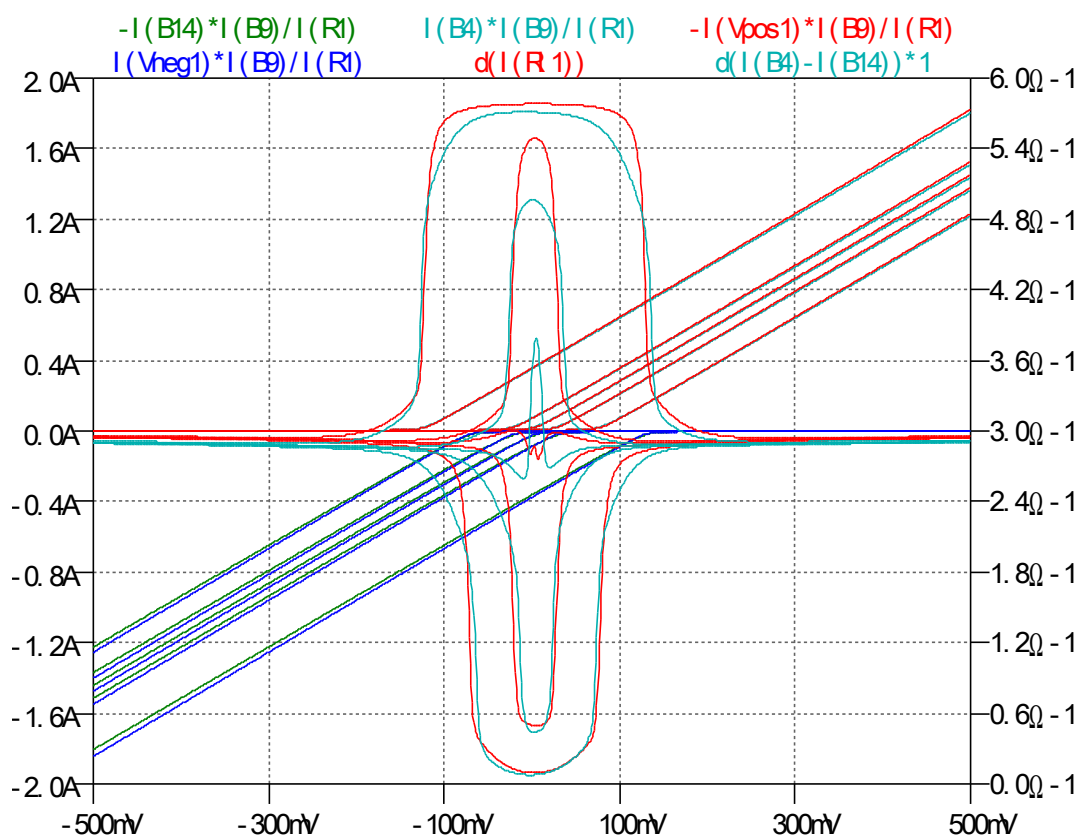


Figure 13.1a. CFP LTspice and equation gain and current plots with $R_e=0.33R$, $R_{be}=33R$ and $V_{sp}=655mV$. V_{sp} is stepped 580mV, 630mV, 655mV, 680mV, 780mV.

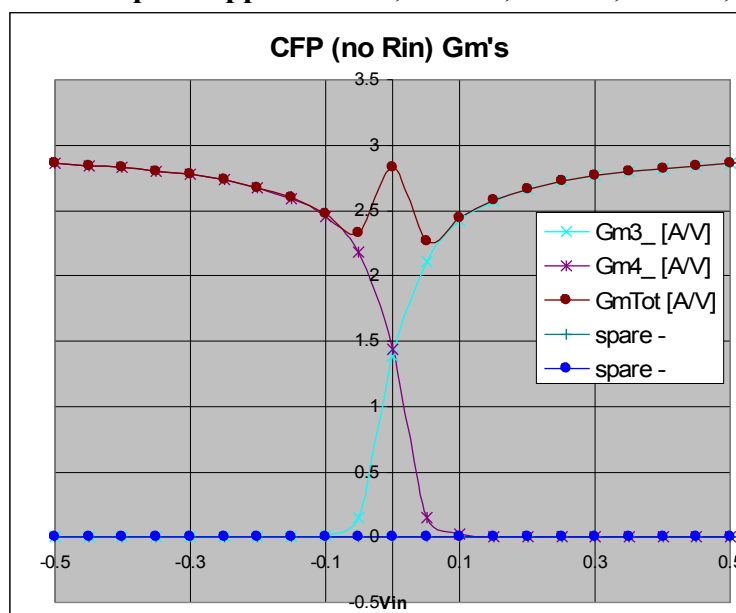


Figure 13.1b. CFP Spreadsheet plot with $R_e=0.33R$, $R_{be}=33R$ and $V_{sp}=655mV$.

It was found that the CFP is sensitive to small variations in bias voltage near optimum bias with $\pm 2mV$ changing the shape of the mid gain wobble from upward to downwards. An offset term is used for fine adjustment of the bias voltage for the equation to align the curves as close as possible.

This high sensitivity to V_{sp} at optimum bias is due to the very small quiescent current through the

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output pair compared to the basic Bipolar Push-Pull output stage in Section 12. With the same emitter resistors of 0.33 ohms the CFP output transistors carry 15mA at optimum bias compared to 80mA for the basic Bipolar Push-Pull output stage. This operating point is only 5mV above the switching voltage (V_{sw} in Section 5) where the output transistors gain begins to rapidly rise.

Weighted distortion are needed to asses the relative merits of the CFP compared to the basic Bipolar Push-Pull output stage and the Darlington output stage [ref 19 L/A Vol 4]. Figure 13.1a shows that more high order distortion is produced by the CFP. Interestingly, removing the base-emitter resistors (Reb1 and Rbe2) eliminates the sharp switching in the CFP as shown in **Figure 13.1c**.

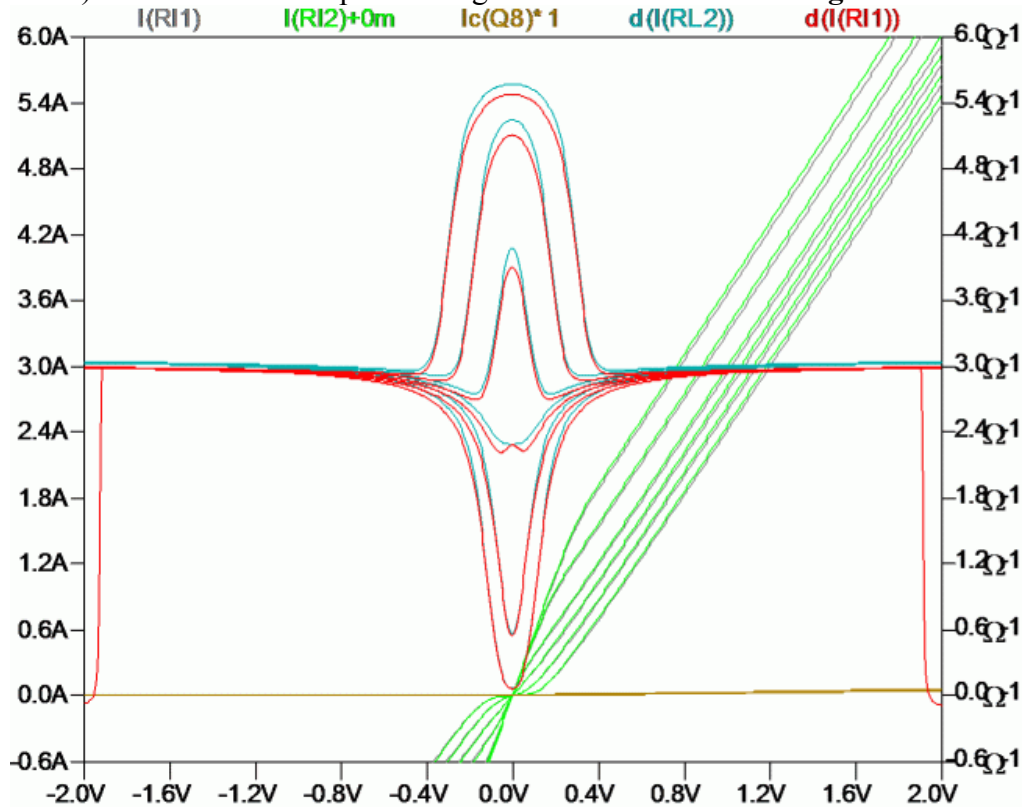


Figure 13.1c. CFP with Reb1 and Rbe2 removed eliminates sharp switching and returns to the shape of a Basic Push-Pull output stage.

Bob Cordell provides an overview of the CFP in his book (p106-108). One major concern with the CFP is high frequency stability and the main cause may be that *both* output transistors are “starved” of current at optimum bias and the F_T of the output transistors fall at this low current even if low value base-emitter resistors are used. In the Darlington output stage there is always one transistor that has enough current to keep the F_T reasonable but in the CFP both output transistors get “starved” passing through the crossover region so the F_T falls and instability can result if the compensation frequency is not lowered.

Figure 13.1d shows a CFP variant that does not use base-emitter resistors for the output transistors. The MOSFET's in the driver transistor emitters allows a relatively high optimum bias of 300mA for 50W and this combination eliminates high order crossover distortion [ref 17 EW+WW Sep 1995]. No problem with the slower turn off of the output transistors was obvious with no base-emitter resistors and the biasing stability was excellent. However, an upgrade to modern high frequency power transistors with low beta-fall will help. Someone said that the base-emitter resistors generally halve the turn-off time and if that is the case then eliminating the base-emitter resistors in a CFP

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halves the usable bandwidth and halves the usable amount of NFB but without the resistors the very high-order crossover distortion is almost eliminated – it appear to be well worth while trade.

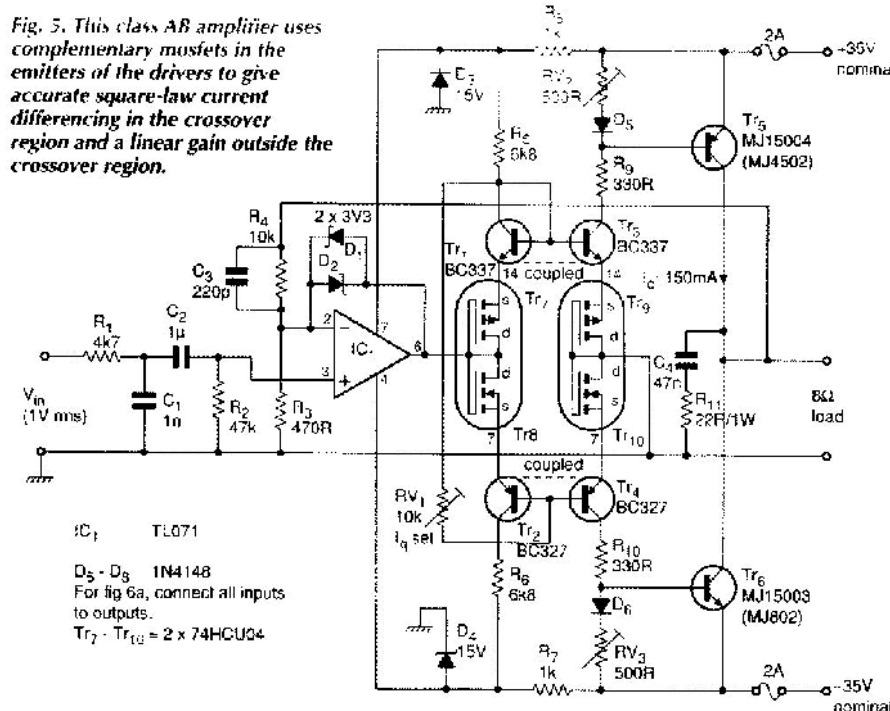


Figure 13.1d. A CFP variant using no base-emitter resistors and a relatively high optimum bias of 300mA for 50W eliminates high order crossover distortion.

14. Push-Pull Darlington output stage

14.1 Push-Pull Common Emitter Darlington output stage

[under construction]

The first (simplest) Push-Pull Darlington circuit is Common Emitter with $R_{in}=0$ and with voltage drive.

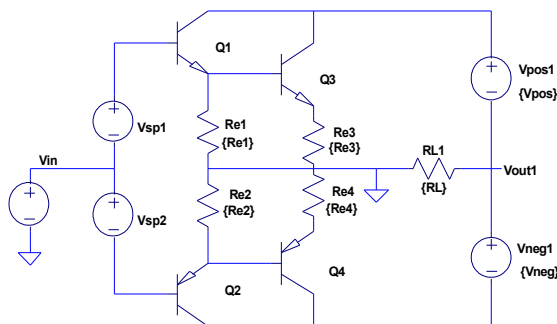


Figure 14.1. Push-pull Common Emitter Darlington output stage with no R_{in}

Figure 14.1 shows an Emitter Follower 'Darlington' output stage with bias voltage generators ($2 \cdot V_{sp}$) and emitter resistors (R_{e1} and R_{e2}). Floating power supplies allow the output current to be summed and fed into a ground referenced load resistor. This simplifies analysis. The load current is $I_L = I_{c3} + I_{c1} - (I_{c4} + I_{c2})$ where I_{c3} flows into the collector of Q3 while I_{c4} flows out of the collector of Q4. Each half of the Darlington pair was solved in Section 7 with constant Beta's and no Early effect.

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The previous solution was $I_{c3} = \frac{N3 \cdot V_t}{R_{eq}} W \left(\exp \left(\frac{A1(V_{sp1} + V_{in} - V_{k3}) - V_{o3}}{N3 \cdot V_t} \right) \right)$ where A1 is the voltage divider ratio formed by Rbp1/Beta1 and Re1 or $A1 = \frac{R_{e1}}{R_{e1} + R_{ep1} + R_{bp1} / \beta_{e1}}$ and $V_{o3} \approx 700m \cdot \ln \left(\frac{1/A1}{1 + \ln(1/A1)} \right)$. Terms for Req, Vk3 and Vo3 were defined in Section 7.

The driver collector current is approximate since it involves Ic3 (which is an approximation) is

$I_{c1} = \frac{N1 \cdot V_t}{R_{e1} + R_{ep1} + R_{bp1} / \beta_{e1}} W \left(\exp \left(\frac{2 \cdot V_{sp1} + V_{in} - V_{k1}}{N1 \cdot V_t} \right) \right) + \frac{I_{c3}}{\beta_{e3}}$ where $V_{k1} = N1 \cdot V_t \cdot \ln \left(\frac{N1 \cdot V_t}{I_{s1}(R_{e1} + R_{ep1} + R_{bp1} / \beta_{e1})} \right)$. Parasitic emitter and base resistances can also be included for Q3 Rep3+Rbp3/Beta added to Re2.

For the complementary transistors the same equations are used but with -Vin in place of +Vin in the two equations above. The power rail currents are IC3+IC1 and the load current is the difference between the power rail currents.

The gm equations are given in the listing below The net gm is used to estimate THD.

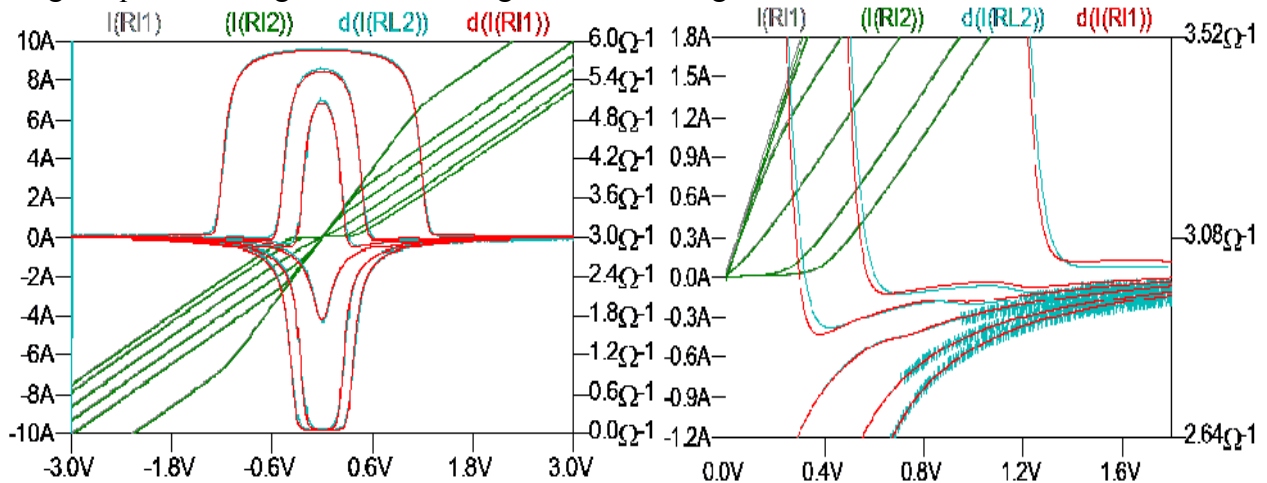
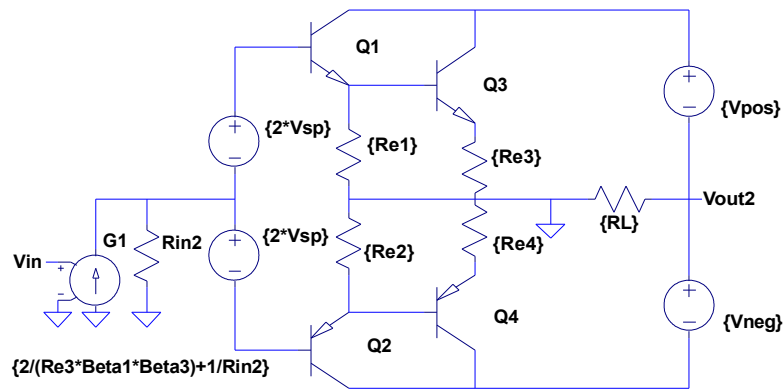


Figure 14.1a. Push-pull Common Emitter output stage voltage drive (Re1=33R, Re2=330m, Red=SPICE). Vsp stepped 480mV, 580mV, 630mV, 680mV, 780mV, 880mV. Right zoom shows detail is replicated by the equation solution

14.2 Darlington Equations for current drive

[under construction]

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In a typical amplifier if the preceding stage is common emitter then it will have a relatively high output resistance and this can be represent by R_{in} shunted to common and a current source driving to the base of Q1. Current source drive can be provided using the Norton equivalent $I_{in} = V_{in}/R_{in}$ and with R_{in} shunted to common. The level of peak current needed is $I_{in_{pk}} = \frac{I_{c2_{pk}}}{\beta_{a1} \cdot \beta_{a3}}$.

The above equations for I_{c1} and I_{c2} with current drive become

$$I_{c3} = \frac{N \cdot V_t}{R_{eq}} W \left(\exp \left(\frac{A1(V_{sp} + I_{in} \cdot R_{in} - V_{k2}) - V_{o2}}{N V_t} \right) \right) \text{ and}$$

$$I_{c1} = \frac{N1 \cdot V_t}{R_{e1} + R_{e1} \parallel R_{bp1} \parallel \beta_{a1}} W \left(\exp \left(\frac{2 \cdot V_{sp} + I_{in} \cdot R_{in} - V_{k1}}{N1 \cdot V_t} \right) \right) + \frac{I_{c3}}{\beta_{a3}} \text{ where terms for } R_{eq}$$

derive from the basic CE Push-Pull output stage with R_{in} included (which involves a 'switch'). I_{c1} terms were been defined above for voltage drive except the $I_{in} \cdot R_{in}$ is the equivalent input voltage.

Figure 14.1b shows the Push-Pull CE Darlington with current drive and R_{in} stepped 1m, 1k and 10k with $R_{e1}=33$ and $R_{e3}=330mR$. This only shows the LTspice plots since equations for the Darlington pair have not been entered into the circuit file (Fig 14s1s2). The current source G is programmed using $\{2/(R_{e3} \cdot \beta_{a1} \cdot \beta_{a3}) + 1/R_{in}\}$ (for Q1/Q3 pair) to give similar peak gains. This plot shows there is significant nonlinearity due to the base-emitter resistors.

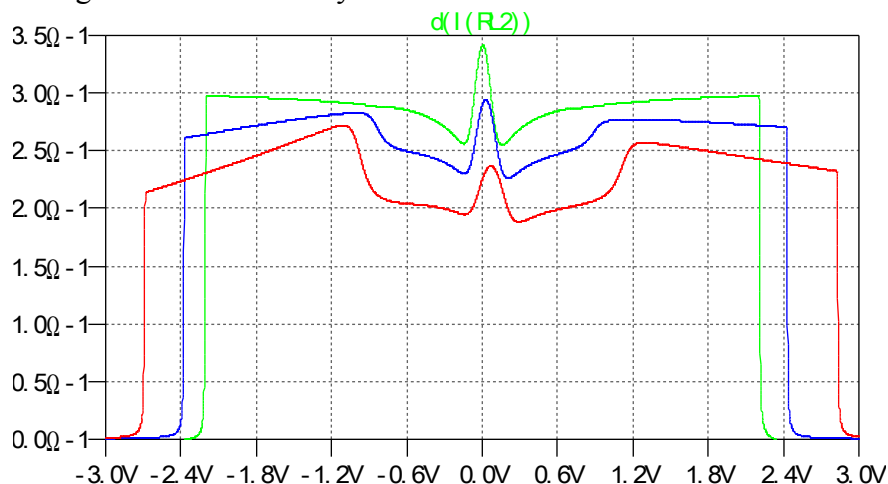


Figure 14.1b. Push-pull Common Emitter output stage with current drive. $R_{e1}=33$, $R_{e3}=330mR$. R_{in} stepped 1m, 1k and 10k, $\beta_{a1}=150$ & $\beta_{a3}=100$. (Red is SPICE).

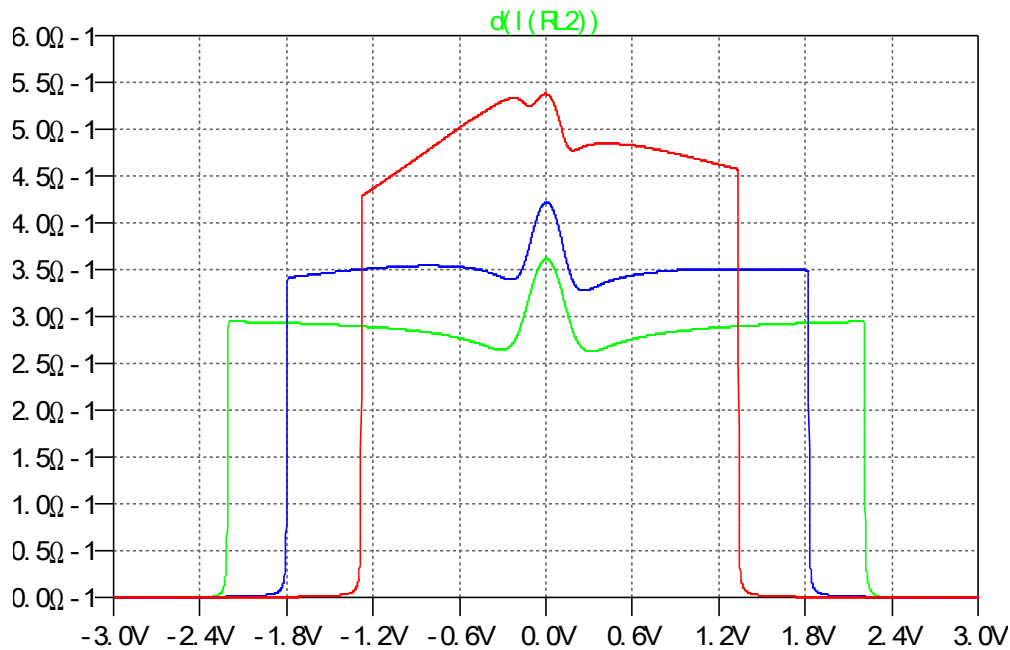


Figure 14.1c. Push-pull Common Emitter output stage with current drive. $R_{e1}=33k$ (for negligible effect), $R_{e3}=330mR$. R_{in} stepped 1m (lower), 1k and 10k (upper), $\beta_{Q1}=150$ & $\beta_{Q3}=100$ & no Beta-fall. (Red is SPICE).

Figure 14.1c shows the same Push-pull Common Emitter output stage with current drive but with $R_{e1}=33k$ to see its effect on linearity when R_{in} is stepped 1m (lower), 1k and 10k (upper). When R_{e1} is made deliberately very high (effectively removed) the effect of a high source resistance can be seen, here $\beta_{Q1} \cdot \beta_{Q3} \cdot R_{e3} \approx 3k\Omega$ (for $Q1/Q3$ pair), so 1k is still partly voltage drive (and not linearity is affected much) but when $R_{in}=10k$ the effect of current drive becomes apparent.

Both Figure 14.1b ($R_{e1}=33R$) and Figure 14.1c ($R_{e1}=33k$) show the the Darlington driver introduces significant nonlinearities when the source resistance is greater than $\beta_{Q1} \cdot \beta_{Q3} \cdot R_{e3}$ (for $Q1/Q3$ pair) for current drive to begin. Changing R_{e1} ($Q1$ emitter resistor) does not provide any relief of nonlinearity when R_{in} is large.

An alternative, Locanthi emitter resistor arrangement, where R_{e1} and R_{e2} are coupled and not connected to the output node (ground or common here), may provide some relief since the driver transistors operate in Class-A and do not turn off after the crossover region for $Q3,4$ is passed (the Locanthi circuit is not included in PAK at this stage).

Another alternative, is to driving the CE output stage with a high current buffer to ensure the output pair are voltage driven as a Basic output stage (Section 12.1). This option is explored in a Linear Audio article [ref 19, L|A Vol.4]. With the floating supply of the Basic CE output stage the buffer driving $Q3,4$ does not need a high voltage supply since only 1-2 volts are needed for the full output current swing. Also only about 200mA is needed to drive an output transistor to 10A – the practical limit for a single 15A power transistor. This means the buffer could be a high current opamp such as the HA5002. A discrete version could use a pair of MOSFET's in class-A to remove the Beta dependency on R_{in} with bipolar drivers.

The standard Darlington is fine for voltage driven output stages (which usually means using Triple Darlington's) but the Darlington pair deteriorates if the source resistance gets too high (ie much

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above $\beta_1 \cdot \beta_3 \cdot R_{e3}$ (for Q1 and Q3 pair)

14.2 Basic push-pull CE amplifier with current drive and Beta-fall

[under construction]

In power amplifiers the current gain of the power transistors can have a significant effect on the distortion generated when the source resistance R_{in} is significant. This section includes the effect of Beta variation at high and low currents using similar equations to Gummel-Poon but modified to include local feedback (degeneration) by the emitter resistance R_e and the source resistance R_{in} .

Fig 14.2 shows the effect of both R_{in} and Beta-fall in circuit Figure 12.1. The effect of Beta-fall is made very obvious when R_{in} is increased. Figure 14.2 is the over-biased case while **Figure 14.2a** shows the underbiased case.

The accuracy of the equation solution is very good for high R_{in} , to better than 1%. At very small R_{in} the equations deviate by 3% (seen in upper Gm and lower IL plots in Figure 14.2).

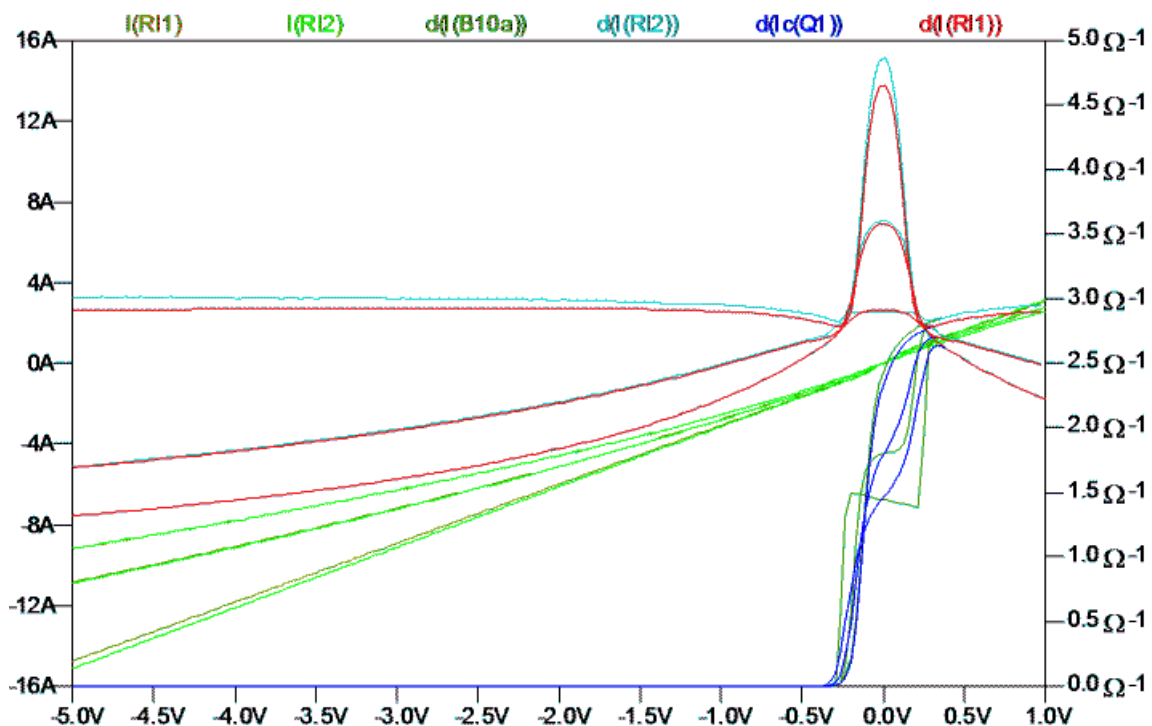


Figure 14.2. Over-biased plots with high current Beta-fall, R_{in} stepped 33mR (U), 33R (M) and 3k3 (L). $I_{KF}=4A$, $V_{sp}=740mV$ ($I_q=280mA$). LTspice Red gm, Ic1 green & gm1 of Q1 blue. At high R_{in} calculated gm1 error prevents low current Beta-fall (a more accurate method is needed).

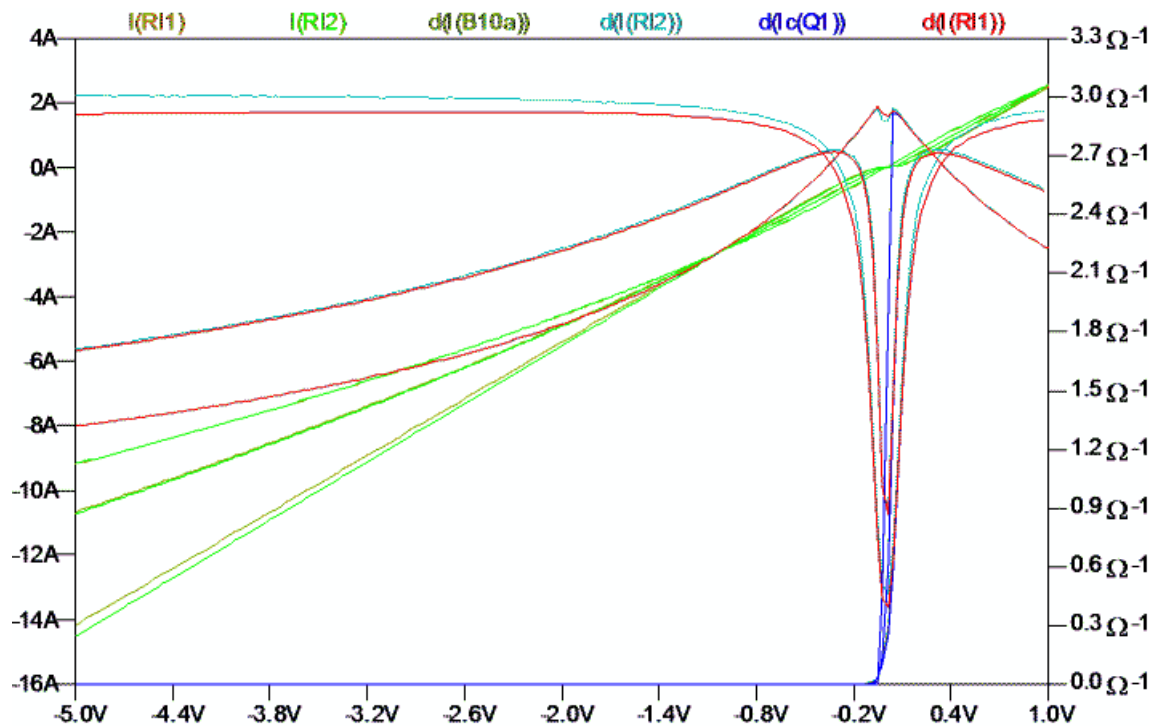


Figure 14.2a. Under-biased with Beta-fall and Rin stepped 33mR (U), 33R (M) and 3k3 (L). IKF=4A, Vsp 540mV (Iq=4mA). Red LTspice gm. High Rin only reduces the gain variation in the crossover region. See crossover region detail below.

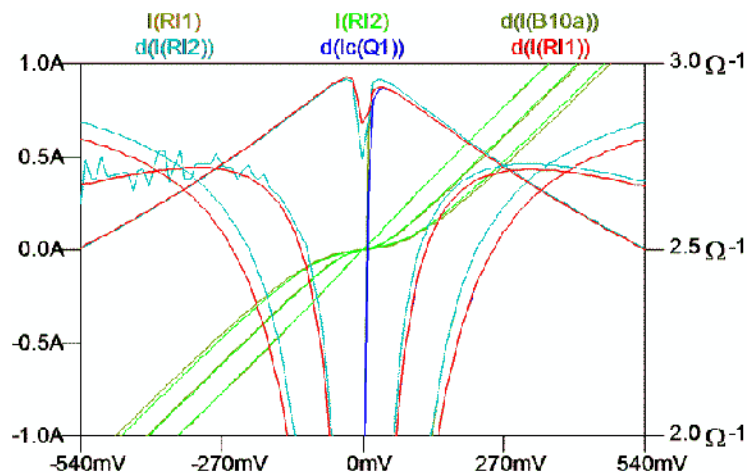


Figure 14.2b. Same as above showing high Rin crossover region detail. With Rin or 3k3 the crossover nonlinearity is almost removed by current drive. Amazing!

There are no Beta-fall equations in the PAK V1.0 circuit files or in the PAK spreadsheet. The above plots were generated from early equations and need updating.

15. THD analysis using Lambert W equations

15.1 Low order Harmonic Distortion calculated from gain error

Section 4.2c (Estimates of Distortion HD2 and HD3) explained how Ed Cherry's equations using trigonometric expansions of a sinewave multiplied by simple nonlinear functions can be used to calculate the 2nd and 3rd harmonic levels from the gain values found at the peak of the sinewave input. Unfortunately these are the only two harmonics that can be calculated directly as independent

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variables from the peak gain deviations [ref 10 Howard Stereophile April 2006].

15.2 Crossover distortion calculated from gain error at peaks

Since the Lambert W solutions provide explicit functions for all the derivatives it may be possible to use the first say m derivatives expressions to solve each harmonic level up to the m^{th} harmonic by using a similar approach to Peter Baxandall in his article 'Audio Power Amplifier Design Pt. 6' [Wireless World Jan 1994, reprint available from LinearAudio.net]. It may be possible now to write a general expression for each harmonic as a function of the input level of a circuit without the high memory workload of FFT. Goal seeking can then be applied in a spreadsheet to minimise high order distortion in a general power amplifier circuit. For a tentative outline of this method see **Section 15.5**.

Until then a simple way has been devised to estimate crossover distortion in Class-AB amplifiers. Crossover distortion creates high order distortion at full output voltage and progressively becomes lower order distortion as the output swing reduces, and eventually creates low-order Class-A distortion when the swing is small enough that the peak current is not more than twice the quiescent current. As mentioned, the Lambert W-function solutions for transistor currents give both current and gain (gm) in explicit form it is possible to estimate the distortion as THD from the distortion energy contained in the deviation from linearity as a fraction of the total energy.

The total energy is related to the average area under the gm curve and since current is the integral of gm area then the current at the peak of the sinewave input is the average area of the nonlinear gm curve. Next, the gm at the peak of the sinewave multiplied by the peak input voltage gives a current that is different to the actual current at the same peak and this arises because of nonlinearity somewhere in the gm curve. The difference between these two current must be related to the distortions RMS current. If we assume there is a constant in this relationship then we can write

$$THD_{est} = SF \cdot \left| \frac{I_{pk} - Gm \cdot Vin_{pk}}{I_{pk}} \right| \quad \text{or} \quad THD_{est} = SF \cdot \left| 1 - \frac{Gm \cdot Vin_{pk}}{I_{pk}} \right|.$$

The absolute value is used to ensure THD is always a positive value (since it is related to ratios of power). The scale factor (SF) can be found using simulations and is approximately [YTBD].

One problem with this estimate for THD is at optimum bias THD_{est} becomes zero and not a finite small value as in actual circuits. The estimate can be improved by adding an offset voltage (HD_0) and vector summing these so the offset does not affect higher distortion estimates. We then use

$$THD_{est} = \sqrt{\left(HD_0^2 + SF \cdot \left(1 - \frac{Gm \cdot Vin_{pk}}{I_{pk}} \right) \right)^2}$$

where HD_0 can be estimated from the area under the zero crossing gain 'blip' (as seen in Figure 12a middle curve) using

$$HD_0 \approx 0.5 V_T \left| 1 - \frac{\#Pr / (Re1 + Re2) / 2}{GmT_0} \right| \quad \text{if } \#Pr / (Re1 + Re2) / 2 > GmT_0 \quad \text{else } HD_0 = 0$$

and term GmT_0 is the total gm at the zero crossing, and $1/Re$ gives the approximate maximum gm at high currents, so when $\#Pr / (Re1 + Re2) / 2 > GmT_0$ the quiescent current is high enough to give a central gain 'blip' in a wingspread plot and thus a null in the original THD_{est} . The factor $0.5V_T$ is an empirical scale factor found to fit the THD minimum distortion at optimum bias and this value is found by simulations.

The above approach can be refined to first calculate HD2 and HD3 distortions using the gain

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deviations at the peaks then subtract these (using RMS means) from the THD_{est} values above to give the remainder of harmonics for the 4th and higher.

15.3 Weighting Harmonic Distortion [Under construction]

Crossover distortion generates a distribution of harmonics that vary in a complex way with the quiescent current and the output swing fraction of the maximum swing before clipping. It has been found that a different scale factor (w-SF) can be applied to the above THD_{est} approach to estimate the weighted distortion. The weighted scale factor w-SF appears to be about 30 times the SF for unweighted THD estimates, but it can vary from circuit to circuit and the weighting method chosen [ref 19 L/A Vol 4]. More work is needed to find general equations for simple weighted THD estimates ($w-THD_{est}$).

The equations used in PAK spreadsheet version 1.0 are only roughly scaled to demonstrate the method – the values should be found from spice simulations from the circuit files provided with PAK.

Figure 15.3a is an example calculated THD and weighted-THD distortions at optimum quiescent current of 90mA. Note THD is calculated for each input voltage step. The weighted-THD is the top curve that is almost flat until small input levels then falls. The unweighted THD rises with reducing level, peaks and then falls at low levels. The weighted-THD correlates better with the level of distortion that we hear.

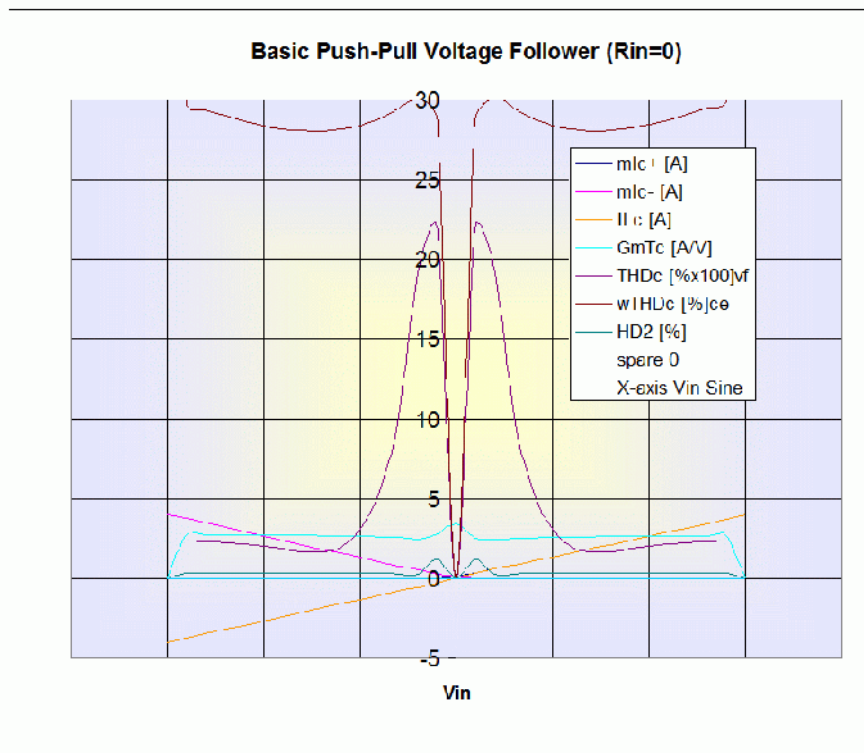


Figure 15.3a shows calculated THD and weighted-THD distortions at optimum quiescent current of 90mA. Note THD is calculated at each input voltage step. SPICE does not plot multiple THD's like this.

15.4 Spreadsheet example – Calculated THD and TLD

The following is an example where crossover distortion is calculated in a Basic output stage with voltage drive. The aim in this case was to calculate the amount of distortion increase when the

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output stage power dissipation changes with RMS input level changes typical of music. This variation is called Thermal Lag Distortion (TLD) due to the temperature lag of the bias voltage generator compared to the output transistor's junction temperature.

The thermal calculations are largely based on Bob Cordell's book Chapter 14 "Output stage thermal design and stability" p277-313 [ref 11].

An example is provided in Sheet 15 of PAK v1.0 spreadsheet (2nd block). There is no related LTspice circuit file with the V1.0 release pack. The spreadsheet has the rows compressed back to the minimum (5 rows) to reduce the spreadsheet file size (since there are so many equations in this example).

The first block in Sheet15 demonstrates the use of THD in a Basic output stage with voltage drive. It differs from sheet 12 in that currents and gains are estimated at both the (+) and (-) sinewave peaks – this is necessary to generate a symmetrical THD plot when the gain plots may be asymmetrical due to transistor mismatch etc. This can be demonstrated if you change Re1 (Re+) to say 0.5 and leave the other at 0.33. The gm plot now has a step in the crossover region but the THD plot remains symmetrical. That's how it needs to be and this requires twice the number of equations in each block to do this.

While we are on THD plots the weighted THD plot is seen were to be basically flat over the range of swept input voltages (ie the full range of output power). Return the emitter resistors back to 0.33 each and you will see the unweighted THD increases quite a lot as the input level falls, and peaks at about 0.1V input (quite a low power), then falls to zero at even lower power (due to Class-A in this region). Now comparing the trend of the weighted THD plot and the unweighted THD plots we find the weighted plot is close to how our ear hears this crossover distortion (being not much affected as the power level falls) and the unweighted THD plot does not correspond with our listening experience and gives an erroneous indication of how good an output stage sounds. For more on weighting of crossover distortion in simulations see the Linear Audio article on this [ref19 LA V.4].

Now to the second block that does the TLD calculations. This is done in two sections; first THD and w-THD are calculated same as the previous block discussed above. This gives the base line distortion before the bias generator is destabilised by a thermal step change.

The second section calculates the thermal model values. You set the music power level (dB) before the step and the step size (dB) in cells EH62 and EI62 respectively. The equations then calculate the power dissipations before and after the step, then the temperature changes and the bias voltage changes. These values depend on the speaker sensitivity (which you can set) and the supply voltage and speaker resistance.

Then the distortion changes are calculated. Also calculated is the amount of negative feedback that you need to provide to ensure the THD is reduced below a threshold level in column HO which is calculated to roughly fit our hearings threshold to the 2nd harmonic of a 50Hz fundamental at different SPL levels.

The weighting factors used for THD and w-THD estimates has not been checked against LTspice distortion simulations for PAK v1.0. The equations are provided as an example of how TLD can be calculated and plotted. For example, the plot provided (once the rows are expanded back to row 102) shows the effect of trying to minimise TLD by choosing the best Vspread.

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Some designers have suggested that TLD can be mitigated by slightly over-biasing the output stage when in steady-state so a thermal step does not lead to severe underbiasing situation [ref 11 Cordell Book p187?]. But if the over-bias is too much then you get more distortion in steady-state than during the occasional step change in music level. And how big (or small) a step change do you need for TLD to raise distortion? And how does our TLD choices change when we use weighted THD rather than unweighted THD? This spreadsheet is able to give answers and insights into to most of these questions.

It was found that the modern TO-264 packages (contact area 317mm^2) have relatively low thermal resistances compared to earlier power transistors of the 1970's to 1990's including the TO-3P TO-247 (183mm^2). This significantly reduces the thermal step size, even to the extent that TLD is almost so small that it now does not need complex simulations or these calculations. But it is always useful to be able to see whether it is or isn't an important consideration in your particular design. So the simulation or calculation exercise is still worthwhile. It gives a feel for what is significant and what isn't for designers.

Some notes on the TLD spreadsheet equations:

Notes on calculations Sheet15 2nd block:

1. 'Vin Sine' (O60) is by an equation from the section where the initial power is set (as 'dBpre' mentioned below). Do not enter a value into this 'Vin Sine' column.
2. 'dBpre' (EH60) Set the dB level for the [heat up] phase of the heatsink. This setting should be less than or equal to the suggested 'dBnoClip' (EH or EI) levels (for sine or music). You can also set $\text{dBpre} = \text{dBnoClip} + \text{dBStep}$ to give a reversed step situation (lower power to higher power).
3. 'MusicPDF' (EM60) is preset at 8dB as recommended minimum but can be changed to say -16dB for classical (orchestral) music to check for differences between music material.
4. 'ScaleHS' (DQ60) Set a scale factor for a practical heatsink eg $\sim 2x$ for 6R loading but determine this by setting RL to 4R and note how Rsa changes, $\text{ScaleHS} = \text{Rsa}(8R)/\text{Rsa}(4R)$. Alternatively, change ScaleHS for acceptable TouchMax.
5. The output transistors are MJL21193/4 and the number of pairs can be set in '#Prs' (U60). This can influence the sizing of the heatsink (see below).
6. 'TouchMax' can be set to ensure the heatsink rating required is not too small that it may burn someone who touches it. Set this high if you want to disable this.
7. 'StableH?' and 'StableC?' determine whether the thermal loop is stable if the temperature sensor is mounted on the power transistors case (ThermalTrak) or on the heatsink (non-ThermalTrak transistors). If it is not stable then you can increase Re for thermal stability (and then maybe increase the number of pairs). Thermal stability is determined directly by the supply voltage (Vpos & Vneg).
8. 'TjhPost' (EY60) and 'TjcPost' (FJ60) are separate calculations for ThermalTrak and non-ThermalTrak transistors. THD calculations are set to non-Thermal Trak devices by using $\text{Vsp} + \text{Hpost}$ in 'Arg+' (GE60) equation. For ThermalTrack distortion calculations instead change to 'Arg+' (GE60) equation to reference $\text{Vsp} + \text{Cpost}$.

15.5 Harmonic analysis of BJT amplifiers using Lambert W

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The following is based on Peter Baxandall's article 'Audio Power Amplifier Design Pt. 6' [Wireless World Jan 1994 reprint from [LinearAudio.net](http://www.linearaudio.net)]. His harmonic distortion analysis of a BJT common emitter amplifier is extended to allow general harmonic distortion analysis for the nth harmonic using the Lambert W-function.

The Lambert W-function can provide an explicit solution for BJT amplifiers (see the PAK project user Guide) and the Lambert W-function allows the nth derivative to be expressed as a polynomial in terms of the solution for the current (in other words, having solved a circuits current using the Lambert W-function all the derivatives can also be written from this equation).

The following are the equations given in Peter Baxandall's Pt.6 article:

$$I_c = I_s \exp\left(\frac{V_{be}}{V_t}\right) \quad (125)$$

where 'I_s' the for base emitter junction diode which determines the collector current.
In his CE circuit

$$v_{out} = -R_L I_q \left(\exp \frac{V_{in}}{V_t} \right) \cdot \exp \left(\beta \frac{v_{out}}{V_t} \right) \quad (126)$$

where β is the feedback factor . To obtain the harmonic levels he proposed using the power series

$$v_{out} = a_1 v_{in} + a_2 v_{in}^2 + a_3 v_{in}^3 + \dots \quad (127)$$

And using Maclaurin's theorem to find the coefficients

$$a_1 = \left. \frac{dv_{in}}{dv_{in}} \right|_{v_{in}=0} \quad a_2 = \left. \frac{d^2 v_{in}}{d v_{in}^2} \right|_{v_{in}=0} \quad \text{etc} \quad (128)$$

These Maclaurin's terms are applied to Eq 2 and equating gives the following coefficients for (3)

$$a_1 = \frac{A_v}{1 - \beta A_v} \quad \text{and} \quad a_2 = \frac{1}{2!} \left(\frac{1}{V_t} \right)^1 \frac{A_v}{(1 - \beta A_v)^3} \quad \text{and} \quad (129)$$

$$a_3 = \frac{1}{3!} \left(\frac{1}{V_t} \right)^2 A_v \left[\frac{1}{(1 - \beta A_v)} + \frac{3|\beta A_v|}{(1 - \beta A_v)^2} - \frac{3(\beta A_v)^2}{(1 - \beta A_v)^3} + \frac{|\beta A_v|^3 + 3|\beta A_v|}{(1 - \beta A_v)^4} - \frac{3(\beta A_v)^2}{(1 - \beta A_v)^5} \right] \quad (130)$$

So we can write the individual harmonic levels if we expand the $(\frac{V_{pk}}{V_m})^m \sin^m \omega t$ terms in (3)

$$v_{out} = b_1 \left(\frac{v_{pk}}{V_m} \right)^1 \sin(\omega_1 t) + b_2 \left(\frac{v_{pk}}{V_m} \right)^2 \sin(2\omega_1 t) + \dots \quad (131)$$

We need to find each b_n so we multiply the relevant derivative values by the relevant harmonic level of $\sin^m \omega t$ and add them all up, the general nth harmonic level is

$$b_n = \sum_{i=0}^m c_i \cdot a_i \quad (132)$$

when m is even only need to use even i , and vice versa and stop at m since there are no higher harmonics from the expansion of $\sin^m \omega t$ (m is an integer).

The coefficients of the expansion of $\sin^m \omega t$ can be expanded and written as

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$$\sin^m \omega t = \sum_{i=0}^m c_i \sin(i \omega t) \quad (133)$$

Where the coefficients c_i have been found using FFT up to $m=100$ (and to the 100th harmonic). They can be arranged into a $m \times m$ sparse matrix of coefficients as a library file and called when harmonic analysis is required.

For audio power amplifiers with 50Hz fundamental we want to weight the harmonics up to around 5kHz or the 100th harmonic. The weighting factor can be $1/4.59 \times n^{2.2}$ for the n^{th} harmonic which can be applied to equation 7 (or equation 8).

The coefficients c_i are all zero for all odd harmonics when power m is even and vice versa coefficients are zero for all even harmonics when m (power) is odd. There are no harmonics above the m^{th} harmonic for power m . Separate tables can therefore used for odd and even harmonics.

Table 1. For odd m powers ($m=1,3,5,7,9$) and there is no dc component

$m \downarrow$	f1	f3	f5	f7	f9	f11
1	1	0	0	0	0	0
3	0.75	-0.25	0	0	0	0
5	0.625	-0.3125	0.0625	0	0	0
7	0.5569	-0.3281	0.1094	-0.01562	0	0
9	0.4922	-0.3281	0.1406	-0.03516	0.003906	0

Table 2. For even m powers ($m=2,4,6,8,10, 20$) where there is also a dc component (f_0)

$m \downarrow$	f0	f2	f4	f6	f8	f10
2	$\frac{1}{2}$	$-j\frac{1}{2}$	0	0	0	0
4	0.375	$-j0.5$	$j0.125$	0	0	0
6	0.3125	$-j0.4687$	$j0.1875$	$-j0.01325$	0	0
8	0.273437	$-j0.4375$	$j0.2187$	$-j0.0625$	$j0.007812$	0
10	0.246094	$-j0.4102$	$+j0.2344$	$-j0.08789$	$+j0.01953$	$-j0.001953$

where the j operator denotes a 90° phase shift.

$m \downarrow$	f0	f2	f4	f6	f8	f10
20	0.176197	$-j3.204\text{e-}01$	$+j2.403\text{e-}01$	$-j1.479\text{e-}01$	$+j7.393\text{e-}02$	$-j2.957\text{e-}02$
cont	-	f12	f14	f16	f18	f20
	-	$+j9.241\text{e-}03$	$-j2.174\text{e-}03$	$+j3.624\text{e-}04$	$-j3.814\text{e-}05$	$+j1.907\text{e-}06$

Figure 1. FFT spectrum for $\sin^{20} \omega t$ for 1kHz fundamental

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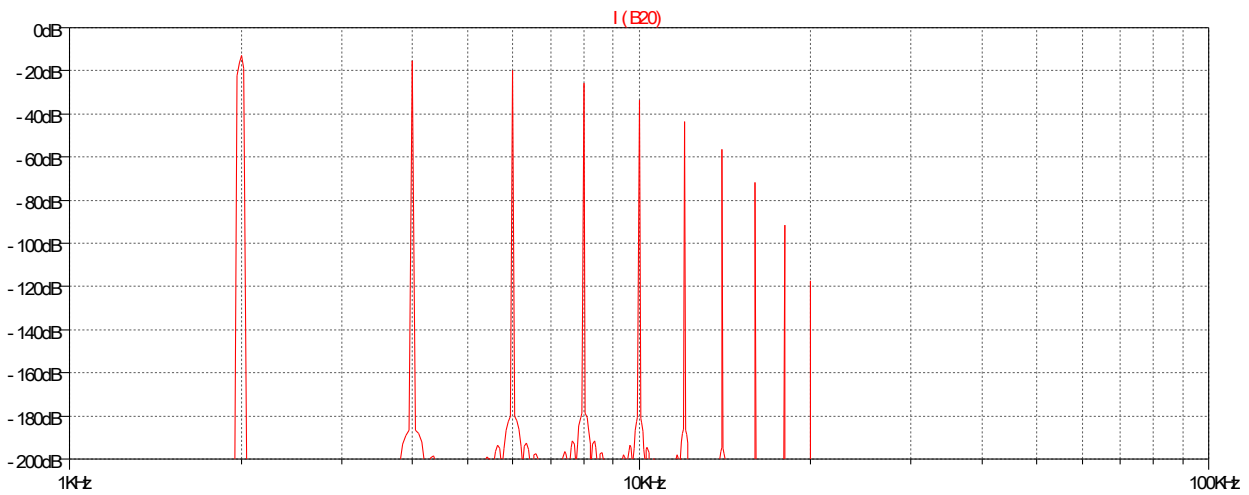
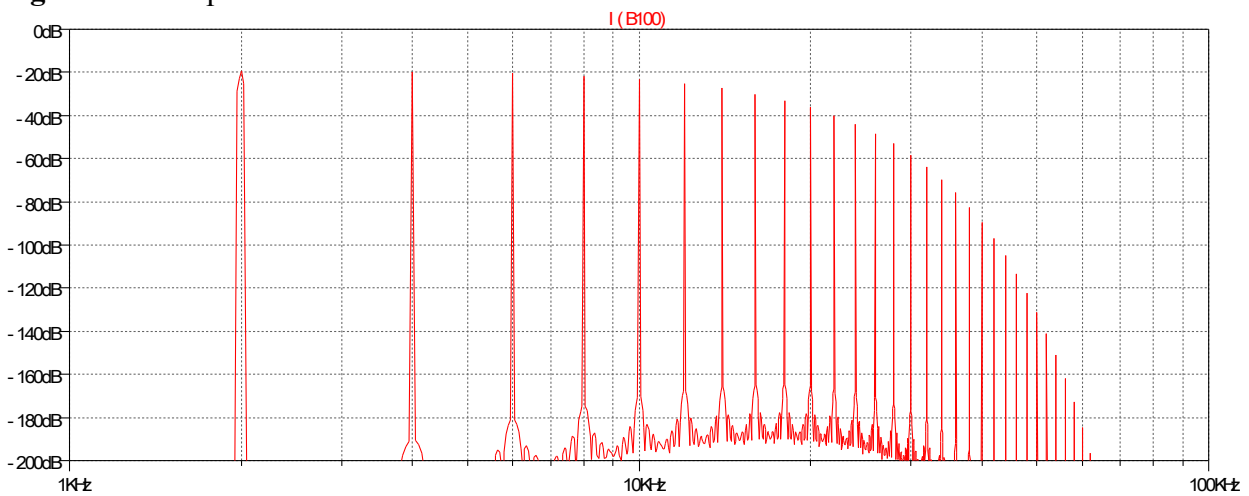


Figure 2. FFT spectrum for $\sin^{100} wt$ for 1kHz fundamental



Note how $\sin^{100} wt$ harmonic levels become vanishingly small by the 50th harmonic and being 20dB reduced by the 20th harmonic then reducing at a fast rate.

Now the application of the Lambert W-function. The Common Emitter circuit has a Lambert W-function solution for Equation 2 which allows all the derivatives to be expressed in explicit form from as a finite series as provided by RM Corless [ref AdvCompMath, RM Corless, On-the-Lambert-W-function, Aug-1996] his equation 3.5, 3.6 as

$$\frac{d^n W(\exp(x))}{dx^n} = \frac{q_n \cdot W(\exp(x))}{(1 + W(\exp(x)))^{2n-1}} \quad n \geq 1 \quad (134)$$

where coefficient q_n is

$$q_n(w) = \sum_{k=0}^{n-1} \binom{n-1}{k} (-1)^k w^{k+1} \quad n \geq 1 \quad (135)$$

The first derivative is $n=1$ (LHS) is compared to Baxandall's a_1 in Equation 5 (RHS)

$$\frac{d W(\exp(x))}{dx} = \frac{q_1 W(\exp(x))}{1 + W(\exp(x))} \leftrightarrow a_1 = \frac{1}{\beta} \frac{\beta A_v}{1 + \beta |A_v|} \quad (136)$$

The second derivative is $n=2$

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$$\frac{d^2 W(\exp(x))}{dx^2} = \frac{q_n W(\exp(x))}{(1 + W(\exp(x)))^3} \leftrightarrow a_2 = \frac{1}{2!} \left(\frac{1}{V_t} \right)^1 \frac{1}{\beta} \frac{\beta A_v}{(1 + \beta |A_v|)^3} \quad (137)$$

The details for coefficient q_n are not sorted as yet but this comparisons show the general form of the Lambert W-solution provides the polynomial for evaluating derivatives for a general BJT circuit. Therefore the distortion for the nth harmonic can now be obtained (in principle) without using specific component values and FFT.

This will be useful with weighted distortion analysis of power amplifier circuits where harmonics from the 7th and above are considered to be non-musical and need to be below the threshold of audibility. It is therefore possible to obtain equations for the level of harmonics from the 7th and above. This may be easiest by calculating the RMS value for the first 6 harmonics and subtracting this from the THD (from gain deviation and peak current) to give the residual for all harmonics from the 7th harmonic and above. The first 6 harmonics only requires 14 c_i coefficients and the first 6 derivatives (a_1 to a_6) which is still doable in a non-macro spreadsheet format. A spreadsheet is useful for plotting specific cases without using a simulator and a MathLab package can be used for symbolic manipulation with the complicated harmonic level equations.

Summary of useful features PAK-1v0-NM

- A simple and flexible framework for evaluating circuits involving bipolar transistors, such as audio power amplifier circuits
- Useful for teaching audio power output stage circuit design and analysis
- Useful for comparing and contrasting different audio power output stage topologies
- Useful for distortion analysis of different audio power output stages
- Useful for thermal analysis of different audio power output stages
- Useful for hybrid circuits using BJT's and MOSFETs and/or valves (tubes)
- Useful for integration into SPICE to overcoming BJT model non-convergence problems.

[end of Chapter]

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SPREADSHEET USER GUIDE

Appendix A

Various topics

Contents:

1. General solution for $p^{ax+b}=cx+d$ or $ax+b=\log_p(cx+d)$.
2. A General solution for $ye^{y^2}=x$ (a call for further info).
3. Proof that a CFP circuit is not solvable using $ye^y=x$ with overall feedback.

A1. Appendix A1

General Lambert W-inverse solution from Wikipedia

2. From http://en.wikipedia.org/wiki/Lambert_W-function the general solution for

$$p^{ax+b}=cx+d \text{ or } ax+b=\log_p(cx+d) \quad (A11)$$

$$x = -W\left(\frac{\frac{-a \ln p}{c} \cdot p^{b-\frac{ad}{c}}}{a \ln p}\right) - \frac{d}{c} \quad (A12)$$

For exponentials $p=e$ so the general solution for $e^{ax+b}=cx+d$ or $ax+b=\ln(cx+d)$

$$\Rightarrow x = \frac{-1}{a} W\left(\frac{-a}{c} \cdot e^{b-\frac{ad}{c}}\right) - \frac{d}{c} \Rightarrow x = \frac{-1}{a} W\left(e^{\frac{bc-ad-c \ln(\frac{-c}{a})}{c}}\right) - \frac{d}{c} \quad (A13)$$

A2. Appendix A2

The page at <http://mathworld.wolfram.com/LambertW-Function.html> includes a reference to Gosper, R. W. Jr. "The Solutions of $ye^{y^2}=x$ or $ye^{(y^2)}=x$ and $ye^y=x$." ACM SIGSAM Bull. 32, 8-10, 1998).

The solution to

$$ye^{y^2}=x \quad (138)$$

may be helpful for solving circuits. In the log form it looks like

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$$y^2 = \ln\left(\frac{x}{y}\right) \quad (139)$$

It would be helpful if someone can obtain details of this solution for this project (see [Feedback](#)).

A3. Appendix A3

Proof that a CFP circuit is not solvable using $ye^y = x$ with overall feedback

A lemma (proof) that the Common Emitter Complementary Feedback Pair in closed loop and with Rbe can't be solved exactly using the standard Lambert W-function $ye^y = x$.

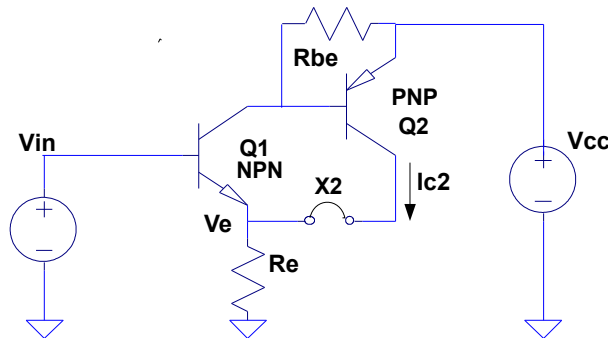


Figure A1. Complementary Feedback Pair closed loop with Rbe for Q2.

With Rbe absent Beta2 for Q2 can be assumed linear. But when Rbe is present the feedback voltage $I_{c2} \cdot R_e$ is now a nonlinear function of I_{c1} . Repeating the steps in Section 5.2 but with this *nonlinear function* of I_{c1} we eventually find that the equations for this cascaded solution being a Lambert-inside-a-Lambert cannot be balanced. It ends up that we get a residual nonlinear term of the form $R_{be} \cdot \ln(I_{c2} \cdot R_{be} / \beta_{2})$ that cannot be removed from the RHS so the equation cannot be balanced and therefore conclude a Lambert W-inverse function cannot be used to solve this circuit in exact form when overall feedback is present around two stages and each stage is only solvable using a Lambert W-inverse.

An exact solution is still available *in open loop* where each stage has its own local feedback loops and their own Lambert W-inverse solutions. But when global feedback is used there is a modulation of the otherwise constant R_{eq} term of a Lambert W solution and this appears to be due to the action of overall feedback (the modulation can be seen in the approximation used for the approximate solution in Section 5.3. This gives an insight for those who need to formulate approximations for circuits involving overall feedback with two or more transistors

The steps involved in the proof are given below. With the NFB link 2 closed we get in the log form $w = \ln(x/e)$

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$$(I_{c1} + I_{s1}) \frac{R_e}{N1 \cdot V_t} = \ln \left(\frac{\exp \left(\frac{V_{in} - I_{c2} \cdot R_e - V_{k1}}{N1 \cdot V_t} \right)}{(I_{c1} + I_{s1}) \frac{R_e}{N1 \cdot V_t}} \right) \text{ and} \quad (A1)$$

$$(I_{c2} + I_{s2}) \frac{R_{be} / (\beta_{2} + 1)}{N2 \cdot V_t} = \ln \left(\frac{\exp \left(\frac{I_{c1} \cdot R_{be} - V_{k2}}{N2 \cdot V_t} \right)}{(I_{c2} + I_{s2}) \frac{R_{be} / (\beta_{2} + 1)}{N2 \cdot V_t}} \right) \quad (A2)$$

we want to move I_{c2} out of Equation (A1) into the LHS of Equation (A2) and then balance Equation (A2) giving I_{c2} in terms of I_{c1} . To do this first use Equation (A2) to solve I_{c1} in terms of I_{c2} . Rearranging into the $w + \log(w)$ form we get

$$(I_{c2} + I_{s2}) \frac{R_{be} / (\beta_{2} + 1)}{N2 \cdot V_t} + \ln \left((I_{c2} + I_{s2}) \frac{R_{be} / (\beta_{2} + 1)}{N2 \cdot V_t} \right) = \left(\frac{I_{c1} \cdot R_{be} - V_{k2}}{N2 \cdot V_t} \right) \text{ so} \quad (A3)$$

$$I_{c1_{CL}} = \frac{N2 \cdot V_t}{R_{be}} \left((I_{c2} + I_{s2}) \frac{R_{be} / (\beta_{2} + 1)}{N2 \cdot V_t} + \ln \left((I_{c2} + I_{s2}) \frac{R_{be} / (\beta_{2} + 1)}{N2 \cdot V_t} \right) - V_{k2} \right) \quad (A4)$$

Substitute $I_{c1_{CL}}$ into Equation (A1) gives

$$(I_{c1_{CL}}(I_{c2}) + I_{s1}) \frac{R_e}{N1 \cdot V_t} = \ln \left(\frac{\exp \left(\frac{V_{in} - I_{c2} \cdot R_e - V_{k1}}{N1 \cdot V_t} \right)}{(I_{c1_{CL}}(I_{c2}) + I_{s1}) \frac{R_e}{N1 \cdot V_t}} \right) \quad (A5)$$

We have now eliminated I_{c1} in (A5) giving a single equation relating I_{c2} in terms of V_{in} .

We note I_{c2} on the RHS now needs to be moved to the LHS and merged with other I_{c2} terms. Then we work at balancing in the $w = \ln(x/e)$ form. Once balanced we can identify w and x terms to solve for I_{c2} in terms of V_{in} .

We can work on expanding (A4) in (A5) only on the LHS (since the denominator is identical)

$$LHS = \left(\left(\frac{N2 \cdot R_e}{N1 \cdot R_{be}} \left(I_{c2} \cdot \frac{R_{be} / (\beta_{2} + 1)}{N2 \cdot V_t} + \ln \left(I_{c2} \cdot \frac{R_{be} / (\beta_{2} + 1)}{N2 \cdot V_t} \right) - V_{k2} \right) \right) + I_{s1} \right) \quad (A6)$$

Where $I_{c2}' = I_{c2} + I_{s2}$ for brevity. Now bringing $-I_{c2} \cdot R_e / (N1 \cdot V_t)$ from the RHS into (A6) gives a modified LHS

(A7)

Let $k = (1 + R_{be} / (N2 \cdot V_t))$. We are trying to balance an equation in the form

$$k \cdot a \cdot I_2 + \ln(b \cdot I_2) = \ln \left(\frac{x}{a \cdot I_2 + \ln(b \cdot I_2)} \right) \quad (A8)$$

where the LHS ' a ' term has been scaled by k . We want a ' $k \cdot a$ ' term on the RHS and since from previous cases we multiply numerator and denominator by some factor we get the form

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$$k \cdot a \cdot I2 + \ln(b \cdot I2) = \ln\left(\frac{k \cdot x}{k \cdot a \cdot I2 + k \cdot \ln(b \cdot I2)}\right) \quad (\text{A9})$$

And now we need to change the LHS to $k \cdot \ln(b \cdot I2)$ by substituting $\ln(b \cdot I2) = \ln(k \cdot b \cdot I2) - \ln(k)$, so

$$k \cdot a \cdot I2 + \ln(k \cdot b \cdot I2) = \ln\left(\frac{k \cdot x}{k \cdot a \cdot I2 + k \cdot \ln(b \cdot I2)}\right) \quad (\text{A10})$$

But this cannot lead to a form that can be expressed as $w = \ln(x/w)$ using linear algebra. Therefore the Lambert W-inverse cannot provide a solution for this circuit in an exact form. The Wikipedia site on the Lambert W-function (at present) shows there is a general solution which does not include a log-inside-a-log (a composite-function or a function-of-a-function).

POWER AMPLIFIER KALKULATOR SPREADSHEET USER GUIDE

Appendix B ***[spare]***

POWER AMPLIFIER KALKULATOR SPREADSHEET USER GUIDE

Appendix C ***[spare]***

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